MACKENZIE PRESBYTERIAN UNIVERSITY

ILTON LUIZ ASSI

THE HYBRIDIZATION CRISIS:
People Analytics as an actor in the resignification of Human Resource Management in the Brazilian Pharmaceutical Industry

São Paulo

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Thesis submitted to the Graduate Program in Business Administration of Mackenzie Presbyterian University, as a partial requirement to obtain the title of Doctor in Business Administration

ADVISOR: Prof. Dr. Almir Martins Vieira

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LIST OF ACRONYMS AND ABBREVIATIONS

4th IR: Fourth Industrial Revolution

ADMS: Automated Decision-Making Systems

AI: Artificial Intelligence

AMO: Skills, Motivation and Opportunity

ANT: Actor-Network Theory

BDQM: Big Data Quality Manager

BPI: Brazilian Pharmaceutical Industry

CHRD: Critical Human Resource Development

DEI: Diversity, Equity and Inclusion

DQ: Data Quality

DQA: Data Quality Assessment

EBM: Evidence-Based Management

GDPL: General Data Protection Law

GO: General Objective

HCM: Human Capital Management

HPWS: High Performance Work Systems

HRBP: Human Resources Business Partner

HRD: Human Resource Development

HRM: Human Resource Management

HRAMM: Human Resource Analytics Maturity Model

IQA: Information Quality Assessment

LLM: Large Language Model

MCP: Model Context Protocol

ML: Machine Learning

NLP: Natural Language Processing

ONA: Organizational Network Analysis

PA: People Analytics

PESTEL: Political, Economic, Social, Technological, Environmental and Legal

PPO: Mandatory Waypoint

RBV: Resource-Based View

RTA: Reflective Thematic Analysis

ROI: Return on Investment

SDHR: Strategic Development of Human Resource

SHRM: Strategic Human Resources Management

SO: Specific Objective

SCRT: Social Construction of Reality Theory

TDQM: Total Data Quality Management

TWI: Training Within Industry

VUCA: Volatility, Uncertainty, Complexity, and Ambiguity

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ABSTRACT

Contemporary Human Resource Management (HRM) is at a critical inflection point driven by the era of hybridization, which consists of the integration of algorithmic tools into its daily operations (Böhmer & Schinnenburg, 2023). This research advances the central argument that People Analytics (PA), propelled by the current algorithmic inevitability, transcends its role as a mere tool to establish itself as a complex organizational sociomaterial actor, capable of actively (re)constructing the reality of HRM. Guided by a challenging dual theoretical framework that articulates the Social Construction of Reality Theory (SCRT) (Berger & Luckmann, 1966) and Actor-Network Theory (ANT) (Callon, 1986; Latour, 2003, 2005), this qualitative, interpretive study analyzed the perspectives of Human Resources Directors and Business Partners from the Brazilian Pharmaceutical Industry (BPI) through semi-structured, in-depth interviews (Flick, 2022), employing Reflective Thematic Analysis (Braun & Clarke, 2020).The research specifically sought to understand how PA and AI are experienced and given meaning, to analyze the changes in HRM processes, and to explore the (im)possibilities of PA/AI adoption, considering the agency of human and non-human actors in a co-construction process. The analysis of the investigated dynamics reveals that the introduction of PA into the HRM of the BPI manifests as a complex process of the Translation of Reality. Stemming from uncertainties and the need to build credibility (problematization), PA positions itself as a key actor capable of redefining the strategic dialogue between HRM and the business, thereby expanding its analytical capacity (Ferrar & Green, 2021).This complex process involves a continuous search for synergies and the inscription of PA with material tools (systems, algorithms, and dashboards) that act as mediators, reconfiguring practices and perceptions within HRM. Additionally, the study identified that the objectivity sought by PA is not inherent in the data but is socially and materially constructed and validated, requiring the institutionalization of this new analytical order. The agency of algorithms is evidenced by the in-depth analysis of their mediating capacity, revealing that PA acts in four distinct ways: in the deviation of goals, in the composition of new goals, in the reversible delegation of competencies, and in the mutual modification of attributes between humans and non-humans. These mediations reconfigure the daily processes of HRM, demanding constant validation and value attribution for their legitimization. The findings culminate in the proposition of a Hybridization Crisis as a sixth evolutionary phase in Larry Greiner's (1998) model of growth crises. This crisis, triggered by massive and profound algorithmization, fundamentally reshapes the challenges of contemporary management. It manifests in the complex governance of human-machine networks, in the dialectics of decision-making, and in the reconfiguration of professional identity. Consequent to this crisis, two phenomena are evidenced: an organizational stupor and an anthropomorphic paradox. The former is a lethargy in the face of algorithmic complexity and exponential evolution. The latter is the process of humanizing AI which, while potentially increasing acceptance, also masks its purely statistical and mathematical algorithmic nature, and may obscure potential risks of objectification and dehumanization of and in the workplace. These findings provide a lens for practical navigation in this new sociomaterial reality, highlighting the need for critical analytical literacy and a critical, ethical, and empathetic humanistic commitment, which requires the continuous maintenance of human agency in and for HRM.

Keywords: People Analytics; Social Construction of Reality; Sociomateriality; Hybridization Crisis; Algorithmization.

RESUMO

A Gestão de Recursos Humanos (GRH) contemporânea atravessa um ponto de inflexão crítico impulsionado pela era da hibridização, que consiste na inserção de ferramentas algorítmicas em seu cotidiano (Böhmer & Schinnenburg, 2023). Esta pesquisa avança o argumento central de que People Analytics (PA), impulsionado pela atual inevitabilidade algorítmica, transcende seu papel de mera ferramenta para se consolidar como um complexo ator sociomaterial organizacional, capaz de (re)construir ativamente a realidade da GRH. Pautado em uma desafiante arquitetura teórica dual, que articula a Teoria da Construção Social da Realidade (SCRT) (Berger & Luckmann, 1966); e a Teoria Ator-Rede (ANT) (Callon, 1986; Latour, 2003, 2005), em um estudo qualitativo e interpretativo analisou as perspectivas dos Diretores e Business Partners de Recursos Humanos da Indústria Farmacêutica Brasileira (IFB) em entrevistas semiestruturadas e em profundidade (Flick, 2022), utilizando-se da Análise Temática Reflexiva (Braun & Clarke, 2020). A pesquisa buscou especificamente compreender como o PA e a IA são experienciados e significados, analisar as mudanças nos processos da GRH e explorar as (im)possibilidades de adoção do PA/IA, considerando o protagonismo de atores humanos e não-humanos em um processo de co-construção. A análise da dinâmica investigada revela que a introdução do PA na GRH da IFB se manifesta como um complexo processo de Tradução da Realidade, que, partindo das incertezas e da necessidade de construção de credibilidade (problematização), o PA se posiciona como um ator protagonista capaz de redefinir o diálogo estratégico da GRH com o negócio expandindo sua capacidade analítica (Ferrar & Green, 2021).Este processo complexo envolve a busca contínua por sinergias e a inscrição do PA com ferramentas materiais (sistemas, algoritmos e dashboards) que atuam como mediadores, reconfigurando práticas e percepções na GRH. Adicionalmente, o estudo identificou que a objetividade buscada pelo PA não é inerente aos dados, mas sim social e materialmente construída e validada, exigindo a institucionalização dessa nova ordem analítica. O protagonismo algorítmico é evidenciado com a análise aprofundada de sua capacidade mediadora, revelando que o PA atua de quatro formas distintas: no desvio de metas, na composição de novas metas, na delegação reversível de competências e na modificação mútua de atributos entre humanos e não-humanos. Essas mediações reconfiguram os processos diários da GRH, exigindo constante validação e atribuição de valor para sua legitimação. Os achados culminam na proposição de uma Crise da Hibridização, como sendo uma sexta fase evolucionária no modelo de crises de Larry Greiner (1998). Esta crise, desencadeada pela algoritmização massiva e profunda, remodela fundamentalmente os desafios da gestão contemporânea, manifestando-se na complexa governança de redes homem-máquina, na dialética da tomada de decisão e na reconfiguração da identidade profissional. Consequentes dessa crise, são evidenciados por; um estado de estupor organizacional e um paradoxo antropomórfico. Sendo o primeiro, uma letargia diante da complexidade e exponencial evolução algorítmica, e o segundo, o processo de humanização da IA, que pode aumentar aceitação, também mascara sua natureza algorítmica, puramente estatística e matemática, assim como pode obscurecer potenciais riscos de objetificação e desumanização do e no trabalho. Esses achados, habilitam uma lente para a navegação prática para esta nova realidade sociomaterial, destacando a necessidade de uma alfabetização analítica crítica e um compromisso humanístico crítico, ético e empático, que requer manutenção contínua do protagonismo humano na e para a GRH.

Palavras-chave: People Analytics; Construção Social da Realidade; Sociomaterialidade; Crise da Hibridização; Algoritmização.

INTRODUCTION

Contemporary organizations navigate a landscape of unprecedented volatility and disruption. Immersed in what is conventionally called the Fourth Industrial Revolution (Schwab, 2016), they are compelled to reconfigure their operations in the midst of recurring crises and, crucially, the management of their human capital. In this context, the HRM function reaches a critical inflection point: the need to evolve to avoid the risk of obsolescence, aligning its conceptions with increasingly complex competitive demands (Bahuguna et al., 2024). This evolution, which moves away from a historically administrative and controlling role to seek a genuinely strategic partnership, finds itself in the advances of Artificial Intelligence (AI) and People Analytics (PA), its most potent and, at the same time, complex catalysts (Fitz-enz, 2009; Suri & Lakhanpal, 2024).

The promise that accompanies the emergence of the PA could be a paradigmatic transformation: consist of the transition from management often guided by intuition and anecdotal experience to a data-driven approach, capable of generating insights to quantify the evolution and/or involution of its actions and align cohesively with the business strategy (Di Lauro et al., 2024). In this movement, Artificial Intelligence (AI) is seen not only as a tool, but as a transforming agent. And data is currently a key strategic asset and, due to its immense potential, is often called "the new oil". However, the path to realizing this promise proves to be winding and multifaceted (Gaur & Riaz, 2019; Hota, 2024; Isson & Harriot, 2016).

Adopting these new analytical technologies is not a simple technical implementation; it is a profound social process confronting the *habitus* (Bourdieu, 2007)*;* of a profession that has historically been less accustomed to purely quantitative cognition and arouses latent skepticism. Thus, fundamental questions emerge about the nature of work, the future of management, and the ethical limits of this new capacity to analyze and modulate human behavior – The datafication. The risk that this intensive datafication may inaugurate a Digital Taylorism or deepen processes of objectification and dehumanization in the workplace is a shadow that hangs over the enthusiastic discourse of innovation, demanding a critical and humanized analysis (Noponen et al., 2023). In this context of profound transformations and challenges confronting contemporary Human Resource Management, this research seeks to achieve an understanding of the following central question: How has People Analytics, a new "organizational actor", influenced and re-signified the social reality of HRM in the Brazilian pharmaceutical industry?"

Given this complexity, this thesis argues that, to understand the real dimension of the change underway, it is necessary to transcend the view of PA/AI as a mere tool. Instead, we propose conceptualizing it as a new, complex sociomaterial organizational actor. An actor who not only reflects or optimizes a pre-existing reality but actively participates in its (re)construction by translating and resignifying the very fabric of HRM. This is a dialectical process, a continuous dance between human intentionality, cultural values, and the performative agency of the algorithms, platforms, and data inserted in this context (Berger & Luckmann, 1966; Latour, 2005).

This investigation is based on dual theoretical architecture to unveil the multiple layers of this intricate interaction. In a critical and productive dialogue, we put the Social Construction of Reality Theory (SCRT) by Peter L. Berger and Thomas Luckmann with the Actor-Network Theory (ANT) by Bruno Latour and Michel Callon. This construction is based on SCRT, which offers the lens to understand how human actors socially constructed, signified, and legitimized (or contested) a new analytical reality. In parallel and additionally, the ANT allows us to open the technological black box (term used by Latour and Callon) to track the agency distributed on the network and recognize the protagonism of non-human actants – the algorithms that classify, the Dashboards that direct the gaze, the data that speak – in the material performance of the new HRM practices (Berger & Luckmann, 1966, 1995; Latour, 1988, 2001, 2003, 2005).

This theoretical-analytical framework is empirically anchored in the dynamic and relevant Brazilian Pharmaceutical Industry (BPI) context. A sector marked by high competitiveness, strict regulation, and a declared focus on innovation and human capital, the BPI proves to be a privileged field to investigate the tensions, challenges, and potentialities that emerge with the arrival of these new analytical actors. Through the experiences and narratives of HR Directors and Business Partners, protagonists of this transformation, we seek to give body and voice to this phenomenon. To this end, this thesis establishes how:

General Objective (GO): To evaluate the potential of People Analytics as a new organizational actor that transforms and resignifies the daily reality of HRM in the Brazilian Pharmaceutical Industry.

This greater purpose unfolds in specific investigations that seek, first of all:

Specific Objective 1 (SO01): To understand how PA and AI are experienced and signified in this context of BPI's HRM;

Specific Objective 2 (SO02): To analyze the daily changes in HRM processes resulting from applying AI and/or PA tools.

Specific Objective 3 (SO03): Explore (im)possibilities of adopting PA/AI in HRM, considering the role of human and non-human actors.

To achieve these goals, this thesis invites readers to an analytical journey woven into three major movements. Initially, to unveil how People Analytics is experienced and signified (SO01), we will dive into the complex Translation of Reality that its arrival precipitates, mapping the arduous negotiation process that begins in uncertainties and advances through the problematization and inscription of new inter- and intra-organizational synergies. Once we understand how this new reality is announced, we will investigate the changes it engenders in everyday processes (SO02) by analyzing the laborious construction of a Required Objectivity. Here, we will examine how new analytic practices seek to become institutionalized and how algorithmic mediation redefines the very nature of decision-making in HRM. Finally, the discussion will consolidate the contributions made along the entire route by reaching the apex of this research and addressing the (im)possibilities of this new era (SO03). We will demonstrate, first, that the introduction of the PA is a complex process of Reality Translation – specific, complex, dual, and even paradoxical – in which HRM seeks to reinvent itself as a center of calculation or to develop an analytical mindset. Next, we will demystify the search for Required Objectivity, revealing it not as a raw datum but as a socio-material performance co-constructed through the detailed mechanisms of algorithmic mediation, an incessant change and resignification. As an inevitable consequence and analytical apex of these processes, the study will culminate in analyzing the emergence of a Hybrid HRM. The analysis of this ecosystem leads us to propose a rereading of the classic model of Greiner (1998) with the addition of a sixth phase: Algorithmization. Everything indicates that this new stage results in a Hybridization Crisis instituted by: Conflict of governance, decision, and identity, designing new and important challenges of management today. The empirical material reveals a possible causality of the hybridization crisis, the current HRM seems to be in a state of stupor, intoxicated by a paralyzing substance: The uncertainty of its daily reality.

This research is justified, in the first place, by its object's relevance and practical urgency. Contemporary HRM is at an inflection point, compelled by an algorithmic inevitability to evolve or risk its obsolescence. However, in practice, the promised analytical revolution of People Analytics turns out to be a complex journey full of challenges. As this

thesis demonstrates, organizations struggle with fragmented systems, a lack of clear governance, and deep cultural resistance, which difficult the link between investment in technology and the effective value generation of this new actor. In addition, this transition is permeated by ethical dilemmas, such as the risks of datafication, objectification, and a potential and real process of dehumanization of work, which demand careful and conscious navigation. Therefore, this study is justified by the pressing need to understand these concrete dynamics, offering an analysis that transcends commercial narratives to investigate the true challenges of this transformation.

To achieve the proposed objectives, this thesis is structured in four chapters. Chapter I is dedicated to constructing the theoretical foundation underpinning this research. It begins with a review of the evolutionary trajectory of Human Resources Management (HRM), from concept of Human Resource Development (HRD)(Swanson, 1995); pass to administrative origins to contemporary strategic perspectives (Armstrong & Taylor, 2023). It then delves into the emergence of People Analytics (PA) and Artificial Intelligence (AI) as transformative forces, addressing their context, value proposition, and intrinsic challenges, such as datafication and objectification (Davenport & Mittal, 2023; Thorpe & Sellar, 2023). The chapter culminates in the exposition of the dual analytical architecture that guides this work, detailing the precepts of the SCRT and the ANT, which, in dialogue, form the lens for the analysis of the phenomenon (Berger & Luckmann, 1966, 1995; Callon, 1986; Latour, 2003, 2005; Law, 1984).

Chapter II exposes the methodological path adopted. The positioning of the research is defined as qualitative, interpretative, and reflective. The procedures for selecting participants – HR directors and business partners working in the Brazilian Pharmaceutical Industry – and for data collection, carried out through in-depth semi-structured interviews, are detailed (Flick, 2022b). Finally, the Reflective Thematic Analysis (RTA) method is described, which guides the treatment and interpretation of the empirical material(Braun & Clarke, 2020b).

Chapter III constitutes the core of the thesis, where the data are analyzed in the light of the theoretical framework. The analysis is structured in three central thematic axes. The first section, The Translation of Reality, investigates how PA is experienced and signified, addressing the initial uncertainties and the process of its constitution as a protagonist actor. The second, The Required Objectivity, examines the complex institutionalization of a new analytic order and algorithmic mediation in HRM's (re)construction. The third section,

Hybridization as a Management Strategy, explores the dynamics of interaction in hybrid daily life and the challenges to the status quo, culminating in the proposition of the Hybridization Crisis.

Chapter IV, entitled Final Remarks, ties the threads of the investigation. This section consolidates the main findings of the research, discusses its theoretical and practical implications, transparently recognizes the study's limits and limitations, and based on them, suggests future research that can expand and deepen the understanding of the phenomenon analyzed here. The commitment to clarity and rigor in the presentation of this research extended to the writing process itself. During the preparation of this work, the author used DeepL in order to improve the readability and language of the manuscript . After using these tolls, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

CHAPTER I

1 Theoretical Foundation

1.1 Human Resource Development (HRD) Studies

There is a remarkable evolutionary trajectory of Human Resources Development (HRD) studies. The field goes from being a transmitter of human knowledge to becoming a partner in the strategy and development of organizations so that, in principle, research in HRD focuses on the transfer of knowledge. However, this perspective has expanded through important historical milestones, such as the work of the Training Within Industry (TWI) agency in the 1940s and the publication of influential works, such as “The Modern Practice of Adult Education” by Malcolm Knowles and his conceptions of the adult learning method, constitutes a basis of great value for the management of human resources and for their development and “Developing Human Resources” by Leonard Nadler, which postulates the term HRD, based on the premises of training, education and organizational development, aiming at its growth and perpetuation (Swanson, 2001).

One avenue for this evolution to be consistent and continuous. HRD researchers developed a threefold agenda: first, human development and empowerment; second, the necessary organizational adjustments under the ages of competitiveness; and finally, social development. This agenda reflects the field's adaptation to the constant changes in the world of work, competitiveness, and cultural values (McGuire, 2011). Thus, the history of this field

has unfolded in distinct phases: a phase of conceptual dissemination (1962-1989), followed by a phase of conceptual growth and application of the same (1990-2007), and, since 2008, a phase of development, with notable contributions coming mainly from the United States and the United Kingdom (Pham-Duc et al., 2023).

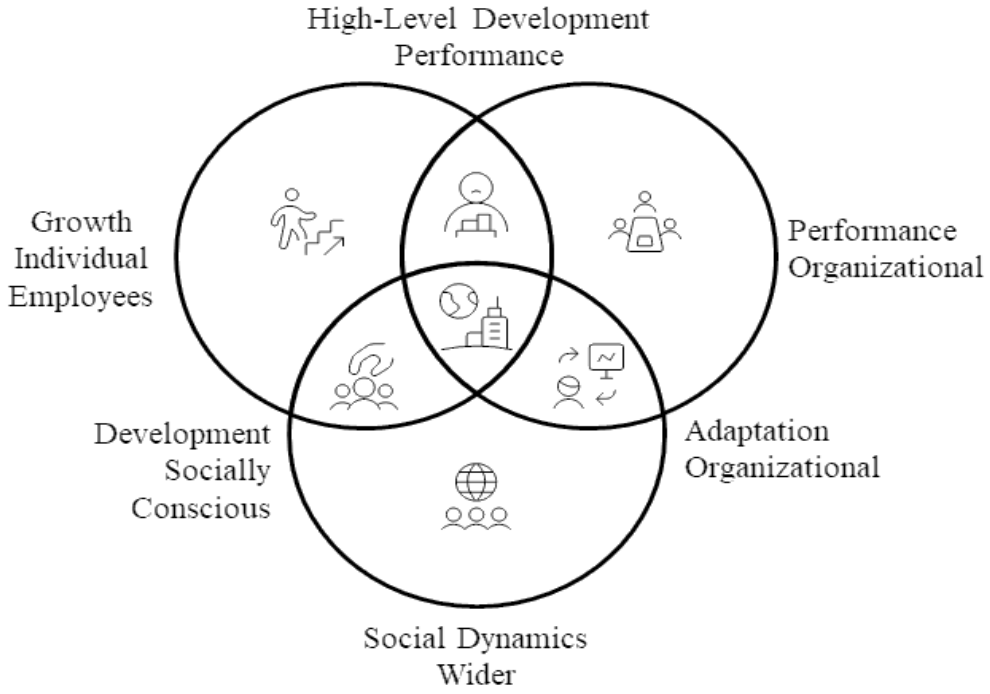
In today's scenarios, high-performing organizations consider HRD to be a key strategic capability. The complex, competitive challenges of adopting and implementing innovations, minimizing rigidity, and using flexibility without losing their characteristics are considered characteristics in which the concept of HRD stands out positively (Richman, 2015). This positive vision proposes a transition from the traditional training model to a learning-focused orientation, which aims to add value to the organization through an engaged workforce and the use of technological capital (Swanson, 2022; Swanson & Holton III, 2008). This incessant need to adapt and position itself strategically leads the field of HRD to revisit and strengthen its own conceptual foundations (Callahan et al., 2015).

Developmental transitions, which produce obstacles to consolidating their influence in professional and organizational environments, require a firmer strategic positioning and continuous adaptation to the changing scenarios of organizations (Ruona, 2016). Shifting from a strict training focus to a more comprehensive approach, which values the learning and development of organizational human and technological capital, requires a solid understanding of the theories that underpin these practices. It is precisely in this effort for legitimacy, relevance, and impact that it becomes vital to understand how HRD is defined, what theoretical bases guide its practice, and its potential to transform organizations(Collins & Callahan, 2022; Kiettikunwong & Narot, 2024; Pham-Duc et al., 2023).

HRD, both as an area of study and as a professional practice, sits at a crucial meeting point: the intersection between individual employee growth, the performance of organizations, and broader societal dynamics. Historically, the field has been dedicated to a multifaceted effort to discover and cultivate human expertise. The goal goes beyond simply improving job performance; conceptually, it is necessary to boost personal development and strengthen the ability of organizations to adapt and consequently compete consistently (Swanson, 2001; Swanson & Holton III, 2008). This pursuit makes HRD a multifaceted discipline. Swanson (2001), for example, designed a theoretical architecture for HRD that unites contributions from cognitive psychology, economic theory, and systems theory. This integrated approach provides HR professionals with a framework for aligning employee goals and behaviors,

designing efficient processes and structures, and optimizing tasks, recognizing HRD as a pillar for advancing people management itself, as shown in Figure 1.

Figure 1: HRD Synergy Core



Source: Research (2025).

This integration aims primarily at the economic development of the organization and its stakeholders, so that HRD uses economic theory, especially under the influence of (Schumpeter, 1934), who highlights the relevance of measurement, as development occurs in specific cycles of expansion and retraction, adjusting to the contextual needs of the organization. Towards such is the creation of metrics systems and the definition of objectives that require constant review and reconstruction, vital elements to ensure that the organization survives and thrives in a competitive market.

In parallel, the theory of systems of Ludwig von Bertalanffy (1972) offers a lens to see the organization as a whole, formed and connected by interconnected parts, enhancing the analysis of integration proposed by HRD studies, which are based on the pillars of human empowerment, structural adjustments of the organization, and social reconfiguration. In this view, the purpose of each component and how they interact can drive performance and create obstacles, underscoring the importance of dynamics between all subsystems.

Complementing this base, the cognitive psychology theory turns its gaze to human capital, which recognizes that people act as essential mediators of productivity and shape the organization's culture, behavior, and environment. Swanson (2001) argues that these three

pillars – economic, systemic, and psychological – are fundamental – because, in concept and practice, HRD is concretized when it enables and empower human knowledge and experience through Organizational Development (OD) and Training and Development (T&D), always focusing on the best and most outstanding performance individual to the collective. In this way, an organizational environment that promotes awareness encourages proactive attitudes – entrepreneurship postulated by Schumpeter – and values social and ethical responsibility emerges as a necessary condition and, simultaneously, as a result of a successful HRD.

Greater and better integration requires a broader and more strategic view of organizational development, as proposed by Garavan (2007) and Garavan and McGuire (2010), which consists of the concept of Strategic Development of Human Resources (SDHR). The concept of SDHR expands the HRD because it proposes a multi-level analysis, which considers the company's mission, the active participation of direct managers, how HR activities complement each other, and the cultivation of harmony with the solidification of culture and values of the institution. From this perspective, SDHR is not limited to optimizing performance; It also acts as an amalgam for the organization to adopt practices of social responsibility, sustainability, and ethics, seeking a balance between human values and business objectives.

Furthermore, Garavan (2007) argues that the SDHR offers a fundamental path to achieve a lasting competitive advantage. This is because the concept enables knowledge and skills aligned with the strategy, integrates internal and external knowledge, and promotes effective change management. Therefore, when applying the concept of SDHR, it requires its manager's attention both to the strategic prioritizations that have been planned and to those that arise along the way, in addition to a focus on stakeholders and the complex interactions between organizational priorities and interests as well as human resource management.

How HRD is conceived, its strategic role is not yet a consensus and faces tensions and criticism. The initial proposal of Garavan (2007) inadvertently attributed to the SDHR a role more limited to implementation than the formulation or participation of and in the organizational strategy, reflecting a functionalist view of HR management. This posture has been increasingly questioned daily (McCartney & Fu, 2021). The Critical Human Resource Development (CHRD) approach gains special relevance in this scenario. The CHRD, explored by authors such as Collins and Callahan (2022) , proposes a more in-depth study of the relations of power, ethics, and social justice issues that permeate HRD practices or its proposal for the evolution of HRD.

Critically, HRD should not be seen only as an instrument to improve organizational performance. Instead, it is necessary to question who benefits from organizational development, what interests are being served or required, and how HRD can help build a fairer, more emancipatory, and integrated work environment. In other words, the concept of CHRD maintains that HRD practices have a dual and ambiguous potential: they can both reinforce structures of control and inequality and promote empowerment and give voice to workers (Kiettikunwong & Narot, 2024; Richman, 2015). Thus, a critical analysis of the HRD suggests that its role should not be restricted to the execution of strategies defined by third parties but should nevertheless actively participate in the creation of strategies that are not only economically viable but also socially just and ethically sound (Collins & Callahan, 2022).

In addition, the current academic view on HRD highlights the importance of managers considering the context's influence. By analyzing the current landscape of HRD theory, points to the key role that culture and the host institutional system play in shaping and effectiveness of HRD initiatives (Jabarkhail, 2023). This contextual view questions the validity of applying universal HRD models without criteria. It underlines the need to adjust theoretical and practical conceptions to local realities and the specific cultural dynamics of each environment. The critique of the undervaluation of the role of culture in the local institutional system, as embroidered by Jabarkhail (2023) in dialogue with Wang and Doty (2022), indicates that a truly dynamic HRD theory needs to be, at its core, context-sensitive. This sensitivity, in turn, echoes the demands of the CHRD, which seeks to understand the particularities of the struggles for justice and recognition in different scenarios, considering in which context these battles are being fought.

Therefore, that the journey through HRD studies shows a vibrant and consistently evolving field. We see a movement that seeks to break its merely technical and functional foundations, embracing a progressively more strategic, self-critical, and sensitive understanding of the environments in which it operates. The essence of HRD lies in cultivating and expanding human and organizational potential aiming at developing both. However, contemporaneity calls on theory and practice to focus on an objective that contemplates broad, higher targets and a vision for the future a high level. Thus, theory and practice are called to dialogue with the complex ethical and social implications, to recognize power dynamics, and to demonstrate a genuine capacity to adapt to the various cultural, institutional, and technological realities. It is this richness, this inherent tension between the strategic and the tactical, the technological and the human, between the universal and the local,

between the digital and the analog, that defines today's HRD and that serves as an essential prelude to the discussion that follows: the evolutionary trajectory of Human Resource Management (HRM) in its search for a role of greater relevance and meaning in organizations (Kiettikunwong & Narot, 2024).

1.2 The Evolution of HRM: From Management to Analytical Strategy

The understanding of HRM in organizations is presented in an extensive evolutionary effort both conceptually and in its practices and was driven by the consistent content of HRD studies. In the early days of HRM, its strategic role was practically disregarded, a posture that was transformed under new conceptual conceptions and significant operational practices. The industrial revolutions played a remarkable role in this evolution, which constituted changes in the organization of work, the evolution of organizational theories, the reconfigurations of power structures, and socio-technological contexts (Caldas et al., 2002). Initially relegated to a predominantly administrative and controlling role, influenced by the paradigms of Taylorism and Fordism, HRM gradually expanded its scope. The advent of the School of Human Relations, for example, has produced a force for change and greater sensitivity to the human aspects of work, followed by an increasing focus on training and development as levers for productivity. This progression has culminated, in recent decades, to consolidate HRM as a strategic partner capable of managing talent and aligning human capital with the organization's primary objectives (Caldas et al., 2002). Table 1 summarizes the main phases and vectors that marked this historical transformation of HRM.

Table 1: The Evolution of HRM: Towards the Analytical-Strategic Perspective

	1920	**1950**	**1970/ 1990**	**2000**	**2010 →**
Theories	Administration Classical	Human Relations Model	Sociotechnical Model, Quality	Competitive Model	Analytical Competition Model
Vectors	Efficiency	Satisfy and Motivate	Engage Compromise Motivate	Competence	Algorithmization, statistics
Focus	Individual	Function	Strategic Collaboration	Partner	Holistic Perspective, Amplitude

Source: Research (2025).

The path of HRM development was not straight or uniform. On the contrary, it unfolded differently, mirroring the singularities of economic, social, technological, and contextual scenarios. Despite this, a common thread stands out: the growing sophistication of

the HR function – due to the increase in organizational complexity – and its continuous search for legitimacy, meaning, and, consequently, for a contribution that makes a difference to the strategic conception of the organization. Harvey (2008) argues that the very essence of capitalism lies in its ability to systematically revolutionize forms of production, organizational structures, and, inevitably, labor management. With this dynamic scenario, HRM faces the constant challenge of evolving and contributing. HRM needs to replace a reactive, "fire-fighting" stance with a proactive stance, capable of responding to changes, anticipating them, and even helping to foster and shape them (Kiettikunwong & Narot, 2024). This proposition is undoubtedly not only impactful but, at the same time, profound. The shift from a production model focused on scale and massification – a widely disseminated and implemented conception – to markets that value personalization, variety, and agility – a process illustrated in Table 2 – required a somewhat radical change in mentality. The focus has shifted from simple cost management to management that seeks to generate value, recognizing that the development of people and the intelligent optimization of processes are keys to success (Autier & Picq, 2005; Bontis & Fitz-enz, 2002; R. Wei et al., 2019).

Table 2 shows the profound differences between markets prioritizing scale and those valuing scope. Through this synthesis, we could visualize and compare, more clearly, the forces that drove HRM in its evolution. In a world dominated by scale, emphasizing mass production, standardization, and rigid hierarchies, people management assumed a predominantly administrative and controlling role. The focus was on the efficiency of operations, discipline, and the equal application of rules – a faithful portrait of HRM's earliest and most functional phases (Storey et al., 2019; Ulrich, 1997).

In sharp contrast, the emergence of a scope-driven market that seeks flexibility, innovation, personalization, and adds value through knowledge and technological evolution brings with it completely different conceptual and practical requirements (Rasmussen et al., 2023; Rasmussen & Ulrich, 2015). The way of working has changed: it has become more autonomous and collaborative and needs greater knowledge transfer, through more robust and integrated communication systems. The structures of companies tend to be built more fluidly – even without eliminating the existing hierarchical structures – demanding new skills from people, both individual and collective, that have come to be seen as essential to meet the need for voracious organizational competitiveness (Farndale & Paauwe, 2018; Zhou & Hong, 2008).

It this transformational movement, which Table 2 seeks to present by comparing aspects such as production, the execution of work, and laws, explains not only the re-design of markets but also fundamentally transformed people's relationship with their work and management, as well as expectations about the role of the individual and the organization in society. As a result, HRM was compelled to evolve to respond to the most complex and emerging strategic demands. And it is precisely this complexity that paved the way for the search for more sophisticated and data-based approaches, essential to understanding and managing the new reality of work and its social relations (Simón & Ferreiro, 2018).

Table 2: Organizational Scale and Scope

	Market based on scale	**Market based on scope**
Production Process and Value Generation	Mass production of homogeneous goods	Production in smaller batches
	Uniformity and Standardization	Flexibility and variety in product types
	Large inventories and inventories	Little or no inventory – on-demand production
	Quality (ex-post) error detection (post hoc)	Process-integrated quality control
	Concealment of defective products	Immediate rejection of defective products
	Increased production time, bottlenecks in processes	Shorter production time, better use of working hours
	Resource-Driven	Demand-driven
	Vertical process integration with few horizontal cases	Subcontracting
	Cost reduction with wage control	Hands-on learning integrated with strategic and long-term planning
Job Execution	Performing a task per worker	Multiple tasks
	Remuneration defined by job definition criteria	Individual payment added to a significant bonus system
	Task Expertise	There are no limits to the accomplishment of tasks
	Little or very specific training	Extensive training program based on the worker's context
	Vertical organization of work	More horizontal organization of work
	Little or no learning in the performance of the work	Greater learning on the job
	Little responsibility of the worker in relation to the work	Worker is co-responsible
	Little or no safety at work	Greater concern for the physical and psychological safety of the worker
Legal Approach	Specific regulations	Deregulation and legal rearrangements
	Rigidity	Flexibility
	Collective bargaining	Individualization - local or organizational negotiations

	Socialization of social welfare	Social security – privatization of collective needs
	International stability - multilateral agreements	International destabilization – rising geopolitical tensions
	Centralization	Decentralization – exponential increase in inter-regional and global competition
	The State/City = Subsidizer	The Entrepreneurial State
	Indirect intervention in the markets	Direct state intervention through acquisitions
	Regional and national policies	Territorial policies to strengthen the third sector
	Investment in R&D by organizations	Investment in R&D carried out by the State
Ideologies	Mass consumption	Personalized consumption
	Modernism	Postmodernism
	Structural Reform	Specialty/ Adaptability/ Flexibility/ Complexity
	Socialization	Individualization; Society of the Self

Source: Research (2025).

In an increasingly volatile and uncertain business landscape and new relational forms, HRM and organizations have realized that the ability to respond quickly and effectively has become vital to their survival and building a healthy environment amid many changing conditions in short periods. This began to require HRM, which was, at the same time, strategic in its vision and flexible in its performance, focused on its human capital (McMackin & Heffernan, 2021). This urgent need served as a catalyst for the emergence of new approaches, both theoretical and practical, to HRM. In this movement, technology has emerged as a fundamental driver of transformation and support for change, redefining essential concepts such as leadership, recruitment and selection, engagement, development, and measuring performance(Pillai et al., 2021). Topics such as stakeholder management (Wright & Steinbach, 2022); the impact of technology on HR 4.0 (Pillai et al., 2021; Sivathanu & Pillai, 2018); Sustainability (Kramar, 2022); and the use of advanced analytical methods Human Resource Analytics (Larsson & Edwards, 2021), became part of the contemporary HRM agenda. The digitalization of business models and the growing influence of artificial intelligence (AI) on people management practices (Minbaeva, 2021) signals a new era where the ability to analyze data and generate actionable insights becomes central.

1.2.1 HRM Strategic Perspectives

The evolutionary trajectory of HRM, which significantly changes its administrative function to its strategic function, has a turning point called Strategic Management of Human Resources (SHRM) (Johnson et al., 2016). SHRM is the materialization of the effort to weave

a strong and cohesive link between human capital and the great strategic ambitions of companies and thus make HR effectively an entity of organizational integration. The concept of SHRM is established as an approach that sees HR administrative activities not only as isolated tasks – and that will not be disregarded for their relevance – but as a planned set of actions that culminate in a process integrated with organizational management.

These actions unfold into different and integrated purposes, which seek to achieve broader corporate objectives, especially in a world marked by technological growth that evolves exponentially (Espindola & Wright, 2021; Wright & Steinbach, 2022). From this perspective, we understand SHRM as a field of study and practice that seeks to equip and innovate HRM, expanding its connections horizontally and vertically.

The goal is to enable HRM to align its policies and practices with the larger strategy and implement them to create sustainable competitive advantages with people as mediators in this process. In addition, SHRM seeks to promote a constant evaluation and review of management processes, ensuring that they are effective and long-lasting, always based on evidence and concrete data (Madanchian & Taherdoost, 2024; Vieira et al., 2023). In order to provide a deeper view without being exhaustive and to capture the depth and dynamism of this field, it is relevant to analyze the different theoretical perspectives that have emerged to explain how HRM can become strategic. These different approaches, which we will detail in the next subsection, reveal how thinking about the connection between people, strategy, and performance has become increasingly complex, according to each temporal context.

1.2.1.1 The Universalist perspective

Contemporary narratives about HRM often emphasize the value of people as a fundamental asset, this idea echoes older thoughts that already challenged purely mechanical views of work. Long before it became a modern jargon, pioneering voices such as E. Wight Bakke (1958), argued that people management should be seen as an inseparable and vital part of general management. Bakke argued that it was necessary to see people as a resource as important as financial capital or the materials used in production. His vision, ahead of his time, is clear in his own words:

> “[...] The view expressed here is that “people” as an organizational resource is equally important to others and that ignorance, neglect, waste, or mishandling of that resource has the same consequences as

> ignorance, neglect, waste, or mishandling of money, materials, or market. (Bakke, 1958, p.7)."

This innovative – and unusual for the time – way of seeing people with strategic importance contrasted sharply with the view that prevailed at the time, which saw HRM only as an operational cog or simply the "personnel department". As the field matured, HRM struggled to break free from this administrative, functionalist label, dreaming of a more active role aligned with the organization's larger goals. In this transformation process, a central challenge was imposed: finding a way to clearly and consistently show how people management practices impacted the company's results was necessary.

The search for a "perfect approach" – a set of HR practices that worked for everyone, solving management dilemmas and driving success – gave rise to what we now know as the universalist perspective (Armstrong & Taylor, 2023; Bassett-Jones, 2023; Huselid, 1995). This approach, also called "best practice theory", conceived the idea that certain HRM policies and actions would be superior in themselves and lead to better performance, regardless of the type or context of the company (Osterman, 1993; Pfeffer, 1994). The rationale behind this proposition was simple: By investing in people through these ideal practices, companies would have a more skilled, motivated, and engaged workforce, which would naturally translate into lasting competitive advantages.

Jeffrey Pfeffer (1994), an influential defender of this perspective, argued that the true source of competitive advantage for companies resided, increasingly, in how they – companies – took care of their employees, as previously postulated by (Bakke, 1958). Pfeffer (1998) identified a set of practices that, if implemented cohesively, could lead organizations to superior performance. Among these practices, frequently mentioned in studies, we find Job security, judicious hiring of new talent, investing in self-managed teams with more autonomy, compensation linked to performance (company or individual), significant investment in training and development, reducing status barriers between hierarchical levels, and transparently sharing financial and performance information with everyone (Pfeffer, 1998).

All these dimensions – which can currently be named in other ways – are present in the challenges of contemporary management, which is often summarized in the AMO framework – suggests that the adoption of these practices boosts the development of employees' competencies (Skills), raises their motivation to deliver the best (Motivation) and creates opportunities for them to use their skills in favor of the company's objectives (Opportunity) (Appelbaum et al., 2001; Appelbaum, 1997; Boselie et al., 2021).

Important works, such as that of Huselid (1995) on High-Performance Work Systems (HPWS), sought to validate this view. They have shown correlations between the integrated adoption of these best practices and success indicators such as productivity, profitability, and market value. The study of Huselid (1995), in particular, was very influential in showing that the impact of these practices was not just a statistical curiosity but something economically relevant to the organization. The great attraction of the universalist perspective was precisely in its apparent simplicity and the promise to offer a clear guide so that HRM could finally show concrete results.

Despite the undeniable contribution of studies such as Huselid (1995) – which were in illuminating the potential of HRM to influence performance and to inspire numerous studies – the universalist perspective soon began to find its critics. Questions did not take long to arise: Are these best practices truly universal? Would it be possible to implement them with the same success in such different companies and cultures? And the relationship between HR practices and performance really that direct? The possibility of an inverse causality was even raised: perhaps the companies with good performance could invest in such best practices and not the other way around, as highlighted in the statements of Guest (1997) and Wright et al. (1999). The idea of a one-size-fits-all solution began to be challenged as evidence grew that an action for company A results in X, but that same action for B can result in Y. This growing awareness of the importance of the particularities and specificities of each organization has opened the door to the development of more refined and environmentally friendly theories, such as the contingency perspective, which we will explore below (Akhter et al., 2022; Arciniega et al., 2021;Rasmussen et al., 2023).

1.2.1.2 The Contingency Perspective

Going beyond the concept of a single “cake recipe” or “silver bullet” that would solve “mine and yours” problems, the contingency perspective emerges in the SHRM scenario. These perspective brought up a central idea: the effectiveness of HRM policies and practices is not something absolute, far from it; it fundamentally depends on how well these practices fit or align with factors specific to the context in which the company operates (Delery & Doty, 1996; Youndt et al., 1996). Therefore, the heart of this approach is the notion of “fit”. The idea is that an organization's performance reaches its peak when the HRM system is in harmony with both the company's business strategy (vertical or external fit) and other HR practices, creating a cohesive set internally (horizontal or internal fit) (Baird & Meshoulam, 1988; Wright & McMahan, 1992).

This way of thinking is grounded in the broader principles of organizational contingency theory, which argues that there is no single ideal way to organize or manage but that the best structures and systems depend on circumstances, both the external environment and internal conditions (Lawrence & Lorsch, 1967; Thompson, 1976). Thus, the contingency perspective at SHRM invites us to focus on identifying which are the most important variables in each situation and, from there, design HR practices that respond in the best possible way to these realities.

The contextual factors that define the choice and success of HRM practices. These are varied and complex, so organizational strategy emerges as one of the most important factors in managerial care. Different ways to compete in the market call for different HRM configurations. Think, for example, of organizations that seek to lead at the lowest cost: they are likely to prioritize HR practices that reinforce efficiency, standardization, and the strict control of personnel costs. On the other hand, companies that compete for innovation or for offering something unique – a focus on differentiation – will tend to adopt practices that encourage creativity, autonomy, the development of special skills, a compensation system that rewards market conquest, and even a willingness to take calculated risks (Bassett-Jones, 2023; Huselid et al., 1997; C. Jiang, 2010). It is clear, therefore, that ensuring fine alignment between business strategy and HR policies is a key ingredient for the success of people and the organization.

No organization sustains itself or survives only with good strategies. Strategic action will depend on the internal characteristics of the organization that exert a marked influence on organizational behavior, as highlighted by Chandler (1962) in his classic studies on the variations between strategy and structure, which defines many of the contours of HRM. More rigid structures – verticalized and/or mechanistic – tend to have more formal and centralized HR practices, while more flexible structures – horizontal and/or organic – do better with more open and decentralized strategic approaches (Liu & He, 2011).

The size of the company, its life cycle, the technology it uses, and the organizational culture are dimensions that impute interests in different proportions, and thus, all act as important moderators for the strategic formulation. For example, a strong and well-established culture can be both an ally and a barrier when implementing certain HR practices, influencing – and sometimes even dictating – the choices and the way people receive new management and/or innovation ideas (Hofstede, 2001; Schein, 2017). In addition, the talent base that the

company has added to its ability to create new strategies from this base is also an important piece in this contingency puzzle (Armstrong & Taylor, 2023; Mintzberg, 1979).

Last, we need to look outside the organization's walls. With all its nuances, the external environment adds extra layers of complexity to people management – externalities (perspective will be addressed). Factors such as the level of uncertainty and the speed of changes in the market, the strength of competition in the sector, the impact of globalization, pressures coming from institutions and laws, and even societal expectations all influence HRM decisions (Boxall & Purcell, 2000; Gannon et al., 2015). In scenarios marked by volatility, uncertainty, complexity, and ambiguity – the famous VUCA world – HRM's ability to foster agility, resilience, and constant learning becomes even more central for survival and success (Schoemaker et al., 2018; Sharma, 2020). The expansion of a company to other countries, for example, poses direct challenges to HRM: It needs to learn to deal with cultural and socioeconomic diversity, adapt practices to different laws, institutions, and regulations, and find the delicate balance between maintaining a global identity and responding to local needs (Bassett-Jones, 2023).

Although the Resource-Based View (RBV) and the theory of dynamic capabilities are not purely contingent theories, they establish a fertile dialogue with this perspective. They suggest that finding the right "fit" of HR practices for each situation may be the key to developing human capital that becomes a lasting source of competitive advantage – that is, capital that is valuable, rare, difficult to imitate, and irreplaceable (Barney, 1991; Teece, 2009; Teece et al., 1997). By emphasizing adequacy and alignment, the contingency perspective undoubtedly represents a significant step forward concerning the simplicity of the universalist approach. However, it also brings challenges. The enormous complexity of identifying and measuring all the important variables and how they interact exponentially, coupled with the practical difficulty of defining and achieving the perfect fit, are real limitations. These difficulties stimulated the search for innovative approaches, such as the configurational perspective, which we will see next.

1.2.1.3 The Configurational Perspective

Going much further and dealing with the complexity that the contingency perspective sometimes fails to provide solutions, and beyond the adjustment of isolated practices, the configurational perspective invites us to look at Strategic Management in a broader way or as a true system (Xi et al., 2022). The added value here is in the affirmation and conviction that the impact of HRM on a company's results does not come from one or another isolated practice

but from the way multiple practices combine, forming “packages” or configurations. These Packages need to be internally coherent, work in synergy – one practice reinforcing the other – and, of course, be aligned with the organization's strategy and reality (Delery & Doty, 1996; Macduffie, 1995). The premise that guides this view is the famous phrase: “The whole is greater than the sum of the parts”. This means that the value created by a well-designed HRM system goes far beyond the simple added effect of each practice. The interconnection, the mutual reinforcement between them, generates the true differential (Gill & Meyer, 2011).

Complementarity is the primary characteristic of the configurational approach. It considers actions in isolation but uses them in specific compositions according to need in a new system that moves, expanding and retracting, in a coordinated harmony. This fine-tuning is even more vital in complex and volatile environments, such as understanding and signifying work, which, like a chameleon, changes and is changed by new technological, political, economic, and social conceptions in an increasing demand for decisions with agility and assertiveness (Kaarsemaker & Poutsma, 2006). In essence, this approach seeks to discover the settings or design, HRM policy ideals, and practices that fit better with certain types of companies or strategies. At the same time, she recognizes that there may be different combinations of practices that lead to success in similar situations, or more specifically, managing the equifinality of each action, which navigates between multiple paths to reach a desired result (Stavrou & Brewster, 2005; Xi et al., 2022).

Within this search for ideal configurations, or equifinality, we can identify two main trails in SHRM research. The first is based on typologies, or ideal types. The idea here is that a limited number of HRM system arrangements prove most effective when aligned with certain strategies or contexts. Miles et al. (1978) present us with a classic example of this attempt to connect strategies and propose the classification of companies into three ideal types: Defender, Explorer, and Analyzing companies – to specific and internally coherent HRM systems (Delery & Doty, 1996). Imagine an Explorer company, always seeking to innovate: its HRM system will probably value actions and, consequently, professionals who present flexibility, the development of diverse skills, autonomy, and rewards focused on the long term. A problem with this perspective is the almost impossibility of institutionalizing policies and practices, and consequently, this behavior can undermine the perception of meaning and its signifiers. On the other hand, a Defender company, focused on efficiency and stability, will tend to a system that prioritizes control, specialization, and predictability – an approach that, in a way, reinforces the more traditional view of command and control, where

innovation tends to be mitigated by resistance to change, as well as by hierarchization and rigidity in and of processes.

The second track within the configurational perspective is the configurational adjustment approach (configurational fit). This lens values the importance of HRM practices being consistent with each other (horizontal fit) and being aligned with strategy and context (vertical fit). The big difference is that it is not tied to pre-defined types. Instead, it dives into the specific reality of each organization, seeking to understand how the synergy and complementarity of a particular set of HR practices, tailored to the situation, help drive performance (Ketchen et al., 1996). Studies such as that of Arthur (1994) illustrate this idea well, which observed steel mills and differentiated HRM systems focused on control from those focused on commitment, showing that these different configurations were linked to different performance results. Likewise, the work of Macduffie (1995), which we have already mentioned, showed how packages of flexible working practices – such as teams, job rotation, and intensive training – generated a positive and synergistic impact on the productivity and quality of the automotive industry.

Looking at the organization as a whole, thinking systemically, and visualizing connections between the various practices and policies reinforces the understanding of the complex organizational fabric more realistically; this is the proposal of the configurational perspective (Garcia-Carbonell et al., 2015). Supported by this lens, we have moved from looking at just simple cause-and-effect relationships to a richer understanding of how HRM systems, in their entirety, can create a lasting competitive advantage with people (Farndale & Paauwe, 2018).

However, taking these ideas to the real world and testing them empirically is not a simple task. There are considerable methodological challenges, such as accurately identifying and measuring these configurations, analyzing interactions that are complex and not always linear, and the need to rely on robust research samples. But, despite these difficulties, there is no doubt that the configurational perspective has deeply enriched the thinking about HRM and has prepared the ground for new approaches to emerge, seeking to further investigate the dynamics of each context and the social processes that are at the heart of people management (Doty et al., 1993).

1.2.3.4 The Contextual Perspective

Evolving the previous discussions, the contextual perspective in the SHRM emerges as a more accurate look, especially when compared to the contingency view. If the contingency perspective has taught us that HRM needs to adapt to the scenario, the contextual perspective invites us to go further. It shows us that context is not just a passive background, a set of external variables to which we react; the other side of this coin is in the context, which presents itself as a dynamic battlefield with many faces, which both impact organizations and is impacted by them (Paauwe, 2009). In this view, SHRM gains agency; it develops the ability to read the environment, negotiate with it, and actively help shape parts of the scenario where it operates. A good example of this is organizations that function as platforms: they not only respond to the market but actively manage supply and demand, redefining not only the market in which they operate but labor relations and the rules of competition themselves (Parker et al., 2017; Poole & Van De Ven, 2021).

From this angle, the context reveals itself as an incredible tangle of influences at different levels, all interconnected – a fabric, a web – susceptible to any small change. At the broadest level, the macro, we find labor laws, the educational systems that train professionals, the cultural norms and values that permeate society, and the regulations and regulations that govern organizational actions. These elements exert pressure – sometimes forcing, sometimes suggesting, sometimes leading to imitation – on how HRM can and should act (DiMaggio, 1995; Dimaggio & Powell, 2021).

Going down a step, at the meso or sectoral level, we look at the characteristics of the organizations, the intensity of the fight with competitors, and the partnership networks that define the game board. Finally, at the micro level, within the organization itself, we have a particular universe: the history of the company, its internal culture, its structure, size, the technologies it uses, the power relations, and the capacity for action of the different actors – managers, employees, unions. This creates a unique environment where HRM strategies are thought out, implemented, and, most importantly, lived in the organizational daily life (Pettigrew, 1987). Recent research reinforces how broader contextual factors – political, economic, social, technological, environmental, and legal (PESTEL) – directly affect how strategies are created and implemented, which makes it critical for HRM always to be attentive and ready to dynamically align with this ever-moving landscape (CARICAD Caribbean Centre for Development Administration, 2024).

Under the contextual lens, SHRM's actions, both inside and outside the company, can have real power to influence and even dictate the rules of the game in the environment in which the organization is inserted. In contrast, the organization itself is transformed. Transformations can happen in many ways: helping to create or expand norms that will manage the market, changing their perception of the role of HR, changing the dynamics of working relationships (whether between colleagues or between companies), and actively shaping organizational culture (Edwards & Edwards, 2016). HRM, in this light, reveals a significant transformative potential; however, it must be recognized that, in practice, its positioning still varies greatly: in some places, it acts as a true strategic partner, actively participating in decisions; in still others, HRM focuses on more administrative and operational functions, a practical example of this contextual power (Donnelly, 2019). In the BPI research area, context analysis is fundamental. The innovations launched by the industry (new concept products, therapies or pharmaceutical technologies) have a reverberating impact on the organization. Consequently, research companies such as the BPI, whose nature is shaped by these events, are required to generate new competencies, forms of communication and practices that support the innovations implemented and create value through the people (Grodal et al., 2024).

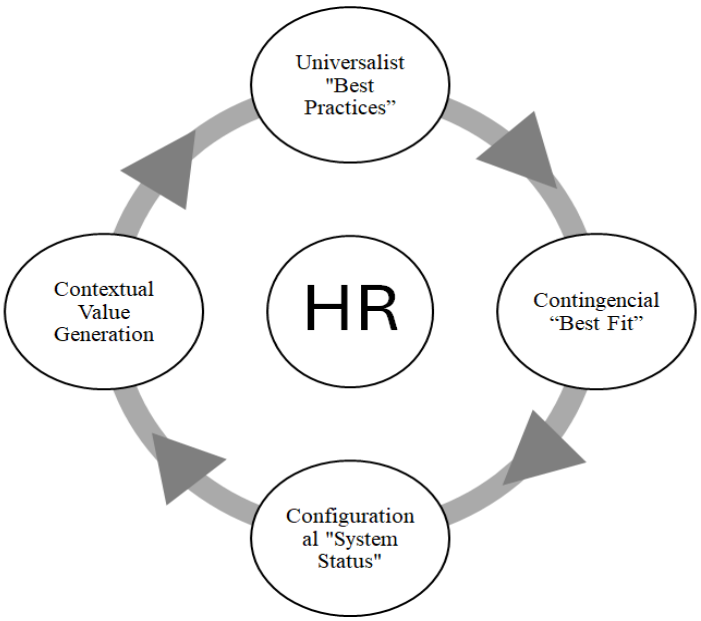
The trends and context demand we are experiencing today, such as the increasing integration of technology into everything we do in HR, the phenomenon of remote work, and the search for more well-being for employees, are perfect examples of this two-way street: they shape the context of SHRM and, at the same time, are shaped by it. The digital age and the so-called 4th Industrial Revolution, postulated by Schwab (2016), represent a change in the scenario so profound that it shakes the organizational foundations.

Artificial intelligence, big data analysis, and digital platforms are not just new tools. These forces alter the nature of work, the skills we need to have, and the relationships between companies and people. SHRM needs to be contextually intelligent and strategically agile to navigate these turbulent and often murky waters (Samarasinghe & Medis, 2020). In this new world, it is not enough to adapt; SHRM needs to take the lead, have an attitude acting proactively to build work environments that are, at the same time, productive, ethical and that promote the development of people in the midst of so many technological transformations (Majam & Jarbandhan, 2022; Nikam et al., 2024).

Figure 2 illustrates this evolutionary cycle of HRM, culminating in a phase that recognizes the centrality of contextual analysis, especially when considering the impact of

new technologies and the 4th Industrial Revolution (Pillai et al., 2021). This perspective challenges the search for universal solutions or static configurations, emphasizing the need for a deep, dynamic, and socially informed understanding of people management, where the ability to adapt and shape the context becomes a key organizational competence(Ferrar & Green, 2021; T. H. Rasmussen et al., 2023).

Figure 2: HR Strategy Evolutionary Cycle



Source: Research (2025)

We witness a field in constant maturation by revisiting, even if in an introductory way, the evolutionary journey of HRM through the theoretical lens of people management strategy. We started from the universalist search for best practices, we moved on to the emphasis on the best fit of the contingency perspective, we explored the systemic view of the packages in the configurational approach, and we arrived at the dynamic and interactive understanding of the contextual perspective. These perspectives offered valuable contributions, building on previous ideas and conceptions and refining our understanding of the intricate composition between people management, organizational strategy, and results. Together, these perspectives paint a picture of the growing sophistication – and complexity – of SHRM. They show how people management has been, step by step, moving from merely an operational function to a field that recognizes the vital importance of strategic alignment, internal coherence, adaptability to the context, and the ability to act and transform. This rich theoretical foundation provides us with the necessary foundation to dive into today's challenges and the new frontiers

of HRM – especially those brought about by promising advanced analytical approaches, which we will explore from now on.

1.3 SHRM and the Influence of Data

When the first computers arrived in Human Resources Departments in the 1970s and 1980s, the tools were, understandably, quite basic. We can imagine them as large systems for recording and processing HR's daily tasks. The main objective at that time was to automate administrative routines that took a lot of time, such as calculating payroll, controlling hours worked, and maintaining employee records. These pioneering systems had a minimal capacity for analysis – they did not offer profound insights – but, for their time, they represented an important evolutionary step. They brought more efficiency to operations and allowed the first time, to centralize basic information about employees in one place (Kavanagh & Johnson, 2018). Table 3, helps to illustrate this evolutionary process of HRM's technological journey.

Table 3: Evolution of HR Tools and Systems

Period	70s/80s	1990	2000	2010	2020
Characteristic	Storage	Integrated Systems	Focus on People	Predictive Thinking	Integration of different sources, real-time analytics, People Analytics Platforms
Tool	Tables (Excel®)	ERP / HRIS	Use of Hardware and Software dedicated to HR	Data-base Integration, Algorithmic Interactions	Algorithmization
Focus	Control	Training & Development Talent Stream	Increase Engagement Decrease Turnover Feedback Process	Wellness, Culture, DEI (Diver, Equid, Inclusion)	Mental Health, Work Hybridization Process, Purpose
Focus Dimension	Demographics Org.Work History Compensation Management Time Tracking	Individual and Team - Beginning of database creation	Stakeholder Satisfaction	Organizational Networks, Communication, Sentiment Analysis	Prol continue on of the Organization, Change by Purpose

Source: Research (2025)

SHRM's journey to become more analytical and evidence-based is deeply intertwined with the evolution of information and communication technologies. The ability to collect, store, process, and, most importantly, analyze data about people and HR processes has gradually transformed professionals and the available tools and redefined what the HR area strategically aims for (Marler & Boudreau, 2017; Strohmeier, 2022). This technological revolution has not only automated tasks, as we have seen, but created new ones, setting the stage for the increasingly data-driven SHRM we see today. However, the evolution of technological tools, from the first information systems (HRIS) to artificial intelligence, is only one side of the coin. For all this technology to generate actual strategic value, it is vital that HR professionals also evolve, developing new competencies and skills (Thite, 2015).

The large storage capacity and the reduction of its costs fulfill part of this long evolutionary path. Another part of this path is the management of this data, which is to understand what the minimum requirements for project are, as well as to think about how protection will be carried out, which now needs to be accompanied by the ability to analyze, interpret, communicate and, fundamentally, transform this data into insights that lead to intelligent strategic decisions (Marler & Boudreau, 2017). This new demand for "data fluency", "data literacy" and analytical thinking is not the same for everyone; it presents itself in different ways, depending on the role that each professional occupies in the HR structure – be it an administrator, a business partner, or a strategic leader (Ulrich et al., 2017).

Realizing that each HR professional needs to learn different things and expand their analytical mindset is a fundamental step to building a Human Resources area that can really thrive in the digital and analytical age. This transition does not just happen with the purchase of new tools; It requires a conscious investment in the development of the team's own people, aligning what each one knows how to do with the new demands that the strategy imposes. Table 4 helps us to visualize this difference, showing the primary learning needs of those who work at the strategic, managerial, and operational levels. It is clear that at the top, the emphasis is on business acumen and the use of data to plan for the future of the workforce. At the management level, the focus shifts to how to use technologies in talent management. Furthermore, on an operational basis, the priority is to ensure that systems work well and comply with the rules. However, which the table also highlights, is the need for a business-oriented and analytical mindset (Khan & Millner, 2023). Even if with different intensities, this way of thinking needs to be present at all levels, as it is what sustains a truly data-driven SHRM.

Table 4: New Skills

Field of Knowledge	Skills	Description
Data-driven	Analytical Thinking	Model business problems into issues and hypotheses through the needs of business leaders.
	Workforce Planning	Act like a strategist so you differentiate between Cost-Based and Competency-Based Workforce Plan
	Data analysis	Be a data scientist. A good start is a deep understanding of Excel®
Be guided by your experience	People-Centered Design	Think like an employee, value their experiences
	Implement Employee Experience	Act like a marketing professional, create relationships with the organization's stakeholders, thus learn the main concepts and processes of marketing.
	Data literacy	Be a technologist, go beyond the basics, so apply AI and Robotic Automation models, even if in a fictitious way or in tests.
Business-focused Competencies	Organizational Acumen	Think like an Executive. Deeply understand the organization, its strategy to understand how they align with HR Management.
	Management of all Stakeholders	Be a diplomat. Establish partnerships, understand your pains, as well as your successes.
	Communicate very well	Be a storyteller. A good story engages and engages faster

Source: Adapted from Ferrar and Green (2021).

According to Azeez Jason Kess-Momoh et al. (2024), as organizations learn to navigate the complexities of a globalized, technology-driven world, SHRM is driven to live with and manage fundamental changes. Innovations in data analytics and artificial intelligence have been the significant catalysts for this transformation in HR functions. These advances, in practice, simplify processes such as recruitment, performance management, and employee engagement, making HRM practices more efficient and effective (Ayuningtias & Wahyuningtyas, 2024).

Efficiency needs to increase, in this way, efficiency can change the organization. For this efficiency to change the organizational reality, it is necessary to change how we think about and manage data, and cloud computing emerges as a key piece in this puzzle. Mallireddy (2022) states that migrating to the cloud can significantly impact agility, security, costs, and

how we understand information management. Integrating Human Capital Management (HCM) software with the cloud, in practice, makes workforce planning and day-to-day management much easier. Modern tools allow HR professionals to access and analyze data in near real-time, which empowers them to make better-informed decisions about managing talent and driving organizational development. This shift to data-driven HR supports strategic planning and helps companies stay competitive in a constantly changing market (Subrahmanyam, 2024). Moreover, the horizon seems even broader: There is intriguing potential for these emerging technologies to begin using innovative analytical perspectives, such as genomics and transhumanism, to personalize employee health and optimize talent in ways we have barely begun to imagine today (Huang et al., 2023).

These technological advances open doors to a future in which HRM can offer highly personalized interventions and policies focused on meeting the needs and improving the performance of each individual. This trend towards customization may even lead us to rethink the idea of "packages" of policies and practices, perhaps moving towards even more individualized solutions. However, this power brings immense responsibility, raising serious questions about ethics and privacy that we cannot ignore (Samarasinghe & Medis, 2020).

Organizations have to face these challenges head-on and develop consistent programs that can balance the desire for innovation with a fundamental respect for the rights and privacy of their employees. Moreover, this adaptation cannot be passive; Companies need to be proactive, embracing technological change while also embracing technological change. Adjust their policies, practices, and strategies (Subrahmanyam, 2024; Subramaniyan et al., 2018). This ultimately means cultivating an innovative organizational culture that encourages everyone to continuously learn and adapt – an idea that finds a strong echo in the statements of Rasmussen and his colleagues (2023).

Suppose organizations choose to act in this way. In that case, they are better prepared for the inevitable technological disruptions that will come, and they tend to collaborate effectively, not just maintain their competitive advantage but also strengthen their values and legitimacy (Belizón & Kieran, 2022; Pongpisutsopa et al., 2020). The convergence between the enormous availability of data – Big Data – and advances in the knowledge and application of artificial Intelligence (AI) and Machine Learning (Machine Learning) tends to work as a powerful catalyst. This concepts, accelerated and deepened the transformation process, pushing SHRM, once and for all, into an era where data is the protagonist (Tambe et al., 2019; Votto et al., 2021).

The ability to dive into these vast and complex oceans of data – whether structured or not – and to find meaningful patterns has opened up a range of possibilities never seen before. With greater analytical capacity, people management can now be more strategic and predictive. This journey is supported by the advancement of tools and technologies, which mirror and drive the evolution of SHRM as a concept. In this vibrant scenario, People Analytics (PA) emerges as a discipline. Focused on using advanced analytical methods to solve people-related business problems, PA is gaining prominence in academia, consulting, and practice (Marler & Boudreau, 2017; Rasmussen & Ulrich, 2015). Equipped with these new capabilities, SHRM can go beyond intuition and experience, and it starts to seek solid foundations in evidence for its decisions, to optimize the use of its resources, to personalize the experience of each employee, and, crucially, to demonstrate in a clear and quantifiable way its value to the organization (Armstrong & Brown, 2019).

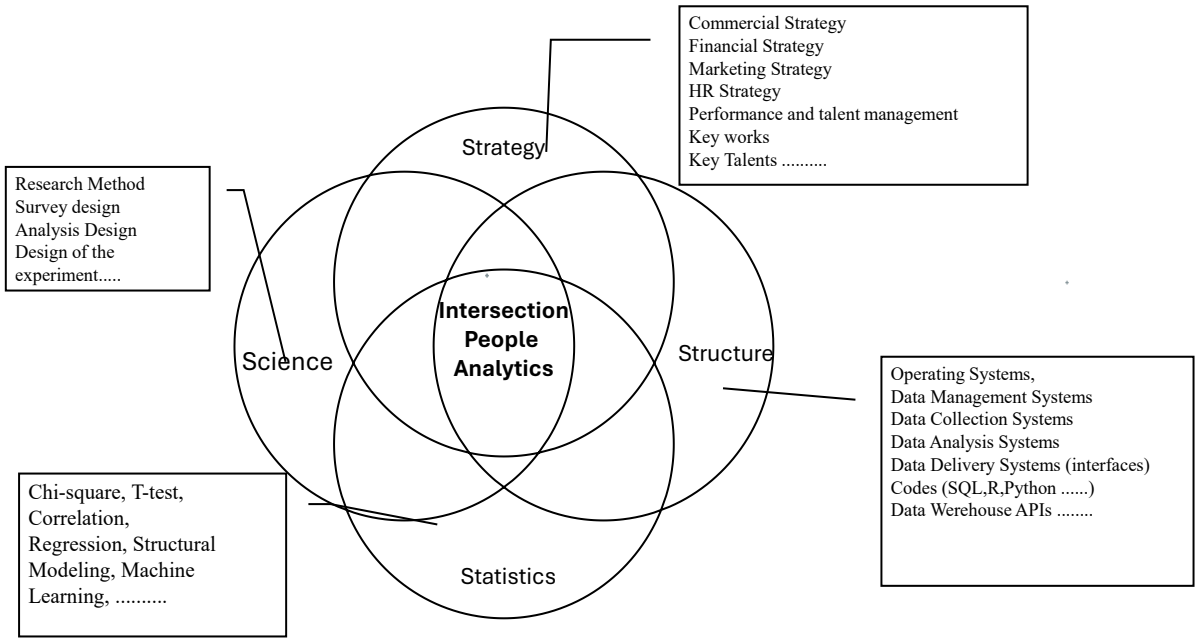
1.4 People Analytics (PA) - The emergence of a new organizational actor

1.4.1 People Analytics - A context of change

The contemporary organizational scenario is characterized by unprecedented volatility. Global companies face a "confluence of disruptions"; managing under the context of the global pandemic COVID-19 and continuous economic, political, and technological challenges are just some of the problems of 21st-century management (Khan & Millner, 2023, p 13). This era, often associated with Industry 4.0 – primarily as a "salvation" – requires profound changes in operations and, crucially, in human capital management (Shukla et al., 2024, p.3).

The pressure for adaptation and resilience puts the Human Resources (HR) function at an inflection point: "[...]either evolve using these innovations or succumbs to possible obsolescence" (Eubanks, 2022, p.xviii). In this context, the adoption of AI emerges not only as a tool on an extensive list to choose from but as an agent of organizational transformation due to its current capacity to build interactions between all organizational actors, and "data consolidates itself as the most strategic asset" (Khan & Millner, 2023, p. 13). PA establishes itself as an indispensable discipline for HRM to build the necessary transformation at the intersection of science, strategy, structure, and statistical tools. Figure 3 represents this ability to interact and iterate.

Figure 3: Intersections for People Analytics



Source: Adapted from West (2019, p 30).

The trajectory of HR, from its origins as a personnel department focused on administrative tasks to the search for a “business impact-driven HR”, has been long (Khan & Millner, 2023, p. 31). However, the speed of current change requires a qualitative leap. The transition, or transformation of HRM to a “people-focused function”(Khan & Millner, 2023, p. 14), and one that deeply integrates business strategy with talent management, tends to be driven by the need to make decisions based on evidence, rather than just intuition or past practices (Ghatak, 2022). PA can be the driver of this transition, defined as the “data-driven approach to managing people at work”, allowing HRM to quantify, predict, and improve the impact of its interventions on business outcomes (Ghatak, 2022, p. 3; Eubanks, 2022, p. 7).

Artificial Intelligence acts as a central catalyst in this evolution. Defined as the ability of computer systems to perform tasks that would typically require human Intelligence, AI is redefining the capabilities of HRM (Eubanks, 2022; Strohmeier, 2022). AI does not aim to replace HR professionals, but rather to “augment their capabilities” (Eubanks, 2022, p. 4), automating repetitive tasks and freeing up time for strategic focus (Behera et al., 2023). The application of AI permeates the entire employee lifecycle since attracting and recruiting; AI tools analyze resumes, identify passive candidates, use Chatbots for initial screening and engagement (Eubanks, 2022; Pandey et al., 2023), and may even conduct preliminary interviews, aiming for greater efficiency and reduction of unconscious biases (Strohmeier, 2022).

On Integration (Onboarding process): AI-powered platforms personalize the onboarding experience, delivering relevant content and connecting new employees with mentors and resources (Eubanks, 2022; Pandey, 2023). AI uses this integration for development and learning: AI recommends courses, creates personalized learning paths, and facilitates the identification of skills gaps, promoting a culture of continuous development essential in Industry 4.0 (Shukla et al., 2024).

All these interactive possibilities can contribute to better and broader performance management and search for solutions for greater engagement, or for sentiment analysis from surveys or internal communications (with the proper ethical precautions) and organizational network analysis (ONA) offer insights on climate and collaboration (Eubanks, 2022; Ghatak, 2022). AI also assists in the identification of talent at risk of leaving, "[…]predictive turnover analytics" (Ghatak, 2022, p. 12), which proactively adopting AI tends to provide a better employee experience such as: With process automation via AI improves the delivery of HR services by offering 24/7 support through Chatbots and personalizing interactions (Eubanks, 2022; Khan & Millner, 2023). The evolution of SHRM to a higher analytical level, like that of PA, is constantly driven by the increasing availability of data and the advancement of extensive statistical technologies, as previously discussed, culminates in the emergence of a field that promises to redefine the HR function fundamentally: the PA (Zander & Zieglmeier, 2023).

Far from being just a new tool or a technological fad, PA represents a shift in analytical conception in HRM, far beyond experience, empowering organizations to make decisions about their talents in a more informed, strategic, and evidence-based way, Figure 3, p. 28, represents the wide range of possibilities for conceptual connections (Marler & Boudreau, 2017; Rasmussen & Ulrich, 2015). As argued by Fitz-enz (1994), demonstrating the value and return on investment (ROI) of HRM initiatives has become a contemporary imperative, and PA emerges as the primary avenue to meet this demand, turning raw data into insights actionable that can drive organizational performance (Chalutz Ben-Gal, 2019; Fitz-Enz & Mattox II, 2014).

Although the terms Human Resource Analytics (HRA), Workforce Analytics (WA), and Talent Analytics (TA) are sometimes used interchangeably, PA is distinguished by its broader scope and strategic focus (Tursunbayeva et al., 2018). While HRA can focus more on the efficiency of HR processes – using horizontal analytics, PA seeks to answer critical business issues through in-depth analysis of people data, covering the entire employee

lifecycle and its interconnections with organizational outcomes, encompassing vertical and cross-cutting analyses(George & Thomas, 2019).

Therefore, PA is "[…]a data-driven methodology for improving people management decisions, with the goal of advancing both individual and organizational success" (Angrave et al., 2016, p. 7). Gal et al. (2020) define PA as computational techniques used to analyze large data sets and inform managerial decision-making, and describe PA as a quantitative, evidence-based, and data-driven approach to HR management. Szlechter and Zangaro (2020) state that PA projects emerge as the primary strategy for workforce management, which, through data analysis using algorithms, unveil individual and organizational interactions present in corporate social networks and other spaces of virtual participation of these business entities. They also argue about the possibilities of using the analysis of feelings and emotions present in the social interactions of the network participant's, now analyzed broadly and forcefully. Therefore, for these researchers, an PA project is like a technique in the field of AI resulting from the analysis and application of big data for the development of practices linked to personnel administration, to measure activity, productivity, efficiency, motivation or implication and satisfaction of all organizational members (Gelbard et al., 2018).

People Analytics refers not to technology but an innovative, quantitative approach. This approach aims to increase the efficiency of core HR functions and optimize employee and organizational performance (Guo et al., 2024; Suri & Lakhanpal, 2024). PA, here, is considered an emerging field not only for its relative novelty and/or for the rapid evolution of its techniques but for the growing, although not yet universal, adoption by companies as a means of deepening organizational knowledge through data. Its rise is inseparable from digital transformation and the perception that human capital is, in fact, a fundamental competitive advantage in the age of knowledge and competitiveness, and now from another perspective of analysis , the organizational change (Isson & Harriot, 2016; Y. Jiang & Akdere, 2022; Ryan, 2020). Facts that do not exempt everyone who has proposed to apply such a method from the challenges inherent to its implementation, which require a sophisticated combination of technology, analytical skills, people who are experts in data analysis, and a data-driven organizational culture and, acceptance of change (Gaur et al., 2024; Tursunbayeva et al., 2018).

The relationship between PA and organizational change is intrinsic. The adoption and implementation of a PA project provides the empirical mechanisms to diagnose the need for change, guide its implementation, and evaluate its results and changes arising from the PA

project. From the perspective of Weick and Quinn (1999), organizational change can be understood as episodic – infrequent, intentional, and often radical – and/or continuous – because it provides an evolutionary, cumulative, and immanent conception, so the complexity that an PA project emanates is represented by its transit between the two theoretical conceptions of organizational change.

For episodic change, which follows the model of defrost-change-refreeze, the PA provides robust data to justify the need to "unfreeze" the status quo, identify hotspots of resistance and opportunities monitor progress during the change phase, and, assess the effectiveness and refreezing of new practices through impact metrics (Ghatak, 2022),. For a continuous change, characterized by the; "[…]constant adaptation and improvisation" postulated by Weick and Quinn (1999, p. 375), this concept encompasses the analytical practices of PA, which can provide feedback in real-time and insights predictive tools, which allow agile adjustments and organizational learning, fostering a culture where evolution is informed by data and not just intuition (Eubanks, 2022).

Expanding this analysis of organizational change, we resort to the "engines" of change of Van De Ven & Poole (1995). This concept can contribute to understanding how PA resonates particularly with the approaches or the engines: Teleological and Evolutionary. At its core, teleological change is an intentional and adaptive process driven by a desired future state and built on an iterative cycle of goal formulation, action implementation, and outcome evaluation (Van de Ven & Poole, 1995, p. 515). An PA project emerges as the ideal analytical infrastructure to operationalize and enhance each phase of this cycle in the sphere of teleological change and an analytical HRM, transforming the way the organization pursues and accepts its organizational goals.

In the formulation phase, the PA moves HR goal-setting from the realm of intuition to that of evidence. "[...] By providing robust baseline data on the current state of the workforce – from turnover and engagement metrics to competency analyses and organizational networks" (Ghatak, 2022, pp. 8-15). An PA project allows leaders to identify the gaps between present reality and future objectives clearly and also use predictive techniques to model scenarios, helping to predict the potential impact of different goals and select those with greater strategic alignment and probability of success in their execution (Eubanks, 2022).

Thus, the PA from a Teleological perspective not only helps to define HR KPIs but ensures that these are strategically relevant and empirically grounded and feasible. During the implementation process, PA tends to function as a navigation system. Instead of implementing

change initiatives in a generalized way – such as a one-size-fits-all training program – PA allows to segment the workforce and target specific interventions to where they are most needed and will have the greatest impact, by conducting tests at various stages of the process. Monitoring tools and dashboards can provide near real-time feedback on change adoption, employee sentiment, and early indicators of success or failure. It makes it possible to make agile adjustments, proactively manage resistance, and continuously optimize actions during the change process rather than waiting for a post-mortem evaluation(Ferrar & Green, 2021).

Last but not least, in the evaluation phase, the PA offers the means to objectively measure the implemented changes' results. PA allows to go beyond subjective impressions, quantifying the impact of HR actions on business metrics – "[…]such as productivity, innovation, customer satisfaction, and, fundamentally, ROI" (Khan & Millner, 2023, p. 32; Ghatak, 2022, p. 5). This ability to demonstrate tangible value justifies HR investments. It generates key insights that feedback into the teleological cycle, informing future rounds of goal formulation and continually refining the organization's ability to manage change effectively and data-driven. The PA, therefore, is not only a support, but an accelerator and optimizer of the teleological engine of change in people management (Cifalinò & Lisi, 2019; Harris et al., 2011).

In the context of the "Evolutionary Engine, where change emerges from a continuous cycle of variation, selection, and retention of practices" (Poole & Van De Ven, 2021; Van De Ven & Poole, 1995, p. 516). PA acts as an analytical nervous system, appropriating the current context of algorithmic inevitability. PA empowers the organization to perceive, interpret, and act on the evolutionary dynamics occurring in its workforce, making the process less random and more specific. Variation, the raw material of evolution, is a fundamental tool for identification and generation in the PA. The PA can identify variations, which can be referenced in literature as patterns that arise organically. As an example, when performance and engagement data analysis highlights teams or leaders who, through their unique practices, consistently achieve superior or inferior results (Brito & Sousa, 2023). In addition to identifying natural variations in organizational and individual outcomes and/or behaviors, PA allows to hypothesize possible variations controlled through pilot programs or A/B testing of new HR policies, wellness programs, or collaboration tools, reinforcing flexibility (Eubanks, 2022). Analysis of feedback data, suggestions, and even network data can reveal emerging innovations at the base of the organization, which represent new approaches to consider in a dynamic network, highlighting previously unnoticed paths (Shukla et al., 2024).

Once identified, these variations enter the selection process, which determines which ones will persist and will be considered and which ones will be disregarded because they are not significant. Here, the PA transforms this process, often subjective or political, into an evidence-based evaluation by providing objective metrics or indications of “aptitude” or “aligned competency” – such as impact on productivity, cost reduction, improved talent retention, or increased engagement (Mahdia, 2024).

PA allows the organization to select the most effective people management practices, making rigorous comparisons between different approaches, and identifying not only what works but where and why it works, considering contextual factors that can influence the success of a practice. Predictive analytics can even help anticipate which variations are most likely to occur in the medium and long term (Rasmussen et al., 2024). Retention in the change process is fundamental, as it ensures that successful practices are preserved, disseminated, and incorporated into the organizational routine, legitimizing them (Poole & Van De Ven, 2021).

The role of PA in this aspect is crucial in measuring the sustainability of these new practices and understanding the mechanisms underlying the success/failure of a practice, facilitating its codification and replication. In addition, the PA monitors the adoption and effectiveness of selected practices as they scale, identifying implementation challenges and ensuring that their value is maintained by demonstrating the continued value of these practices through data in pursuit of analytical excellence (Ferrar & Green, 2021). PA amplifies "retention" in memory and organizational processes. In this way, PA does not eliminate the evolutionary nature of change but informs and guides it, appropriating a contextual perspective. It allows the organization to; “[...] learn from yourself faster and more effectively, transforming change management in human capital from a reactive and subjective process to a strategic and measurable discipline”, ensuring that the evolution of HR practices is aligned with the organization's needs and strategic objectives (Shukla et al., 2024, p. 15).

Thus, PA presents its flexibility and complexity by offering a new possibility of managing organizational change. Under a consolidated conceptual lens, PA tends to offer a guide to organizational change processes when approached by episodic, continuous, teleological, and evolutionary change of variation, selection, and retention practices within the organization (Van de Ven & Poole, 1995; Weick and Queen, 1999).

It becomes to recognize that the PA function itself is not a static entity; on the contrary, the implementation and sophistication of a PA project within a company itself follows a development trajectory that can be seen as an intrinsic evolutionary process of building its

foundations, data governance, methodological rigidity, and management of interests of different stakeholders (Ferrar & Green, 2021).

It needs to mature for the PA to reach its full potential as a catalyst for strategic and informed change. This journey must reflect the organization's growing ability to collect, analyze, and, most importantly, translate people's data into insights actionable and business value (Gaur et al., 2024; Sharma, 2020). Therefore, understanding the stages of this evolution, often outlined in maturity models, becomes essential to assess organizational readiness and chart the path toward truly data-driven people management (Ghatak, 2022).

This evolution is often described through maturity models, which describe the stages of development that organizations can take to enhance their HR analytics capabilities. For example, the HR Analytics Maturity Model (HRAMM) provides a framework for organizations to evaluate and improve their analytics capabilities, emphasizing the importance of aligning IT and HR strategies (Rigamonti et al., 2024). As organizations progress through these maturity levels, they can replace anecdotal decision-making with anecdotal decision-making. Insights data-driven actionable capabilities, thereby maximizing the impact of HR activities, such as selection, evaluation, and compensation, on business performance (Madhani, 2023).

Therefore, the maturity of HR analytics is linked to the development of creative problem-solving capabilities, which mediate the relationship between the use of analytics and organizational performance, highlighting that for such a proposition, the role of data quality and staff experience as moderators are critical (Thakur et al., 2024). As HR analytics continues to evolve, it is expected to become an established discipline with a significant influence on operational and strategic decision-making, potentially leading to its integration into a core human resource analytics function, transcending business units per se (van den Heuvel & Bondarouk, 2017). The consolidation of PA as a strategic organizational actor does not occur uniformly or instantaneously. On the contrary, organizations typically progress and regress between different stages of maturity in developing and enhancing their HRM analytical capabilities.

Even though each maturity level corresponds to and depends on a specific technological sophistication, this process is not linear. It will depend directly on the specific objectives of each PA project (Suri & Lakhanpal, 2024). To illustrate this evolutionary trajectory Table 5 presents a synthesis of the levels of development of analytical capacity in People Analytics, detailing the main focus, the level of analysis, examples of activities,

standard technologies, challenges, and the potential organizational impact in each phase. Analytical maturity levels impact HRM by enhancing decision-making processes, aligning HR strategies with business objectives, and improving organizational performance. As HR analytics matures, it moves from traditional methods to more statistically sophisticated approaches that use advanced technologies as a means or mediator of increasingly deep and scientifically developed organizational learning in pursuit of analytical excellence (Ayuningtias & Wahyuningtyas, 2024; Ferrar & Green, 2021).

Table 5: Analytical Maturity

Stages	Main Focus	Level of Analysis	Examples of Activities	Technologies and Tools	Common Challenges	Organizational Potential
Stage 1: Operational and Reporting	Collection and organization of basic HR data.	Descriptive	Generation of headcount, turnover, absenteeism reports; Basic dashboards with HR metrics.	Spreadsheets (Excel), Basic HR Information Systems (HRIS).	Data quality and consistency; Lack of standardization; Excessive focus on vanity metrics.	Improvement in HR administrative efficiency; Basic visibility into the state of the workforce.
Stage 2: Diagnostic Analytics	Understand the causes of results and trends.	Diagnostic	Analysis of causes of turnover in a specific department; Investigation of factors that impact employee engagement; Comparison of performance between different groups.	BI tools (Tableau, Power BI), basic statistical packages.	Need for initial analytical skills; Integration of data from different sources; Cultural resistance.	Identification of problems and opportunities for improvement in HR; Basis for more informed decision-making in HR.
Stage 3: Predictive Analytics	Predict future outcomes based on historical data.	Predictive	Modeling of key employee turnover risk; Prediction of future hiring needs; Identification of candidates with the highest probability of success.	Advanced statistical software (R, Python), Machine Learning platforms.	Availability of large volumes of quality data; Advanced analytical skills and data scientists.	Anticipation of human capital challenges; Optimization of HR processes (recruitment, retention); Proactive resource allocation.
Stage 4: Prescriptive Analytics	Recommend actions to optimize future results.	Prescritive	Simulation of the impact of different HR interventions (training, benefits) on business results; Optimization of succession plans; Personalized development recommendations.	Optimization algorithms, AI, simulation.	Complexity of the models; Need for continuous validation; Acceptance and implementation of recommendations.	Data-driven strategic decision-making; Direct impact on business results; HR as a strategic partner.
Stage 5: Integrated and Strategic People Analytics	PA as a central part of organizational strategy.	Strategic Integrated	PA informs all talent and business decisions; Data-driven culture throughout the organization; Continuous innovation in evidence-based people management practices.	Integrated ecosystem of analytics technologies, advanced AI.	Robust data governance; Strong leadership and data culture; Ethical and privacy issues.	Sustainable competitive advantage through human capital; Organizational agility; Innovation in HR.

Source: Adapted from Davenport & Harris (2007).

1.4.2 The Strategic Value Proposition of PA

The PA's key value proposition lies in its ability to transform HRM from a role often perceived as administrative or reactive into a proactive strategic partner capable of influencing business decisions (Jamison, 2020; Saling & Do, 2020). First, PA tends to improve evidence-based decision-making (Marr, 2018). PA replaces intuition, anecdotal experience, or generic "best practices" with data analysis specific to the organization's context. Allows talent decisions – from recruitment and selection to development, engagement, compensation, workforce retention, and planning – to be more objective, practical, and aligned with strategic goals (Gaur et al., 2019; van den Heuvel & Bondarouk, 2017).

Second, HRM transformation, through PA as a strategic value proposition, is directly proportional to the progression of analytical maturity of each organization, which will consequently result in a unique and specific strategic impact (R. V. McCarthy et al., 2022; T. H. Rasmussen et al., 2024), therefore, the added value of PA intensifies as organizations advance in their implemented analytical practice. In order to characterize this evolutionary process of analysis capacity, the classical model of Davenport and Harris (2007).

Table 5 describes this progression promptly: (1st) Descriptive Analysis ("What happened?"): Reports and dashboards that summarize historical data (turnover rate, time to fill vacancies). (2nd) Diagnostic Analysis (Why did it happen?): Investigation of the causes and factors that influence the observed results (why did turnover increase in a given department?). (3rd) Predictive Analytics (What is likely to happen?): Using statistical models and machine learning to predict future outcomes (which employees are most likely to leave the company in the next six months; which candidates have the highest potential for high performance). (4th) Prescriptive Analysis (What should we do about it?): Generating recommendations on the best actions to take to achieve desired outcomes or mitigate risk (which interventions are most effective at retaining key talent; how to optimize training investments to maximize the development of critical skills). The ability to operate at both the predictive and prescriptive levels is where the PA demonstrates its most significant strategic potential, enabling organizations to anticipate challenges and opportunities and act proactively (Ayuningtias & Wahyuningtyas, 2024; van den Heuvel & Bondarouk, 2017)

However, progression through these analytical stages and the effective attainment of strategic value through PA is not merely a matter of technological adoption. As Gerald Kane (2019) warns in his discussion of the technology fallacy, the belief that simply implementing

new digital tools will solve organizational challenges is a common misconception. True transformation and the ability to leverage the power of the PA resides fundamentally in the people, processes, and organizational culture. Kane (2019) argues that “digital maturity” – an organization's ability to capitalize on the opportunities offered by new technology – is achieved through changes in organizational dynamics, fostering a more agile, risk-tolerant, and experimental culture and leadership capable of driving this transformation.

In the context of PA, this means that, in addition to the acquisition of sophisticated or hyped analytical platforms, it is necessary to develop the analytical skills of HR professionals – or the team that will be responsible for – to redesign processes to integrate the Insights of PA in decision-making and to cultivate an organization-wide mindset that values and acts on evidence generated from people data (Kane, 2019; Rasmussen et al., 2024). Ignoring these organizational and cultural aspects in favor of an exclusive focus on technology can lead to frustrated investments and underutilization of the strategic potential of People Analytics, reinforcing that technology is an enabler, but fundamental transformation is driven by organizational strategy and the human capacity for adaptation and innovation (Konovalova et al., 2021; Mahdia, 2024).

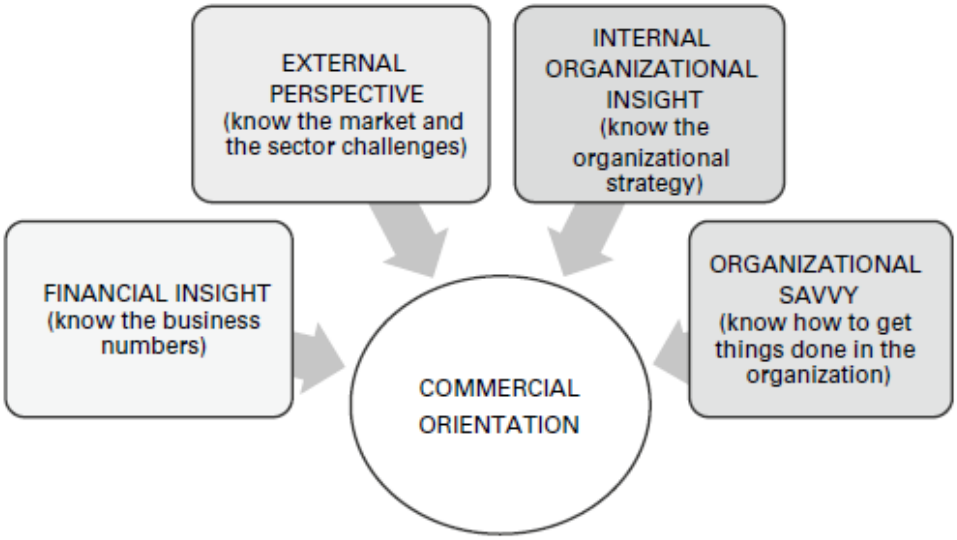
Though, what can be considered to develop an analytical mindset? Commercial orientation emerges as the central pillar of this transformation, being defined as the set of skills that enables professionals and the HR function to contribute directly to the organization's financial results, whether through increasing revenues, optimizing profitability, or generating cash, culminating in increased value for stakeholders. Business leaders emphasize that while PA is a powerful tool for generating workforce insights, its true potential is only unlocked when handled by professionals with this commercially oriented mindset, which is a fundamental prerequisite for any successful analytics journey (Khan & Millner, 2023).

Surveys, such as the Corporate Research Forum, identify key qualities that differentiate executives with strong business orientations (Ragas & Culp, 2021). These orientations, include commercial acuity, which acts as a lens for analyzing situations and determining actions; forecasting, allowing to anticipate trends and visualize both the big picture and the details; the prioritization of the business to the detriment of the function; a relentless focus on continuous improvement; firmness of purpose and post-decision resilience; and the ability to persuade. In addition, the research by Khan and Millner (2023, p. 85), illustrated in Figure 4, groups these characteristics into four essential themes that underpin commercial orientation in HR: Strategic Awareness (deep understanding of the business,

market, and customers), Financial and Commercial Acumen (understanding of numbers, ROI, and value levers), External Orientation and Curiosity (the ability to look outside the organization and question the status quo) and the Impact on Stakeholders (ability to understand needs, build relationships and influence).

The internalization and development of these topics are, therefore, imperative steps for HR to use data analysis not as an end but as a means to add strategic and measurable value to the organization.

Figure 4: What is Trade Orientation.



Source: Khan & Millner (2023, p.85)

Therefore, PA needs a strategic purpose, so that there is a direct connection with the business results – a project. The PA makes it possible to quantify the impact of HRM initiatives on business performance indicators, as shown by the returns on investment (ROI) case studies, with significant results in areas such as Reducing key talent turnover – PA helped IBM save approximately $300 million by predicting exit risk and proactively intervening; Increased engagement and its correlation to profitability: as in the case of Best Buy, where a slight increase in engagement resulted in substantial operating income gains, the optimization of training programs, and improved recruitment effectiveness (Chalutz Ben-Gal, 2019; Cifalinò & Lisi, 2019; Ferrar & Green, 2021; Fitz-enz, 1994).

By deeply understanding the factors that drive employee performance, engagement, and well-being, the PA allows organizations to create strategies that contemplate general assumptions with specific actions, such as the definition of talent for the organization –

customizing policies and practical actions (Huang et al., 2023). These new determinations include everything from identifying critical skills for the future, which are more effective and personalized, to developing individualized career plans to creating a more inclusive work environment that promotes a positive employee experience, and co-opts the organization for decisions that are increasingly integrated with data (Falletta & Combs, 2021).

1.4.3 Decision Making with AI

Integrating AI into decision-making processes is one of contemporary organizations most significant transformations. In several sectors, including HRM, there is a growing adoption of algorithms and Automated Decision Making Systems (ADMS), often with the promise of greater efficiency, quality, and fairness in choices (Koutsikouri et al., 2024). This transition raises the expectation that Advanced Distribution Management Systems[1] (ADMS) can overcome human limitations, such as biases, conflicts of interest, or fatigue, by relying on vast volumes of data (Lepri et al., 2021).

However, this seductive perspective proves to be problematic and simplistic, especially when applied to intrinsically human domains, such as people management. Critical studies warn of emerging challenges, such as the tendency to overvalue the amount of data to the detriment of details and the complexity of the context. The assumption that automation ensures objectivity ignores that algorithms can incorporate and amplify existing biases (Eubanks, 2022). Most importantly, it underestimates the continuing relevance of human judgment—the ability to interpret nuances, consider ethical implications, and apply accumulated professional experience (Koutsikouri et al., 2024).

Therefore, the discussion about AI and decision-making should not focus on replacing the human with the machine but rather on understanding the human-ML augmentation, which can be translated as: The amplification of the ability of humans and AI to work together on a task. Shrestha et al. (2019) consider amplification in decision-making – man-machine – as a hybrid structure, either as sequential decision-making between AI and humans or as sequential decision-making between humans and AI. The former allows organizations to utilize the strengths of AI and humans, often sequentially, where humans make the final, authoritative

[1]ADMS is a high-tech set of software and hardware solutions used by electric power distribution companies. Its main function is to monitor, control, and optimize the operation of the electricity distribution network. It integrates various tools to improve operational efficiency, supply reliability, and responsiveness to outages

decision based on the information provided by AI. The second follows the opposite sequence, where humans provide the initial information, usually in a small sample, and AI is used to assist in decision-making, often with predictive modeling (Teodorescu et al., 2021). It is essential to investigate how to ensure meaningful human agency (meaningful human agency) in these systems, ensuring that HR professionals can interact, question, intervene, and, ultimately, take responsibility for decisions made with the help of AI (Koutsikouri et al., 2024; Wagner, 2019).

The effectiveness of human-AI collaboration in decision-making, while dependent on factors such as trust, training, and human agency, cannot be understood solely in the vacuum of individual interaction or task. It is imperative to situate this collaboration within the larger context in which it operates: the organization as a complex and intelligent system. Artificial Intelligence is increasingly being integrated not only as a one-off tool but as a "[…] strategic and integral part of products and operations in sectors that inherently deal with high complexity, such as healthcare and aviation" (Troqe et al., 2024, p. 173). HRM, with its web of social, cultural, and political interactions and its direct impact on organizational performance, is no exception to this complex nature.

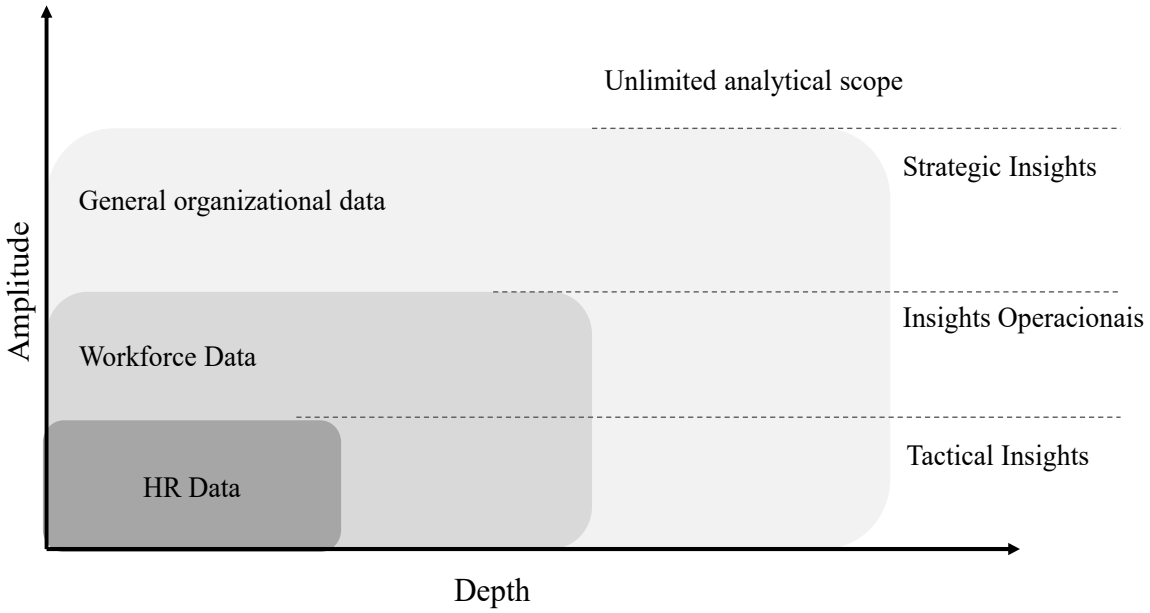
The introduction of AI solutions, particularly those based on Machine Learning (ML), to process large volumes of data and support human decisions in HR brings with it both benefits and challenges amplified by the systems perspective (Shrestha et al., 2019; Troqe et al., 2024). If, on the one hand, AI can help identify patterns and insights on an organizational scale that would be impossible for human analysis, such as modeling the impact of compensation policies on talent retention in different business units, on the other hand, the risks associated with its implementation also acquire a systemic dimension – antithesis. As stated by Troqe et al. (2024, p. 173).

> "[…]These advances [in AI] potentially provide many societal benefits, such as better-informed decisions and better healthcare for society, but they are also associated with several potential challenges. These involve a continuous transformation following the evolution of intelligence, the systems lifecycle, interpretability, trust, data management, and quality, to name a few[…]".

In the context of HRM, this means that a seemingly siloed AI decision like a candidate's recommendation can have cascading and non-linear effects across the entire organizational system – affecting team dynamics, culture, innovation, and ultimately, the

business strategy itself. The complexity lies in the fact that these interactions are difficult to predict and model completely, even with AI. Therefore, AI-powered decision-making in HR requires not only Human collaboration at the task level but also human oversight informed by systems theory, capable of anticipating, monitoring, and mitigating potential unintended consequences across the organization, reaffirming the need for human agency that transcends the operational and reaches the strategic(Baumgart et al., 2023). The proposition explained in Figure 5.

Figure 5: Analytical Scope with Data



Source: Adapted from Khan & Millner (2023, p.67)

As Shollo and Galliers (2024) point out, despite the explosion of data and investment in AI technologies, many organizations struggle to become truly data-driven and reap the promised benefits, as evidenced in Figure 5, achieving unlimited analytical scope. There is a missing link between data/AI availability and effective organizational decision-making. This missing link is the process of building Insights. It is not a purely technical process but rather an activity of meaning-making sensemaking profoundly human (Weick, 1995).

AI can identify correlations and patterns in large volumes of data, but turning those patterns into actionable insights requires interpretation, contextualization, and applying tacit knowledge and business acumen, which we discussed earlier (Khan & Millner, 2023). It means understanding what data means in the specific context of the organization's strategy, culture, and operating environment and, from there, formulating concrete actions. Shollo and Galliers (2024) argue that taking advantage of business intelligence and analytics (BI) and

artificial intelligence (AI) is not a simple task, and success depends on this translation capacity. In HR, this implies that a predictive model of turnover only becomes actionable when professionals can interpret why certain groups are at risk and develop targeted and feasible interventions, going beyond the simple number presented by the algorithm.

1.4.3.1 Machine Learning (ML) in HRM: From Standard to Prediction

HRM's transition to a data-driven approach and Insights is intrinsically linked to adopting Machine Learning (ML) techniques. Defined as the ability of algorithmic systems to "learn from data and find patterns without being explicitly programmed" (Rosett & Hagerty, 2021, p. 175), ML provides HRM with a statistical arsenal to navigate the complexity of people data across the organization and extract predictive and explanatory value. Within this vast field, Supervised Learning, which operates with labeled data to predict known outcomes, is particularly relevant. As detailed by Keith McNulty (2021), Regression Modeling stands out as a robust and interpretable set of supervised learning tools widely applicable to PA challenges. These techniques allow not only to describe relationships, but, to explain and predict outcomes of interest to people management. Understanding these techniques is vital for HR professionals to be able to dialogue with data scientists, interpret results, and base their strategic decisions on quantitative evidence.

One of the pillars of Supervised Learning is Linear Regression. According to McNulty (2021), this technique is ideal for understanding how variables such as experience and training influence continuous results such as salary or productivity. “[...]Its simplicity and interpretability [...]" makes it an essential starting point for quantifying the impact of HR factors, although it requires careful verification of its assumptions (McNulty, 2021, pp. 99-105). For categorical results – so common in HR, hiring, turnover, KPI evaluations – Logistic Regression is the tool of choice (McNulty, 2021). It models the probability of an event occurring, allowing HR to act proactively. Whether it is predicting the chance of an employee leaving (Binomial), understanding the choice of a benefit (Multinomial), or modeling performance levels (Ordinal), these techniques offer valuable insights into human behavior in the organization.

However, ML in HRM is not limited to predicting known outcomes. Unsupervised Learning comes into play when the goal is to "[...]discover hidden structures or patterns in data without pre-defined labels [...]" (Rosett & Hagerty, 2021, p. 178). Techniques such as clustering allow to segment the workforce to customize policies or identify emerging themes in research. Dimensionality Reduction assists in simplifying complex data sets, making

analysis easier. Additionally, although less common, the system of Reinforcement Learning, which learns through rewards, seeks – with reinforcement – to point to future applications in career optimization and training, which is fundamental for organizations in the process of merging or downsizing (Rosett & Hagerty, 2021).

According to Eubanks (2022), unlike traditional computing, machine learning algorithms identify patterns and generate models, thus incrementally improving their performance in performing specific tasks. This learning capability is achieved through iterative exposure to data, where the algorithm refines its parameters based on error corrections and feedback obtained during training, which is critical to its ability to generalize to new or invisible data. In supervised learning, the system is provided with labeled datasets, each input matched with an expected output. The algorithm learns to map inputs and outputs, minimizing the error between predicted and actual outcomes. On the other hand, unsupervised Learning involves algorithms that work without pre-assigned labels. Here, the system finds hidden structures or patterns directly from the data. While unsupervised learning remains an active research area, current commercial applications predominantly utilize supervised methods due to their stability and predictability in production environments.

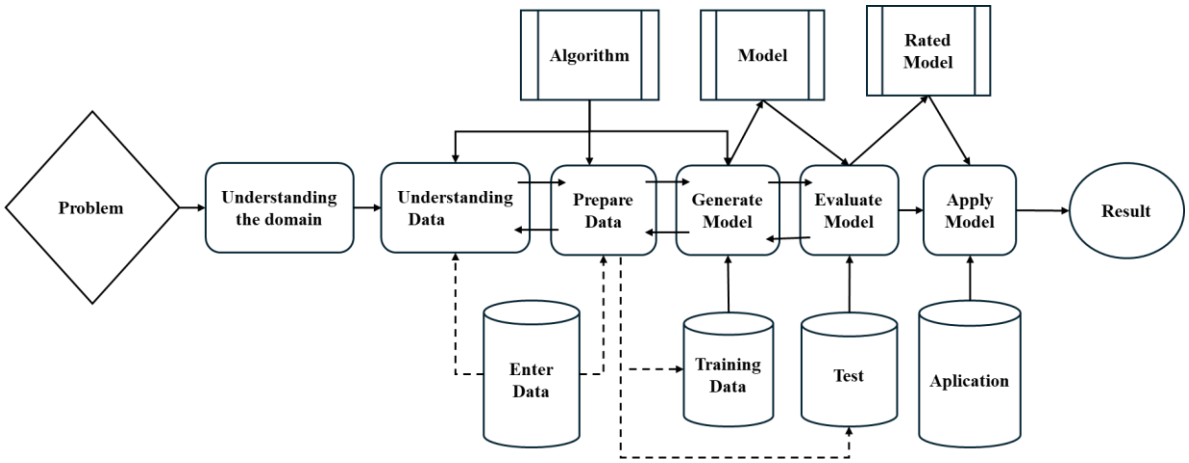
Advanced machine learning architectures often incorporate iterative methods such as gradient descent, regularization techniques to prevent overfitting, and hyperparameter tuning to enhance predictive performance. The adaptability of these models makes them highly effective for a wide range of applications. For example, in HRM, supervised models can analyze historical employee data to predict future attrition – exit intent – while unsupervised clustering methods can uncover latent categories in candidate profiles or group feedback into coherent themes with minimal human interference.

Despite its transformative power, machine learning has several limitations and concerns. The quality of a machine learning model is intrinsically linked to the quality and representativeness of the data used in training. Bias in training data can lead to skewed predictions, thus raising ethical and practical questions about fairness and accountability. Additionally, the interpretability of complex, high-dimensional models can be challenging, which necessitates the development of explainable AI techniques to ensure that stakeholders understand the underlying decision-making processes (Eubanks, 2022; McNulty, 2021).

In short, Machine Learning, ranging from interpretable regressions (McNulty, 2021; Eubanks, 2022) to the discovery of patterns with unsupervised learning (Rosett & Hagerty, 2021), equips HRM with a diverse set of tools to dissect the complexity of human relationships

at work. They are essential for transforming raw data into explanatory and predictive insights. However, its successful application requires a combination of technical expertise, a deep understanding of the HR context, and an unwavering commitment to ethical interpretation and responsible decision-making, ensuring that technology serves to empower and support people in the organization in a process summarized in Figure 6.

Figure 6: AI Framework in HR



Source: Adapted from Strohmeier (2022, p 26)

1.4.3.2 Integration of NLP into SHRM Decision-Making

The integration of Natural Language Processing (NLP) into Strategic Human Resource Management (SHRM) significantly improves HR decision-making by leveraging data-driven insights and optimizing various HR functions and their interrelationships. NLP facilitates the analysis of unstructured data, such as employee and/or customer feedback, allowing HR departments to perform sentiment analysis, topic modeling, and keyword extraction to understand employee sentiments better and identify areas for improvement, thereby promoting a more engaged workforce (Jamuna, 2024).

Understanding how to work with NPL is a capability fundamental for predictive analysis on employee turnover, what is the level of employee and/or customer satisfaction through analysis of internal communications, allowing for more informed decision-making in recruitment and retention strategies, relationship building, and strategic alignment of policies and practices (Saxena et al., 2024). In addition, NLP helps automate recruitment processes, reduce administrative burdens, and personalize employee onboarding and training, which not only cuts costs but also improves the employee experience (Islam, 2024). The use of NLP in human resource management also extends to extracting skills and matching them to the

required skills and competencies of the position, which are essential for effective talent management and ensuring that organizations have the right skills to achieve strategic goals (Otani et al., 2024). By integrating NLP with machine learning, HR departments can efficiently retrieve and classify HR data, streamlining processes and improving the accuracy of decisions(W. Li & Zhou, 2024).

Li and Zhou (2024), present a structured method under two main components: classification and data extraction. The classification component optimizes the organization of HR information into relevant categories, including job titles, competencies, and performance grades. Meanwhile, the data extraction component enhances the ability to search for HR data quickly and accurately, thereby reducing the time and effort required for the HR team to obtain the necessary information. This dual algorithmic function produces efficiency and accuracy of information for HRM, as the algorithm is evaluated in various datasets derived from real HR environments, demonstrating its effectiveness in simplifying HRM processes. The results indicate that the method significantly reduces manual work and increases the accuracy of HR-related decisions. This efficiency is a goal for businesses looking to improve employee relations and streamline HR procedures.

In the specific application of Li and Zhou (2024), algorithmic technology is designed to be compatible with corporate HR offices and assessed educational environments, adhering to global university standards. This compliance ensures that the method can be widely adopted across different industries, making it a versatile tool for both academic and commercial applications. More broadly, but with a significant contribution to HRM using NLP, Otani et al. (2024) provide comprehensive research on the application of NLP techniques to various HR tasks, categorizing these tasks into five broad areas of research: language comprehension, information extraction, retrieval and recommendation, generation and interaction, and fairness and bias. This categorization helps organize existing research and identify impactful opportunities for further exploration in the HR domain. Otani et al. (2024), discuss that specialized tasks of the algorithm, such as job understanding and skill extraction, contribute to broader HR goals, such as improving job recommendation systems and increasing the fairness and transparency of HR processes. In doing so, they underscore the importance of intermediate tasks in improving system performance in downstream applications.

Despite challenges such as data privacy issues and algorithmic biases, the strategic implementation of NLP in SHRM can lead to increased organizational performance and competitive advantages by optimizing talent management and aligning HR practices with

organizational objectives (Wassell & Bouchard, 2020). Otani et al. (2024) address the critical issue of biases that can be reflected in data-driven systems used for HR applications. These biases often stem from stereotypes, proxies of sensitive attributes, and external barriers such as accessibility to technology.

The research emphasizes the importance of identifying and mitigating these biases to ensure that NLP techniques are applied fairly in real-world HR scenarios, for example, specifically, the association between occupations or sensitive roles and attributes, which has been a significant focus in the context of text representation and generation. The authors warn that word embeddings often associate gender pronouns with specific job titles, such as linking she to nurse and he to doctor, contribute to producing biases or reinforcing them with the use of NLP so that several methods of rejection have been proposed to address these issues, demonstrating ongoing efforts to reduce gender biases in NLP systems (Otani et al., 2024).

The introduction of AI systems into organizational decision-making processes transcends the mere optimization of tasks, triggering profound reconfigurations in work dynamics and, in the occupational identities of employees. In fact, AI systems "[…]fundamentally transform employees' work processes, tasks, and responsibilities[…]" (Mayer & Strich, 2024, p. 305), which are key components of how professionals perceive themselves. Identity and occupational meaning reflect employees' responses to the questions: Who am I (as a member of a specific organization)? And what do I do? How do I contribute to? As explored by Mayer and Strich (2024), the way AI is perceived and its impact on this professional identity is not monolithic.

Rather, it is significantly shaped by contextual dimensions, including "[...]the skill level of the employees; the type of decision-making practice; and the impact of decision-making practice [...]" (Mayer & Strich, 2024, p. 308). It is observed, therefore, that highly qualified professionals, who "[...]define themselves through their decision-making competence, creativity and autonomy [...]", can experience the automation of their main decision-making activities as a threat, perceiving themselves "[...]degraded to 'data entry assistants[...]". On the other hand, for workers with lower specific qualifications, the same automation can represent a vector of empowerment, as in the case of service employees who were able to take on the role of loan advisors after AI automated the central decision (Mayer & Strich, 2024, pp. 313-314).

1.4.4 PA as a Vector of Organizational Integration

Network theory is positioned as a discipline that elucidates the complexity of organizational structures, interdependencies, and flows. This approach challenges traditional reductionist views by emphasizing that a company's value and performance are deeply rooted in the social, technical, and relational connections that constitute its network dynamics. Networks are not merely static diagrams but are composed of nodes – actors and resources – and links whose configuration and density significantly influence the flow of information, decision-making processes, and organizational innovation (Ujwary-Gil, 2019).

The growing sophistication of PA positions is beyond a mere HRM optimization tool, revealing its potential as a dynamic and active builder vector of organizational integration. From the constructionist perspective, integration is not conceived as a static organizational attribute, but as a continuous social process, emerging from interactions, information flows, and the negotiation of meanings and signifiers among the members of the organization (Berger & Luckmann, 1966). The PA, by intervening in the way these processes are understood and managed, not only reflects the organizational reality but actively participates in its co-creation, influencing cohesion, collaboration, and strategic alignment, using diverse AI tools(Dreković et al., 2023; Kumar, 2024).

In the context of PA, a central theme for this integrative capacity is the application/combination of techniques such as Organizational Network Analysis (ONA) (McCuloh et al., 2013; Ujwary-Gil, 2019). The ONA allows us to unveil the "[...]microstructure of interpersonal relationships[...]" (Granovetter, 1973, p. 1361); Mapping how individuals and groups connect, share knowledge, and exert influence. In this context, Mark Granovetter's seminal theory of the strength of weak ties offers insights tends to collaborate with the greatest knowledge of these networks. Granovetter (1983; 1973) argued that strong bonds – dense and frequent relationships with close individuals, such as family members – are important for local cohesion and emotional support. Its counterpart, weak ties – more distant and less frequent connections with acquaintances or work partners – are vital for the diffusion of new and non-redundant information and for the integration of different social groupings.

Weak ties, therefore, act as bridges or links that connect network segments that would otherwise remain isolated, thus facilitating information mobility and integration on a broader scale of the organization. The PA, by identifying the nature and distribution of these ties, allows HRM to intervene strategically to "[...]cultivate an optimal pattern of ties, both strong

and weak, in line with organizational goals and strategies [...]" (Ujwary-Gil, 2019, p. 22); whether to strengthen the cohesion of specific teams or to promote innovation and Interdepartmental collaboration by fostering weak ties. The visualization and analysis of these interaction networks, provided by the PA, also become instrumental in actively constructing a collaborative organizational culture and, consequently, a new social reality.

Arena, Hines and Golden (2023) emphasize that employees' day-to-day interactions are critical to sharing and shaping the norms and behaviors that make up culture. By identifying communication patterns, informational bottlenecks, or core individuals who act as disseminators of culture – or, conversely, as blockers of integration – the PA offers leadership and HRM the ability to design interventions that not only react to the existing culture but actively shape it towards greater openness, trust, and synergy among the various parts of the organization. perspective corroborated by Fontana et al. (2023). This process of diagnosis and intervention in internal social networks illustrates how the PA helps in the management and development of the company's social capital, understood as the resources embedded in the relationships (Adler & Kwon, 2002).

It reinforces the premise that one of the primary mechanisms by which the PA fosters integration is strengthening HRM's vertical strategic alignment (Barends et al., 2014; Pfeffer & Sutton, 2006). By converting workforce data into actionable intelligence, PA offers an empirical basis for dynamically constructing and adjusting HRM strategies in dialogue with the organization's broader goals. This process, inherently linked to the principles of Evidence-Based Management (EBM), allows decisions about talent, development, and organizational design to be informed by a deeper understanding of their implications and contributions to corporate strategy (Rousseau, 2006).

The PA reveals and enhances integration on the horizontal plane by analyzing collaboration and communication networks. By making these connection structures visible, the PA provides Insights that can lead to targeted interventions to strengthen weak ties, optimize knowledge sharing, and break down functional silos(Granovetter, 1973). Arena et al. (2023) posit that understanding how employee interactions shape norms and behaviors is central to cultivating the desired culture, especially in the context of hybrid work. PA, through ONA, offers a means to do so. In a predominantly hybrid organizational environment, the authors suggest a "3C's" structure as a conceptual model to cultivate organizational culture in a hybrid work environment. This framework comprises three interrelated dimensions: cultural

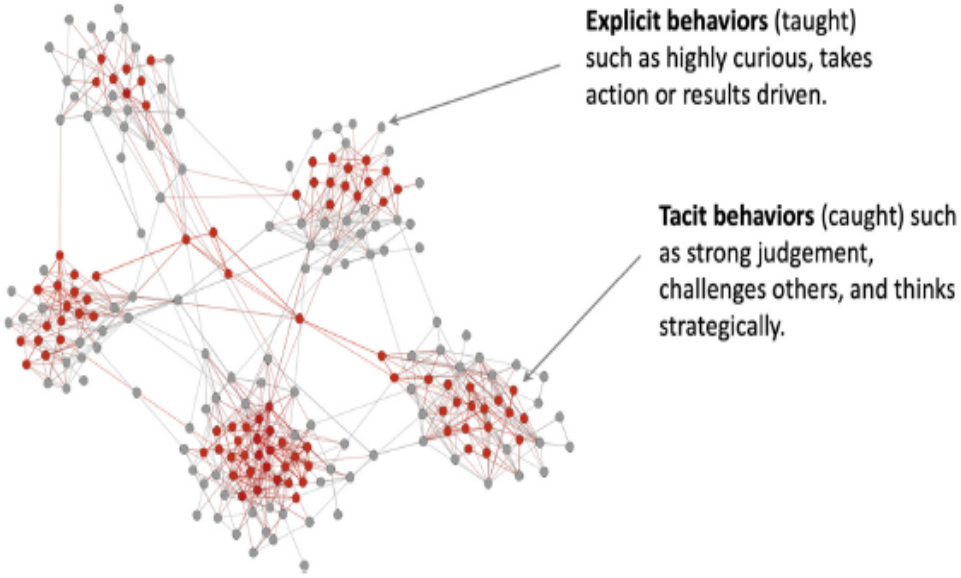
clusters, cultural contagion, and context issues, each representing mechanisms by which cultural behaviors are formed, transmitted, and shaped within the organization.

The first component, Culture Clusters, posits that organizational culture is not homogeneous but exists in discrete pockets or clusters within an organization's social network. These groups comprise localized employee interactions with similar behaviors, values, or cultural norms. The clustering effect is attributed to the interaction of informal day-to-day interactions that take place in small groups rather than a top-down imposition of leadership. Data visualizations from network analytics illustrate that certain cultural attributes, such as being results-driven, operating at speed, or exhibiting strong relational trends, tend to cluster into specific network parts. This highlights that culture evolves through localized reinforcement and repeated social encounters, particularly critical in hybrid environments where physical proximity can be limited (Arena et al., 2023).

The second component, cultural contagion, examines how cultural attributes spread from individual to individual through social connections. Similar to epidemiological models of contagion, research shows that an employee's cultural behavior influences not only their immediate contacts but also extends to colleagues with two or three grades removed from the network. This contagion effect is quantified by transmission percentages - such as relationship orientation being 40% more likely between direct connections and decreasing with distance from the network - and it is observed that it is maintained in face-to-face and remote environments(Arena et al., 2023, p.5-6). Strong ties are emphasized, as mutual and frequent interactions intensify the spread of behaviors and ensure that key cultural practices are embedded throughout the organization.

The third component, a matter of context, emphasizes the influence of situational factors on the formation of cultural groups and the diffusion of cultural attributes. The organizational context, including the purpose of specific groups, the network's density, and the network's overall energy, critically determines the effectiveness of cultural transmission. For example, different teams (such as services vs. engineering) may inherently value different cultural attributes, and dense silos within the network may facilitate or inhibit the spread of new cultural behaviors, depending on their alignment with existing norms. The research also indicates that contextual factors are especially pertinent in hybrid work cycles, where changes in connectivity patterns can lead to changes in cluster formation and contagion rates (Arena et al., 2023, p.5-6).

Figure 7: ONA Identification of Behavior



Source: Arena et al., (2023, p.4)

Figure 7 reveals the clustering of red nodes, which suggests that the behaviors taught are more likely to be centrally located, implying that they are commonly shared and explicitly encouraged within the group. Gray knots – explicit behaviors – are more dispersed, they are marginal. While tacit behaviors – the red knots – are developed individually and may not be evenly distributed across the network, they are picked up as strong judgments, challenging others, and strategic thinking. The network layout, under the ONA analysis, helps to understand how explicit and tacit knowledge is situated in social or organizational structures, affecting learning and the propagation of behavior (Arena et al., 2023).

This ability to map and influence social interactions positions the PA as a participant in the construction of organizational culture and social cohesion – a non-human organizational actor because, without this data-based analytical construction and extensive statistical combinations, such detections could not be performed by a human actor. The organizational social construction, understood here not as something monolithic but as a set of shared and continuously negotiated meanings, can be diagnosed and influenced through the Insights of the PA/ONA on engagement, communication, and collaboration. By identifying the factors that promote a positive, inclusive, and psychologically safe work environment and by allowing interventions to be more targeted and data-driven, the PA contributes to building a more integrated and resilient social fabric (Davenport, 2019; Wang & Katsamakas, 2019).

In this dynamic, the PA begins to operate as a mirror of the organization and as an actor that actively (re)configures its relational structures and processes. By selecting, processing, and presenting data, PA algorithms and systems mediate how the organizational reality is perceived and interpreted by its members (Tyagi & Pandita, 2022). They participate in defining "problems", legitimizing specific "solutions", and assigning value to the diverse contributions of human capital and organizational knowledge, influencing resource allocation and the design of interventions(Chang & Ke, 2024). This performance aligns with the Resource-Based View (RBV), which recognizes human capital and organizational capabilities – such as integrating and reconfiguring resources – as sources of competitive advantage (Barney, 1991; Teece et al., 1997). PA, by providing the means to understand and optimize these intangible resources and their interconnections, PA becomes an essential component of these dynamic capabilities, utilizing increasingly robust connective networks, mitigating the organizational silo effect (Ujwary-Gil, 2019).

1.4.5 People Analytics Challenges

According to what has been discussed in the previous sections, PA exerts the power to generate processes of significant change in organizations through new or innovative mental perspectives of top management, which are being fostered by academia and outside it (Tursunbayeva et al., 2018). The underutilization of PA adoption for specialists – academics and consultants – is a symptom of cause and effect due to cognitive and practical complexity, and PA is still a developing field of knowledge, which generally generates skepticism and a particular behavior of waiting for maturation (Rasmussen et al., 2024). As evidenced by (Minbaeva, 2021), this complexity and multiplicity of applications can lead to lethargy in the adoption of this innovative process (PA), as well as its disruptive tendency in SHRM (Sharma, 2020), enhanced by global megatrends such as the resignification and trend of flexibilization of work, the digitalization of business models, and the rapid and forceful insertion of artificial intelligence in organizational processes called algorithmic inevitability (Morgenstern et al., 2021). On the other hand, excessive enthusiasm can cause disruptive processes because the new HRM must be convinced that it can be transformed, and when it does, be aware that this process will reverberate in the organization as a whole. Table 6 seeks to summarize these challenges in a concise way.

Table 6: Disruptive Processes in HRM

Megatrends applied to HR	Disruption in HR	What paradigm shifts will be needed between HR research and practice to address potential disruptions?	
		Knowledge	Core Issues
Resignification of work	Working relationships will change in time, space, task type	It will be necessary to have a broader frame of reference, focus between actors and required actions, consider the context, as well as the business model	How will top management and HR be affected when the employee lifecycle is getting shorter and shorter? How to delineate, time, space and tasks of increasingly digitized and interconnected employees? How can leaders commit and engage their subordinates in an era of Big Data?
Digitisation	The organizational value proposition managed by HR must be reinvented	Understanding of HR conceptual change processes, where the customer is the focus	By enhancing organizational ambiguity and ambidexterity, how does HR tend to manage its consequences? How should the need for a "resumption" of the valorization of Human Capital be conducted beyond rhetoric through HR in digitalized businesses?
Algorithmic Inevitability	Manage the pressing need for AI insertion in HR processes, as well as its opportunities and challenges	A new data literacy process that contemplates all stakeholders co-opted by this algorithmic inevitability.	What can and should be automated? Should HR be responsible for this process? How to build credibility and legitimacy in AI-managed processes? How to strengthen organizational compliance, as well as ethics?

Source: Adapted from Minbaeva (2021).

One of the main challenges of PA in HRM is establishing a clear, quantifiable business issue that aligns with organizational goals, as this determines the direction and focus of analysis efforts, as well as the search for results (Loscher, 2021). Establishing a business issue in HRM means establishing a solid database and selecting appropriate analytical methods; consequently, the organizations – HRM, aligning everyone's interests and Stakeholders – must decide whether they want to focus on in-depth, case-specific analysis or create a reporting system focused on efficiency and scalability. Each approach requires different strategies and organizational structures for successful implementation, which can complicate the HRM integration process with an analytical perspective, and the different intra- and inter-organizational interests (Loscher, 2021). In addition, PA demands an environment that welcomes change as a challenge to be overcome. Organizational resistance to change, as widely discussed by Weick and Quinn (1999), can lead to misalignment in the core of the AP project, obfuscating business objectives and the measurement of their impact on

organizational performance, which can be caused by a lack of data quality and availability, insufficient technological infrastructure, and limited analytical skills among HR personnel (McCartney & Fu, 2024). This complexity tends to generate major conflicts in the organization, as it impacts the organizational dimensions – employees and the organization itself in a broad way. Table 7 seeks to elucidate these risks in each dimension and proposes specific recommendations.

Table 7: PA Risks.

Dimension	Risks	Recommendations
Employees	Enhance prejudice and discrimination of the psychological or social profile	Transparency and fairness
	Model Behavior	Proportionality, Protection, Consent Guidelines and Data Rights
	Reducing people in numbers	
	Create a situation of inconvenience and/or insecurity for performance	Inclusion of all organizational stakeholders in relation to organizational culture
	Threat to privacy or autonomy	
Organizational	There is no history of legal precedent and risk analysis strategies are not applicable	Legal compliance and formal guidelines
	Feeling of surveillance, monitoring and collection of sensitive data from people	
	Regulatory compliance	Business model matters
	The competence of HR to judge what is acceptable and what is possible	

Source: Adapted from Tursunbayeva et al. (2021).

Integrated interests, a PA project demands data availability and quality, which are essential (Hassenstein & Vanella, 2022). Organizations often struggle to access reliable, relevant data for meaningful insights (Okatta et al., 2024). Concerns about data privacy and security further complicate implementation, requiring robust measures to protect employees' confidential information(Holz et al., 2024b). Integrating advanced technologies into HR practices introduces ethical dilemmas and the risk of algorithmic bias, which can undermine employee trust and organizational values (Cho et al., 2023; Rehman, 2023). In addition, there is a significant skills gap in analytical capabilities in HR departments, requiring investment in

training and development to build the necessary expertise (Holz et al., 2024). Addressing these challenges, therefore, involves developing strategic frameworks, fostering a culture of continuous learning, and establishing ethical guidelines to ensure that PA projects add value while maintaining organizational integrity (Zubair & Erdoğdu, 2024).

1.4.5.1 Introduction to the Fundamentals of Data Quality

The efficacy and transformative potential of the PA in contemporary HRM are intrinsically linked to a robust foundation: data quality. Ignoring this critical dimension compromises the validity of the insights generated, undermines the credibility of strategic decisions, and can perpetuate inequalities. The specialized literature, including the contributions of Hassenstein and Vanella (2022); Jones (2024); Y. W. Lee et al. (2006), converges on a set of assumptions that define, measure, and guide the journey towards excellence in data quality.

1.4.5.1.1 Data Quality: Beyond the Technique

Data is currently a valuable asset and should be treated as such. So much so that we can say that for organizations that encompass industry 4.0 technologies, a concept postulated by Klaus Schwab (2016), that organizations will be integrating sources of three dimensions that will determine the 4 IR, which are: the physical dimension, the digital dimension and the biological dimension. Specifically in our context, the digital dimension encompasses; IoT – the internet of things; AI – the algorithmization process discussed here; The expansion of digital platforms, as we currently see and to which the new HRM is directly linked, and finally, the concept of blockchain. Thus, Kumar et al. (2020, p. 766), state that "[...]Data is a new oil and plays a vital role in the smart factory, providing new meanings [...]".

Strong et al. (1997) highlight the importance of the work of IT professionals, since they deal with intrinsic problems of accessibility and quality of organizational data. The provision of high-quality data – for Strong and collaborators – is about how much that data will contribute to the goal of your analysis. Therefore, it is to differentiate between technical accessibility – data modeling – and the broader concern about how to access such user or consumer data. The key, therefore, is to determine what constitutes a given with or without quality (Hassenstein & Vanella, 2022). Therefore, data quality transcends mere technical correctness, being fundamentally defined as fit for purpose. This implies that data must not only be free of flaws but, satisfy the specific needs of its users and the application context.

In this conception, trust emerges as a vital component; Data is only truly quality if decision-makers trust it to support actions and strategies (Mayer-Schönberger & Cukier, 2013). The nature of data quality is undeniably multidimensional. Multidimensionality is postulated by Hassenstein and Vanella (2022) from two perspectives: (1st) User-centered characterizing this dimension with the categories: Intrinsic, Contextual, Representational, and Accessible; And the (2nd) dimension as generalists characterized by: Inherent and System Dependent, detailing key dimensions such as Accessibility, Accuracy, Completeness, Consistency, Timeliness, Relevance, Reliability, Timeliness and Validity. Jones (2024), reinforces this view, highlighting Utility, Precision, Validity, Uniqueness, Completeness, Consistency and Timeliness, emphasizing that the optimal level for each dimension varies according to the intended use.

Lee et al. (2006), in turn, operationalize the dimensional characterizations of the data, through their evaluation methodologies. To manage and improve data quality, a systematic and continuous evaluation is indispensable and this process allows diagnosing problems, justifying investments and monitoring progress. Hassenstein and Vanella (2022) emphasize the importance of FAIR principles and Good Scientific Practices, as well as a formal Data Quality Assessment (DQA) process. Jones (2024) proposes a multi-pronged approach to measuring quality, including collecting consumer feedback on trust in data, running tests and continuous profiling through data observability tools, and treating data issues as formal incidents to increase visibility and a sense of urgency.

More specifically, Lee et al. (2009) offer a detailed methodological framework, which includes: First, the subjective evaluation of the data**:** Carried out through research such as the *Information Quality Assessment (IQA)*, which captures the perceptions of *Stakeholders* (collectors, custodians and consumers – the "3Cs"); According to the objective evaluation of the data**:** Using quantifiable metrics, including adherence to Codd's (1990) integrity constraints. Codd's (1990) integrity constraints consist of entity integrity, referential integrity, domain integrity, and column integrity. Codd introduced a fifth, general-purpose category, which he called business rules. This category is intended to capture health rules that are specific to an organization. The metrics developed to measure adherence to Codd's first four health constraints can be viewed as task-independent metrics. They reflect the states of the data without contextual knowledge of their application; The third methodological step encompasses comparative and diagnostic approaches: This approach seeks to cross-reference subjective and objective evaluations to identify discrepancies and perform gap analysis

against *Benchmarks* or between different organizational roles. Finally, the fourth methodological step of Lee et al. (2009) consists of verifying the integrity of the data that are embedded in the process: A process that aims to integrate the integrity analysis into the Total Data Quality Management (TDQM*)*: define, measure, analyze, and improve, in a continuous, recursive process (Codd, 1990).

If data is "the new oil", in the era of big data and AI, for an PA project, high data quality has gone from being a differentiator to becoming a strategic imperative. According to Jones (2024, p. 5-9), even the most sophisticated AI models critically depend on quality data for their effectiveness, and an inferior model with superior data will always outperform a superior model with inferior data. For the PA, this translates into the ability to generate reliable predictive and prescriptive insights about human capital. Neglecting data quality comes with high and multifaceted costs – financial, operational, and reputational.

Jones (2024) points to annual losses in the millions of dollars and significant impacts on companies' revenues. Lee et al. (2009, p.29) already indicated that data quality problems could cost the "[...]companies between 8% and 12% of their revenue, affecting customer satisfaction and competitive advantage [...]". In addition, legal and regulatory compliance such as the Sarbanes-Oxley Act makes data quality a matter of corporate governance and risk (Lee et al. 2009). The proposition that data quality is imperative today transcends types and areas of study. Thu Nguyen, Hong-Tri Nguyen, and Tu-Anh Nguyen-Hoang (2025), reinforce and deepen the issue of the centrality of data quality, taking as an analytical basis the implications in the educational field. The authors argue that, especially in sectors such as education – and, by extension, in any data-driven domain such as HRM – data quality has profound, though often underexplored, implications. The survey systematically covers everything from requirements analysis and standards development to the implementation of tools for monitoring and improving Data Quality (DQ) in Big Data workflows.

This focus resonates directly with the logic of PA project implementation, underlining that the impact of QD on analytical results is a key issue, and that the absence of effective data quality management represents a primary obstacle to achieving the value promised by PA. The identification by Nguyen et al. (2025) of challenges inherent to the volume, variety, speed and veracity of Big Data, corroborates the premise about data as an instrument, providing an overview of the technical and conceptual difficulties that need to be overcome for data to serve as a reliable instrument for PA, that is, that this organizational actor contributes consistently to the result.

Additionally, the work of Nguyen et al. (2025) advances by proposing a multi-stage Big Data Quality Management (BDQM) framework, which emphasizes data quality assessment, centralized governance, and, improvement techniques enhanced by AI and distributed computing. This proposition reinforces the pressing need for the "new HRM" to build or constitute a deeper analytical mindset, aiming at an environment in constant hybridization where humans and algorithms collaborate to ensure informational reliability. The authors' emphasis on the need to establish standardized evaluation standards, metrics, and methodologies for measuring and evaluating DQ, as well as the exploration of techniques and tools for their improvement—including automated data cleaning, enrichment, and monitoring—offers a practical and technologically informed counterpoint to the challenges of datafication. The view that data quality frameworks can be significantly improved by AI aligns with a new sociological perspective of "algorithmic inevitability", and that data quality management itself tends to become an increasingly sophisticated socio-technical practice mediated by algorithms (Dasari & Sujatha, 2023).

The purpose of enterprise data governance emerges as an indispensable function. According to Madhavan (2024), enterprise data governance has emerged as a critical function for organizations looking to maximize the value of their data assets while mitigating risks associated with data management. This role becomes even more pressing at HRM, where data is inherently sensitive and decisions directly impact employees' lives and careers. As proposed by Madhavan (2024), it is articulated around key components, including: Clear policies, well-defined roles and responsibilities, and the establishment of data quality standards, which is a unilateral decision of management, a proposition reinforced by Alhassan et al. (2016), under the starting point of data governance is the establishment of high-level policies and standards for data management across the organization. For the PA, this implies specific guidelines on the collection, use, storage, and disposal of employee data, always aligned with the strategic objectives of HRM and the organization as a whole. The creation of a data governance policy document that outlines the organization's approach to data management is a key step.

The objective is to maintain clarity in role assignments. Madhavan (2024) and Alhassan et al. (2016), identify key roles such as *Data Owners* – senior executives responsible for strategic decisions about data assets; *Data Stewards* – experts who oversee the day-to-day management of data in your domain – in a PA project it could be HRBPs or PA specialists; *Data Custodians* – IT professionals responsible for the technical aspects, and a cross-functional Data Governance Committee. Clearly designating these roles ensures

accountability and ownership of data throughout its lifecycle. Panian (2010) points out that the data stewards to maintaining the quality and meaning of data, acting as a bridge between IT and the business areas, promoting organizational integration and iteration.

Data stewards must be mindful of data legal compliance. With the growing and complex tangle of data protection regulations, such as the LGPD (Law No. 13,709, 2018) in Brazil and the GDPR in the European Union, it is one of the main drivers of the creation of data governance (Voigt & von dem Bussche, 2017). Madhavan (2024) states that successful data governance, in the context of big data, fundamentally depends on compliance with regulatory standards. The GDPR, for example, establishes principles such as legality, fairness and transparency, purpose limitation, data minimization, accuracy, storage limitation, integrity and confidentiality, and accountability (Voigt & von dem Bussche, 2017).

For the PA, this implies challenges such as obtaining organizational and individual consent for the use of employee data in predictive analysis or profiling, which uses specific data. Another aspect of this management is data minimization, which means collecting and processing only the data strictly necessary for the specific purposes of the proposed PA project. Another important challenge lies in algorithmic transparency**,** that must ensure that employees understand how their data is used in PA algorithms and how decisions are made, a considerable challenge given the "black box" nature of many AI models (Burrell & Fourcade, 2021). A PA project must be guided by the premise that data has holders. Therefore, the use of organizational data must ensure that that employees can exercise their rights of access, rectification, opposition and, in certain cases, portability or deletion of their data (Kalpokas, 2019).

In-depth analysis of data quality, as explored earlier, reveals its multidimensional nature and its irreplaceable role as a foundation for any PA initiative that aims for strategic relevance and reliability. Having established the premise that data, beyond its technical correctness, must be fit for purpose, inspire trust, and be managed under robust and ethically conscious governance, a subsequent and equally issue emerges: the very process of transforming the complex nuances of human behavior and social dynamics into quantifiable data. This process, known as datafication, represents one of the challenges and, at the same time, one of the cornerstones in the consolidation of the PA as a new organizational actor, capable of quantifying the social. The next section will delve into the implications, challenges, and potentialities inherent in this endeavor, investigating how HRM navigates the tension between numerical objectivity and the subjectivity of human experience in the workplace.

1.4.5.2 Datafication, quantifying the social

The emergence of PA as an active organizational actor in HRM intrinsically depends on a fundamental and transformative process: datafication. It is important, initially, to establish the conceptual distinction between digitization and datafication, terms often used interchangeably, but which encompass different meanings and implications for organizational analysis. Digitalization, as elucidated by Manresa et al. (2024), refers primarily to incorporating and integrating digital technologies into an organization's processes and operations, intending to transform how value is created and captured. This process – digitalization – encompasses the implementation of advanced technologies and the consequent organizational changes aimed at optimizing operational efficiency, the quality of products or services, delivery times, and associated costs. Going beyond the simple automation of pre-existing tasks, digitalization, as pointed out by Cöster et al.(2023), has the potential to sustain and broaden the understanding of the complex interconnections that permeate the organizational structure, influencing everything from the management of goals and strategies to risk assessments, production processes, commercialization and, fundamentally for the scope of this thesis, people management.

Distinct from digitalization, datafication, a term coined by, Viktor Mayer-Schönberger and Kenneth Cukier in 2013, refers to transforming social action or facets of real life into quantified data. This conversion allows for the subsequent analysis and use of this data to inform, characterize, and optimize decision-making processes (Thorpe & Sellar, 2023). It is a progressive and continuous transformation phenomenon that has deeply impacted organizations, governments, individuals, and society. Datafication is consolidated on digital platforms that (re)structure, drive, and give new meanings to contemporary social and commercial practices. This occurs, to a large extent, through digital transactions that make the collection, analysis, and use of data defining characteristics of today's society (Kalpokas, 2019; Mayer-Schönberger & Cukier, 2013).

Although relatively recent, this term describes a practice with deep historical roots which acquires new dimensions and urgency in the age of Big Data and AI. Datafication refers to the "[…]transformation of aspects of life and work into digital data for subsequent analysis and monetization[…]" (Goniwada, 2023, p.1); allowing social phenomena, behaviors, interactions, and even individual characteristics to be represented numerically, making them amenable to algorithmic processing and statistical analysis. For HRM, datafication opens up the promise of more objective, evidence-based, and predictive management, but it also raises

a critical debate about its epistemological, ethical, and social implications (Granulo et al., 2023).

The underlying premise of datafication in HRM is that by quantifying aspects of human capital – from performance and engagement to potential and relationship networks – organizations can achieve deeper insights, optimize decision-making processes, and ultimately achieve greater efficiency and competitive advantage. Gobble (2017, p. 59) notes that the "[…]wave of digitalization[…]" transforming research and development will also reach across organizational functions, including HR, driving the need to turn "[…]HR practices into data that can be analyzed". This movement is echoed in the views of Mayer-Schönberger and Cukier (2013), who argue that the ability to generate and analyze vast amounts of data fundamentally changes how we understand and interact with the world. In the context of HRM, this transformation entails "[…]seeing[…]" and managing employees through a new quantitative lens, where "[…]metrics and analytics become the primary language for understanding and evaluating human value[…]"(Rousseau & Barends, 2011, p. 223).

However, datafication is not a neutral or merely technical process. How Mejias and Couldry (2019, p. 2) argue that datafication is a "[…]new form of colonialism[…]" insofar as it extracts value from social life, converting it into capital through processes of data appropriation. While the analogy with colonialism may seem strong for an organization's internal context, it alerts us to the power dynamics and potential expropriation of personal information when the boundaries between the personal and the professional become increasingly blurred under the gaze of datafication. The very definition of what is given and what is "[…]personal information[…]" becomes a field of dispute, as he points out Mai (2016, p. 193), which questions whether privacy can be adequately protected when personal information is continuously dated and aggregated into systems of Big Data.

The ambition of quantification or datafication imputes limits and epistemological implications for HRM. In an era of "algorithmic inevitability" and the exponential supply of new data management technologies, it is almost natural that there is a particular ambition to quantify the social within the scope of HRM, which, although seductive, faces significant epistemological challenges. Translating complex and multifaceted phenomena such as "engagement, organizational culture, or leadership potential" into numerical metrics is, by nature, an act of simplification and reduction (Algazo et al., 2024; Henninger, 2024). Cristina Alaimo and Jannis Kallinikos (2024, p.35) argue that digital data, despite its apparent

objectivity, are "[…]constructs that result from specific choices about what to measure, how to measure, and how to interpret measurements[…]". These choices are often influenced by theoretical assumptions, technological limitations, and the organization's goals, meaning the data never speaks for itself (Alaimo & Kallinikos, 2024).

Applying AI and machine learning (ML) algorithms to analyze this datafied data intensifies these issues. Goniwada (2023) describes how AI and ML are used to identify patterns, make predictions, and automate decisions from the data collected. However, suppose the input data reflects historical biases or is an incomplete representation of reality. In that case, algorithms can perpetuate or even amplify these biases, leading to discriminatory decisions in recruitment, promotion, and compensation (O'Neil, 2016). Galliers et al. (2017, p 186) warn of society's "[…]seemingly tacit and unconscious acquiescence to this increasingly prevalent phenomenon[…]",of algorithmic decision-making, often without a clear understanding of how these decisions are reached.

For the PA, this implies the need for a healthy skepticism regarding the apparent objectivity of numbers, a commitment to the transparency and explainability of analytical models, and their implications for organizational social relations. The Theory of the Social Construction of Reality (Berger & Luckmann, 1966) offers a valuable framework for understanding how metrics and categories created through datafication can become objectified realities within the organization, shaping perceptions of employees and influencing management practices. The ANT in turn, allows us to analyze how the datafication systems themselves – the algorithms, the databases, the interfaces – act as non-human "actants" that mediate and (re)configure what is considered real and manageable in the field of HRM (Latour, 2005).

By unveiling their employees relational and social content through data and datafication, organizations tend to confer greater power and control. From a historical perspective, Cieslik and Margocsy (2022, p 353) demonstrate how the; "[…]extraction, accumulation, and commoditization of personal data and resources by colonial empires, national governments, and commercial organizations[…]", served as mechanisms of power and control. Although the contemporary organizational context differs, the fundamental dynamic of transforming information into a controllable asset remains.

Datafication can lead to new forms of surveillance and monitoring in the workplace, where every click, every interaction, and even the tone of voice can be turned into data points for analysis (Zuboff, 2019). Lavazza and Farina (2023, p 844) discuss how the growing

"infosphere," fueled by datafication, has "[…]enormous consequences on workplaces, workers, as well as the decision-making processes necessary for the functioning of future societies." They argue that integrating biological and social domains into an infosphere suitable for AI technologies will require "extensive datafication," as data is the basis of the logical-mathematical codes and models that drive and guide AI. In the context of HRM, this can translate into systems that monitor past performance and predict future behaviors, such as the likelihood of an employee leaving the company or their potential for developing new competencies (Lavazza & Farina, 2023).

While potentially beneficial for workforce planning, this predictive capability raises ethical questions about individual autonomy and the risk of management by anticipation that can limit employee opportunities based on algorithmically calculated risk or potential profiles (Jansen et al., 2023). Data governance, discussed in the previous section, becomes a way not only to ensure the quality and security of data but also to establish the ethical limits of its use and to protect the rights of employees in an increasingly datafied environment (Tursunbayeva et al., 2021).

Datafication is profoundly reshaping the practice of HRM. Gobble (2017, p. 60) suggests that HR is moving toward a model where:"[…]data-driven insights inform talent strategy and operational decision-making[…]". HR professionals must develop new analytical competencies and a critical understanding of datafication tools and techniques. However, there is a risk that an overemphasis on quantification could lead to a dehumanization of people management, where the complexity and individuality of employees are obscured by averages, scores, and control panels (Brison et al., 2022). Dysfunction that tends to resistance to datafication, can arise, both on the part of employees, who may fear surveillance and loss of privacy as well as by HR professionals themselves, who may see the quantitative approach as a threat to their traditional expertise based on intuition and interpersonal relationships (Mai, 2016).

Overcoming this resistance requires transparency, clear communication about the goals and benefits of PA, and the involvement of employees in the design and implementation of datafication systems (Rigamonti et al., 2024; Vargas, 2015). Ultimately, datafication in HRM is a dialectical process of social resignification. It offers the potential for more informed, equitable, and effective talent management. However, it also carries the risk of intensifying control, exacerbating biases, and reducing the human experience to mere data points. The ability of organizations to navigate this tension, harnessing the benefits of datafication while

mitigating its dangers, will depend on an ongoing commitment to critical reflection, ethical governance, and valuing the human dimension at work (Mayer-Schönberger & Cukier, 2013; Pellegrino et al., 2019).

1.4.5.3 Objectification and Dehumanization of Data

The rapid integration of AI into work environments has represented a transformative force in modern society, not only as an extension of previous industrial changes, but as a distinct revolution as termed the 4th Industrial Revolution (IR) (Schwab, 2016). (Li; Hou and Wu (2017), Pillai et al. (2021) corroborate the thought of fusion between digital, physical and biological and considers the 4th IR as a new organizational software that will have a strong impact on all organizations with different forms and intensities. These impacts can be described as:

Table 8: Cause and effect of the 4th IR

Causes	Effects
Smart Industry	Adoption of algorithmic systems that generate greater productivity, agility, physical safety for workers, lower cost
Customization	Custom processes based on clusters, predictive processes
Integration	Greater knowledge, greater satisfaction of all stakeholders
Adjusting business models	From linear design to exponential design
SHRM	New Skills, Skills, and Continuous Qualification

Source: Research (2025).

This argument situates AI within the broader socioeconomic paradigm shift, in which conventional human-centric processes are increasingly replaced by algorithm-driven systems (Johnson, 2021; Reinhold & Reuter, 2022). Datafication tends to represent a reductionist, synthetic approach that contributes to self-objectification, whereby workers begin to internalize a data-driven external perspective that objectifies and dehumanizes their capabilities and worth (Alessandro et al., 2025).

Organizations that integrate AI technologies into their broader recruitment and work environments must address one of the key implications, which is to carefully design AI systems to avoid negative psychological consequences on job seekers and their employees. Empirical findings indicate that interactions with AI recruiters can decrease self-efficacy, increase self-objectification, and reduce beliefs in candidates' free will. The results suggest that the deployment of AI in recruitment processes can have unintended adverse impacts on candidates' self-perception, potentially affecting their overall job performance and well-being,

which also extend to broader perceptions of threat, objectification and dehumanisation (Alessandro et al., 2025; Wu & Wang, 2025).

Therefore, the knowledge and/or perception of being treated as an object, or being objectified, goes back to the Kantian definition of demotion of a person, a being with humanity, to the status of an object (Belmi & Schroeder, 2020). Although often associated with sexual objectification, as highlighted by Wu and Wang (2025), there is a need for broader conceptualization to understand its impact on various social dynamics, including labor relations. In the organizational environment, objectification manifests itself when individuals are perceived and treated more as means to achieve organizational ends – resources to be managed – than as human beings with agency, emotions and experiences of their own. This phenomenon is exacerbated in work contexts because these often encourage more calculative and strategic thinking, where decisions are made based on the computation of costs and benefits (Belmi & Schroeder, 2020).

The concept of objectification is related to this research, as it seeks to understand how AI tools or projects change the organizational daily life of contemporary HRM, which is increasingly hybridized. In this regard, the contributions of Iddo Landau (2007), provides an in-depth comparison of two distinct but interrelated accounts of objectification. The first, which is juxtaposed with Martin Buber's relational theory - where objectification is understood mainly in terms of I-thou versus I-it relations, the second perspective, uses the contemporary ethical-legalist notion of objectification (Landau, 2007).

The analysis elucidates how Buber's structure, which is imbued with a religious-spiritual and emotionally connective orientation, contrasts sharply with the contemporary approach that emphasizes ethical and moral issues, such as the denial of autonomy, instrumentality, and other characteristics that result in the devaluation of human subjects. A central contribution of Iddo Landau's Article lies in the detailed exploration of the differentiated characteristics that make up each account of objectification. The contemporary account, based heavily on the influential work of Martha C. Nussbaum (1995), is characterized by elements such as denial of value, rationality, and subjectivity. In contrast, Buber's approach emphasizes relational characteristics such as non-reciprocity and the Partial nature of reciprocity in I-thou and I-it relationships (Landau, 2007).

The proposed discussion is of great importance because it critically evaluates these disparate attributes, postulating that while some aspects of Buber's analysis may be incomplete for the modern framework (e.g., its inclusion of aspects such as relating through rules and past

experiences), other features—most notably the concept of self-objectification—could be successfully adapted to enhance contemporary theory. This cross-fertilization illuminates how the strengths of each perspective can mutually enrich the other (Landau, 2007).

Furthermore, Landau (2007) portrays the seven central characteristics of objectification, applicable to the work context, and some examples outside it, which tend to be enhanced by datafication. Characteristics such as instrumentality, that is, using a person only to satisfy purposes or needs, are considered by many to be the essential characteristic and is reflected in the term human resources. Other facets include treating the person as devoid of agency – unable to act or plan autonomously, or the person devoid of experience – unable to feel emotions or pain, or devoid of autonomy – restricting their freedom of choice, or more forcefully, as a property – such as a commodity, which may be fungible – interchangeable with others, and, a violable entity – disregarding its physical well-being, as highlighted in Table 9 (Belmi & Schroeder, 2020).

Table 9: Characteristics of Objectificationn

Characteristics of objectification: Treating a person as	Definition	Example in the Context of Work	Example in the Non-Work Context
Instrumental	Using a person only to satisfy their purpose or need	Talking to a coworker just because they have a resource you need	Talking to a person only to satisfy your own sexual needs
Lack of Agency	Treating a person as if they cannot act, plan, and exercise self-control on their own	Doubting a coworker's ability to competently perform their part of the job	Doubting your partner's ability to competently care for the home or children
Lack of Experience	Treating a person as if they can't feel pain, pleasure, or emotions	Forgetting that a coworker is hungry and expecting him to work during lunch	Forgetting that a migo gets hungry and waiting for him to run his errands at lunchtime
Lack of autonomy	Reduce a person's freedom of choice	Selecting Retirement Plan or Employee Stock Options	Making fidelity decisions for your family
Property	Commodifying a person	Forced child labor	Child trafficking
Being Fungible (that can be run" or "that can be spent)	Treating a person as an asset interchangeable with another person	Replace one employee with another with similar attributes	Replacing a sexual partner with a putro with similar attributes
Being Violable	Disregarding a person's physical well-being	Failing to provide workers with fair work standards, such as meals, breaks, and overtime	Domestic Abuse

Note. Some of the above examples may exist in work contexts or not (e.g., sexual objectification is also found in the workplace), but we categorize the examples based on where they seem to occur most often, based on previous research (see main text for more details). Some of the examples above may also reflect multiple characteristics of objectification (e.g., expecting employees to work during lunch is an example of treating employees as if they have no experience and violating their physical well-being).

Source: From Belmi & Schroeder (2020, p. 2).

Datafication, by transforming aspects of life and work into digital data for analysis and monetization, can intensify each of these facets, for example, by treating employees as interchangeable data profiles or by making decisions about them based on algorithms that ignore their subjectivity (Goniwada, 2023). Academic attention has grown in recent years around the concept of organizational dehumanization (OD), defined as the employee's perception of being treated as a tool or an object by his or her organization (Alaimo, 2022). This interest is justified by the common nature of the phenomenon and its negative implications both for employees, such as decreased job satisfaction and consequent reduction in performance, and for organizations in achieving their goals (Brison et al., 2022). Studies suggest that OD can be induced by formal bureaucratic rules and procedures, with the red tape – bureaucracies, unnecessary, ineffective and onerous organizational rules – an important driver of this perception of dehumanization by employees (Lagios et al., 2023; Stinglhamber et al., 2023).

In the contemporary context, this perception of being treated as an object or instrument gains new nuances with the growing prominence of datafication (Mejias & Couldry, 2019). Datafication involves a process of abstraction, selection, and transformation of everyday reality, of individuals, and organizations. Rob Kitchin (2014) argues, data does not exist naturally, but emerges from interactions, which through a process of capture, of what is taken from interactions between people, things and processes are stored and analyzed. For analytical conceptions, specialists are needed so that the capture, storage and analysis are carried out, according to the specific objectives of each organization. Still, for Kitchin (2014), without these specialists, the results of data analysis can be poor, especially when dealing with confidential data about human behavior. This highlights that expert involvement is not only in articulating problems, but also in evaluating the results, emphasizing that data scientists are often equipped to explain the “what” of the data, but not the “why” of it.

Datafication practices are described as a means of accessing and monitoring people's behavior, making it a new accepted scientific approach to understanding the social behavior of the individual in the organizational context (Mejias & Couldry, 2019). This new way of interpreting the world involves a vision of datafication processes that, from the perspective of Shoshana Zuboff (2019, 2021), represents a new stage of surveillance capitalism, where human experience becomes the raw material to produce behavioral data used to influence and even predict our actions. Thus, when applied to the work environment, datafication, especially

through the use of PA projects, is directed to the quantitative analysis of people's conduct and psychological constitution, aiming, understanding, developing, codifying and standardizing their behavior and character (Weiskopf & Hansen, 2023b).

Datafication, as a data-driven approach, can inadvertently promote objectification and dehumanization by reducing the complexity of the human experience to measurable data points (Schweitzer & De Cremer, 2023). The underlying logic of prediction and control that guides datafication and the application of AI can lead to a treatment of individuals as entities to be managed and optimized simply based on the metrics and/or outcome data that they may present, rather than the conception that humans are complex subjects with subjectivity, values and intrinsic needs. The desire to replace human tasks with artificial intelligence, opposing technical controls and the rational action of machines to human affection, is a distinctive mark of our era, characterized by exponential algorithmic interference and predominantly data-based governance (Gal et al., 2020).

This type of behavior, combined with the adoption of an PA project, can create a vicious cycle of ethical challenges (Bar-Gil et al., 2024; Gaur & Gupta, 2022), highlighting algorithmic opacity (Burrell, 2016; Giermindl et al., 2022; O'Neil, 2016), a and/or with comprehensive datafication, expand the practice of nudging – a form of impulse or subtle influence of the human decision-making process – which can limit people's ability to cultivate their subjectivity, their creativity as a human virtue (Parmer, 2023; Sunstein, 2014).

This generalized datafication privileges an evidence-based truth, even if decontextualized, to the detriment of other moral values such as empathy, wisdom, or justice. Reality, particularly moral reality, is always richer and more complex than any concrete data-based representation (Noponen et al., 2023). In other words, these negative effects are more likely to manifest themselves when organizations frame PA as an epistemologically superior technology to its human counterparts (Gal et al., 2020).

Thus, if, on the one hand, datafication and the use of PA have the potential for a more competitive and analytical HRM (Davenport et al., 2010; Mahdia, 2024); on the other hand, can contribute to the perception of dehumanization and organizational objectification, where the employee feels like an easily replaceable tool or cog whose role is limited to applying rules, now codified and automated by algorithms. This reduction of the individual to a set of data represents a form of objectification, in which subjective and contextual aspects are devalued or ignored in favor of quantifiable metrics (Vidhya & Sangeetha, 2024).

The conception that we are socially pressured by the advent of the growing algorithmic inevitability, driven by datafication, inserts the PA not only as a tool, but as a new organizational actor with the ability to shape the daily reality and its work relations in a broad context, where it can be said that robots are here! (Williams, 2020, p. 51-75). To understand the risks, challenges, and development opportunities arising from the insertion of AI in our daily lives, it is essential to investigate how the hybridizing aspects of human-machine interaction manifest themselves and how the PA acts as an organizational actor. In this sense, to go beyond the mere description of phenomena, examining the knowledge practices and associations – human and non-human – that build this new organizational reality. It is necessary to analyze how knowledge about work and workers is produced, validated and put into circulation through these algorithmic systems. This leads us to deepen the review of the literature on the sociology of knowledge, seeking its integration with ANT, as proposed by Latour (2005), to analyze the complex web of agency and interdependence in this dataficated scenario.

1.5 The Sociology of Knowledge

The sociology of knowledge, articulated by Peter L. Berger and Thomas Luckmann in their seminal work The Social Construction of Reality (1966), represents a fundamental shift in understanding how knowledge and reality are intertwined through social processes. This approach posits that reality is not an objective entity but is socially constructed through human interaction and shared meanings, a concept that has become fundamental in social constructivism(Abels & König, 2010).

Berger and Luckmann's theory draws on earlier philosophical and sociological traditions, notably those of Max Scheler and Karl Mannheim, which emphasized the social origins of knowledge and the influence of social structures on ideational realms (Whiteley & Seyd, 1996). As particularities of the concept of Sociology of Knowledge, it can be inferred that the “social” comprises interactions and iterations, conviviality, and broad and unrestricted communication between the parts, which enables the constitution of reality. Therefore, society results from a socially constructed reality – relative and symbolic – and it is therefore up to the sociology of knowledge to understand how such a process is constructed. Furthermore, the constructed society tends to be presented to the individual in a dual way: a) objectified – based on a process of institutionalization and/or legitimation, and b) subjectivized – based on the capacity for internalization and/or identification(Lopes & Thomé, 2021; Segre, 2016).

For Berger and Luckmann (1966), the process of institutionalization is central to the construction of social reality. It is explained with the consideration that repeated human actions when habitualized and typified, crystallize into institutions that govern behavior and provide stability to social interactions. These institutions, ranging from legal norms to cultural rituals, are historically contingent and objectively real, as evidenced by their ability to survive and shape the subjective experiences of individuals, thus the dialectical process of externalization, objectification, and internalization forms the methodological backbone of its argumentative construction, demonstrating that the shared accumulation of knowledge transmitted through language and social practices sustains society's objective existence.

Nevertheless, the work diverges from traditional epistemology. It focuses less on the mechanisms of perception and more on how social affiliations shape what is considered authentic or real (Whiteley & Seyd, 1996). This perspective underscores the reciprocal relationship between knowledge and social reality, suggesting that what is known and how it is known are deeply rooted in social contexts and interactions (McCarthy, 2019). Thus, the sociology of knowledge, as redefined by Berger and Luckmann, has influenced several fields, including sociology, anthropology, and even literary criticism, providing a framework for analyzing how social dynamics influence the construction and dissemination of knowledge (Da Silva Moreira, 2022; Whiteley & Seyd, 1996).

From the aspect of legitimation, the Sociology of Knowledge has been fundamental, and academic discussions on the legitimation of social realities have broadened the scope of how cognitive and normative interpretations of meaning are essential in understanding the processes of social evaluation (Nicolae, 2025). Despite its transformative impact, some critiques suggest that its theory's assumptions may not fully align with contemporary social realities, indicating the need for continued evolution in the field (Sica et al., 2016). However, Berger and Luckmann's sociology of knowledge remains a cornerstone in understanding the complex interaction between society and knowledge, emphasizing the constructed nature of reality through social processes and interactions (Fraser & Turcan, 2025).

1.5.1 The Sociology of Knowledge: opinions and beliefs in modern societies

The sociology of knowledge significantly influences the formation of opinions and beliefs in modern societies by examining the interaction between social conditions and the ideational realm, which includes knowledge, ideas, and mindsets. As articulated by early theorists, this subfield of sociology posits that knowledge and reality are mutually constituted

in social contexts, meaning that truths and beliefs are not independent of historical and social processes (Fraser & Turcan, 2025;McCarthy, 2019).

The sociology of knowledge explores how intellectual products and processes are linked to socio-historical contexts, material and cultural conditions, power relations, and language, thus shaping society's ideologies, beliefs, and values (Camic, 2000). In modern societies, which are increasingly characterized as knowledge societies, the role of knowledge has ceased to be a mere economic asset to become a central element in social relations, influencing political agendas and societal norms and values (Stehr, 2023). This shift underscores the importance of understanding how knowledge is socially constructed and distributed through institutions such as schools and media, which play a role in shaping public opinion and belief systems.

In southeastern Europe, social networks and cognitive frameworks further illustrate how public opinion is shaped by ingrained beliefs and communication dynamics, emphasizing the need for transparent and ethical communication to build public trust (Stojcevski et al., 2024). Furthermore, the sociology of knowledge emphasizes the functional and adaptive characteristics of ways of knowing, suggesting that beliefs and opinions are shaped by social conditions and serve specific social functions, such as maintaining social order or facilitating change and innovation (Fuller, 1998). This perspective highlights the complex interrelationships between knowledge, power, and social and economic interests, illustrating how the construction of beliefs is a product and a contributor to the social structures in which they arise (Stehr, 2023). Concerning the verb “to build” the interrelations, from the perspective of Leeds-Hurwitz (2009 p. 891-895), the work highlights how emotions, meanings, and beliefs are materialized and brought into existence through social construction.

For social constructions to remain active, social actors must continually maintain and repair them, preventing the forgetting or dissolution of the relationships and interactions that make up the collective. Social relations need to be updated or relocated due to the dynamic nature of people, which can lead to changes in social constructions, requiring the revision, discarding, or recreation of certain aspects. In other words, social construction is an ongoing process that requires effort and attention to stay relevant and meaningful. As initially conceptualized by Max Scheler, this subfield focuses on the collective nature of knowledge and its distribution through social institutions, highlighting that all mental acts are sociologically co-conditioned by societal structures (Berger & Luckmann, 1966, 2003; Fraser & Turcan, 2025). The sociology of knowledge is not merely a historical analysis of ideas but

seeks to understand the social functions and conditions that give rise to different forms of knowledge, as argued by Robert K. Merton, who emphasized the social responsibility of truths and their relationship to historical contexts (Merton, 1938).

It contrasts traditional epistemology, presenting knowledge as a collective achievement, shaped by social processes and not just individual cognition (Bloor, 2004). Federica Isabella Malfatti corroborates this proposition by arguing that understanding can be attributed to collective entities, which can reach levels of understanding beyond the sum of the understandings of the individual members. The sociology of knowledge also examines the interplay between social conditions and ideas, as seen in the works of Karl Mannheim and others, who have explored the epistemological and political implications of the social origins of knowledge (Malfatti, 2022).

Therefore, knowledge and social reality exist in a reciprocal relationship and is reinforced, as proposed by Berger and Luckmann (1966), indicating that social realities are constructed and communicated through significant systems (McCarthy, 2019). In fact, for Berger and Luckmann (1966), daily conduct, or our daily life, is directly being conducted and possibly seduced by our knowledge, also that the intrinsic quality of phenomena – which do not depend on our actions for their existence – at the same time affect us, they are affected by us. Thus, contemporaneously, it is necessary to discuss the argument of Berger and Luckmann (1966) that social beings constitute a network of associations – a collective – with an infinity of actors encompassed by a multiplicity of beliefs, values, and meanings, producing diverse knowledge, at different levels and spheres of reality (Nahapiet & Ghoshal, 1998).

As a means of disseminating everyday knowledge, common sense knowledge is an indispensable resource by which individuals orient themselves in their social environment. Therefore, the cognitive structure of everyday life is not chaotic. However, it is imbued with patterns and routines that are recognizable to individuals and that individuals learn and reproduce through social interaction. These routines structure personal biographies and allow for the integration of discrete episodes into a coherent narrative of a person's life. In this way, all interaction and habitual practice contribute to the collective construction of what is recognized as reality in daily life (Berger & Luckmann, 1966).

These interactions result in a process of intersubjectivity inherent to everyday experience. This shows that the reality perceived by an individual is continuously negotiated in the whole of interactions, where shared language and mutual expectations provide the means for the reintegration of personal experiences into the collective order. This systemic

process underlines the principle that the seemingly “given” character of everyday existence results from complex, socially mediated transformations that involve individual subjectivity and community validation (Berger & Luckmann, 1966).

Intersubjectivity is not merely a philosophical concern but a sociological problem, as highlighted by Schutz, who emphasized the importance of understanding the intersubjective world of everyday life beyond the traditional subjective-objective scheme (Segre, 2016). This shared understanding is essential for co-action and linguistic communication, allowing individuals to describe and interact with the physical and social world objectively and truthfully from a third-person perspective (Praetorius, 2009). In this regard, the sociology of knowledge examines how social and group origins influence the ideational realm, including the mutual constitution of knowledge and social reality (McCarthy, 2019). This shared understanding is to reducing complexity and bringing order to the contingent world as individuals navigate various social life worlds with distinct cultural grammar (Malfatti, 2022).

In addition, the sociology of knowledge also explores the interaction between social conditions and ideas, tracing how factors, socio-historical contexts and power relations shape intellectual products and processes (Camic, 2000). Intersubjectivity, therefore, is not only a conviction lived in everyday contexts but also a critical element in the broader sociocultural and epistemological structures that underpin the sociology of knowledge, therefore, how society becomes possible (Vaitkus, 1990). Thus, intersubjectivity in the sociology of knowledge is characterized by its role in facilitating shared understanding and communication, which are essential for the functioning of social life and the development of knowledge systems while transcending and affecting it. Such symbolic universes define the intersubjective relations that integrate social reality (Shadurski, 2021).

1.5.2 Symbolic Interactionism

Symbolic interactionism, a sociological perspective that emerged in the early twentieth century, emphasizes the role of individual agency and the meanings that people attach to their interactions, collectively shaping society (Fletcher, 2023). Scholars such as George Mead and Herbert Blumer have notably advanced this perspective, who have highlighted the importance of symbols and communication in social life (Vaitkus, 1990). Berger and Luckmann's seminal work aligns with symbolic interactionism by proposing that reality is socially constructed through processes of habitualization and institutionalization, where repeated interactions solidify into social norms and structures (Loenhoff, 2011).

While symbolic interactionism has traditionally struggled to explain larger social structures due to its focus on individual interactions as approached by Fletcher (2024), Berger and Luckmann provided a framework that reconciled this by conceptualizing social structures as emergent from these interactions, thus appealing to later interactionist scholars for their consideration and development. This synthesis of micro-level interactions and macro-level structures has been fundamental to understanding the dynamic nature of social realities, particularly in contexts such as race, gender, and political discourse (Hoyer, 2022). Despite criticism of the limitations of symbolic interactionism in addressing social structures, the integration of Berger's and Luckmann's ideas has enriched the theoretical landscape, offering a comprehensive view of how individual actions contribute to the broader social fabric (Gergen, 2023).

Dingwall (2024) traces the significant European influences about symbolic interactionism. At first, German philosophy, especially Immanuel Kant and his response to David Hume's skepticism, the Kantian idea that we act in the world through pre-existing categories (culture) influenced Mead and Park. Georg Simmel is highlighted for carrying forward the Kantian program of understanding society as a human construction based on experience, emphasizing sociation.

In a second moment, Albion Small's Scottish Enlightenment recognized the influence of Adam Smith, arguing that modern sociology seeks to realize the program implicit in Smith's moral philosophy. Concepts such as Smith's impartial spectator are seen as precursors to Cooley's mirror self and Mead's generalized other. The third moment, contemplates the classical heritage of symbolic interactionism, and its even older roots in Stoic Philosophy, which emphasized the indeterminacy of the world and the role of human cognition in its stabilization. Therefore, symbolic interactionism cannot be considered a marginal novelty, but heir to lengthy discussions about human beings and their relationships. So it is best visualized as a “jagged bush,” full of false starts, dead ends, and competing branches, rather than a linear, ineluctable march toward the present (Dingwall, 2024)

1.5.3 Intersubjectivity and Typifications in the Formation of Social Reality

According to Berger and Luckmann (1966), these social structures are inserted in two important dimensions: Space and time. The space is about how we physically interact with other people, in our daily lives inevitably have a spatial structure. This structure refers to how things are organized in the space around us, and it's important to note that our personal space,

or “manipulative zone,” overlaps with other people's spaces. This overlap means that our spatial experiences are shared with others, giving them a social aspect. For example, when you sit in a classroom, your personal space intersects with your peers, creating a shared environment. Our body's natural rhythms influence our personal time experience, but we also rely on shared, socially awake time to coordinate with others, so, our time is coordinated with the time of others, constantly shared and negotiated. Understanding time helps us navigate the world and interact with the people around us.

For Berger and Luckmann (1966), the primary focus is on the temporal structure or how time is organized in our daily lives. Time is a fundamental part of how we experience the world. Thus, “[...] Every individual is aware of the inner flow of time, which in turn is based on the physiological rhythms of the organism, although he does not identify with these.” (Berger & Luckmann, 2003, p.44). Even if, in many situations, we do not perceive such flows of change, our consciousness, or perception, is always aware of the passage of time. This is called “temporality,” which means the quality of being related to time. Our thoughts and experiences are always happening in a sequence, one after the other, which is why they are ordered temporarily, as the authors state: “[…]It is easily seen that the temporal structure of everyday life is extremely complex because the different levels of empirically present temporality must be continuously correlated” (Berger & Luckmann, 2003, p. 45).

Intersubjectivity can be analyzed under two aspects. The first is intrasubjective temporality, which refers to how each experiences time internally, each person has an “internal flow of time,” which represents the subjective perception of the passage of time in their mind. The body's natural rhythms, such as heartbeat or breathing, influence this inner time, but it is not identical to these physical rhythms. For example, when waiting for an exciting event, time may pass slowly, even if the heartbeat remains stable. Therefore, intrasubjective temporality refers to the individual's internal experience of time, shaped by personal consciousness and subjective interpretation of the reality being experienced. This dimension is influenced by the phenomenological insights of Alfred Schutz, with whom Berger and Luckmann studied, emphasizing the pre-reflective and lived experience of time (Abels, 2004; Rosenthal, 2016).

The second aspect is beyond our personal experience of time; there is also a shared experience of time in society, known as intersubjective temporality. It involves the shared understanding and synchronization of time between individuals within a society. This is evident in how social structures and interactions are organized temporally, allowing for a collective experience of time essential for maintaining social order (Abels, 2020).

This is when everyone in a community or society agrees, such as the time shown on clocks or calendars. This shared time, called “standard time”, helps people coordinate activities and communicate effectively (Berger & Luckmann, 1966, 2003). The interplay between these dimensions is crucial to understanding how individuals internalize social norms and how these norms are continually reproduced through socialization processes (Abels & König, 2010). All interactions are and will be loaded with typifications. These refer to individuals mental constructions or characterizations to categorize and understand the world around them. These are essentially the “images in our heads” about what a typical example of something looks like, and they play a significant role in how people navigate and understand their social environments (Burr, 2024; Gergen, 2023).

1.5.4 Social construction as a dialectical process

Berger and Luckmann argue that reality is constructed through a dialectical process involving three stages: externalization, objectification, and internalization. In this process, typifications are formed and reinforced through social interactions and become part of the shared social knowledge that individuals use to interpret their experiences (Berger & Luckmann, 1966). Binder (2016) emphasizes that typifications are not static; they are continuously shaped and reshaped through social interactions and are essential for maintaining a particular social order, as they help individuals predict and understand the behavior of others, thus facilitating calmer social interactions.

In addition, the typifications are embedded in language and culture, which are products of social construction. As a system of signs, language provides how typifications are communicated and shared within a society, thus contributing to forming a collective reality (Segre, 2016). This process underscores the dual nature of reality as subjective and objective, where objective social structures and vice versa shape subjective experiences. And the importance of emphasizing that what is considered real is deeply rooted in social processes and interactions (McCarthy, 2019). Through typifications, individuals and societies create a shared understanding of the world, influencing behavior, social organization, and meaning making (Maines, 2000).

Language, through its structures such as metaphors, metonymies, and idioms, influences the way individuals perceive and interpret the environment, thereby shaping the worldview and impacting the cultural, social, and social realities developed and institutionalized by leadership (Nastenko et al., 2024). Berger and Luckmann emphasize the

dialectical process of externalization, objectification, and internalization, in which language is fundamental in externalizing subjective experiences into a shared social reality, which is then internalized by individuals as objective truth. This process is further supported by the theory of linguistic relativity, which posits that different languages divide reality in unique ways, thus influencing speakers' perception of the world (Nastenko et al., 2024).

As Nastenko et al (2024) claim, the interaction between language and cognition is central, as it facilitates communication and builds social identities and cultural norms. This perspective highlights the importance of language in the sociology of knowledge, as it regulates behavior in everyday life and maintains the continuity of shared reality. Therefore, linguistic structures are not merely communication tools but are fundamental to the construction and perception of reality, influencing how individuals understand and interact with the world around them. Statements that are summarized below:

> "[…]Everyday life is above all the life with language, and through it, in which I participate with my fellow human beings. The understanding of language is therefore essential for my understanding of the reality of everyday life. [...] In this way, language is able to become the objective repository of vast accumulations of meanings and experiences, which it can then preserve over time and transmit to succeeding generations"(Berger & Luckmann, 2003, p.57).

1.5.5 Origins of institutionalization

For Berger and Luckmann (1966, 2003), institutionalization is presented as a central argument to explain how recurrent human activities gradually transform into socially established structures. Consequently, that all human activity is inherently subject to habitualization, which can be defined as: Actions performed repeatedly become routinized and, eventually, consolidate into standardized behaviors.

This process of habitualization is considered the first step towards institutionalization, as each repeated action begins to take on a standardized form that allows individuals to save effort through repetitive performance, providing cognitive relief by minimizing decision-making demands. The authors point out that even solitary individuals, through the consistent repetition of their own patterns of behavior, lay the groundwork for institutional phenomena, indicating that institutionalization is not exclusively a product of complex social

arrangements, but stems from the fundamental human tendency to establish routines, as follows:

> "These processes of habit formation precede all institutionalization, in fact they can be applied to a hypothetical solitary individual, detached from any social interaction. [...] Empirically, the most important part of the habit formation of human activity is coextensive with the institutionalization of the latter. The question then becomes to know how institutions originate. Institutionalization occurs whenever there is a reciprocal typification of habitual actions by types of actors. To put it differently, any of these typifications is an institution" (Berger & Luckmann, 2003, p. 78-79).

The process of reciprocal typification means that when two or more individuals interact, they begin to interpret, classify, and assign meanings to each other's repeated behaviors. These reciprocal typifications become common, shared understandings that regulate subsequent interactions, thus cementing a socially recognized order. The emergence of these reciprocal typifications marks the genesis of institutions – they provide stability and predictability – by structuring individual actions into a coherent collective pattern. By visualizing the institution as the result of these processes of typification, there is emphasize that the objective reality of the social order is not predetermined but is progressively constructed and continuously reaffirmed through the intersubjective exchange of shared types and routines (Berger & Luckmann, 1966, 2003).

The concept of reciprocal typifications involves the transformation of social habits into obligations, which are then objectified as institutional facts characterized by unquestionably shared typifications. This transformation is of significant importance for the emergence of institutions, as it involves the recognition and communication of meanings that are solidified into social norms and rules. Institutions, therefore, emerge as exogenous devices that influence individual actions and incorporate intersubjective and constitutive relationships between individuals and the social structures they inhabit. Perspectives that underscore the importance of understanding the social dynamics and structural constraints that facilitate the emergence of institutions, which are often assumed rather than explained in economic models (Andrews, 2012; Burr, 2024; Francis, 2024).

1.5.6 The Roles

For Berger and Luckmann (2003), roles are defined as the socially objectified typifications of conduct that emerge from the reciprocal habitualization of actions. They are not merely isolated examples of behavior but are portions of a broader, collectively shared stock of knowledge that determines how specific actions are recognized and institutionalized. In addition, the process of institutionalization is significantly influenced by individual roles and interactions. Thus, the construction and/or identification of individual competencies and interactions are fundamental in the creation, maintenance, and disruption of institutions, as seen in the successful implementation of a service platform in the Italian public sector, which facilitated innovation through the interaction of public administrations, small and medium-sized enterprises with citizens (Sabini & Muzio, 2012).

In community-based initiatives, interaction factors, such as leadership that crosses boundaries and social capital linkage, are more influential in acquiring government support than formal institutionalization factors, indicating the importance of interorganizational interactions. The postmodern condition further complicates the institutionalization process, as the multiplication of social roles and the reflexive management of the self lead to an individual identity crisis that institutions must face (Igalla et al., 2021). Interactions within institutions are constituted for maintaining social order and assigning identities or agency, demonstrating the strong link between the order of interaction and institutional structures. The integration of role theory with symbolic interactionism addresses the critiques of the latter's microfocus, providing a framework that considers macro-sociological issues, such as power dynamics and social structures, without losing sight of the agency and interpretive processes of the individual (Callero, 2003). This approach allows for a nuanced analysis of how roles are not only internalized but also executed in various social contexts, influencing both personal identity and broader social standards (Efendi et al., 2024; Maines, 2000; Sabini & Muzio, 2012).

The institutionalization of managerial innovations in governments highlights the central role of actors in the invention and maintenance of institutions, highlighting the tensions inherent in the process. The development of non-state regulatory organizations like GLOBALGAP illustrates how early institutionalization, driven by interactions with multiple actors, can shape organizational identity and objectives (Casey, 2019). A great example of the American presidency shows how individual interests and environmental factors interact to achieve stability and value, with government activity having the most significant impact

(Ragsdale & Theis, 1997). Wherefore, the institutionalization of forecasting practices within governments and/or organizations reveals the challenges of aligning forecasting with the formulation/action of applicable policies and practices due to the different institutional practices and structures, emphasizing the need for a differentiated understanding of these interactions, regardless of their dimension of analysis (Burr, 2024; Efendi et al., 2024; Maines, 2000; Sabini & Muzio, 2012).

Therefore, this objectification of behavior in roles is achieved through linguistic coding and dialectics. This process of role formation is embedded in the dual structure of externalization and internalization. Given that, as individuals engage in habitual behavior, they begin to perceive themselves and others as embodiments of specific socially assigned patterns. The resulting roles are then communicated and reinforced through language, which provides the framework for a common understanding and ensures normative expectations. In this sense, roles can facilitate immediate and concrete interactions between actors and, at the same time, incorporate these interactions into a system of historically and culturally mediated meaning that is subject to communal verification and continuity (Berger & Luckmann, 1966).

In addition, roles are not inherent in human nature. They emerge and solidify through the process of habitualization and, mainly, institutionalization. When reciprocal actions between individuals are repeated and crystallized in expected patterns associated with positions – the interaction between those who teach and those who learn in the school environment –, these patterns become objectified and become the "role of the teacher" and the "role of student". The structure of the role becomes a "social fact," apparently external to the individuals who occupy it at a given moment, as Berger and Luckmann state:

> "[…]Papers can be reified in the same way as institutions. The sector of self-consciousness that has been objectified in a role is then also apprehended as an inevitable fatality, and the individual can deny any responsibility. The paradigmatic formula of this kind of reification is the proposition: "I have no choice in this matter, I have to act in this way because of my position," as husband, father, general, archbishop, chairman of the board, bandit or executioner, whatever the case may be. This means that the reification of roles narrows the subjective distance that the individual can establish between himself and the role he plays. The distance implied in all objectification remains, of course,

> but the distance caused by disidentification is reduced to the point of disappearing" (Berger & Luckmann, 2003, p. 125, authors quotes).

1.5.7 Breadth and the Determinants of Institutionalization

The degree to which institutionalization permeates social life is fundamentally determined by the extent of structures of shared relevance within a society; therefore, a determinant of the institutionalization process lies in the comparative question suggested by Berger and Luckmann (2003, p.106); "[...] How big is the sector of institutionalized activity compared to the non-institutionalized?" Thus, the degree to which social practices become institutionalized is determined by the scope and broad acceptance – generalization – of significant social elements or understandings – significant structures.

Likewise, institutionalization extends to everyday life, where institutions such as family and health play a fundamental role in forming individual experiences and adaptations to new environments (Brett, 2024). From another aspect, legislative institutionalization is affected by political structures, such as the presidential system and electoral processes, with the era of democracy being a positive determinant, or when members of society share numerous common concerns and problems, social actions quickly become institutionalized. This institutionalization manifests as extensive and continuous, approaching an ideal type in which almost all actions are included in a universal normative framework (Brett et al., 2016).

On the other hand, when shared relevance is limited, institutionalization appears in a restricted way, with social practices compartmentalized into isolated and specific domains of functions (silos). This dichotomy sets the stage for understanding the diversity of institutional manifestations in societies. It emphasizes the importance of factors such as the division of labor and the available economic surplus in modulating the breadth of institutionalization (Berger & Luckmann, 2003).

Compartmentalization leads us to contrast two heuristic extremes of institutionalization: a society in which all social actions are institutionalized – producing a monolithic common stock of knowledge – and another in which institutionalization is limited to isolated problem areas, resulting in highly fragmented domains of social practice. While no empirical society fully adheres to either model, these extremes provide a theoretical framework for examining variations in institutional scope. In more primitive societies, where common problems predominate and social interactions are less differentiated, the process approaches total institutionalization. In modern and complex societies, the increased

specialization generated by a refined division of labor leads to a more segmented institutional scenario, where specific knowledge of a function proliferates, and distinct sub universes of meaning emerge (Bahuguna et al., 2024).

Although institutionalization stabilizes social interactions by reducing uncertainty and promoting predictability, it remains inherently dynamic and susceptible to historical modification. Thus, specific domains may undergo deinstitutionalization as social practices evolve and normative structures are renegotiated. This framework suggests that institutions, which are socially constructed entities, can be deconstructed or deinstitutionalized through changes in these social processes and interactions. The process of deinstitutionalization tends to involve the reduction or transformation of institutional structures, often to promote autonomy and independence, as seen in the context of services for people with disabilities (Stiker, 2023).

This dynamic quality is fundamental to understanding the transformation of the social order between generations, for example, as well as the dialectical process in which society is externalized, objectified, and subsequently internalized by individuals through the mechanisms of socialization and cultural transmission. For example, changes in the economic surplus or changes in the division of labor can lead to an adaptation of preexisting institutions, thus reconfiguring the common "stock of knowledge" that sustains objective social reality, which refers the authors to the writings of Marx, Durkheim, and Weber (Berger & Luckmann, 1966).

The evolution of contemporary societies towards greater complexity tends to be marked by institutional segmentation and a more elaborate division of labor. This leads to an inevitable distribution of specialized knowledge among the different strata and groups of roles. While the need for integrative meanings and the legitimation of activities seek to maintain social coherence, this specialization itself favors the emergence of distinct sub-universes of meaning. These sub-universes represent partial realities, often with their language and logic, accessible only to their inner members. The prevalence and nature of these sub-universes vary according to social structure. However, their existence points to a social reality that is not monolithic but rather composed of multiple domains of meaning. Understand the internal dynamics and relationships between these sub-universes (Berger & Luckmann, 1966, 1995)

1.5.8 Legitimacy

Legitimation refers to how knowledge, institutions, and social practices gain acceptance and validity within a society. It is the mechanism by which specific ideas, norms, or structures are justified and normalized, thus ensuring their persistence and stability. Berger and Luckmann emphasize that legitimation is not merely passive acceptance but an active process that integrates cognitive facts and normative values, providing a moral and epistemological basis for social reality (Cvetkovic, 2016).

Under the lens of the sociology of knowledge, legitimation is intimately linked to the construction of reality. It is through legitimation that knowledge becomes objectified, which means that it is transformed from individual experiences into shared collective understandings that are taken as objective reality (Mustafai, 2019). Specifically, Berger and Luckmann (2003, p.127) define: "[…]Legitimation as a process is better defined by saying that it is an objectification of meaning of the second order. […] Legitimation produces new meanings, which serve to integrate the meanings already linked to disparate institutional processes[…]".

Berger and Luckmann emphasize the dialectical relationship between the individual and society in the process of legitimation. While individuals internalize legitimized knowledge through socialization, they also contribute to the reproduction and transformation of this knowledge through their interactions and practices (McCarthy, 2019). Thus, the role of legitimation is inextricably linked to institutionalization, as it provides the necessary validation for establishing and maintaining institutions. Berger and Luckmann argue that institutionalization without legitimation would result in a lack of moral and cognitive validation, leading to the eventual erosion of the institution(Cvetkovic, 2016; Fraser & Turcan, 2025).

For Berger and Luckmann, legitimation encompasses the objective and subjective aspects, as it has a double function. Legitimation makes first-order objectification – basic ideas or practices – objectively available and subjectively plausible. Objectively available means these ideas are available for all to see and understand. Subjectively plausible means that people believe and accept these ideas as actual or reasonable. For example, the idea of a father also being a military commander may be objectively available in the rules of a society. However, it becomes subjectively plausible when individuals accept and understand this dual role. Therefore, "[...] knowledge precedes values in the legitimation of institutions"(Berger & Luckmann, 2003, p. 124, authors' quotation marks).

Berger and Luckmann postulate that, legitimation functions as the underlying process by which institutions acquire lasting objective validity and are maintained as parts of an integrated social order, "[...] It is important to understand that legitimation has a cognitive and normative element" (Berger & Luckmann 2003, p.124). As a categorization of levels of evolution of legitimation, Berger and Luckmann point out that at its incipient level, legitimation arises from the basic transmission of discrete institutionalized meanings through linguistic objectification of human experience.

These pre-theoretical legitimations are formed when common-sense systems, everyday language, and ritual practices convey what everyone knows about the institutional order, providing immediate pragmatic stability and normative certainty to socially established roles and routines. This level of legitimacy ensures that institutions are not seen merely as arbitrary aggregations of behavior but are imbued with shared meanings that guide everyday conduct. At the incipient level or first level, legitimization ensures that institutions are maintained steadily through daily social routines, allowing individuals to play habitual roles taken for granted in society(Berger & Luckmann, 1966, 2003).

Theoretically, theoretical possibilities are presented in a second moment of legitimation, but in a still rudimentary way. There are already explanatory systems relating some sets of objective meanings. Such systems or "schemes" are generally more pragmatic, action-focused, and focused on everyday practices. As the institutional order becomes more complex, legitimation evolves into a multi-layered process – a third level. With the development of specialized theories of legitimation administered by dedicated legitimators, the process begins to transcend its pragmatic roots. At this advanced level, legitimation is no longer just about providing immediate practical guidance; it becomes pure theory an abstract, reflexive integration of discrete institutional practices into a coherent cosmic or symbolic totality. This stage is characterized by the creation of symbolic universes, where institutions are represented as portion of a larger metaphysical or mythological order. Therefore, they are often entrusted to a group of specialists to transmit knowledge under rigorous initiation procedures. These representations not only ensure the persistence of the institutional order but also offer a framework for understanding its historical continuity and its role in maintaining the universe (Berger & Luckmann, 1966, 2003).

This level (the third level), which is then characterized by reflexive and theoretical formulations, creates a comprehensive structure that integrates the plurality of institutional processes. By establishing a symbolic totality, advanced legitimation justifies the existing

institutional order and transforms it into an objective reality that seems timeless and beyond individual control. Thus, legitimation operates on the pragmatic and reflective planes. The lower level ensures immediate social consensus and functional coherence, while the upper level provides intellectual and symbolic integration that reinforces the institutional order over time. These layered processes ensure that institutions are continually reproduced in daily life and maintained as part of a larger, stable, interconnected social cosmos, ultimately imposing an overarching normative logic on individual and collective behavior (Berger & Luckmann, 1966, 2003).

Recently, Fraser and Turcan (2025) seek, in a way, a "reconstruction" of the understanding of the construction of social reality. The authors posit that these two processes, while inherently connected, are analytically distinct, allowing for a nuanced understanding of how social artifacts embed each other in their respective communities. The approach begins with the abstraction of the social world into three sets: U (the universe of all social artifacts), L (the set of legitimized artifacts), and I (the set of institutionalized artifacts). By framing the analysis in this way, the authors have set the stage for a granular exploration of the conditions under which social artifacts acquire legitimacy, institutionalize, or go through both processes simultaneously.

Whereas each artifact has a certain status, such status is enhanced as it moves from peripheral positions to more widely accepted positions in the community, and institutionalization refers to the normative incorporation of these artifacts into the social structure. According to the approach described, a social artifact can experience different combinations of these processes: it can be legitimized without full institutionalization, institutionalized with an associated legitimation process, or even simultaneously subject to both processes in mutual causal emergence. The structure, therefore, allows for multiple trajectories, where the time, sequence, and interaction of these two fundamental forces create distinct patterns of incorporation of elements within the social environment (Fraser & Turcan, 2025).

Fraser and Turcan (2025) critically evaluate the theory proposed by Berger and Luckmann, emphasizing its criteria of adequacy, feasibility, modifiability, and relevance within the sociology of knowledge. They articulate that typology not only corresponds to observable social phenomena but also extends traditional models by accommodating the complexities introduced by legitimation and institutionalization. This evaluation reinforces the novelty of the approach in capturing the dynamics within stable and unstable, or liquid

social conditions of today. It affirms the typology of explaining the processes by which new and traditional social artifacts become legitimate and institutionalized Figure 8.

The framework is positioned as a flexible but rigorous means of understanding the interplay between legitimation's cognitive and sociopolitical dimensions. This duality allows the theory to capture differentiated trajectories, in which the interactions between innate subjectivities and overarching institutional mechanisms result in diverse patterns of social construction. Consequently, the discussion underscores that the model explains not only simple cases of legitimation or institutionalization but also more complex phenomena, such as mutual causal emergence or suppression, which play yours roles in real-world scenarios, such as the changing landscapes of political power or cultural practices (Fraser & Turcan, 2025).

More specifically, Fraser and Turcan (2025), Figure 8, explains that for a better understanding of the objectification of meaning, the authors present the decomposition of this process into two axes: Vertical axis, Legitimation, 1st Order of the objectification of Meaning, representing the process by which an innovation, new practice or new entity gains acceptance. It is thus the first step for something “new” to present itself as appropriate, desirable, and worthy of support. That is, “Is this valid”?. The Horizontal Axis, Institutionalization, 2nd Order of Objectification of Meaning. This is a more advanced stage in which innovation, or legitimized practice, becomes an integral and stable part of the social structure. That is, “Is this established?”

The interaction between the different levels of legitimation and institutionalization generates four quadrants, each describing a specific state in constructing social reality. In quadrant, I Instantiating – Mutual Emergence, it is a “becoming” moment characterized by fluidity and the formation of new understandings and shared practices. An example could be the initial emergence of an online community around a new interest, where the first interactions and rules begin to form and gain validity among participants In quadrant II, Realizing – Legitimizing towards institutionalization, by which the validity of the new social reality is actively used to build lasting structures and norms. Think of the evolution of a social movement that, after gaining broad public support (legitimization), begins to create formal organizations, procedures, and laws to ensure its continuity and impact (institutionalization). In quadrant IV, Aspiring – Legitimizing from institutionalization, he describes how he reaffirms that institutions exist but need to fight to (re)conquer the belief and support of individuals. An example can be an old law that is still in force (institutionalized) but that most

of the population no longer considers fair or relevant (low legitimacy), leading to debates about its reform or repeal. In quadrant III, legitimation and institutionalization are in an advanced weakening state. A "mutual causal suppression" occurs, where the lack of belief in the validity of the institution and the dismantling of its structures feed each other, leading to its disappearance. It is the fate of institutions, beliefs, or practices that become obsolete or are actively rejected by society.

Figure 8: Reconstructing the Sociology of Knowledge

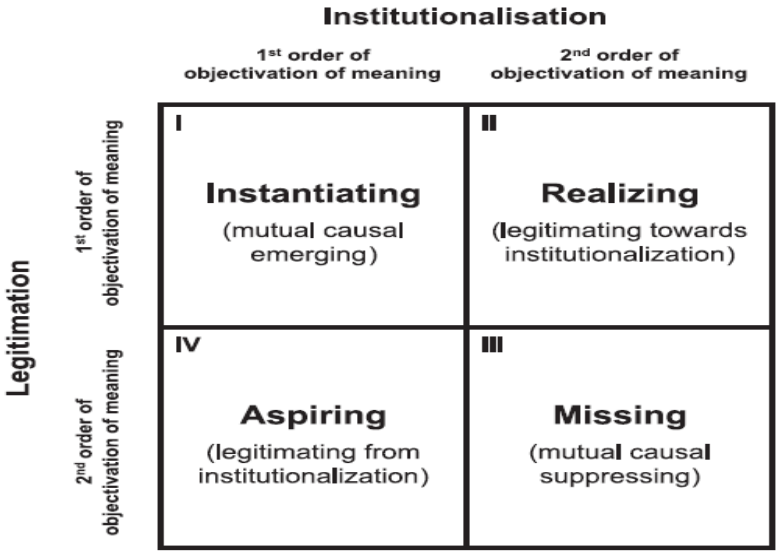


FIGURE 3 | Typology of the social construction of reality. *Note:* (I) Mutual causal emergence, coming into being. (II) Legitimating toward institutionalization. (III) Mutual causal suppression, unrealized potential. (IV) Legitimating from institutionalization.

Source: Fraser & Turcan (2025, p. 4)

Still, for Stefan Nicolae (2025, pp. 81-87), the theme of legitimation needs to be expanded and combined with the theory of justification on the way to the sociology of evaluation. Legitimation and justification serve as two closely interrelated but analytically distinct functions in the construction, maintenance, and contestation of social value. At its core, legitimation explains why particular practices, roles, and states of being are given authority, stability, and coherence. It operates by embedding evaluation into an intersubjectively shared system of classification and memory that, once established, confers a normative reality on institutions and practices, effectively constraining alternative interpretations of value. In contrast, justification concerns the explicit discursive practices actors employ when they defend or contest a given state of value. Justification is accomplished through "value testing," in which individuals invoke overarching principles or reference value orders to validate their actions and decisions.

This process is dynamic and situational, as the actors must negotiate divergences and articulated why one evaluation is preferable to another within a given symbolic universe. Boltanski and Thévenot's model posits that justification occurs when actors, faced with conflicts or ambiguities, mobilize higher common principles that transcend personal interests or situational specificities. These discursive practices serve not only to legitimize individual actions but also to reestablish or reconfigure the system of implicit relevance that structures collective understandings of value. Thus, agreements reached through justification help contain normative disputes and stabilize the shared order of value (Nicolae, 2025) .

Whereas legitimation provides the general structural and historical basis for why a set of values is accepted as the structure of reality, justification works on a more immediate practical level. It offers a mechanism for critiquing, revising, or reinforcing the prevailing order by explicitly referencing symbolically mediated criteria. Together, these processes highlight the dual nature of assessment: It is both an emerging body of constructed knowledge and a practice embedded in everyday interactions. The interplay between legitimation and justification, therefore, not only stabilizes but also continuously negotiates the boundaries of what is considered valuable, ensuring that social orders remain responsive to change while maintaining coherence over time (Nicolae, 2025).

1.6 Sociology of Knowledge Amplified by the Sociology of Associations

The quest to understand the nature of social reality, the processes that shape it, and the agency that constitutes it has been a central endeavor in the social sciences and organizational studies. There are two theoretical perspectives that offer particularly insightful – though distinctly formulated – views on the non-essentialist nature of reality are the SCRT, as articulated by Peter L. Berger and Thomas Luckmann (1966), and ANT, associated prominently with Bruno Latour (2005), Michel Callon (1986).

This research, which takes a qualitative, interpretive, and reflective approach to analyzing the transformation of HRM by PA, requires not only presenting these theories individually but, examining them in critical dialogue, exploring their intercessions, complementarities, and notably their ontological and epistemological tensions. This comparative analysis provides the conceptual framework necessary for a multifaceted reading of the phenomenon under study, recognizing organizational change's symbolic, institutional, material, and relational dimensions.

Both theories share a fundamental critique of deterministic or universalist views of reality, arguing that what we consider "real" – whether social, technical, or natural – is largely a contingent product of specific processes. However, the nature of these processes, the constituents of this reality, and the locus of agency are conceived in radically different ways, which makes their contrast particularly illuminating.

1.6.1 Reality as Human Edification: The Theory of Social Construction

The SCRT posits that social reality is a human artifact (Berger & Luckmann, 1966). Far from being an external and immutable structure that simply constrains individuals, the social order is understood as the result of a continuous and dialectical human activity. The authors describe this dialectic in three fundamental moments: 1st Moment externalization: Human beings, in their interaction with the environment and with others, constantly project meanings, ideas, norms, and artifacts, so human action is inherently expressive and world-creating. 2nd Moment the objectification: The products of human externalization—ideas, tools, social relations, or institutions—acquire an existence that seems independent of their creators. Certain interactions and practices solidify through habituation (repeated actions that become patterns) and reciprocal typification (mutual categorization of behaviors and roles).

Institutionalization occurs when these reciprocal typifications are shared and passed down through the generations, acquiring a facticity – a quality of being real and objective, which confronts the individual as something out there (Berger & Luckmann, 1966). Language is the primary vehicle of this objectification and transmission, providing the categories and narratives that make social reality coherent and shareable (Alvesson & Kärreman, 2000). In the 3rd Moment, internalization occurs: The objectified reality is apprehended by individuals as if it were a single reality tout court. Through primary and secondary socialization, the individual internalizes the social world and its typifications, which begin to shape their identity, perception of reality and actions (Berger & Luckmann, 1966).

The social world becomes his subjective world. Additionally, the core of SCRT lies in the primacy of human interaction and the construction of shared meaning. Social reality is, in essence, a reality of meanings, symbols, and institutions that emerge from human activity. Although Berger and Luckmann (1966) acknowledge the existence of a "pre-social" or "natural" reality, their analytical focus is on the layer of reality that is specifically socially built by human interaction. Agency, in this perspective, is predominantly human, residing in the ability of individuals to externalize, negotiate meanings, and reproduce (or, to a lesser

extent, challenge) the objectified structures. The social order, for them, is a precarious conquest, constantly threatened by anomie, but sustained by the processes of legitimation that justify institutions and the knowledge that corresponds to them. Reality is socially defined as real and which, because it is internalized, becomes real in its consequences for individuals, as he highlights Merton (1995), on Thomas' Theorem, aligned with the constructionist logic of social interpretation.

Therefore, SCRT is the basis for interpretative and qualitative approaches. It directs the researcher to investigate how social actors define and experience their reality, how meanings are negotiated, how institutions are created and maintained through everyday interaction, and how individuals internalize these structures (Schwandt, 1994). Reflexivity in SCRT implies recognizing that the researcher's knowledge of social reality is also a social construction shaped by its context and conceptual lenses. Under this lens, Karen Ross and Peiwei Li (2025), argue that recognition and reflexivity are pieces of a big puzzle to understanding the meaning and validity of research results, as it involves recognizing the social and relational contexts in which knowledge is produced. This aligns with social constructionist epistemology, which posits that knowledge is co-constructed through interactions between researchers and participants, as demonstrated in studies involving marginalized groups such as street-connected children and migrant women in Bolivia (Ross & Li, 2025; Exeni, 2015).

Sheila McNamee also highlights the importance of reflexivity in research, suggesting that researchers should engage with various discourses and perspectives to ensure that local and situated practices are adequately represented and ethically rigid (Mcnamee, 2009). This approach is supported by Marie-Louise Österlind and colleagues, who advocate prioritizing respondents' perspectives to facilitate agent change, thereby recognizing the power dynamics inherent in research interactions between researcher and researched (Österlind et al., 2023).

SCRT offers a valuable perspective on how knowledge and institutions are socially constructed, Actor-Network Theory (ANT) proposes a complementary and contrasting approach to analyzing the formation of social reality. ANT expands the scope of analysis beyond human actors, also considering the active role of non-human elements in the constitution of the society. This perspective is particularly relevant when examining complex technological phenomena such as People Analytics, where algorithms, data, and computational systems interact with human actors in intricate ways. By adopting the principles of ANT, we can understand heterogeneous relationships that perform new

organizational realities, thus offering a more comprehensive view of the ongoing transformations in human resources management; such theoretical integration will be addressed below(Latour, 2001, 2005).

1.6.2 Reality as Network Performance: The Actor-Network Theory

Actor-network theory (ANT) offers a distinctly different ontology and epistemology. While SCRT focuses on constructing a social reality by and for humans, ANT is interested in the association and performance of heterogeneous realities by a multiplicity of human and non-human actants (Latour, 2005). The central principle of ANT is generalized symmetry or the principle of symmetrical symmetry (Callon, 1986; Latour, 2005). This rigorous methodological principle requires the researcher to use the same language and the same type of explanation when describing the actions and interactions of human and non-human entities. Thus, it is not a matter of attributing consciousness or intentionality to inanimate objects but of recognizing and describing their capacity to make a difference, alter the state of things, imposing resistance, or enable actions within a network. An actant is any entity – be it a scientist, a mollusk, a technology, a law, a data, or an algorithm – that acts or is made to act, modifying the behavior of other actants (Callon, 1986).

The accumulation cycle describes the dynamic process by which calculation centers are built and maintained. It involves bringing elements from the (distant) field to the center, transforming them into forms that allow calculation and comparison (often through inscriptions such as data, reports, graphs…..). Dashoborads, and then use the analytical power generated in the center to act back on the far field – implicit in using PA/AI for decision-making (Prasetyaningtyas et al., 2022).

This cycle highlights that a center's strength and capacity of influence are not inherent but result from a continuous work of collection, translation – in the Latourian sense – of mobilization and reorientation of interests and resource manipulation(Callon, 1986; Latour, 2005). The concept of inscription, therefore, becomes crucial here because it is the material representations – which, in conjunction with the application of data analysis of the people of the organization as a whole – that make what is distant and complex something present and with the possibility of modeling and deep analysis, at the center of calculation the HRM.

PA can carry out this in-depth analysis – a mechanism for building and solidifying this calculation center, it aims to identify challenges (such as system fragmentations), expand the capacity of decisions that emerge from this attempt to build a calculation center for the “new

HRM". From the perspective of ANT, reality is not something pre-existing to be discovered or represented, nor is it a symbolic layer constructed by humans. The reality is the provisional effect and performance of heterogeneous actant networks. Networks are not mere metaphors; they are analytical descriptions of the concrete associations that connect actants and allow something to "be real or happen". Thus, the stability of a network – and, consequently, of the reality it entails – does not derive from symbolic coherence or internalization by subjects but from the ability to orchestrate and maintain together the dispersed agencies of the multiple actors (Law, 1992).

In the same way that SCRT has different moments of understanding social reality – as already mentioned – the fundamental process in ANT is translation (Callon, 1986). Through translation, the actors seek to coordinate, align, and control each other, establishing equivalences and passages. It must be considered that the strength of an actor or a network derives, in large part, from its ability to accumulate and mobilize these heterogeneous allies effectively. Therefore, the "centers of calculation" – the new HRM – are the strategic points where familiarity with what is far away is built and where the ability to act "at a distance" is forged, often by overcoming uncertainties and creating synergies (Callon, 1986; Latour, 2005).

Callon (1986) describes the translation in four moments, described here in an introductory way: The 1st Moment is the problematization: One or more actors define a problem. It is necessary to identify and present a "flow" as the "obligatory passage" of interaction to the solution, making other actors relevant to your project, or so that such flow is not only understood but identified as a form of solution. Problematizing tends to involve actions such as critically examining and questioning the norms, definitions, and practices established in the world.

This process – problematization – can serve as a tool for critical analysis of social transformation, as seen in the work of Santos Junior and da Silva, who use ANT to challenge gender discrimination and violence against women, encouraging reflective and critical stances on gender relations through educational actions(Junior & da Silva, 2022). From another perspective, such as in educational research, problematization builds categories and structures that allow a deeper understanding of reality, avoiding predefined content and encouraging openness and critical thinking (Madrigal Segura, 2024). More specifically, in information systems design research, problematization addresses implicit and underdeveloped problems, enhancing understanding and innovation in design processes (Nielsen, 2020).

Diego Márquez Arancibia (2022) emphasizes that the problematization is dual. The dual nature of problematization can be analyzed as a verb or a noun. This duality is significant, as it highlights the action-oriented aspect of problematization (of X) and its role as an object of investigation (he the problematized X). This distinction is important to understanding how Foucault's philosophy engages with historical and political issues, reinforcing the interaction between these dimensions rather than presenting them as methodologically incoherent.

In addition, Arancibia (2022) discusses the role of problematization in Foucault's critical ontology, emphasizing its function in the analysis of the present through the lens of historical problematizations. This approach allows for a critical examination of how certain practices and discourses have been shaped and how they continue to influence contemporary thought and politics. The article highlights the importance of problematization in revealing the contingent nature of our current truths and practices, thus contributing to a critical understanding of the present. Therefore, problematization can be understood as a process that uses the historical perspective to mediate – in the Latourian sense – the present and aligning future interests.

The 2nd Moment the interest (interessement): The actants now try to seduce or block others from joining the proposed network, breaking their previous associations. Furthermore, it is a critical phase in which actors try to strengthen ties with other actors in roles defined during the previous problematization phase. This process involves negotiation, conflict, and alignment as actors work to stabilize the network, ensuring the participation of other entities, human and non-human, in the network's objectives (Rivera & Cox, 2016); which are explained by Latour (2005, p.50) as:

> "[…]Relating to one or another group is an endless process made up of uncertain, fragile, controversial and changeable ties. [...] We must consider social aggregates as realities constituted by "individuals", "organizations", "classes", "roles", "life trajectories", "discursive fields", "selfish genes"..... [...]"

The interest is characterized by using various strategies and devices to align the interests of different actors, ensuring their commitment to the network's objectives (Best & Walters, 2013). This phase is crucial for the network's success, as it involves the creation of alliances and negotiating roles, which are essential for the stability and functionality of the network. The concept of interest highlights the dynamic and often contested nature of network formation, in which actors must continually negotiate and renegotiate their positions,

relationships, and power within the network (Andrade & Ekundayo, 2011). This phase is about ensuring participation and managing the tensions and conflicts that arise as actors with diverse interests and motivations come together. By focusing on the interest phase, ANT provides a nuanced understanding of how networks are built and maintained, emphasizing the importance of human and non-human actors in shaping social and technological systems (Benqatla et al., 2017).

The third moment is the moment of registration (enrollment), which occurs when the actors accept – or are forced to accept – the roles assigned to them in the network, forming alliances and coalitions. The application focuses specifically on stabilizing the network and ensuring the participation of actors through the alignment of their interests, which is pillar for the sustainability and functionality of the network (Ahmed, 2013). In practical applications, such as the adoption of collaborative technologies or the translation of literary works, the application involves overcoming challenges related to interface design, individual motivations, and the socio-technical context, highlighting the dynamic interaction between actors and the network (Leng & Xian, 2024). The application's success depends on effective communication, trust, and coordination between actors, which are essential for the resilience and adaptability of the network (Ahmed, 2013). ANT's emphasis on the relational and process-oriented nature of power dynamics further underscores the importance of inscription in establishing and maintaining network hegemony, as it seeks to describe how power is achieved rather than assumed. For Callon (1986), the inscription is not a mere labeling of actors but a dynamic process through which provisional definitions are transformed into concrete affiliations and operational functions during ongoing negotiations. In this phase, the actors' provisional reports, previously constructed during the problematization and interest stages, are submitted to greater stabilization through a series of interactions that seek to convert a provisional agreement into a set of mutually recognized roles.

The 4th Moment is mobilization, when the main actant, or group, seeks to ensure that the actants involved effectively represent the entities or collectives they intend to mobilize, consolidating the network; Therefore, mobilization addresses the issue of representation by questioning whether the selected intermediaries can legitimately speak for the heterogeneous groups – that is, scallops, fishermen and scientific colleagues – that the researchers intend to represent. This analysis is critical because the entire translation process depends not only on the formation of alliances but also on the credibility of the voice of these actors in encapsulating the interests and behaviors of broader, often silent populations (Callon, 1986).

ANT's approach to mobilization is distinctive in its anti-essentialist stance, rejecting the notion of fixed agency and seeing agency distributed across a network of heterogeneous actors, including social and technical elements (Best & Walters, 2013). This perspective allows for a nuanced understanding of how networks are built and maintained, highlighting the importance of human and non-human actors in the translation and mobilization process(Vitale, 2023). The framework's application extends beyond the traditional social sciences, influencing fields such as design education, public policy, and gender management, where the translation process is used to navigate complex socio-technical environments and facilitate participatory structures (Tietjen et al., 2018).

The agency, in ANT, is distributed. It is not an inherent property of humans or objects but a relational effect emerging from the associations within the network (Latour, 2005; Orlikowski, 2000). A technical object, such as software, does not have agency in itself but acquires agency to the extent that it is translated, associated with other actors – data, users, infrastructure –, and participates in the performance of an action or process. The rigid distinction between the social and the material is undone; reality is inherently socio-material (Orlikowski & Scott, 2008; Scott & Orlikowski, 2025). Thus, the ANT proposes following the actants (Latour, 2005). It is an empirical and descriptive approach that seeks to map the associations and translation processes that constitute networks and perform realities. Empirical ANT requires substantial interpretive work to discern the relevant actants, trace the translation chains, and analyze how the network acts in a particular order or reality. Reflexivity, in ANT, involves being transparent about one's own methodological choices and the role of the researcher in the description and construction of the analytical network itself (Law, 1992).

1.6.3 Theoretical Intercessions: Similarities, Differences and Productive Tensions

By putting SCRT and ANT in dialogue, we identify qualified similarities and fundamental differences that generate a productive analytical tension for this qualitative, interpretive and reflective research.

1.6.1.1 Similarities:

1.6.1.1.1 Rejection of Essentialism and Determinism

Both theories agree that reality is not fixed, given, or predetermined by transcendent structures – natural, social, or technological. Both see it as something actively brought into existence – constructed, performed or activated (Berger & Luckmann, 1966; Latour, 2005).

Rejecting essentialism and determinism in theories of social construction significantly alters our understanding of reality by emphasizing the fluid and dynamic nature of social constructions at the expense of fixed, inherent truths.

Essentialism posits that social groups have intrinsic characteristics, while determinism suggests that these characteristics dictate outcomes. In contrast, social constructionism argues that reality is shaped by cultural and social interactions, challenging the notion of objective truths and highlighting the role of human agency in constructing social norms and identities, as in gender issues (Bartels et al., 2024). This perspective suggests that what is perceived as "real" depends on social interactions and cultural contexts rather than being predetermined or universally applicable. The implications of this change(s) are profound, as it allows for a more nuanced understanding of identity, particularly in areas such as gender and sexuality, where essentialist views have historically marginalized non-conforming identities (Bartels et al., 2024; Stein, 2021).

By acknowledging the socially constructed nature of these identities, social constructionism provides a framework for challenging existing power structures and advocating for more inclusive and equitable social policies (Bartels et al., 2024; Martignani, 2016; Mustafai, 2019). In addition, this approach encourages interdisciplinary dialogue by integrating Insights from sociology, psychology, and philosophy to explore how reality is continually negotiated and redefined through social processes(Mireanu, 2022). The rejection of essentialism and determinism, therefore, opens up possibilities for reimagining and/or reframing social realities, promoting a critical examination of the assumptions underlying social norms and promoting a more flexible and inclusive understanding of the human experience (Gergen, 2023).

1.6.1.1.2 Process Focus

Both theories focus on how reality and order are constituted and maintained. For Berger and Luckmann (1966), it is the dialectic externalization-objectification-internalization and institutionalization. For ANT (Callon, 1986), it is translation and networking. Both frameworks emphasize the dynamic and ongoing nature of the construction of reality, albeit through different lenses. ANT highlights the procedural nature of scientific and technological developments by focusing on the interactions and networks between human and non-human actors, termed actants (Collin, 2011; Tan, 2023). This approach underscores reality construction's fluidity and relational aspects, where agency is distributed in a network rather than being human-centered (Tan, 2023). Similarly, Berger and Luckmann's SCR theory posits

that social reality is continuously constructed through habitualization, institutionalization, and legitimation, transforming subjective understandings into objective social structures (Mustafai, 2019). Another similarity is that both also emphasize the role of social interactions and shared practices in shaping reality, suggesting that reality is not merely a static entity but a product of ongoing social processes (Gergen, 2023)

1.6.1.1.3 Stability as a Contingent Achievement

For both, the apparent solidity of reality is an achievement that requires continuous work. For Berger and Luckmann (1966), the objectivity of institutions is maintained by legitimation and internalization. ANT network stability depends on successfully maintaining associations and translations through its shared focus on social phenomena' dynamic and constructed nature (Callon, 1986; Latour, 2005).

ANT posits that social order is not an inherent property but emerges from networks of interactions between human and non-human actors. Stability is achieved through the successful black box of these networks, making them appear natural and unchanging (Couldry, 2008). This aligns with Berger and Luckmann's view that reality is socially constructed through continuous processes of negotiation and interaction, where social facts are not given but are the results of collective human activity and knowledge (McCarthy, 2019).

The two theoretical frameworks emphasize the contingent nature of social reality, where stability is a temporary state achieved through complex montages and negotiations. Latour's ANT suggests that what is perceived as a stable social phenomenon are, in fact, “stabilized states of affairs”, resulting from the resolution of controversies and the establishment of networks that unite various entities in an everyday world (Latour, 2001). This resonates with the idea of social constructionism that social norms and institutions are the results of historical and cultural processes rather than fixed entities. In addition, ANT's focus on the role of non-human actors and materiality in shaping social reality expands on the traditional human-centered approach to social constructionism, suggesting that objects, technologies, and inscriptions play fundamentals roles in constructing and stabilizing social facts (Best & Walters, 2013).

The notion of contingency in ANT, as explored through collective experiments, highlights the indeterminate and evolving nature of social reality, where new forms of agency and social order can emerge from disruptions and uncertainties. This reflects the social constructionist view that reality is continually constructed and reconstructed through human

interaction and interpretation, emphasizing social life's fluid and contingent nature. Thus, both the ANT and the social construction of reality emphasize the importance of understanding the processes and interactions that produce and stabilize social phenomena, highlighting the interaction between stability and contingency in the continuous construction of reality (Gergen, 2023; Tan, 2023).

However, to qualify these similarities. Although both speak of construction or constitution, the nature of what is being built or constituted is radically different. SCRT focuses on constructing meaning and symbolic/institutional order by humans. ANT focuses on the performance of realities by the heterogeneous association of actants. In other words, Berger and Luckmann postulate that reality is constructed through a dialectical process involving externalization, objectification, and internalization, which are influenced by the structures of social power, which can be alternated by or between political groups, bureaucratic groups, and groups of specific interests. This structure emphasizes reality's subjective and objective dimensions, where power is a constructed and constitutive element of social interactions (Dreher & Lang, 2019).

In a complementary perspective, Latour's and Callon's in a sociology of translation/association focus on the material and relational aspects of construction, where human and non-human entities are involved in networks that shape and stabilize social realities. For ANT, there is an emergence in the role of materiality and the agency of objects in the construction of social phenomena. Voegtli and Fillieule (2012) trace the formation, diversification, and normalization of the cause of AIDS in Switzerland. Particular emphasis is given to the interactions between three different entities: the medical field, the public authorities, and the associative space, the latter being understood as the place where individual and collective actors compete for the definition of the cause. The authors argue that the main phases of the structuring of the struggle, the pace of state intervention, and the creation of a multi-organizational field can only be understood if we adopt a configurational perspective attentive to how, in a given context and under the effect of specific constraints, key actors interact strategically and contribute to the transformation of their environment and their chances of achieving their objectives.

This approach considers the changing sociobiological characteristics of those who have committed themselves to the cause, suggesting that reality is socially constructed and materially constituted. Thus, the sociology of translation further deepens the discussion on how actors negotiate and redefine roles and relationships within networks, thus constituting

new social orders (Voegtli & Fillieule, 2012). These perspectives underscore that construction involves the dynamic interaction of social, material, and relational elements. At the same time constitution refers to the processes by which these elements are stabilized and institutionalized in a given context.

1.6.1.2 Fundamental Differences

1.6.1.2.1 From the ontological perspective: Anthropocentrism vs. Generalized Symmetry

The fundamental ontological distinction between SCRT and ANT lies in their radically different stances regarding the human's centrality and non-human's agency. This section deepens the discussion on the anthropocentrism inherent to SCRT and the principle of generalized symmetry proposed by ANT, showing how these perspectives shape the understanding of the constitution of reality (Berger & Luckmann, 1966; Latour, 2001, 2005).

As previously outlined, SCRT, articulated by Berger and Luckmann (1966), is explicitly anthropocentric. From this perspective, social reality is an eminently human enterprise constructed and sustained through interactions, shared meanings, and institutionalization processes mediated by language and human consciousness. The focus is on how individuals and groups create, maintain, and transmit the symbolic structures that give meaning to the world (Burr, 2024). Although the material world and objects can influence or provide the setting for social action, they are not endowed with intrinsic agency in the constitution of this social reality; Their relevance is derived from the meanings that humans attribute to them. Therefore, the sociality of reality and the social processes of its construction are the core of the analysis. This centrality of the human implies that agency – the ability to act and produce effects – is seen primarily as an attribute of conscious and intentional human subjects (Knoblauch & Wilke, 2016).

In direct contrast, the ANT proposes a radical break with this anthropocentric view through the principle of generalized symmetry (Callon, 1986; Latour, 2005; Law, 1992). This methodological principle requires the analyst to use the same vocabulary and the same explanatory framework when describing the actions and contributions of both human and non-human entities (objects, technologies, texts, microbes) in the construction and stabilization of any phenomenon (Akrich & Latour, 1992; Latour, 2005).

ANT does not attribute intentionality or consciousness to non-humans in the same way it does to humans but recognizes their ability to "[…]make a difference", mediate, resist,

enable, and configure actions and interactions within a heterogeneous network. As researchers in the area state, in ANT, "[…]objects, ideas, processes and any other relevant factors are seen as important in the creation of social situations as human ones[…]" and one should not "[…]change registers when moving from the technical to the social aspects of the problem studied[…]" (Callon, 1986, p 199; Gergen, 2018; Holstein & Cohen, 2025; Knoblauch & Wilke, 2016).

The reality, for ANT, is not a purely social – human – domain superimposed on a passive material world but is the performative and contingent result of the associations between these various "actants". The social is not a substance or a specific type of material but a "movement", a kind of connection between things that are not themselves social (Latour, 2005, p. 64-65). The ontology of ANT is, therefore, flat, refusing hierarchies to be *a priori* between humans and non-humans and seeking to trace the connections and transformations (translations) that constitute the world (Law, 1992). This approach challenges the dualistic separation between the social and the technical, the human and the material, proposing in its place an inherently socio-material reality (Costa Júnior et al., 2024; Orlikowski, 2007; Sandberg et al., 2025)

1.6.1.2.2 Nature of Reality and Construction: Symbol/Institution vs. Performance/Association

It is necessary to introduce a distinction between SCRT and ANT concerning the nature of reality and the process of "construction." As noted by Berger and Luckmann, social reality is primarily symbolic and institutional – it resides in intersubjectivity and objectified structures of meaning. For ANT, reality is performative and relational – it is what networks of actants do, the way they associate and translate (Law, 1992). From ANT's perspective, the concept of construction is less about building a system of shared meanings and more about forging and maintaining the connections that make something real in its performance. This differentiated approach to reality and its genesis deserves in-depth detail to elucidate the nuances of each perspective and substantiate the analytical richness that the joint use of these theoretical lenses can offer to your research.

From the perspective of Berger and Luckmann (1966), social reality is fundamentally a product of continuous human activity, a complex symbolic and institutional edifice. The process of social construction of reality unfolds dialectically in three interconnected moments. First, externalization, through which human beings, through their physical and mental activity,

project meanings into the world, creating culture and institutions as an anthropological necessity to shape their existence (Berger & Luckmann, 1966).

This is followed by objectification, the moment in which the products of this externalization – whether they are ideas, norms, institutions, or artifacts – acquire a facticity, being apprehended as an external and objective reality, apparently independent of their creators. This process is vitally sustained by the habitualization of actions, the reciprocal typification of behaviors and roles, and, centrally, institutionalization, where reciprocal typifications of habitualized actions are consolidated as patterns of conduct (Berger & Luckmann, 1966). Language emerges here as the primary vehicle of objectification, typifying experiences and conferring on them an anonymous and objective character, while symbolic universes integrate different provinces of meaning, legitimizing the institutional order in its entirety.

Internalization completes the dialectical cycle, with the objectified social reality being introjected into the individual's consciousness, primarily via socialization, shaping the individual's identity and perception of his/her world (Berger & Luckmann, 1966). Construction, from the perspective of SCRT, is, therefore, a dynamic process of building, maintaining, and transmitting a social order that, although a human product, confronts individuals with the force of an objective reality, focusing on how shared meanings and institutions stabilize everyday life.

On the other hand, the ANT, associated with authors such as Latour (2005), Callon (1986), and Law (1992), proposes a vision of reality and its "construction" – or, more precisely, its "enactment" or "assemblage" – which distances itself from the emphasis on internalized symbolic systems. For ANT, reality is not a pre-existing structure or a system of stable meanings. However, an effect or a contingent performance is continuously produced by heterogeneous networks of actants, both human and non-human. The nature of reality is, therefore, eminently performative and relational. Entities do not have intrinsic properties outside of their relations but acquire their form, attributes, and agency by connecting and interacting with other entities in the network (Latour, 2005; Law & Mol, 2002). For example, demonstrates in its practical ontology that a disease such as atherosclerosis is performed in multiple ways in different contexts, suggesting that reality is enacted in specific practices. Consequently, the focus of ANT is on the concrete work of forging and maintaining connections that make something real through its performance, where the stability of a phenomenon derives from the robustness of the network of actants that sustains it (Law, 1992).

The divergences between SCRT and ANT in conceptualizing reality and its construction paradoxically offers a powerful justification for its complementary use in this research – despite the challenges and risks. SCRT is particularly valuable for understanding social reality's symbolic and institutional dimensions on a more macro and experiential level. It illuminates how overarching narratives, symbolic universes, and institutions are established, legitimized, and internalized, providing individuals with a sense of order and objective reality and allowing them to analyze how human actors experience and make sense of these structures. As a complementary substance, ANT, with its focus on performance and heterogeneous associations, allows us to "open the black box" of these institutionalized realities, investigating how, concretely, they are assembled, maintained, and transformed in everyday life. ANT directs attention to the small and often invisible work of connecting actants, translating interests, inscribing practices on material artifacts – such as the algorithms and PA/AI systems, focus this research – and managing the controversies inherent in these processes, revealing materiality and distributed agency that can be underestimated by an analysis purely focused on human intersubjectivity (Berger & Luckmann, 1966, 1995; Callon, 1986; Latour, 2005; Law, 1984).

Therefore, combining the two theories allows for a richer and more multifaceted analysis of the current research. SCRT can help understand how HRM practitioners and other organizational actors socially construct the meaning of PA/AI tools, how they perceive them within their symbolic universes, and how practices around these tools become (or do not) institutionalized and legitimized. The ANT can then detail how these PA/AI tools, as non-human actors, actively perform in the network, how they translate the objectives of humans, how they are associated with other actors (data, infrastructures, other humans), and how this socio-material network engages new realities in HRM, modifying processes, relationships and the very nature of people management work. By utilizing both lenses, this research can capture both the dimension of shared meanings and institutional structures that shape perception and action (SCRT) and the dynamic, relational, and performative processes through which these realities are continually constructed and reconstructed with the active participation of human and non-human elements. This strengthens the deeper understanding of how the social reality of HRM is simultaneously a symbolic edifice and a socio-material performance in constantly becoming (Berger & Luckmann, 1966; Latour, 2005).

1.6.1.2.3 - Agency: Human vs. Human Distributed

SCRT locates an agency primarily in human subjects. ANT sees the agency as distributed across the heterogeneous Network; any actant who makes a difference in the course of action has relational agency (Callon, 1986; Latour, 2005). Objects, data, algorithms, and regulations are considered actants with the capacity to act and modify the network, not mere instruments or symbols of human action.

In the SCRT, the agency is primarily, if not exclusively, located in human subjects. The ability to act is associated with awareness, intentionality, the ability to create and attribute meaning, symbolic interaction, and the ability to construct and maintain social and institutional structures. It is the human being, as a social being endowed with consciousness, who externalizes his activity, participates in the objectification of social reality, and internalizes the world, being, therefore, the fundamental engine of the construction and reproduction of the social order (Berger & Luckmann, 1966).

The material world, objects, and technologies are, in this perspective, seen mostly as products of human agency, tools to be used, or elements of the environment that impose constraints or affordances for human action, but not as entities that possess agency in the same sense or on the same plane as human beings. Their relevance derives from the meaning attributed to them by humans and their function within socially constructed practices. Agency is conceived as something that emanates from the human individual (or from interacting human groups) and is exercised over the world(Alvesson & Kärreman, 2000).

In marked contrast, the ANT (Latour, 2005; Callon, 1986; Law, 1992) proposes a radically different conception of agency, which is seen as distributed and relational. For ANT, agency is not an intrinsic or innate property of a particular type of entity (such as the human being). Instead, the agency is an effect that emerges from the associations and relationships forged within a heterogeneous network (Latour, 2005). Any entity – human or non-human – that participates in a network and can make a difference in a process or the form of association is considered an actant and, in this sense, has agency (Callon, 1986).

Agency, in ANT, is not synonymous with human intentionality, conscience, or rationality. A non-human object can exert agency by its simple materiality and capacities (Schatzki, 2025). A closed door exerts agency by preventing passage; a ramp exercises agency by allowing wheelchair access; an algorithm exerts agency by classifying data in a way that influences a human decision; a database exerts agency by retaining or releasing information,

conditioning what can be known or done (Akrich & Latour, 1992; Law & Mol, 2002). These non-human actants are not merely passive instruments in the hands of human agents; They are active anticipants who, by being associated in networks, allow ,or prevent, specific actions and certain realities to be performed. Agency is seen as something that is accomplished by a Network between actants and not something that originates In fashion a single point (such as the human subject) and propagates outwards (Latour, 2005).

The human agency vs. distributed agency distinction directly reflects the different ontologies. The anthropocentrism of SCRT naturally leads to centering agency on the human, the creator and signifier of the social. The flat and symmetrical ontology of ANT, by treating humans and non-humans as analytically equal, leads to the conclusion that the ability to make a difference (agency) must be traced across the entire network of associations, regardless of the nature of the entity (Callon, 1986; Law, 1992). This difference has profound implications for analyzing complex organizational phenomena such as the implementation and impact of PA. An analysis based on SCRT would tend to focus on the agency of HRM practitioners, managers, and employees – how they decide to use PA, how they interpret its results, how they negotiate its meaning and legitimacy, and how their actions (or inactions) shape the incorporation of HRM into organizational practices. The agency would be in the human sphere, in conscious decision-making and social interaction about technology and data (Scott & Orlikowski, 2025).

An analysis based on the ANT tends to show the agency in a distributed way in the PA network itself. The raw data would have agency by revealing patterns or biases; the PA software would have agency in structuring the collection and analysis; algorithms would have agency in classifying and predicting; the dashboards would have agency by focusing attention on certain indicators; and data protection regulations (such as the LGPD) would have agency in restricting or allowing certain associations and uses of the data. So that, the agency would not only be in the PA analysts or managers who use the reports but in the ability of the entire PA network to make things happen – such as identifying candidates, flagging turnover risks, or redefining what high potential means – and thus, to enact in (Bräuner-Osborne et al., 2001). About the competitive enactment cycle, that implies a new reality in HRM, show how performance – the result of the action network – modifies mental models – the human/cognitive sphere, illustrating this mutual influence that involves agency at multiple points in the cycle (Chakraborty & Biswas, 2021; Frey & Osborne, 2013).

Understanding agency as distributed in the socio-material network allows us to analyze the capacity of PA to transform and re-signify HRM does not depend only on human acceptance or the social construction of its legitimacy but, on its intrinsic capacity as a network of heterogeneous actants to perform actions and cause effects that humans alone could not, that is, the union of two apparently distant theorical conceptions. PA agency lies in its ability to mobilize data, algorithms, software, and people into new associations that alter the routines, decisions, and, ultimately, the practical reality of HRM. Therefore, the transition from the idea of agency as a purely human locus to a distributed agency in the heterogeneous network constitutes an ontological watershed that requires an analytical approach capable of tracing the sources of agency in all relevant actants and analyzing how their interaction in networks engages organizational reality in ways that go beyond the mere construction of human meaning or intention (Berger & Luckmann, 1966; Callon, 1986; Latour, 2003; Law, 1984).

The joint adoption of these theoretical lenses offers a multifaceted analytical framework to investigate the resignifications that adopting PA – with its apparent socio-material composition and distributed agency – an organizational actor – can generate in the everyday reality of contemporary HRM. This approach not only expands the possibility of analyzing the social and institutional (by SCRT) and material, relational, and performative (by ANT) dimensions of the phenomenon but, enables an in-depth reading of power dynamics and processes of enactment that can reconfigure HRM (Bräuner-Osborne et al., 2001). The research of Bräuner-Osborne et al (2001) contemplates this relational search, considering the "agency" of a specific human protein, which has the ability to transmit signals from the inside to the outside of human cells.

It is based on this theoretical foundation that the analysis of the subsequent empirical results will be conducted, seeking to unveil how this complex interaction between human social construction and the performance of the socio-material network of People Analytics manifests itself in practice, transforming and resignifying the reality of HRM in the Brazilian pharmaceutical industry, through this new organizational actor.

1.7 The Analytic Architecture: SCRT-ANT Dual Approach to PA

Understanding the transformative capacity of PA, in this research considered one conjunction AI tools, the daily reality of HRM requires a theoretical framework that transcends views purely managerial, technological, or even solely social. This study argues that PA(AI) is not merely a tool or a set of metrics but emerges as a complex socio-material

organizational actor; with the potential to reconfigure practices, relationships, and the very meanings that constitute the reality of HRM (Sandberg et al., 2025; Scott & Orlikowski, 2025). To capture this complexity, this research adopts a dual analytic architecture, placing in critical dialogue the SCRT of Berger and Luckmann (1966) and ANT of Latour (2005) and Callon (1986).

This combination is justified by the need to apprehend both the phenomenon's symbolic-institutional dimension and the material-performative dimension. SCRT offers indispensable conceptual tools to investigate how PA is socially signified, legitimized, or contested, and internalized by human actors, shaping their perceptions and actions. However, an analysis limited to SCRT would risk underestimating the agency of the non-human elements—the data, algorithms, and platforms—that make up PA and their ability to actively perform and reconfigure organizational reality in material and relational ways(Berger & Luckmann, 1966).

This is where ANT becomes essential. By adopting the principle of generalized symmetry and focusing on the translation processes and the agency distributed in heterogeneous networks, the ANT allows tracking of how PA(IA), as a socio-material network, enacts new realities in HRM. It forces us to consider how algorithms classify, how dashboards direct attention, and how databases enable or restrict certain actions beyond mere human interpretation (Latour, 2001).

Therefore, the productive tension between these two perspectives is central to this qualitative, interpretative, and reflective investigation. SCRT anchors the analysis in the lived experience and construction of meaning by human actors. At the same time, ANT expands the look to the web of relationships and performances that involve actors of all natures. It is at the intersection of these lenses that we seek to respond to the objectives of this research: to evaluate the transformative and resignifying potential of PA in the daily reality of HRM in the Brazilian pharmaceutical industry, identifying how it is experienced and meaningful, analyzing the changes it promotes and exploring the (im)possibilities of its adoption considering the protagonism of human and non-human actors. This dual architecture thus allows for a critical and multilevel analysis of the emergence of a new analytical reality in contemporary HRM.

1.7.1 The Lens of SCRT: PA and the Social Construction of HRM Reality

Applying the perspective of SCRT (Berger & Luckmann, 1966), we understand the daily reality of HRM at the BPI as a social construction sustained by shared meanings, habitual practices, and legitimized institutions. The introduction of PA represents the entry of a new set of possible externalizations – new tools, metrics, discourses on efficiency, and strategy – that seek to objectify and institutionalize themselves in this social field. Research under this lens focuses on how this "new reality proposition" is negotiated. The first dimension is objectification and institutionalization: How do the metrics and insights generated by PA acquire (or not) the status of objective and reliable knowledge within the organization? How do routines around the use of data (dashboard review, discussions based on KPIs) become (or not) institutionalized practices, replacing or complementing previous forms of management based on experience or intuition? Language it becomes an important path when creating new categories (turnover risk, and readiness for promotion based on data) that come to define the reality of employees (Berger & Luckmann, 1966).

The second dimension, legitimation, proposes a reflection on the process by which PA gains (or loses) validity and acceptance? What arguments (cognitive and normative) are mobilized to justify its use and decisions? How does the organization deal with the second order of objectification by integrating the meanings derived from PA with the preexisting values and norms in the organizational culture, especially in an industry that values human care? Resistance can be seen as a failure in this process of legitimation (Berger & Luckmann, 1966).

The third dimension is located in internalization and typification: How do HRM professionals and managers internalize (or resist internalizing) the analytical logic proposed by PA? How does this affect their professional identities and their perception of the "reality" of work and performance? What new employee typifications, for example; high potential identified by algorithm vs. human evaluation, emerge and how do they impact everyday interactions? SCRT directs the analysis to the subjective experience of these actors and to the way they navigate and make sense of this changing reality. SCRT allows us to analyze PA as a phenomenon that challenges and seeks to reconfigure the symbolic and institutional order of HRM, focusing on the social processes of negotiation of meaning, establishment of legitimacy, and molding of subjectivities (Berger & Luckmann, 1966).

1.7.2 The Lens of ANT: PA a Network in HRM

The ANT (Latour, 2005; Callon, 1986) offers a complementary perspective by shifting the focus from constructing meaning to performing reality through heterogeneous networks. PA is conceived as one of these networks, composed of a myriad of human and non-human actants who enact HRM in a new way through their associations and interactions. Under this lens, the analysis focuses on Network Mapping: Identifying the relevant actors in the HRM-specific PA network at BPI: the types of data collected, the systems (HRIS, PA platforms, ERPs such as SAP or Oracle), the specific algorithms used (for recruitment, sentiment analysis, turnover prediction), the dashboards, HR analysts, managers, employees, and even regulatory actors such as the LGPD. Distributed Agency: Analyze how the agency is distributed in this network. Recognizing non-human agency is vital to understanding how PA operates materially. Translation Processes: Tracking the moments of translation (Callon, 1986) in the implementation and use of PA:

- *Problematitization:* How was PA defined as a solution to which HRM-specific problems at BPI?
- *Interessement:* What strategies were used to align (or failed to align) the interests of managers, HR, IT, and employees with the PA project?
- *Enrollment:* How were roles defined and accepted (or resisted)? Have the systems "agreed" to provide the data? Have users "agreed" to use the tool as prescribed?
- *Mobilization:* Are PA results (reports, predictions) accepted as reliable representatives of the reality they seek to describe? Where do controversies and failures in representation occur?
- Reality Performance: Describe how the operation of the PA network performs or *engages* the reality of HRM. How does the network "make" performance transparent? How does it "make" recruitment faster (or more biased)? ANT seeks to describe what the network *does* and how it stabilizes (or destabilizes) itself when it does.

ANT, therefore, allows for a detailed analysis of the socio-material infrastructure of AR and how its operations actively reconfigure HRM practices and outcomes, going beyond the perception or meaning attributed by humans.

1.7.3 Intercession: The Socio-materiality and the Significance of PA

The analytical strength of this research lies in the critical and complementary articulation of SCRT and ANT. Using only one of the lenses would offer a partial view of the complex phenomenon that is the introduction of BP in HRM. SCRT, with its anthropocentric focus, illuminates aspects that ANT can not to perceived how: The subjective experience of human actors in the face of technology, the complex negotiation of meanings, the processes of social legitimation that confer (or deny) validity to PA, and how this new analytical reality is internalized (or resisted) based on preexisting cultural and institutional values.

She (SCRT) reminds us that no matter how objective data may seem, its meaning and impact are always socially mediated. On the other hand, with its symmetrical ontology and focus on performance, ANT corrects the potential blindness of SCRT to the material agency of nonhumans. It demonstrates that AP is not just a set of symbols or ideas to be interpreted but an active network that makes, performs, and materially reconfigures the possibilities of action and knowledge. PA's ability to transform HRM derives from its material and performative inscription and social acceptance and significance (Berger & Luckmann, 1966; Callon, 1986; Latour, 2005; Law, 1992).

The productive tension between approaches becomes a powerful analytical tool. It allows us to investigate, for example, how the performance of an algorithm – perhaps generating unexpected or biased results – interacts with the processes of human interpretation and legitimation. Or how the social construction of PA as a “threat” can lead to resistance and non-use of the tool, preventing network stabilization (Berger & Luckmann, 1966; Callon, 1986; Latour, 2005; Law, 1992).

Returning to the example of sales force management at BPI: the ANT allows mapping the network that engages performance through metrics and dashboards. The SCRT, in turn, analyzes how this new performed reality is experienced by the salesperson, how he negotiates its meaning (Do these numbers reflect my real work?), how the typifications of the good salesperson are socially reconstructed, and how the legitimacy of this new evaluation system is established (or contested) in the organizational culture. The combination of the lenses allows you to see the dynamic interaction: network performance shapes social construction, and social construction influences network stability and evolution. This dual analytical architecture, which recognizes both the social construction of meaning and the socio-material performance of the network, offers a robust and reflective path to investigate how PA, as a complex organizational actor, is effectively transforming and resignifying the everyday reality

of HRM in the Brazilian pharmaceutical industry. It enables us to move beyond simplistic visions and capture this ongoing change process's richness, tensions, and contingencies.

CHAPTER II

2. Methodological Procedures

This chapter will be dedicated to the exposition of the methods selected for the execution of the research. Initially, the first section will discuss, in a non-exhaustive way, the philosophical currents that will underline the study. Subsequently, the second section will detail the chosen method to elucidate the research question and achieve the established objectives. In the subsequent section, the type of research to be conducted will be emphasized, as well as the nature of the social reality that will be the object of investigation.

2.1 Methodological Positioning

The current investigation employs a qualitative research methodology, which is particularly suited for endeavors that aim to gain a deep and nuanced understanding of complex social phenomena, specifically through the lens of participant's perspectives and lived experiences. Qualitative research is inherently interpretive, emphasizing the importance of the meanings its participants attach to their everyday realities and experiences rather than seeking the quantification of data or statistical generalization of findings across broader populations. This methodological approach facilitates an immersive exploration, which is rich and contextually grounded in the subject under investigation, recognizing and valuing the dimensions of subjectivity and intersubjectivity as valid and legitimate contributions to the body of scientific knowledge. The choice of methods and techniques for research is intrinsically linked to the researcher's worldview and conception of knowledge. This thesis is based on an epistemological position that guides understanding the phenomena investigated and how the data are interpreted (Denzin & Lincoln, 2011; Flick, 2022b).

2.1.1 Philosophical foundation of the research

The research is based on a philosophical foundation that intertwines the principles of the SCRT and the ANT, culminating in the adoption of Reflective Thematic Analysis (RTA) as an analytical strategy. SCRT, as articulated by Berger and Luckmann (1966), posits that social reality is a product of human activity constructed and maintained through dialectical processes of externalization, objectification, and internalization. From this perspective, language and intersubjectivity are bases for constructing a social world that, although created

by humans, presents itself to individuals as an objective and coercive reality. Agency here, is predominantly human, residing in the ability of individuals to create, negotiate, and internalize meanings and institutions.

In addition, the ANT, notably in the works of Latour (2005), expands this understanding by introducing the principle of generalized symmetry, which urges us to analyze the contributions of human and non-human actors with the same analytical lens. For ANT, reality is not just a human symbolic construct but a continuous performance of heterogeneous networks, where agency is distributed and relational, emerging from the associations between multiple actors, in these context – people, technologies, data, or algorithms. The translation process is central, describing how actors align (or fail to align) interests and mobilize others to form and stabilize these socio-material networks (Callon, 1986).

The articulation of these two theoretical lenses – SCRT and ANT – constitutes the dual and challenging analytical architecture of this thesis, allowing an in-depth investigation of how PA, understood as a socio-material organizational actor, influences and resignifies the reality of HRM. While SCRT illuminates the construction of meanings, the legitimation of new practices, and the subjective experience of HRM professionals in the face of the emergence of PA. The ANT focuses on the concrete performance of this network, the active role of algorithms, data, and platforms, and the translation processes that materially reconfigure practices and relationships. This dual approach recognizes that the reality of HRM is simultaneously a symbolic edifice and a socio-material performance(Berger & Luckmann, 1966; Latour, 2005).

It is on this complex epistemological foundation that Braun and Clarke's (2020) Reflective Thematic Analysis (RTA) is justified as the most appropriate analytical strategy. The ANT, emphasizing engaged subjectivity and the researcher's positionality, is intrinsically aligned with the constructionist view that knowledge is situated and co-constructed. The analytical process is not seen as a discovery of pre-existing themes but as an interpretative and reflective act, where the researcher, informed by his theoretical and epistemological baggage – specifically SCRT and ANT – plays an active and central role in generating and naming themes from the narrative data. The flexibility of the RTA thus allows us to explore both the meanings constructed by the participants, as well as the evidence of the performances and mediations of the non-human actors, which emerge in their speeches, enabling a rich and nuanced analysis of the transformation of the social reality of HRM by the introduction of PA.

Adopting this constructionist view has direct implications for the design and conduct of this research. The recognition that research questions are constructive engines that actively shape responses rather than merely extract them informs the reflective posture needed during interviews and analysis. The initial questions of the interviews are detailed in APPENDIX – A. The questions have a complex architecture that can incorporate assumptions and outline topical and action agendas. Therefore, the participant's answers are understood as intrinsically linked to the questions that provoked them, requiring that the analysis never detaches the participants' speech from the dialogical context in which it was produced(Berger & Luckmann, 1966; Braun & Clarke, 2020b; Latour, 2005).

This perspective also requires critical attention to the role of the researcher himself. The way participants are recruited under a specific category is already a constructive act that makes that category salient and encourages participants to formulate their answers in terms of it. In addition, the practice of reflexivity, while fundamental, is viewed through a critical lens. The conversations are held in a controlled environment, and of the interviewees baggage and positions are not treated as literal and transparent descriptions but as content that also participate in the argumentation and construction of the credibility of the research, managing the researcher's participation and interest in the result. This awareness is what underlies and justifies the choice for a Reflective Thematic Analysis (RTA), which places the researcher's engaged and conscious subjectivity at the center of the analytical process (Potter & Robles, 2022).

2.1.2 The role of the researcher

In line with the constructionist foundation adopted in this thesis, the researcher is not conceived as a neutral observer or a mere collector of pre-existing data. Despite this, it is recognized that the researcher plays an active and central role in co-construction of the investigated reality and its knowledge. This premise implies that the research's findings, or insights are not extracted passively, but are shaped by the interaction between the researcher, the participants and the context of the study itself. The language used, the questions asked, and the interpretations made are integral parts of the process of giving meaning and meaning to the phenomenon analyzed (Potter & Robles, 2022).

The constructionist perspective, as detailed by Potter and Robles (2022), illuminates several facets of this active participation of the researcher. From the early stages, methodological choices are already constructive acts, as mentioned earlier: the way

participants are recruited and categorized, for example, influences who participates and how their responses can be guided. Consequently, the answers obtained are understood as "versions" of reality, oriented towards action within the specific context of the interview, and not as direct mirrors of internal experiences or truths. Even the final writing of the research, including the thematic analyses, is an act of construction, where the researcher selects, organizes, and presents the data in a particular way to argue in favor of particular interpretations.

Given this ineluctable implication of the researcher in the research process, reflexivity emerges not as an option but as a methodological and ethical imperativeness (Alvesson et al., 2022). Reflexivity involves a continuous process of critical self-examination of how subjectivity, assumptions, values, personal and professional trajectory, and the researcher's interest and participation can influence interaction with participants, data interpretation, and results presentation. It is, therefore, a rigorous scrutiny of the power dynamics and interpretative choices that permeate the entire investigation (Braun & Clarke, 2020). In this thesis, reflective practice was an integral and continuous component. It manifested itself first in constructing the theoretical complementarity that the context demanded.

Specifically, in the exercise of understanding that the approaches of Berger & Luckmann could not offer a sufficient conceptual framework, given the phenomenon's complexity to be analyzed. The ANT, in turn, offers this possibility of conceptual deepening, but by itself, it imposes great complexity. A particular and specific evaluation of the similarities and differences between the two concepts was carried out to overcome this challenge, resulting in a specific section in this thesis. From these new premises – of conceptual alignment – the semi-structured interview script was created APPENDIX - A, aligning the personal and conceptual questions in search of the general objective of the thesis. The construction of this initial pre-field phase was exhaustively discussed with the advisor, who reflected in the result as a "peer debriefing". This moment of interaction was fundamental to eliminating rough edges, confronting interpretations, and unveiling potential biases. In addition, the researcher has extensive professional experience at the BPI, which increased interest in the theme and its practical consequences. The specificities of the BPI, as characterized by its strongly hierarchical structures, HRM is, at its core, centered on its human capital. Facts that enabled my interactions with the participants(Flick, 2022b).

If, on the one hand, the PA theme is still expanding in academia, in commercial or executive literature, the theme is widely and systematically disseminated on specialized

platforms. This fact demanded a new effort from the researcher, which consisted of reflecting and separating what was conceptually solid and what was being linked only with commercial interests. This analysis contributed to the distinction between the “hypes” of any innovation and the approximation of the core of the thesis, with the necessary relevance for the practice of the new HRM. This commitment to reflexivity aims not to eliminate subjectivity – which would be inconsistent with the constructionist perspective – but to make it transparent and manageable, strengthening the credibility and depth of the analysis presented.

2.2 Research Design

2.2.1 General plan

2.2.1.1 Nature of qualitative research

The present investigation adopts a qualitative approach to exploring the phenomenon under study. Qualitative research, at its core, is distinguished by its commitment to an in-depth understanding of human experiences, the meanings that individuals attribute to their experiences, and the social contexts in which these experiences unfold (Denzin & Lincoln, 2011; Flick, 2022a). Unlike approaches that seek quantification, measurement of variables, or statistical generalization, qualitative research focuses on the richness of details, the interpretation of narratives, and the elucidation of the complexity inherent to social phenomena (Creswell, 2010).

The choice of a qualitative methodology is fully justified because of the objectives outlined in this thesis. The general objective was to: To evaluate the potential of People Analytics as a new organizational actor that transforms and resignifies the daily reality of HRM in the Brazilian Pharmaceutical Industry. To this end, It was imperative to: Specific Objective 1 (SO01): To understand how PA and AI are experienced and signified in this context of BPI 's HRM; Specific Objective 2 (SO02): To analyze the daily changes in HRM processes resulting from applying AI and/or PA tools. Specific Objective 3 (SO03): Explore (im)possibilities of adopting PA/AI in HRM, considering the role of human and non-human actors. Such objectives require an immersion in the participants' perspectives, a dense analysis of their perceptions, and an understanding of the nuances that would be difficult to capture by standardized instruments or purely numerical analyses. Qualitative research offers the necessary tools to unveil the how and why of these transformations, going beyond the simple verification of their occurrence (Flick, 2022).

Furthermore, the qualitative nature of the research intrinsically dialogues with the social constructionist and interpretative philosophical foundation adopted – detailed in section 2.1.1. If the reality of HRM and the interventions of the PA/IA in this reality are understood as socially constructed phenomena and continuously (re)negotiated by the actors involved – whether human or non-human – the qualitative approach presents itself as the most fruitful methodological path. It allowed us to access and interpret the multiple "versions" of reality that emerge from participants' discourses and practices, focusing on the processes of signification and socio-material dynamics that constitute the phenomenon of PA at BPI. (Potter & Robles, 2022).

2.3 Participants

The careful selection of participants is a fundamental pillar in qualitative research, as the depth and richness of the data collected depend intrinsically on the experiences, perspectives, and narratives of the individuals chosen to be part of the study. This section details the profile of the participants in this investigation, the criteria that guided their selection, and the sampling strategy employed.

2.3.1 Definition, Selection and Characterization of Participants

To meet the objectives of this thesis, we sought to include participants with direct experience and practical knowledge about HRM and who were active in their functions. The group of participating interviewees comprised 10 (ten) HR professionals. The participants were directors and/or HR Business Partners (BP) of the BPI. The choice for this profile is justified: First, for its decision-making power in its function, in creating, developing, changing, or excluding HRM policies and practices; Secondly, for its connection to the upper echelon. These characteristics translate the capacity for interactions in both the horizontal and vertical dimensions.

This more holistic view of Board and BP positions tends to provide Insights in-depth information on the strategic needs of the organization, as well as unveil how the PA is being experienced, signified, and implemented (SO01), the changes that it (PA) brings to the daily processes of HRM (SO02) and the dynamics of interaction between human and non-human actors in the adoption of these technologies (SO03). These professionals are the ones who are at the forefront of the analytical transformation in HRM, making their perspectives indispensable for the present investigation. The selection of participants did not aim at statistical representativeness but rather at the depth and relevance of the information that could

be obtained, in line with this research's qualitative and exploratory nature. Therefore, individuals who, due to their positions and experiences, could offer rich and diversified narratives about the phenomenon analyzed were sought. Table 10 summarizes the characterization of each participant (Flick, 2022a).

Table 10: Classification of Participants

Interviewee	Gender	Age group	Function	Time in Function	Subordinate Quantit.	Answers to th	Superior Lease
01	M	41-50	Diretor	Above 10 years	11-30	CEO	Brazil Plant
02	M	31-40	HRBP	5,1 and 10 years	1-10	CEO	Brazil Plant
03	F	31-40	HRBP	2.1 to 5 years	NA	CEO	Brazil Plant
04	F	41-50	HRBP	2.1 to 5 years	31-50	CEO	International Plant
05	F	31-40	HRBP	0 to 2 years	1-10	CEO	Brazil Plant
06	F	60+	Diretor	Above 10 years	50-60	CEO	Brazil Plant
07	F	51-60	Diretor	Above 10 years	50-60	CEO	Brazil Plant
08	F	41-50	Diretor	Above 10 years	50-60	CEO	Brazil Plant
09	F	41-50	Diretor	2.1 to 5 years	1-10	CEO	Brazil Plant
10	M	51-60	Diretor	Above 10 years	50-60	CEO	Brazil Plant

Source: Research (2025)

A relevant characteristic of this list of participants is that those selected did not have preexisting social or professional relationships with each other. This absence of previous interconnection between the interviewees was considered a methodologically significant factor, as it tends to minimize the potential for mutual influence or coordinated narratives, which could arise in groups with a history of consolidated interactions. Such a condition favors the emergence of more individualized and potentially more diversified perspectives and constructions of reality about the evaluated, enriching the range of each version and each experience collected (Potter & Robles, 2022). By ensuring that each report represented a unique experience and interpretation, we sought to expand the heterogeneity of the data, a valuable aspect for a reflective thematic analysis that aims to capture the complexity and multiplicity of facets of the object of study (Braun & Clarke, 2020).

Additionally, although the confidentiality of the data is, in the first instance, a strict ethical safeguard ensured by the researcher to each participant individually, the lack of social ties between the interviewees may, secondarily, contribute to a perceived environment of greater security for the sharing of sensitive information. The guarantee of anonymity and the promise that reports would be treated in a de-identified and aggregated way were paramount (Flick, 2022). The absence of mutual knowledge among the participants reduces, for example, the fear that their identities or the specificities of their reports could be inferred or recognized

by other members of the group of interviewees, even if indirectly, thus fostering a space of greater openness and authenticity in the interviews This care aims to protect the participants and, at the same time, ensure the reliability and depth of the information that supports this thesis (Creswell & Poth, 2016)..

This individual strength of each participant is based on experience. 50% of the participants have more than 10 years of experience in the function, and only one interviewee has not yet completed two years of function (10%). This data does not diminish their contributions because, as E05 reports, "[...] The daily intensity is such that it seems much longer to me". Another important aspect is the number of subordinates of each participant. 60% of the interviewees had large teams of subordinates. Only E03, that the structural configuration of the organization – decentralized – does not provide a team for this interviewee here in Brazil, with all HR being geographically spread and mirrored around the globe, according to the expertise of each country, for the contribution to the organizational whole. With regard to education, it is observed that all the interviewees had an academic degree with an MBA in areas related to HRM, specifically HR. All participants reported direct or indirect involvement with the concept or with tools that use AI in their organizations. As already highlighted, some PA projects are in practical use.

In addition to the individual profile of the interviewees, the understanding of the investigated phenomenon requires an analysis of the organizational context in which their experiences materialize. The characteristics of the participating organizations – such as their size, type of capital, management model, and market segment – are elements that can significantly influence the adoption, implementation, and perceived impacts of these new analytical tools. Table 11, Characteristics of Organizations, summarizes the main characteristics of the respective companies represented in this study, whose data were collected to ensure the anonymity and confidentiality of each organization.

Table 11: Characteristics of Organizations

Organization	Capital	Company Size	Type Industry	Management	HR Size	Market	Nº Func. Brazil
01	Multintional	Big	Manufactires/ Markets	Professional	Up to 20 people	Dermatological	Above 1000
02	Multintional	Medium	Manufactires/ Markets	Professional	Up to 20 people	Biological	up to 500
03	Multintional	Medium	Manufactires/ Markets	Professional	Single management	Carcinogenic	up to 500
04	Multintional	Medium	Manufactires/ Markets	Professional	Above 50 people	General Line	up to 500
05	Multintional	Big	Manufactires/ Markets	Professional	Above 50 people	General Line	Above 1000
06	National	Big	Manufactires/ Markets	Familiar	Above 50 people	General Line	Above 1000
07	National	Big	Manufactires/ Markets	Familiar	Above 50 people	General Line	Above 1000
08	National	Big	Manufactires/ Markets	Familiar	Above 50 people	General Line	Above 1000
09	Multintional	Medium	Manufactires/ Markets	Professional	Up to 20 people	Dermatological	up to 500
10	Multintional	Big	Manufactires/ Markets	Professional	Above 50 people	General Line	500 to 1000

Source: Research (2025)

The analysis of Table 11 reveals a diversified panorama. There is a predominance observed organizations – 70% (7 out of 10) – have multinational capital, consequently 30% are national capital. Regarding the size of the company, there is a distribution between large companies – 60% and medium 40% with the total number of employees in Brazil varying considerably, from companies with up to 500 employees 40%, with one organization located between 500 and 1000 employees, and 50% of the participating organizations are registered as being organizations above 1000 employees.

A notable point of homogeneity is the Industry Type, where all companies (10 out of 10) are characterized by manufacturing and selling their products, even if, in a few cases, the factory is not installed here in Brazil, such as Organizations 2, 3, and 4. Regarding the management model, the majority (70%) adopt a professionalized management, and 30% operate under a family-run model. The size of the HR department also varies, with some organizations having physical and face-to-face structures of up to 20 people (30%), 60% with teams above 50 people, and one case with a single-person structure dedicated to HR here in Brazil. Finally, the markets in which it operates demonstrate specialization, covering segments such as Dermatological, Biological Products, Carcinogen Development, and the General Line. National organizations generally cover the latter market.

This diversity in the profile of the participating organizations is methodologically valuable. It allows the research to explore how the different capital configurations, sizes, management models, and HR structures can modulate the implementation and meanings attributed to the PA in the BPI. For example, pressures, available resources, and organizational culture tend to vary substantially between a large, professionally managed multinational and a national family-run company, directly impacting strategies and challenges related to adopting analytical technologies in HRM (Flick, 2022). The characterization of these contextual variables is, therefore, essential for a richer and more nuanced interpretation of the findings, contributing to the understanding of the complexity of the phenomenon in different scenarios within the same industrial sector (Creswell & Poth, 2016).

2.4 Data Collection

Data collection was carefully planned to capture the richness and depth of the experiences and meanings attributed to the participants. The main instrument for obtaining these narratives was the semi-structured in-depth interviews. This technique is aligned with the constructionist perspective by allowing a flexible dialogue and the co-construction of

understandings between researcher and participant (Kvale & Brinkmann, 2009). The choice for this type of interview sought not only to obtain answers to predefined questions but also to explore contexts, nuances, and individual perspectives in a detailed and contextualized way (Flick, 2022).

2.4.1 Detailed Data Collection Procedures

The empirical data collection process took place between November 2024 and January 2025. Each stage, from the initial contact to the conduct of the interviews, was guided by ethical principles and the objective of creating an environment conducive to the participants' sharing their experiences in an open and reflective way. Each initial contact was made first by e-mail and or WhatsApp, and later, a telephone contact was made so that the contextualization of the research and adequacy of the interviewee's agenda was made.

The interviews were conducted in person, which aimed to facilitate the construction of rapport and allow a more attentive observation of the nuances of non-verbal communication, enriching the understanding of the interview context (Flick, 2022). The meetings took place predominantly in the offices of each interviewee. Only E01, the interview was conducted in a specific place, outside his office, for reasons beyond the researcher's control. Another exception – E04 – her interview was conducted virtually, as the interviewee had been promoted to the position of General Director for Latin America and was in the United States at the time of the research, a fact that did not harm or reduce the quality of the material produced, on the contrary, her experience and details of her project was another milestone of the research.

A relevant observation about the context of the BPI organizations participating in this study concerns the configuration of their workplaces. It can be seen that the offices are primarily located in modern corporate facilities, located in prestigious business centers in the city of São Paulo, Brazil. Characterized by their advanced technological apparatus, these spaces seek to represent the sector's dynamism and investment, fostering an organizational environment conducive to adopting innovations and integrating advanced technologies into its management processes. All meetings were held in private rooms, which ensured confidentiality and comfort, as well as privacy and total interaction between interviewee and researcher.

Face-to-face interviews, which are carried out in environments with which the interviewee is familiar, provide an environment of greater confidence and perception of

greater control of the interviewee; that is, it mitigates the pressure of being questioned and the fear of not generating the expected answers. Therefore, all interviews were unruffled, uninterrupted “conversations” in an atmosphere of genuine co-creation among all participants. So much so that, in all interviews, the scheduled time was used efficiently, constituting a period in the interviewees daily lives of relaxation and sharing(Creswell & Creswell, 2021; Flick, 2022b).

The interviews conducted in this investigation lasted an average of 62 minutes each. All interviews began with the express request for authorization to record the interviewee. To ensure the faithful and integral capture of the participants' narratives, all sessions were recorded in audio through the use of a professional digital recorder, model Tascam DR40 X. The choice of this specific equipment was due to its ability to provide high-definition sound capture. This condition proved to be fundamental to ensure the subsequent elaboration of accurate and detailed transcriptions. This fidelity in transcription is an essential pillar for the quality and rigor of the qualitative analysis of the data that followed. Thus, all transcriptions were performed manually; that is, the researcher interacted with the recorded audio as many times as necessary so that all the nuances were captured. In each transcription, at least 11 hours of work were invested.

This transcription process, carried out manually by the researcher, who dedicated an average of eleven hours per interview, had transcended a mere technical task of converting audio into text. The repeated and attentive listening of the recordings, seeking to capture not only the explicit verbal content but also the nuances, intonations, and significant pauses, constituted a first and deep dive into the experiential universe of the participants. As Braun and Clarke (2006, 2020) recommended, this immersion and intimate engagement with the material are characteristic of the initial phase of familiarization with the data, a fundamental and indispensable step in RTA. With the transcriptions completed and this initial familiarization established, the material was adequately prepared to be submitted to the systematic and interpretative analysis that will be detailed in the following section.

2.5 Data Recording and Analysis

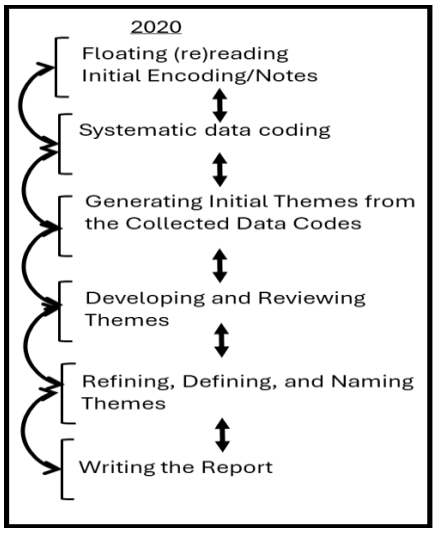
With the conclusion of the collection stage and the literal transcription of the ten interviews carried out, a robust and detailed textual corpus was obtained, ready to be submitted to a systematic and interpretative analytical process. The transcribed interviews produced a significant body for analysis, totaling 197 pages of written content, an average of 19.7 pages

per interview; the phase of the research RTA was chosen according to the guidelines proposed by Braun and Clarke (2020). This approach was chosen for its methodological flexibility and, fundamentally, for its alignment with the social constructionist perspective that underlies this study, allowing an in-depth exploration of the meanings and realities constructed by the participants in their narratives about PA. RTA values the active and reflective role of the researcher as the primary analytical instrument, engaging in a deep dialogue with data to identify, analyze, and report patterns of meaning (themes) in a theoretically informed and contextually sensitive way.

To facilitate the management, organization, and coding of the vast volume of textual data from the transcriptions, the Computer-Aided Qualitative Data Analysis (CAQDAS) software NVivo, version 15®, was used. The license was purchased with funds from the researcher, directly from the official software vendor. It is important to note that NVivo 15® was used as an operational support tool, assisting in the systematic organization of the material, in the segmentation of data into codes, in the exploration of relationships between different codes and themes, and the maintenance of an auditable record of the analytical process (Bazeley & Jackson, 2013). Specifically, this version of NVivo 15® has an AI tool added to its license. However, it is reiterated that the interpretative task of identifying, constructing, and refining the themes remained entirely under the intellectual responsibility of the researcher in a continuous reflective engagement with the research corpus.

The analytical process, guided by the principles of RTA, unfolded in interconnected and sometimes recursive phases, which will be detailed below. This methodological path sought to ensure rigor in the analysis and an interpretation that was faithful to the richness and complexity of the experiences shared by the participants, always in dialogue with the theoretical references and objectives of this thesis. However, it is essential to emphasize that these phases, although presented sequentially for didactic clarity and structuring of the methodological report, were not seen as a rigidly linear or prescriptive process. Instead, in keeping with RTA's intrinsically reflective and organic nature, the analytical journey involved a movement back and forth. The deep engagement with the data, the analytical decision-making, and the interpretative construction of the themes required, at various times, a return to previous phases, a refinement of codes, or a reevaluation of emerging themes, always in the light of the concept (Braun & Clarke, 2020). These phases are summarized in Figure 9

Figure 9: Phases of the RTA



Source: Adapted from Braun & Clarke (2020).

2.5.1 Familiarization

The analytical journey of this research began with the exporatory phase to familiarization with the data. This process aims at a deep and comprehensive immersion in the collected textual corpus (Braun & Clarke, 2006, 2020). As mentioned, after being meticulously transcribed in full by the researcher, the ten interviews resulted in a considerable volume of material, with an average of 19 pages, and an average of 8,000 words per transcription in Word format. This rich dataset was then carefully imported into the NVivo 15® software environment. Within NVivo's data management system, each transcribed interview was designated as an individual “case”, representing the narrative of each of the ten participants. Similarly, the contextual information and the organizations characteristics were also structured as cases in the software, allowing future relational and contextual analyses.

However, from RTA perspective, familiarization transcends this initial organization, demanding an active and repeated engagement with the full content of each transcript (Braun & Clarke, 2020). Thus, this phase involved the careful reading and rereading of all the interviews, a process in which I sought not only to apprehend the manifest content of the speeches but also to begin to identify nuances, emphases, possible contradictions, and the particularities of the language and tone of each participant. As already explained, the very act of manual transcription, with repeated audio listening, constituted the first and most valuable moment of this immersion, allowing an initial connection with the voices and stories shared.

During the subsequent readings, with the material already organized in NVivo, initial notes were made directly in the documents or through “memos” associated with the cases. These notes recorded preliminary thoughts, potential ideas for codes, observations on particularly rich or intriguing passages, and reflections on how the content of an interview connected or differed from others. This process of active reading and annotation was not, at this point, aimed at formal coding but rather to deepen intimate knowledge of the dataset as a whole, identifying the breadth and depth of the available material and setting the stage for the subsequent initial code generation phase (Braun & Clarke, 2020b). For example, the memoranda of interviewees E01 and E02 regarding the organizational structure necessary for a consistent advancement of analysis with PA:

> Memorandum 1 E01: The interviewee makes it very clear that his organizational structure in relation to data analysis and provision falls far short of his need or a minimum ideal. This statement is supported by comments on his previous professional experiences in other laboratories. Therefore, this characteristic of lack of structure is not widespread.
>
> Memorandum 1 E02: It is clear to the interviewer that the organization's desire to present itself as an innovative organization, which is present throughout its physical structure, office location, access all digitized, technical modernity of the physical structure. This reinforces the paradox between what is presented and what is actually constituted, the use of organizational aesthetics to reinforce apparent innovation, rather than as a resource for organizational development. Hierarchical structures are evident, so social interactions are basically carried out in a formal setting and.

Therefore, the primary goal of this familiarization phase was to achieve a holistic and detailed understanding of each individual narrative and the corpus of data in its entirety. This understanding was progressively built by exhaustive reading and rereading of the transcripts and by the continuous recording of reflections and initial observations, materialized in analytical memorandums as in the examples highlighted. Examples of these notes include the perception of the structural lag in data analysis reported by Interviewee 01, who contrasted this reality with his previous experiences, or the identification of an apparent paradox in the

organization of Interviewee 02, where an aesthetic of modernity and technological innovation seemed to coexist with less evident organizational development practices. In addition, observations about the broader context, such as the formality of social interactions in markedly hierarchical environments, were woven. This exercise of recording “my comparative reflections”, as mentioned earlier, aimed to capture the first intuitions about the dynamics at play, including the congruence – or lack thereof – between the manifest discourse of the participants or the organizations and the practices experienced or inferred from the reports (Braun & Clarke, 2020).

2.5.2 Generation of Initial Codes

After the intensive familiarization phase, the subsequent step consisted of generating initial codes, marking the transition from a holistic immersion to a more segmented and detailed analysis of the data corpus. This process involved the systematic reading of each transcript, during which concise labels – the codes – were identified and assigned to specific pieces of data, words, sentences, or entire paragraphs, which proved significant in answering the research questions of this thesis. As a result of this first analytical scan, the ten interviews originated an initial set of 128 preliminary codes, applied to data segments of varying lengths, and that appeared with different degrees of recurrence among the participants(Braun & Clarke, 2020a, 2020b).

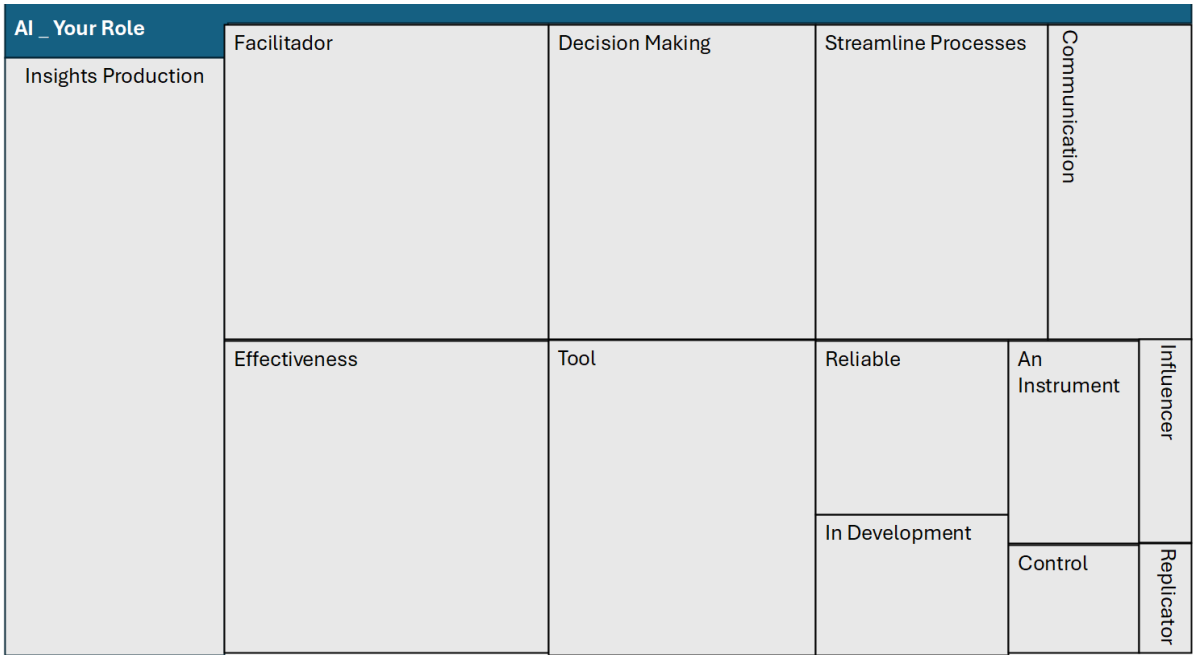
Initial codification, in RTA, is intentionally an inclusive and thorough process oriented toward capturing the maximum diversity of meanings and potential patterns pertinent to the scope of the study without a premature concern with the final structure of the themes (Braun & Clarke, 2020a). This first effort to name and categorize relevant aspects of narrative content to initiate the organization and reduction of the vast textual material, paving the way for the later identification of broader thematic patterns.

The codes were developed as fundamental interpretive units, anchors or conceptual hooks to unravel participants' experiences and perceptions of the role and challenges of AI and PA at BPI. Using the NVivo 15® software resources, each code was methodically associated with the respective text segments in the transcriptions of the interviewees' "cases." The approach to generating these codes was predominantly inductive, allowing them to emerge from the participant's speeches. This was reflected in the creation of codes that, initially, were kept close to the language of the interviewees, that is, the codes were being generated freely – still without worrying about how and where they would be located.

Although the RTA advocates this openness to data, it is inherent to the process that the researcher's theoretical background – in this case, familiarity with SCRT, ANT, and the specific literature on PA – sensitizes the gaze to certain aspects, subtly influencing what is identified as significant (Braun & Clarke, 2020, 2020a).

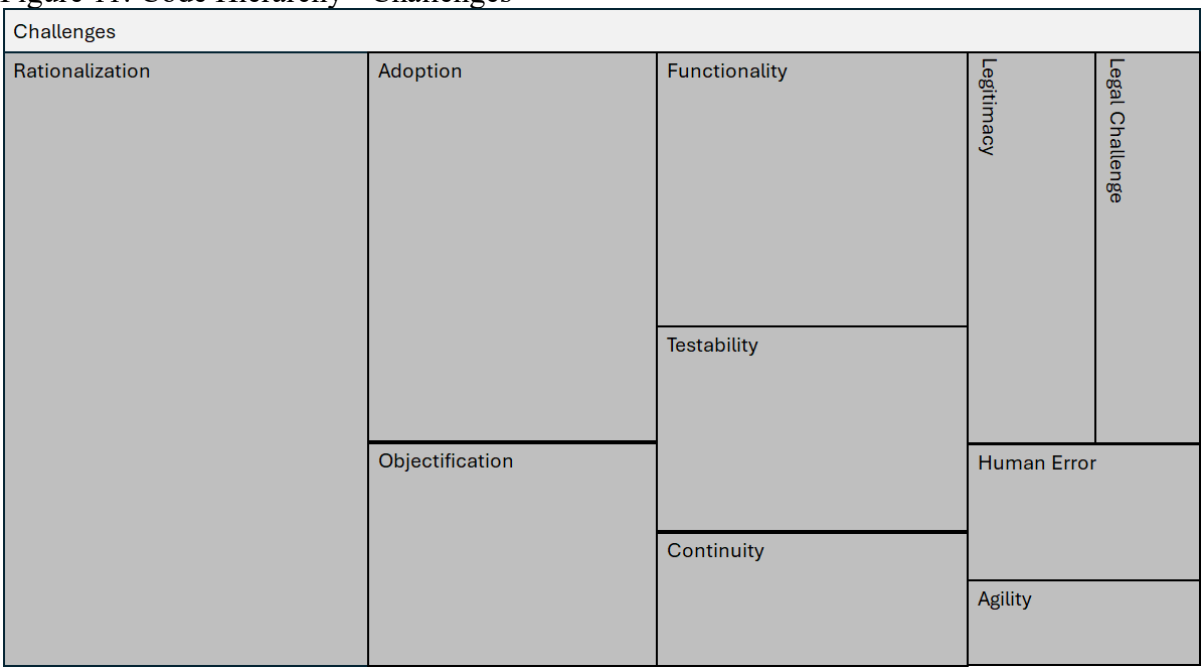
NVivo 15® played an instrumental role in enabling a systematic and organized recording of these encodings, facilitating the retrieval of the data snippets and the visualization of how these initial codes could begin to cluster around broader concepts. This preliminary organization of the codes, for example, allowed the visualization of emerging hierarchical structures, such as the one presented in Figure 10, where the conceptual node “AI its role” begins to articulate with dimensions such as “Production of Insights”, “Facilitator” and “Decision Making”, and, similarly, Figure 11: Code Hierarchy - Challenges which explores the “Challenges” such as “Rationalization”, “Challenge of adoption” and “Objectification”, indicating the first steps in the aggregation of meanings that would lead to the construction of the themes (Bazeley & Jackson, 2013).

Figure 10: Code Hierarchy AI Your Role



Source: Research (2025) – Figures generated by NVivo15®

Figure 11: Code Hierarchy - Challenges



Source: Research (2025) - Figures generated by NVivo15®

The richness of the initial codification lies in its ability to identify significant aspects of the data that, although granular, point to broader understandings of the phenomenon, which is an eminently interpretative process. Initial reflections recorded during the familiarization phase, such as the examples of the memoranda for E01 and E02, directly informed the generation of some of these codes. For example, Memorandum 1 E01, which highlighted the respondent's perception of the deficiency in the data analysis structure in his current organization, comparing it with previous experiences and signaling that such a deficiency was not widespread in the sector, led to the creation of codes such as Organizational Structure or Analytical Maturity. Similarly, Memorandum 1 E02, which pointed to a paradox between organizational aesthetics, as a mirror of the organization's innovation (modern physical structure, digitalization) and possibly less bold organizational development practices, suggesting the use of aesthetics to reinforce an “apparent innovation”, inspired codes such as Innovative Organization. This vast and detailed set of initial codes, the product of a systematic, reflexive and interpretative engagement with the data, established the solid and indispensable basis for the progression to the subsequent phase of the Reflective Thematic Analysis: the active search and construction of candidate themes(Braun & Clarke, 2006, 2020b, 2020a).

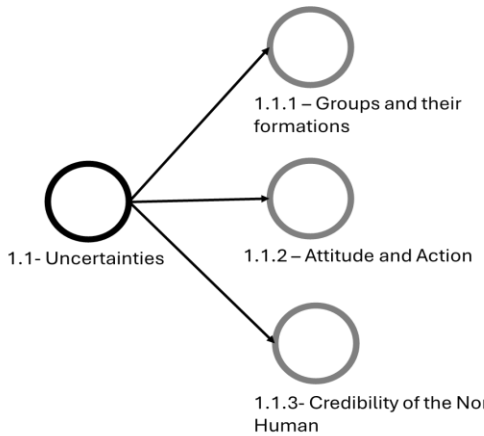
2.5.3 Search by themes

In this phase of the RTA, a “testing” movement operates, where the researcher seeks to ensure that the arguments and narratives of the participants, now represented by the codes,

align with their deepest convictions and the central meanings that emerge from the data set. This process demands an intensified dedication on the researcher's part, as it requires a return to transcriptions, which must be accessed more judicious and analytically than in the initial coding. It is not just a matter of rereading but of engaging in detailed scrutiny, where all the signs of the participant's argumentative process – both explicit statements, justifications, and clear lines of reasoning, as well as implicit elements such as hesitations, emphases, contradictions, or even the absence of certain discourses (the lack of them) – are considered for the construction of the initial categories that will give rise to the themes (Braun & Clarke, 2020b, 2020a).

This close attention to the multiple layers of discourse is critical to discern whether an observed pattern reflects a genuine, recurring conviction or a more superficial or isolated manifestation. Identifying these argumentative structures and underlying convictions allows the researcher to begin to group related codes into more robust and meaningful initial categories, representing the first sketches of the themes that will be refined, later refined. The NVivo 15® software allows the organization and hierarchical visualization of nodes (Figure 12), as in the example of the category “Uncertainties” and its descendants “Groups and their formations, Attitude and Action, and Credibility of the Non-Human (IA)”, helps in this process of identifying how codes can coalesce around central ideas, forming the basis for the candidate themes. These initial categories are, therefore, the result of careful interpretation and an effort to capture the essence of the participants experiences and meanings about the phenomenon investigated.

Figure 12: Category Formation



Source: Research (2025) Image generated by NVivo 15®

The identification of “Uncertainties” as a central node, unfolding into aspects such as “Groups and their formations”, “Attitude and Action”, and “Credibility of the Non-Human (IA)”, highlights the complexity inherent in the introduction and perception of an PA project, for the HRM of the BPI use the social lens. These initial categories, although provisional, signal how the participants construct and negotiate the reality of a changing organizational environment, where the emergence of new non-human actors and new analytical logic generate ambiguities and questions. The Uncertainties category, for example, can range from doubts about the return on investment in PA (E01, E04) and the prioritization of these tools over other demands (E07) to hesitation around the full capacity and real impact of AI on HRM, reflecting a phase in which the analytical reality is still in the process of objectification and legitimation(Berger & Luckmann, 1966; Braun & Clarke, 2020b; Latour, 2005).

2.5.4 Review of the themes

With these preliminary code groupings and the initial categories outlined, the researcher moves on to the fourth phase of the RTA: Reviewing the preliminary themes (Braun & Clarke, 2006, 2020). This stage as it thoroughly scrutinizes the categories that emerged from the previous phase. The objective here is to ensure that the proposed themes are coherent and internally consistent and accurately and richly capture the meanings present in the dataset concerning the research question. This review process operates on two interdependent levels.

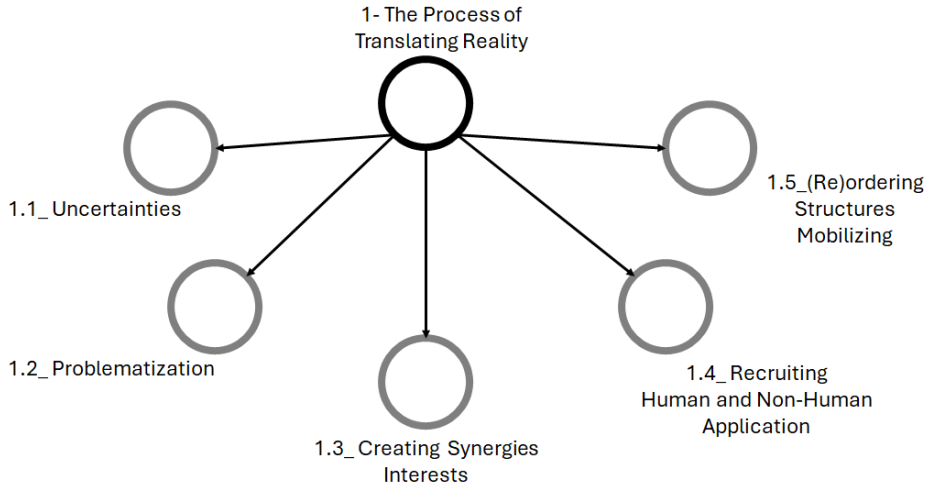
At the first level, the researcher revisits the coded data extracts allocated to each initial category. The central question is: Does the data support the proposed theme? It examines whether the extracts form a coherent pattern and whether the theme captures the essence of these data convincingly. Some data extracts do not fit well, indicating that the theme needs to be refined or that the extract would fit better into another theme. In other cases, the data set within a theme may be very diverse, suggesting that the candidate theme is too vague or complex and needs to be broken down into separate themes. It is a moment of "fine-tuning," where the researcher ensures that each theme has a clear and distinct narrative, well-grounded in the data that compose it (Braun & Clarke, 2020).

At the second level, the themes are reviewed in relation to the dataset as a whole and the research question. The researcher evaluates whether the preliminary thematic map – the set of candidate themes and their interrelationships – adequately represents the story told by the data. The question is: Do the themes cover the data's most salient and relevant aspects in relation to the study's objectives? Are there significant overlaps between the themes that could

be merged? Are there important gaps in the data that are not being adequately represented by current issues. This phase also involves considering the "validity" of the themes, not in the positivist sense, but in the sense of how well they capture the meanings and experiences of the participants, and whether the analysis offers a plausible and insightful interpretation of the phenomenon. The review of themes is, therefore, an iterative and reflective process, where the researcher may need to return to previous phases, recode data, discard candidate themes that do not hold up, merge similar themes, or divide comprehensive themes, always in a constant dialogue with the data and the objectives of the research.

At the end of this meticulous review phase, the researcher aims to achieve a set of well-defined, internally homogeneous, and externally heterogeneous themes that offer a rich and representative representation of the patterns of meaning relevant to the research question. In the context of the present research on transforming HRM into the BPI by the emergence of PA/AI, this thematic purification and consolidation process can culminate, for example, in articulating a central and multifaceted theme such as "The Process of Reality Translation". This theme, as illustrated in Figure 13, does not present itself in a monolithic way but unfolds into interconnected sub-themes or categories – such as "Uncertainties," "Problematization (protagonism)," "Creating Synergies (interests)", "Registration (recruiting humans and non-humans)" and "(Re)Ordering Structures (mobilizing)" – each capturing a specific dimension of how the reality of HRM is perceived, negotiated and reconfigured by the introduction and mediation of new analytical actors. The identification and refinement of a theme of this nature, with its constituent components, demonstrate RTA's ability to go beyond the superficial description to build an analytical interpretation that unveils the complexity of the phenomenon studied, preparing the ground for the subsequent phase of definition and final naming of the themes and the preparation of the research report.

Figure 13: A Theme in the Making



Source: Research (2025) – image generated by NVivo 15®

2.5.5 Refining, Defining, and Renaming Themes

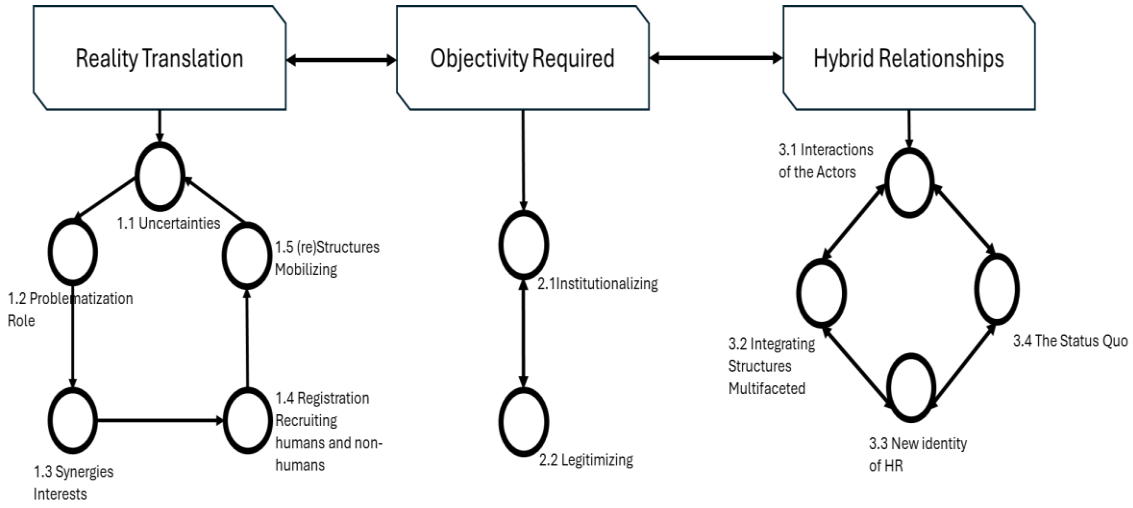
The fifth phase of the RTA assumes a central role in the consolidation of the analysis. At this point, the researcher is dedicated to refining the essence of each topic that demonstrated robustness and relevance during the review. The aim is to clearly articulate the scope and meaning of each theme, defining what it encompasses and, equally important, what it does not. For each final theme, a detailed analysis is elaborated that captures its intrinsic narrative, exploring its nuances and the way it manifests itself through the participants' data (Braun & Clarke, 2006, 2020, 2020a).

Fundamentally, the themes are rigorously aligned with the research objectives at this stage. In the context of this thesis, this means that each consolidated theme should directly contribute to responding to the general objective (GO)and, simultaneously, to address the specific objectives (SO01, SO02, SO03) that investigate how PA is experienced and signified, the procedural changes it entails, and the (im)possibilities of its adoption considering the protagonism of the actants. At this stage, the relationships between the codes, as well as the relationships between the categories, are practically finalized, presented in APPENDIX B.

The thematic structure that emerges outlines the analytical axes of the sections: 3.1 The Translation of Reality in HRM with PA, 3.2 Objectivity Required by PA, and 3.3 Hybridization as a Management Strategy in Everyday Relationships represent more than a simple organization of ideas. It constitutes a new form of analysis, a robust interpretative

framework that allows a critical and reflective dissection of the organizational reality based on the interviews conducted represented in Figure 14.

Figure 14: Defining Themes



Source: Research (2025)

Build the name of the themes consists of relevant task at this stage. The names chosen should be concise yet evocative, capturing the topic's essence in an immediately understandable and analytically potent way. They should be more than mere descriptive labels; they must carry an interpretative weight that signals the contribution of the theme to the global understanding of the phenomenon. Thus, creating a refined and significantly named thematic structure, such as the one glimpsed in the figure, becomes the skeleton of the final analysis. It is through this structure that the researcher will weave his argument, using empirical data (extracts from the interviews) to illustrate, sustain, and deepen the interpretation of each theme, offering a consistent and robust view of the complex interaction between PA/AI, human actors and the social reality of HRM in the Brazilian pharmaceutical industry. This process, informed by the theoretical lens of SCRT and ANT, ensures that analysis transcends the surface of the data to reveal the underlying dynamics of meaning-making, distributed agency, and organizational transformation (Braun & Clarke, 2006, 2020, 2020a).

2.5.6 Use of data for reporting

We move on to its sixth and final phase: the production of the report. This stage represents the culmination of the analytical process, where the interpretive work carried out in the previous phases transforms into a cohesive, compelling, and academically rigorous narrative. What has been accomplished up to this point – the immersion in the data, the generation of codes, the search, review, definition, and naming of robust themes aligned with

the research objectives – establishes the foundation for the report. The final themes, exemplified by their respective subcategories and detailed analyses, become the structural pillars of the analysis and discussion chapters (Braun & Clarke, 2006, 2020).

Therefore, the task of effectively communicating the results of this analysis lies ahead. Using data for the report, under RTA guidelines, goes far beyond a simple presentation of findings. It involves stitching together an analytical argumentation where the identified themes are explored in depth, and the extracts from the interviews are strategically employed not only as illustrations but as vivid evidence that supports the researcher's interpretation (Braun & Clarke, 2021). Each theme will be presented and discussed, demonstrating how it contributes to answering the central question of the research and the specific objectives outlined, so the report will seek to articulate the "story" that the data tells. In this final phase, the analytical and reflective voice of the researcher is prominent. The report offers an engaged and theoretically informed interpretation, explaining the meaning and implications of the identified themes. Thus, the use of data in the final report aims to build a persuasive and well-founded argument, which not only presents the results of the research but also demonstrates the rigor of the analytical process and contributes significantly to the understanding of the investigated phenomenon, revealing the underlying dynamics of meaning-making, distributed agency, and organizational transformation in the context of contemporary HRM.

Seeking to meet the criteria of rigor that govern contemporary qualitative research, this manuscript employs a carefully structured methodological and analytical design. The quality approach adopted is inspired by the perspectives of authors such as Flick (2022) and Creswell (2010), focusing both on the application of validation strategies to ensure the reliability of the data and on the maintenance of reflexivity and adequacy throughout the research process. The credibility of the findings is fostered, first and foremost, by prolonged and persistent engagement with the field and the data.

This is manifested in conducting ten semi-structured in-depth interviews, with an average of 62 minutes each, and in the exhaustive process of manual literal transcription, which required approximately eleven hours per interview, allowing a deep immersion in the participant's narratives. In addition, the systematic application of the six phases of the RTA, from familiarization with the data to the definition and naming of the themes, represents persistent observation and continuous scrutiny of the empirical material. The "peer debriefing" with the advisor of this thesis, as explained in the section on the role of the researcher, served as a vital mechanism for the critical questioning of interpretations and the

mitigation of biases. Transferability, in turn, is addressed by the detailing of the profile of the participants and their organizations (Tables 10 and 11) and by the detailed exposition of the data collection and analysis procedures in Chapter II (Method), allowing readers to ascertain the relevance of the findings to other scenarios (Flick, 2022b; Lincoln & Guba, 1985).

The stability (dependability) of the research is supported by creating a robust and transparent audit trail, embodied in the detailed explanation of the methodological path in Chapter II. The detailed description of each phase of the RTA, from familiarization with the data through the generation of codes, search, review, definition, and naming of the themes to the future production of the report, together with the use of NVivo 15® software for organization and management of data and codes, it allows the traceability of analytical decisions. This systematic record ensures that the interpretive process is amenable to scrutiny in a interactive and iterative process.

The objectivity (confirmability) of the findings, in turn, is reinforced by this same audit trail, but also, by the explicit commitment to reflexivity, detailed in the section “The role of the researcher” and practiced throughout the RTA (Lincoln & Guba, 1985). This reflective exercise aims to become aware of and manage the researcher's assumptions and subjectivity, ensuring that the interpretations and themes developed are firmly anchored in the participant's narratives and supported by direct evidence from the data, as recommended by the ANT guidelines for preparing the report. This methodological cycle, which intertwines deep engagement, systematic analysis, and continuous reflexivity, provides the basis of reliability on which the data analysis will be presented below is based (Flick, 2022).

CHAPTER III

3. Data in the Light of Theories

We proceed to analyze the empirical content – the interviews – from the perspective of the theoretical lenses presented above (SCRT and ANT) to understand and answer the research question, and the understanding of how this actor alters the organizational daily life.

3.1 The Translation of Reality in HRM with PA

Contemporary HRM is undergoing a process of profound transformation, driven by increasing digitalization, datafication, and the emergence of analytical tools, commonly grouped under the term PA or AI applied to HRM (Dasari & Sujatha, 2023; Dasari & Devi, 2024; Thorpe & Sellar, 2023). These new technologies promise to revolutionize the way

organizations manage their human capital by offering data-driven insights to optimize processes, make more assertive decisions, and align HRM with business strategy(Ayuningtias & Wahyuningtyas, 2024; Ioakeimidou et al., 2023). However, introducing and adopting these tools do not occur in a social or technical vacuum but rather in specific organizational contexts permeated by cultures, structures, power relations, and preexisting meanings (Jansen et al., 2023; Khan & Millner, 2023).

More specifically, the BPI is characterized by an environment of high competitiveness, high regulation and generally with a strategic focus on differentiation. Differentiation is rooted in creating unique value propositions that resonate with consumers, increasing customer loyalty and ensuring a lasting competitive advantage. Differentiation can be achieved through various means, including product and technology innovation, quality enhancements, brand management, and developing unique intangible assets such as patents and strong R&D networks (Singh et al., 2022). Another feature is the strict regulation of innovative technologies, such as nanomedicine and additive manufacturing, which require up-to-date regulatory pathways to address their unique complexities and ensure they meet quality standards without stifling innovation, thereby promoting greater competitiveness (Eubanks, 2022).

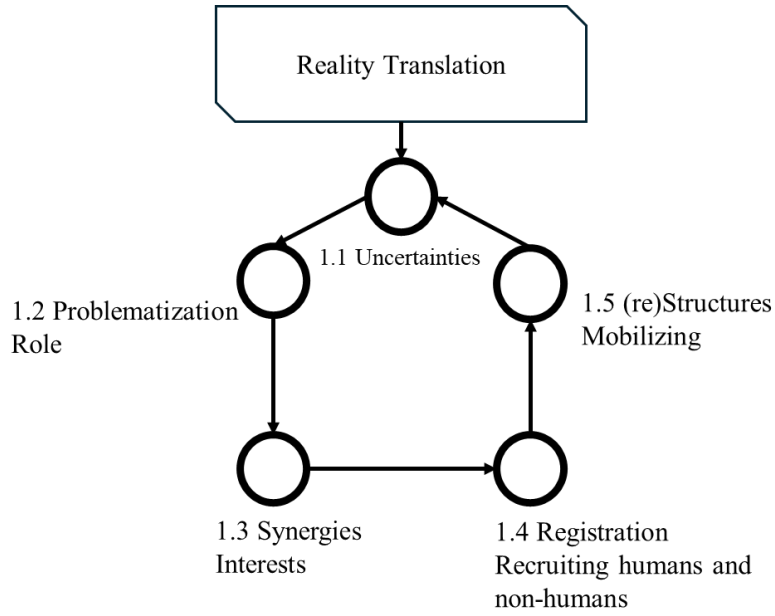
BPI emphasizes innovation in products and services, as well as HRM policies and practices. There is a continuous effort to update, modernize, and focus on people, so the BPI represents a fascinating scenario to investigate PA(IA). But, paradoxically, at the same time that the BPI invests in technology for research, development, and production, like many traditional organizations, it may present significant resistance and specific challenges to the incorporation of new management logic, especially in the area of HRM, often perceived from a more administrative – functionalist – perspective, not meeting the current demand for a greater analytical and quantitative emphasis on HRM, as stated by interviewee(E02):

> “[...] I think it's super cool (the adoption of new technologies), it's because maybe, as I said [sic], maybe the thinking of HR professionals is still from the old personnel department."

To meet the SO01 – to understand how the PA is experienced and signified in the BPI’s HRM, the concept of Reality Translation is a starting point to explore how the PA is redefining what counts as valid knowledge (uncertainties), how this process is changing the daily life of HRM (reordering), the effort that is being made to produce synergies to face the new intra and interorganizational relationships, as well as how the legitimate action of this new actor – non-

human – is being perceived in the HRM of the BPI. The analysis will be developed around four thematic categories: Uncertainties and Problematization, Synergies and Inscription, and Reordering Structures. Each section details how the PA is experienced and signified by the Directors and Business Partner HR (HRBPs), articulating the interviewees' reports with the concepts of CSRT and ANT. (Berger & Luckmann, 1966; Callon, 1986; Latour, 2005). Figure 2 outlines the proposal for the analysis of this theme, which hides in its apparent linearity, considerable complexity, and recursion.

Figure 15:Reality Translation



Source: Research (2025)

3.1.1. The Construction of Need: Uncertainties and Problematization of the PA

The introduction of the PA in the HRM of the BPI is not presented as a linear or automatically accepted process. On the contrary, the interviews reveal that the emergence of this new analytical reality often begins in a field of ambiguities and questions, where its need and value need to be actively constructed. This subtopic explores the initial phases of this construction, marked by "Uncertainties" and "Problematization".

3.1.1.1 Navigating Uncertainty: The Need to Build Credibility

The initial experience with the PA is often marked by uncertainty. Managers' report doubts about the return on investment, such as:

> "E01 – And if you make this investment for this condition, do you have a return back too or not? So I think that's what's important to understand."

> “E04 – So it's very *business-driven.* That is one point. Second point, which is less controversial, is difficult to prove. Because you have to be very comfortable with an ROI that is not very clear. So you are an ROI, so the culture, again, is a culture that represents your society, but has changed the result of the organization [...]”

In addition, the reports reinforce that there is uncertainty about what priority should be given to these tools in the face of other demands:

> “E07 – We are going to calculate how much we would spend per vacancy, if it were done by consultancy. We will see the total amount. Then we will see what the cost of our team is. Then we'll see how much a LinkedIn costs for this group to work on recruiter. And we'll see how much the premium costs if I want to pay individual for you. It's impossible for the guy to say that he won't pay me for the technology based on the savings I'm bringing to him [...]”

And even about the understanding and potential of PA in HRM. Uncertainties that tend to be a limiting factor for its adoption and/or expansion of more robust analytical projects, as one interviewee points out, questioning the lack of investment E01:

> “But then cost, then I think it is also very related to cost and how much the company wants to invest at this point. [...]And if you make this investment for this condition, do you have a return back too or not? So I think that's what's important to understand.”

These uncertainties reflect, from the perspective of the CSRT, a phase in which the “analytical reality” is not yet objectified; its contours, rules and values are fluid and subject to negotiation. The "*habitus*", less focused on data, one-dimensional, in the social aspect still prevails. In this scenario, the PA is meant as something that demands the active construction of credibility. For E04; “[...] It is not enough for technology to exist; it is necessary to convince stakeholders – top management, peers, the HR team itself – of its importance.”

3.1.1.2 The Habitus and Resistance to the Analytic Mindset in HRM

The *habitus* is understood as a system of dispositions incorporated throughout individuals' social trajectory that tends to shape their perception and response to new social practices and the adoption of technologies. So that the intricate relationship among *habitus*, *field*, and *capital* unfolds in a truly dialectical dynamic. We can observe that *habitus* is not only forged from experiences lived within a specific field, but also, acts upon that very field, shaping it and being shaped by it. Concurrently, the field serves as the stage where *habitus* expresses itself in all its complexity, a space of constant transformations largely driven by the incessant pursuit and contention for different forms of capital. It is precisely this capital that, ultimately, confers power and distinction upon actors within this social arena, solidifying itself not only as a result of incorporated dispositions but also as an engine that feeds them back and redefines them (Bourdieu, 2007)

At HRM, many professionals were socialized – or have your *habitus* – in an organizational model that prioritizes more subjective and relational aspects of people management, leaving an approach based on data and quantitative analysis in the background. This socialization occurs at different levels: (a) Academic and professional training – The educational trajectory and the training received tend to emphasize interpersonal and behavioral skills, with less focus on quantitative metrics, statistics, and predictive analysis. This reinforces a *habitus* that favors a more intuitive model of decision-making. (b) Organizational practices – Companies traditionally structured under a more operational and less strategic management model reinforce the decision-making process based on past experiences, making it difficult to transition to a data-based HRM approach. (c) Organizational culture – The perception that HR should act as a "guardian of human relations" – not only is it valid, but also a pillar of HRM – it is a fact that directly impacts the adoption and/or expansion of PA projects or data-driven decision-making processes. As indicated by the analyzed interviews and the academic literature, they reinforce the uncertainties about the possibility that technology can "dehumanize" management (E07, E10,Baldissarri et al., 2017; Lagios et al., 2023).

The resistance identified in some interviews can be interpreted as a conflict between *habitus*, field, and capital in new organizational practices. As Bourdieu (2007) states, structural changes can cause tensions when social agents do not have dispositions compatible with the field's new demands. In the case of PA, the adoption of these tools requires a reconfiguration of *habitus* field, and capital, HR professionals', shifting their traditional view

to a more analytical and data-driven posture (Khan, 2024). Thus, overcoming this resistance depends on reframing processes, in which professionals are gradually exposed to the analytical mindset through training, practical experience, and cultural adjustments within organizations. Investment in the development of analytical skills can be seen as a means of expanding the *habitus*, reducing barriers to the institutionalization of the PA in the BPI's HRM (Babu & Govindappa, 2022; Marler & Boudreau, 2017; Singhraul; Prof. Budheshwar Prasad & Anuragi, 2024).

One manager reports the need to "[…]convince my HR team so that they could buy the idea[…]" before taking an analytical tool project (PULSE) to the board (E01). This need for convincing is linked to the perception that HR, traditionally, may not possess or value an analytical cognition, as reported by:

> E02 – "[...]but I think it's still incipient, I think there's still a lot to see, the artificial intelligence too, the chat GPT too, I'm starting to use it now, it's been a month since they granted a license".
>
> E01 reinforces that: "[...] We have to think of a different person's profile to be able to dream [with PA], but I think this has to stay within the people area".
>
> E06 concludes by stating that: "[...]In general, HR management is not very friendly with numbers. Right?"

Consistently, the comments reflect the perceptions of a skills gap and of a historical difficulty in HR in dealing with numbers, and as a proposed solution, the need to build a standard indicator panel to standardize basic concepts such as turnover (E06). From ANT's point of view, this phase of creating an indicator panel may correspond to the beginning of problematization. The PA proponents – whether HR managers, external consultants, or technology itself as a solution in search of a problem – need to define the situation so that their intervention becomes indispensable. The uncertainty of the other actors is an obstacle to be managed.

Building credibility is an essential part of the interessement process, seeking to air and align the interests of other actors with the analytical project (McCartney & Fu, 2024). Forming groups or alliances is a strategy observed to strengthen this initial construction (E01; E04;

E07; E10). The difficulty lies in how to translate the abstract potentialities of the PA into convincing arguments for different audiences, as emphasized by E01: "[...] I think it will be important for you to also take the other side, saying, [the HR guy] brought your data, help you?" or, especially in organizational cultures where, as mentioned, "it has always been like this" (E07), a statement that characterizes a significant resistance to change (Weick & Quinn, 1999).

Uncertainties manifest in both family-controlled companies and those with professional management, applying equally to national and multinational companies. BPI has in both corporate configurations a characteristic of rigid hierarchy and optimization of human resources. Therefore, even if HRM presents an innovation project – regardless of the level of complexity – there is great difficulty in building "groups of notables" (Latour, 2005); because, when the organization is family-owned, it is common for the "owner not to approve" and to take radical measures of resistance, as was the case of interviewee E07. In this specific case (E07), the interviewee makes it very clear that her intention of innovation was to: First implement a more efficient and effective recruitment and selection process that would minimize an exhaustive process of primary analysis of documents – which usually revolved around the case of hundreds – and free up precise time for other tasks. Second, these partial automation are in no way objectifying human capital (Schweitzer & De Cremer, 2023), but streamlining processes that were still carried out manually (McIver et al., 2018).

In a not-so-radical way, but responding to agility, but with the optimization of resources, the so-called group can be used in another country, as portrayed with great emphasis by the interviewees (E04, E05, E07). The problem with this optimization is that in the case of the globalized organization, the sponsor is not local and is responsible for a much greater demand that it can meet, as follows in the statement in the E05 interview:

> "[...]I have *(an analytical support),* but this person is sitting in Costa Rica, I don't know the name of this person, I have to open a request, a ticket in the system, write everything I need, wait for someone to pick up my request, get in line, and ...., if God wants to be a nice person who will look for me by chat so I can explain a little better what I expect, Do you understand?" (italics mine)

This statement shows not only the challenge of innovation in Brazilian HRM, but how this structure undermines or wears out, with sinuous contours, the entrepreneurial impetus of the HR professional, and the support search depends on a myriad of conditions for which neither one nor the other has outstanding "agency" and explicit leadership. Latour (2005, p.54-55) portrays this condition clearly:

> “The formations of groups leave many more traces in their wake than already established connections [...]. Groups are not silent things, but a temporary product of a constant rumor made by millions of voices [...]. One thinks of the mass of speeches and writings accumulated to outline an extraordinary set: *Homo Economicus"* (author's italics).

However, such uncertainties should be overcome constructively. This is a more positive view of how technological processes tend to integrate rather than drive away, as portrayed in the E08 interview: “[...]There is a preconception that technology will drive away the people's area of people. So, there is a lot of these preconceptions. And then the first step was to try to break this paradigm. I'm not saying that the people management area will not have this interaction, these conversations. But they have to be value-generating conversations.”

Another aspect of these “value-generating conversations”, highlighted in the previous statement, increases uncertainties and controversies, as in interview E07: “[…]I have been a partner, like this. It's a lot of listening and partnership to try to understand their vision and always bring the other side, right? How can this impact the organization and how we engage, because otherwise we are born with a failed project, right?” The feeling of incompleteness and lack of breadth in the conceptions and applications of theories in organizational practices are clearly "claimed" in the interview E10:

> “[...] Look, I went on a human resources development journey, we hired a company, you know, to do this development journey like the people there, right? And even the topics related to artificial intelligence that we don't end up not having that much content, you know? So I even expected more....., more, it was a bit of a disappointment ..... A little..... [sic].

This important and forceful statement, portrayed in interview E10, can be approached in two dimensions. The first dimension is the desire to implement new possibilities of interaction, capacity for analysis, and professional, individual, and group development. As a second dimension, it evidences the feeling of frustration, not only in their speeches but in their pauses and expressions. Revealing feelings for making an investment that does not result in the desired development, and on the other hand, the process does not eliminate its uncertainties (Gaur et al., 2024;Sharma, 2020; Shrivastava et al., 2018).

It is slowly revealed that this list of different propositions or entities and their intimate nature can become intermediaries or mediators of a flow. Which tends to limit or inhibit not only the adoption of innovations but also their development, as discussed in interview E01; "[...] we do not understand the value that this [PA] would have, how it can help, how we can better convince the organization with the data". For the ANT, the force of inertia will not carry the spectacle forward (Latour, 2005). In addition, the concept of intermediary and mediator tends to provide a greater understanding of the impact of the mediator. According to Bruno Latour, an intermediary is a carrier of meaning or force without transforming it. Mediators, on the other hand, tend to assume multiple entities; that is, it can be worth one or none, as well as several. "[…]What goes into them never defines exactly what goes out[…]", so mediators transform, translate, distort and modify the meaning (Latour, 2005., p. 65).

Mediators play a critical role in the formation and maintenance of groups. They are not only passive elements but actively contribute to creating and sustaining social bonds. This is essential to understanding how groups are not static entities but require continuous effort and interaction. Depending on the entity's nature, it can assume the role of intermediary or mediator. To illustrate, a computer in perfect working order is an intermediary; however, when broken or without proper functioning, it becomes a mediator, exposing all its incomprehensible complexity between hardware and software. Therefore, the introduction of AI-managed systems, or specific PA projects, goes from intermediaries to mediators and back to intermediary – in intensity and speed – depending on how the entity or non-human actor is enabled in its technological power, in its competencies in uniting distinct characteristics, in its ability to expand or reduce the scope analyzed. Therefore, how is PA being utilized as an organizational actor or a non-human entity that makes decisions or can enable better decisions for humans, as evidenced in interview E05:

> “[...] I'm being specific to HR, so it's a huge difficulty in this sense, and then in addition, how are decision-making made based on this information? Now we know that with this they will start making decisions based on data[...] in the next 10 years that we have this definition of competencies of the future, one among the five is Data Driven Decision Making, it is very much this capability that I commented with you, of Data-Driven Decision”.

We reinforce at this point that this flow of uncertainties does not provide solid support for institutionalization, much less to initiate a process of legitimation for this new actor to integrate the sociology of knowledge. This process generates confusion, because without the value, belief and perception of programmed and/or already initiated actions, there will be no valid results applicable to the organizational reality (Berger & Luckmann, 1966), as the concerned E02 statement:

> “[...]As a business partner, we are making the right decision, we are giving the right information, we are supporting the business in the right decision and I think it brought security, first, to take away this perception, ah, I think it's coming out, I think it's low salary, I think it's having the target, no, here are facts, and when you're in a business area, what they want to see is the result, right[sic]"

3.1.2 The PA as a Protagonist

When and if the initial uncertainty of this recurring process is overcome or managed, the translation process advances to a more explicit problematization phase, as defined by Callon (1986). Here, PA is meant to solve several problems identified or built upon in HRM. The interviews are rich in examples of how the PA is experienced as an “organizational actor” capable of conferring a new status and a new capacity for action to HR. It is necessary to align the proposition of the actor protagonist with a more insightful analytical mentality. A recurring significance is that the PA can take HRM out of a reactive or merely operational position, giving it greater strategic prominence (McMackin & Heffernan, 2021).

As E04 comments: “[...] … yes, I think so....., I think we are starting to be perceived as an area that generates value, that has a ‘seat at the table,’ that talks to the business, that is proactive if we are ahead of the significant changes [...]”

This involves building an analytical mindset (Suri & Lakhanpal, 2024); as highlighted in interview E10:

> “[…]There was a lot of adjustment, they did several pilots.... This story, right, but you know that one thing that the pharmaceutical company taught us, for me a lot, was the issue of the pilot, right? And I learned this at the time of productivity, that we did a segmentation project since it was a project kind of imported from the United States from the headquarters and it didn't work out that we didn't do a pilot, we didn't make the adjustments involving people here, we had a very gradual process. That was an advantage, because we were implementing, adjusting, or giving a pilot, so that was it in that sense[…] [sic].”

The data and consequently the analyses carried out will generate information (*outputs),* which are seen as the necessary language to dialogue with other areas, especially finance, marketing and senior management, “[...] but the issue is the development of the HR team itself, because HR acts a lot as support for other areas, but I say, HR itself has to take care of HR” (E02). One manager says: “[…]you from HR will talk to finance, you take data, information[…]” (E01); Other highlights that having data; “[…]brings security to decision-making and that HR, it is going on the path of supporting the areas in the correct decision” (E06). The experience of presenting data on turnover to demystify perceptions and influence decisions (E02), illustrates this search for protagonism through analysis.

The lack of this mentality is seen as an obstacle: “[HRM] is still very much in the operational [...] I see a difficulty in the areas of Human Resources today, to really get out of the operational a little[…]” (E01,E07). Interviewee E06 was blunt about the protagonism of HRM: “[...] There is none! This is a very strong word, but it gave a lot of visibility and a lot of recognition”

We can infer that there is a leading role of the following actors: Data, Humans, Applied and or Shared Knowledge in the resignification of everyday reality. The protagonism of data*:* It is referred to as a broad process that highlights it as an actor that (re)signifies the role of HRM. Two unanimities emerge among the interviewees. The first is the conception that data is a strategic instrument. The second is in the conception that data is a means of integration. About data being a strategic instrument, the terms information, communication, security, conciseness, and best decisions are recurrent in the interviews. Examples of how the data has provided you with more clarity on a problem, such as: "[..] To whom are we losing our talents, and if we are really losing talents? (E02)". A question answered with the necessary propriety and specificity, thanks to the data that the operating system (Visier®) has provided.

The second conception, data is a means of integration. The data provides HRM with greater proximity to other organizational areas, being in the integration of projects or from the perspective of business partners (BP), which contributes to behavior analysis, performance, creation of metrics, and all other fundamental processes of HRM. More broadly, data integrated because they tend to provide consistency to evaluations of goals (E01), internal and external communications (E02), presentations focused on results (E03), the concept of justice (E04), the critical analysis of the data itself (E05), organizational equity (E06), as an alert of evolution itself, which until then in a way remained far from HRM finance (07), as a means of counterpart knowledge (E08), a means of generating discursive credibility (E09) and, finally, a driver of recognition and knowledge of people and their human capital respectively (E10).

Using the previous example – how turnover is characterized (E02) – the data show the confidence with which the answers were given. E02's satisfaction in communicating with his CEO, in a pragmatic, punctual and assertive way, was evident in his expressions and in the level of detail provided. This leads us to a clear and evident overcoming of the second uncertainty proposed by Latour (2005, p 71) – Action is assumed. For this case, the CEO's proposition was that the organization could be losing its talents to a specific competitor. Thus, "social" is centered specifically on the ability of other market players – other laboratories – to focus on their sales force. E02's satisfaction is twofold. First to provide robust data that the losses were not talent – only 1 in 16 employees were classified as talent. Secondly, it was to identify that the 16 people were all for different companies.

Therefore, the ability to have and provide a holistic view with robustness generates legitimacy and security. Thus, an PA project is also meant as the solution to the fragmentation

of information and the lack of an integrated view of the organization and HRM processes, because “[...] we are not going to collect the data by hand” (E01). ERP systems tend to provide a holistic view, which can quickly and efficiently centralize information about the organization, such as Workday, Vizier, SAP, among others. As E04 states: “[...] Oh, so I want to know, I don't know, the inclusion and diversity point, it's just a click.” The lack of a “sponsor,” a “project owner”, or adequate “technical support” is experienced as barriers to achieving this integrated vision. The premise represented in the speech of E01:

> “[…]There is no sponsor to feed this information, and more than feeding is building this panel of indicators, which would be like the airplane pilot, that he has a cockpit and someone is missing to build this cockpit, we are seeing a cockpit well... [...]But then again, if I tell the truth, is this system running properly? No, because there is no owner, like, there are several owners, several people putting information, but his governance there, like, is not cool, nowadays his governance is not cool, it is not good, and it is an interesting topic that, and again, it is a prioritization topic within HR, I will tell you about my reality, As there won't be one... How can you talk, I'll say things”.

The same situation is posed in E02 in a very simple and direct answer: “[...] no, I don't have anyone”, In addition to this lack of effective and active governance, the need to have a single system is expressed: “I think we should have one tool and not multiple tools” (E01, E02, E03, E05, E07, E08). HRM's proposition of a broad view of the organization is fragmented – or frustrated – by the adoption of parallel tools, reported in the interviews as the need to deal with multiple data sources and multiple systems. The justification of the interviewees cited for this situation – the non-use of the organization's standard ERP, whether SAP or Oracle – is that the organizational management does not “pay” the analytical extension of these systems to HRM at cost. As information, SAP® has a specific People Analytics system called SAP® Success Factors HCM, which only 1 out of 10 interviewees stated uses this system.

In general, the PA is meant as the link that connects HRM to the business strategy (E01, E04, E06). It is experienced as a means to generate value (E03), promote innovation

(E04), and ensure that HR actions (recruitment, development, compensation) are aligned with organizational objectives (E07, E09, E10). Technology is seen as an essential enabler because, for E06: “[…]We can optimize time in the analyses, what you used to do sometimes, two days, three days, four days, you do it today in hours. And it is much more optimized [...]”. AI helps to make “more assertive decisions”. Under this same lens, E02 states that: “[...] it eliminates much time that you have to think about [...]”, focusing on time optimization. For E10, when discussing the role of technology, he states: “[...] I think technology can perhaps take away more and more work less linked to people management [...] and leave the leader for what is important [...]”, focusing on freeing up time for activities of greater value.

From the CSRT point of view, this Problematization phase corresponds to an intense work of externalization and objectification. New categories of analysis (HR KPIs, predictive metrics), new tools (dashboards, integrated systems) and new routines (data-driven meetings) are proposed and are beginning to be seen as objective elements of organizational reality. The ANT describes this as the definition of the Mandatory Waypoint (PPO): to be strategic, to have a holistic view, to create value, HR “needs to” go through the datafication process that PA tends to provide. In addition, the “new HRM” will be required to seek solutions – or go through PPOs – for problems such as the management of people's data, balancing rationality with the subjectivity that human relations require, training on this new lens, with the aim of minimizing inefficiencies, as well as legal and ethical management of this complex process, because each PPO, it encompasses and requires multiple interactions between different actants, in an exponential cycle.

3.1.2.1 The Translation of Interests in the Implementation of the PA

The ANT approach allows us to analyze how the implementation of PA in HRM at BPI evolves through successive moments of translation of interests. This process involves transforming the actors' perception of a phenomenon, adjusting expectations, mobilizing alliances, and redefining organizational trajectories. HRM needs to balance initial interest and uncertainty. The interviewees, E2, E05, E06, and E07, initially faced difficulties in obtaining support for the adoption of PA within the organization, these resistances reflect a typical scenario of the first moment of translation, where the project's proponent needs to align its interests with the interests of the other actors.

Latour (1988) points out that, at this stage, the success of the implementation depends on the ability of the translator to convince others that the proposed project meets their needs. The interviewees highlight the absence of an internal sponsor and the perception of technology

as a costly and complex tool, making mobilization difficult. In the case of interviewee E02, he states that there is no possibility of implementing a PA project in the organization at this time because any request in this direction is forwarded to a person located in Canada and that the project will depend on his availability. A similar context is observed in the E05 report, where its support is in Costa Rica, which manage the global demand about HRM.

At another time, HRM begins the work of blockages and structural resistance. The lack of engagement from the top reflects the second moment of the translation, in which the conventional paths to adopting the technology encounter barriers. Latour (2005) emphasizes that when faced with changes, organizational actors tend to reject innovations that are not perceived as necessary or destabilize consolidated practices. E01's interviews report that, at first, the organization's leaders questioned the technology's return on investment (ROI) and expressed distrust about its applicability in HRM, making it difficult to legitimize (Chalutz Ben-Gal, 2019). A fact that does not prevent E01 from using resources from his department to implement the PULSE project will be addressed in a subsequent section.

Challenges are inevitable, and shortcuts and strategic reformulation are necessary. The transition to the third moment of translation occurs when E01 and E09 reformulate the project approach, presenting its feasibility in a way that is more aligned with the company's strategic objectives. The first does not use its IT department, hires an external partner, and still uses data from “new quantitative surveys of its sales force”, which causes him frustration due to the low response rate. In the case of E09, the company had requested global integration, which did not occur, so a new perspective for local strategy was needed. These adjustments are related to the need to translate the interests of internal actors in a more accessible and pragmatic way (Latour, 1988, 2005). Rather than focusing on the technical sophistication of PA, respondents were clear, choosing to highlight the ability to reduce operational costs and improve the accuracy of performance analyses. This adjustment strengthens its position and generates an opening for negotiation.

Mobilization is a vector for sustaining the network. Once the first supports are conquered, the PA goes through a stabilization phase, corresponding to the fourth moment of translation. Here, according to Latour (2005), technology is no longer seen only as an experimental tool, but as a functional solution to preexisting organizational issues. E01, E04 and E09 report that, after adjusting the way the PA is applied and presented, some leaders began to perceive value in the tool, increasing its acceptance. The inclusion of internal and

external performance indicators and benchmarks, as well as greater involvement of the top management reinforces the legitimacy of the project.

The project becomes indispensable. The fifth moment marks the institutionalization of the PA as an essential resource aligned with the company's objectives. When an organizational actor becomes indispensable, it integrates deeply into the network, making its presence an obligatory crossing point (Latour, 1988). In the case of E04, after strategic adjustments and proof of the value of the tool, the PA project is no longer treated as uncertain experimentation and is to be incorporated as an official analytical practice, evidencing the success of the translation of interests.

3.1.3 Weaving the Analytical Network: Synergies and PA Enrollment

Once the need for the PA begins to be established, the focus shifts to actively constructing the socio-technical network that will sustain it. This involves creating synergies between different actors and processes and ensuring the durable inscription of technology in everyday practices (Callon, 1986; Latour, 2011; Law, 1984). If established - or in the process of being established - through problematization, the construction of the new analytical reality advances to phases in which it is actively sought to create connections, align interests, and materialize technology in everyday practices. This section seeks to explore the categories “Synergies and Inscription”, analyzing how the socio-technical network of the PA is woven and stabilized or contested, encompassing inter- and intra-organizational connections, based on the interviewees' reports.

Pharmaceutical companies often follow technological trends established by industry leaders, seeking to minimize risks and strengthen their competitive position, increasing their differentials in products and services. According to Serra et al. (2021), mimetic isomorphism can be observed in the standardization of analytical processes and in implementing AI systems to optimize talent management and organizational productivity. This trend is driven by the need to meet the expectations of the market and especially regulatory bodies, ensuring that the company is perceived as innovative and aligned with best practices. Therefore, benchmarking plays a key role in this process, as it allows pharmaceutical companies to compare their practices with those of competitors and adjust their strategies to not fall behind in the technological race. This practice is no longer recommended in recent studies because, with the maturation of these new technologies, it is recommended to implement unique projects according to the specificities of each organization (Rasmussen et al., 2024).

Studies indicate that adopting AI and PA in people management has been primarily influenced by institutional pressures and the search for compliance with global standards. This means that even if some companies do not have an immediate need for PA, they adopt these technologies to avoid the perception of delay, or they address this challenge with small-scale and micro-level insertions to avoid obsolescence in policies and practices (Wassell & Bouchard, 2020).

3.1.3.1 Seeking Synergies and Integration: The Challenge of Alignment

The PA is often presented as a promise of integration, an actor capable of breaking down silos and promoting collaboration. The idea of democratizing access to data is central, as exemplified by the use of the Great Place to Work (E1) dashboard or the Workday system (E1, E3) for managers to access information from their teams and assume co-responsibility for people management. The interviewee (E1) emphasizes:"[…]Technology there democratizes a little access to information and decision-making. [...]he is a good example of data, information, that the owner is not only in HR [...] each board, each manager, each manager, he has access to the training of his team, of his area, so that he can make decisions as well" (E01). The interviewee (E6) also highlights the importance of shared, data-driven goals to force interdepartmental collaboration, even if it faces resistance linked to a lack of trust.

Therefore, PA can be a vehicle that catalyzes synergies, an actor capable of promoting greater integration between HRM and the other areas of the organization; therefore, "[...]They need to understand that what brought them here does not take them where they need to go in terms of the future" (E07). The desired experience is that data works as a "[…]means of integration" (E04), creating a common language and facilitating communication (E02, E05, E07, E08, E10). This democratization of access to data is seen as fundamental to promote "co-responsibility" (E06, E08, E10), for people management, taking HR out of the role of sole guardian of this information and engaging line managers, as each one has an internal flow, breaking down interdepartmental barriers (Alaghbari et al., 2024).

However, the creation of such synergies faces considerable obstacles. The fragmentation of systems is a recurring theme. Interviewee E2 describes the difficulty of reconciling Workday and ADP. E5 points out the inconsistency between Workday and the CLIC tool. E08 laments the lack of centralization and conversation between existing data. This technical fragmentation makes it challenging to create a single and integrated view, that is, from the Latourian perspective, symmetry is constantly and routinely impaired (Latour,

2005). In addition, human barriers persist. In the E01 interview, he notes that "[…]The day to day kills people [...] no one has '[sic][sic],' everyone wants things ready here[…]" and that even with accessible systems, managers prefer to ask HR. In the E09 interview, the difficulty of integration with suppliers that use different systems is pointed out. This resistance or inertia makes it difficult for actors to enroll (ANT) in the network: They do not assume the roles or use the prescribed tools. From the point of view of SCRT, this indicates a failure to internalize the new analytical reality; Actors continue to operate within old logic and priorities, maintaining organizational silos. Latour's (2005) analysis of the difficulty of connecting the "local" and the "global" resonates here: global PA initiatives often fail to effectively connect with local practices and resistances.

Therefore, the standard knowledge base – common sense – that data could provide does not automatically translate into coordinated action or a lasting process of co-responsibility, as the actors still operate within different logics and priorities, the organizational "silos" (E02, E06). The attempt to use shared goals based on data is a clear strategy to force this internalization, alignment, agency, and legitimization. (E06, E10, E04).

3.1.3.2 Inscribed Analytical Reality: The Materiality and Agency of the PA

The "Translation of Reality" promoted by the PA is consolidated through its inscription in material and durable elements that can be represented by: software systems, algorithms, standardized reports, visual dashboards, new work routines. It is through these inscriptions that the PA is experienced in everyday life and begins to exert its influence. The interviewees describe the use of several tools that materialize this inscription: Workday (E01, E02, E03 and E05), SAP (E04, E10), Vizier (E02); ADP – a specific internal provider of HR information for this interviewee (E02); CLIC – Customized tool (E05); PULSE – PA tool provided by a third part (E01), Phoenix a third tool used by the interviewee's HRM (E05), SuccessFactors (E04);, Power BI (E07), in addition to generative AI tools such as ChatGPT or Copilot (E01, E02, E03, E07, E09 and E10) and specific platforms such as Vagas .com (E02, E03, E06 and E07), Gupy – AI platform specialized in providing services to HR Gupy.io, Genially – Gamification-based training platform), LinkedRH – HRM service platform LinkedRH (E07); UKG – American HRM Platform (E09).

These inscriptions are significant because they confer an agency to PA how non-human actor. The system "tells you what you interacted with the most", "was able to indicate [...] risks of losing", "already makes a filter" (E01), "generates admission", "generates KPIs" (E02). This influencing action of the non-human is central to the experience of PA: technology

is not just a passive repository of data, but an actor that actively processes, analyzes, filters, recommends, and sometimes decides. Reported experience with using Copilot in Teams to find policies (E10) or ChatGPT to summarize articles or draft announcements (E03, E08), illustrates how generative AI inscribes itself in everyday tasks, promising efficiency and effectiveness (Soonthodu & Shetty, 2022).

Successful enrollment depends on “value attribution in technology adoption” (E01, E05, E06, E07, E08). Statements can be portrayed as:

> “[...] it has to add value, it has to really be practical, which I try to fight a lot with the teams, when we work there will be some process to defend, it is to do it simple, practical, objective (E01);
>
> “[...]having to stop these heated discussions, which I love, which I'm super passionate about, and open a system that I don't feel comfortable with, that I don't even know where to start, it consumes me hours, that I could be doing things that I think are much cooler and that I believe add much more value (E05);"
>
> “[...] it is more to see the talent panel a little, to see and see how we are developing these talents, internal use versus the loss of these talents (E06)";
>
> “[...]It's no use wanting to do things just because we want to and because it's cool. I think we need to maybe show this team as a whole the value it has (E07)";
>
> “[...] I think it will bring a need to generate value from everything that artificial intelligence offers us, because information is reachable, democratic, analysis available, but from that, how do we generate value? I believe we will have to reinvent (E08)"

The reports provide a deeper discussion about how this process of interaction – human-technology – is being perceived. The contributions start from the perception of functionality (E01) to a broader proposition; about the democratization and generation of “social value” these innovations can produce (E08). Therefore, when you realize that using the system, analyzing the dashboard, or following the algorithm's recommendation, tends to be a counterpart that brings concrete benefits, whether in terms of time savings, decision quality, or strategic alignment, interactions become easier and without deviations. When this value is

not realized, or when the cost (in terms of effort, time, complexity) outweighs the benefit, enrollment fails and technology is underutilized or abandoned, due to lack of legitimacy (Berger & Luckmann, 1966).

The fact that the PULSE project was interrupted, despite its predictive merits (E01), indicates that this actor (service provider for E01) interrupted its interaction through a process of re-evaluation of its existence – a service perceived as having high added value by the market – and was absorbed by an actor with greater economic power, and not because of its inefficiency and lack of results. What Latour (2011, p.168) calls "translation of interest: Moment 1 – I want what you want". In this context, Latour refers to a simple search for convergence of interests – portrayed by the contracting of the services of the third party by E01 – something that corresponds to the idea of "capturing", which consists of adopting positions on less controversial arguments, with simpler "black boxes" (reiterate that this term is from Latour), and only then addressing the most relevant controversies – portrayed in the passage about the interruption of services for a reason particular to the provider and not to the contractor (Latour, 1988). However, he warns that this search for convergence can excite followers and that it cannot be perceived as a "silver bullet" for all problems, its opposite being also a truth.

Fundamentally, the inscription of the PA redefines the very nature of decision-making in HRM, moving it (or attempting to move it) to a data-driven logic (E01, E02, E06, E10). "You can [...] having much harder and more raw data, you will be able to do much more personalized things" (E01, E10). "With more data, more information, more statistics, more argumentation based on numbers, HR has more credibility" (E09). However, this inscription does not eliminate human agency. On the contrary, it reconfigures it. There is "evidence of human protagonism" (E01, E01, E03, E04, E05, E06, E07, E08, E09, E10) in the need to feed the systems with correct data, to ask the right questions as E02 reports: "[...]I need to stop and think, first, what data is important? What data do I want to show? What data will be important for decision-making? Because there is a lot of data, what do I do with it?" Also, interpret the results critically to contextualize the analyses (E06, E10), and, ultimately, make the final decision, which may even contradict the algorithm's suggestion, since the human intelligence behind it, our subjectivity, remains the main element (E03, E06, E07, E10).

From the ANT, registration is the moment when the socio-technical network stabilizes (provisionally) in durable formats. Systems and algorithms become spokespersons for the interests and definitions negotiated in the previous phases (Gutiérrez, 2023). For SCRT, these

inscriptions are the maximum objectification of the new analytical reality. Dashboards become facts, KPIs become the objective measure of performance, and data-driven decision-making becomes an institutionalized practice, shaping the way individuals internalize and reproduce this reality in their daily lives (Berger & Luckmann, 1966).

3.1.4 The New Order: Mobilization and Reordering of Structures

Enrolling the PA in systems and routines is a technical step, but even though the organizations could be technically capable it does not guarantee the consolidation of the new analytical reality. The final phase of the translation process involves the continuous mobilization of actors (human and non-human) to ensure that the network works as planned and the necessary reordering of organizational, cognitive, and cultural structures that may offer resistance. This section addresses the challenges and dynamics involved in this phase of (de)stabilization.

Analytical routines and systems are critical to the success of data-driven organizations, turning vast amounts of data into actionable insights that enhance decision-making, operational efficiency, and competitive advantage. Organizations that use advanced analytics, to predict trends, optimize operations, and personalize customer experiences, drive innovation and growth (Huang et al., 2023). The shift from experience-based to evidence-based decision-making, facilitated by data analytics tools, enables organizations to make informed decisions quickly, reducing risk and improving adaptability to market changes (Gaur et al., 2019).

A data-driven culture, characterized by robust data quality, governance, and cross-functional collaboration, is essential to integrate these analytics systems effectively. In industries such as oil, data analytics thinking is used to identify problems and extract value from data. This process requires collaboration between domain experts and data scientists to maximize strengths and and minimize communication costs (Wahdaniah et al., 2023).

Moreover, an analytical management model framework, which includes components such as strategic positioning and transformative leadership, provides a structured approach to leveraging data for strategic decision-making, generating sustainable competitive advantages. Despite challenges such as cybersecurity threats and the complexity of technology integration, the benefits of adopting routines and analytics systems in data-driven organizations are substantial, enabling them to navigate complexities and achieve long-term success in dynamic environments (Abdul-Azeez et al., 2024).

3.1.4.1 Structural, Cognitive and Cultural Challenges in Reorganization

The implementation of the PA often collides with the organization's preexisting structures, generating friction and requiring significant reorganization (Madanchian & Taherdoost, 2024). The fragmented nature of existing people management systems can represent significant barriers to the adoption of PA, a fact stated in the interviews, E01, E02, E03, E05, E06, E07, E08 and E09. The statements – almost unanimously – portray a certain lack of analytical prioritization offered by the organization to HRM. This implies not only the difficulty of access, which such decentralization of data provides in the search and compilation of information based on dispersed data, but also in the maintenance of the integrity and consistency of these data, as portrayed as a structural dilemma of HR systems in the interviews:

> “[...]so we migrated from SAP to Workday, which is our HR information system, all people's information should be in this system, and what we have been suffering is precisely the creation of this Pipo Analytical system that we can call, because before in SAP we already had[...] (E01)"

> “[...] Today Workday® and Vizier® are already integrated, so we enter Workday®, in fact the Visier® data is extracted within Workday®, when you enter Workday® there is already an icon that you access Vizier®. (E02)"

> “[...]So, for example, we have a Workday system, which is a very robust super-system, right? Now, how do we use and extract information in our favor? I think it's still very limited. And then not enough, apparently with not enough Workday®, a tool is developed in addition, which I don't know if it's a general tool, or if it's a customized tool by organization, for our organization called CLIC, which still has it, which then you can extract other reports, which probably pulls a lot of information from Workday itself. And then you start to have difficulties in cross-referencing information, so you pull something that should come out, 103, one tool and another, and it doesn't come out, because then you start to generate code complexities, which then have slight differences, then you can't match 100% of the information [...] (E05)"

Organizations often struggle with data integrity, ethical considerations, and the rapid pace of technological change, which are dimensions that must be worked on to maximizing the benefits of PA (Holz et al., 2024a). While the fragmentation of HR systems presents challenges, it also underscores the importance of a strategic and well-coordinated approach to PA adoption, ensuring that organizations can harness their full potential to drive performance and innovation(McCartney & Fu, 2024; Rehman, 2023).

In relation to physical and functional structures, this fragmentation of systems represents a significant structural barrier, challenges reported in all interviews. The existence of "multiple systems (HRIS)" (E05, E02), often not integrated, makes it difficult to mobilize data and create a unified view, as highlighted by (E01): “[...] Because we and our colleagues, each one is creating their own cockpit, I have my cockpit, so you start to have cockpit that you don't talk to, but each one has their own”.

The proposition of functional structure as barrier highlights how departmental silos inhibit collaboration – diluting “interests”, fragmenting the process of “problematization”, hindering “enrollment and mobilization”, which is essential for (PA) to realize its full integrative potential. Without this calculation center, the challenges in applying the principle of symmetry are exponentialized, a fundamental principle of ANT. The principle of generalized symmetry states that all actors—human and nonhuman—should be analyzed equally, without predetermined hierarchies that favor one over the other—a distributed power (Law, 1984). By extending Bloor's symmetry to encompass all actors, Bruno Latour aims to offer a more comprehensive and equitable framework for analyzing complex networks, thus promoting a deeper understanding of their dynamics.

A solution for HRM to avoid/minimize the challenge of system and structure fragmentation is provided by (Latour, 2011), when he presents a third moment of the translation process, which consists of demonstrating that small deviations can keep the interests of the actors convergent. Therefore, it is necessary in the first instance to recognize the block(s) in the interaction flow. This recognition tents to contribute for the construction and control of new avenues of interaction. This process tends to elucidate the PPO, and figure out where the controversy lies, give up on innovation or dilute the technical and relational barriers to AI adoption or PA design, and close that "new black box" and create a new analytical perspective for HRM, which will provide insights for the entire organization.

An example of this process in practice is in interview E01 in the implementation of PULSE (Project 01), as well as in the promotion of interviewee E04 and her involvement with

a functional mobility project, which involves the organization in a global way (Project 02), and finally, E09 with a local project that encompasses the organization's turnover analysis (Project 03). The examples are useful and relevant because: First for their breadth, second for their focus, third for the challenges and fourth for the changes they generated and that are expected. Regarding the amplitude, projects 01 and 03 can be classified as initial projects, of lesser amplitude, as their analyses will result in local actions. Even though the organizations are multinational companies, therefore, responding to an international direction, the Directors chose not to exceed the Brazilian territorial limits. On the other hand, project 02 expands the analytical conception to another level. The project involves all organizational units worldwide. The demand was initiated by “global”, which determines the involvement of all its multinational subsidiaries.

From Latour's theoretical perspective, the translation process of each project occurs at a specific moment. Project 01 is a local project, of initiative and planning and resources exclusive to its creator. Therefore, we can frame this effort in the moment of "translation 2 – I want; Why don't you want to?" (Latour, 2011, p. 172). In relation to Latour's analysis of the network of actors, this moment constitutes a fundamental process by which actors (human and non-human) are mobilized and engaged in a project, transforming their interests, identities and actions to align themselves into a common goal – to provide greater and better analytical capacity. The object of analysis in this case is the engagement of the employees of the organization's sales force.

With the objectives aligned and consequently moment 1 of the translation – I want what you want (Latour, 2011). Moment 2 is focused on a dynamic of association and displacement, where connections are established between heterogeneous elements (people, instruments, facts, theories, institutions) so that they act cohesively in favor of an objective as highlighted by E01. Moment two of the translation describes a specific modality of this process that seeks reinterpretation and explicit channeling of the interests of the actors to be mobilized. The core of this strategy lies in directing the action and objectives of others so that they follow the path proposed by the actor translator, instead of the actor translator adapting to the already existing paths of the others (Latour, 2011).

This approach becomes particularly relevant when the original interests or conventional trajectories of action of potential allies encounter impasses at the moment, allowing the proponent to present its path as an alternative or even as the only possible way to achieve, albeit in a reconfigured way, the desired objectives. Latour (2011) uses an example

from the automobile industry. In which the board's interest in “new and more efficient cars” went through the first translation when it was equated with “fuel cells”.

What is configured as the second translation is the conviction, exercised by the research director, that the enigma for fuel cells lay in the “behavior of electrodes in catalysis”. The original interest (efficient car) is displaced and redefined, coming to mean “research on the efficient battery model” (the third translation, which is the transitive result of the chain). This exemplifies the channeling of the board's interest into the specific research trajectory of engineers and scientists(Latour, 2011).

This same logic can be applied in the example of E01 and its project 01. The objective of analyzing the engagement of the sales force was shifted to a larger project of analysis of results that involved other actors, including the organization's president, in a project to talk to the president. E01 comments that his difficulty was specifically in increasing the level of participation of those involved. Therefore, paradoxically, a project that seeks to understand and foster engagement results in participation of between 35 and 40% of its target audience. In addition, the lack of resources, adequate “planning”, and organizational priority are often cited as obstacles that prevent the investment and development necessary to reorder these structures according to the statements that are summarized as: “Where am I going to put my resources, my money to be able to invest and use this because even within HR, this is not as valued as it should be” (E07, E05, E02, E08). Despite this minor misfortune for E01, with the data collected and analyzed, a reordering of the monthly board meetings was carried out, with discussions, which were much more effective and focused, and with the PULSE data – which were always shared – consistently inserted in the presentations of strategic plans according to the needs of each area (E01).

With an inverse dynamic, project 02 is portrayed with much greater enthusiasm by E04. The demand is from senior management, which wants a broad vision of how to be more effective and punctual in relation to the functional mobility demands of its employees, at a global level. The belief and value of this premise is more easily institutionalized and mobilized for action. The registration and integration of the participants (human, financial, technical and cognitive resources) all entities are focused on ensuring that the process is executed with the greatest possible precision and efficiency (E04). From the perspective of Latour (2011, p.187), this project can be allocated in a moment 5 – "Translation 5 – Becoming indispensable".

Within Latour's analytical framework, particularly in *Science in Action*, the process of translation 5 – becoming indispensable – is the central mechanism by which scientists and

engineers, who are operating as "*fact-builders*", are able to mobilize and sustain the buy-in of a diverse network of actors, both human and non-human, to validate and disseminate their claims and artifacts. The challenge inherent in this task, of the fact-builder, lies in the simultaneous need to (1) inscribe, register, enlist, recruit (*"enrol*") others to participate in the construction of the fact, and (2) to control the behavior of these others so that the claim is transmitted without being radically transformed or abandoned. Actors are inherently multi-conductors, capable of deflecting, modifying, ignoring or rejecting claims (Latour, 2003). The first four translation strategies (Translation 1 to Translation 4) represent different tactics for navigating this complex dynamic, displacing or reinterpreting interests to associate others with the proponent's project.

Translation five is not so much an active strategy to initially mobilize others, but rather the state or end result of the successful application of previous strategies and the work of keeping interested groups in line as he postulates Latour (2003). This perspective is particularly relevant for career development, where careers are seen as precarious effects of networks rather than as the result of individual strategic actions. In HR management, translation becomes a lens for addressing the paradoxes that arise when top-down HR strategies collide with local practices (Noponen et al., 2023). For example, in service organizations such as hospitals and hotels, HR staff and managers develop translation skills to adapt strategic decisions to local conditions, thereby facilitating effective strategy implementation (Sandoff & Widell, 2015). This approach not only enhances the strategic integration of HRM systems but also supports the creation of reflective spaces to evaluate the performative agency of design within organizations (Johnson & McHattie, 2014).

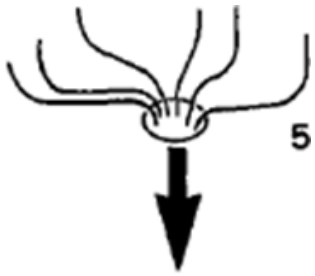
The central features of Translation Five are: (1) Elimination of the Need for Active Strategies: At this stage, the proposer no longer needs to actively strive to convince, negotiate, or deflect interests. Previous strategies – catering to explicit interests (Translation One), arguing that usual paths are blocked (Translation Two), offering shortcuts (Translation Three), or subtly rearranging interests (Translation Four) – become redundant. (2) The Others in Flux: Instead of the proposer meeting the others or persuading them to take detours, the others flow effortlessly toward the proposer. They rush to buy, use, believe, and participate in disseminating claims and artifacts. (3) Reversal of the Negotiation Dynamic: In Translation Five, the others are the ones who make the movement, the supplication, the commitment, and the negotiation. They are forced to follow the path of the proposer. This is in direct contrast to Translation One, where the proposer follows the path of others. (4) Binding and Hegemonic

Position: Geographically, achieving Translation Five means that anything one wants to do requires going through the proponent's position. Linguistically, “[…]the proponent's version translates all the others, acquiring a kind of hegemony. The proponent's interest becomes the interest of others: "Whatever you want, you also want it” (Latour, 1988, p.121). (5) A Routine Black Box Status: Claims or artifacts achieve routine and unchallenged black box status in the hands of all. They become indispensable. In this case, the process is institutionalized and legitimized through shared values, beliefs, and objectives (Berger & Luckmann, 1966).

It is paramount to highlight that the strength and capacity to influence an actor or network, such as the PA in HRM, derive precisely from this ability to accumulate and mobilize heterogeneous allies – controverses – (data, algorithms, systems, people, regulations) effectively and efficiently. When the PA reaches the Translation Five stage, it has become so crucial that other actors in the organization, such as managers and other business areas, realize that they need to go through this analytical network to achieve their own goals (such as making hiring decisions, evaluating performance, managing global talent), as highlighted by interviewee E01. This need to go through the PA to access the information accumulated, processed, and transformed into calculable insights demonstrates that it (or the HRM that operates it) has managed to establish itself as the strategic point where familiarity with what is distant and complex (how turnover generates (E02 and E09), sales force performance (E10) or global mobility (E04) is built and where the ability to act effectively “at a distance” in the domain of HRM is forged. Therefore, the success in Translation Five, as evidenced by the reports on how data provides security and credibility (E07) and becomes indispensable for decision-making (E01, E02, E03, E04, E05, and E09), reflects the success in building and maintaining the PA as an effective calculation center in the everyday reality of HRM at the BPI.

In the light of the ANT, the two projects (E01 and E04) go through the process of translation and its five movements. However, each one with different challenges. It is explicit from the reports of the interviewees when confronted with the literature that the breadth of social impact for the organization assumes different proportions in intensity and value, even though both constitute a means to each desired end. Such differences portray the capacity of each project to institutionalize and legitimize new values, goals, beliefs, and roles in a new social construction of reality; as Berger and Luckmann proposed in their seminal work (1966), the second project in principle already has legitimizing and institutionalizing characteristics due to the hegemonic force of the power of its sponsor or governance (Henninger, 2024).

Figure 16: Becoming Indispensable: Translation 5



Source: Latour (1998, p.199)

This section is dedicated to the presentation and analysis of the Reality Translation of PA and AI in HRM, exploring how these tools emerge and are signified in the daily lives of organizations, particularly in the BPI. Under the lens of the SCRT and the ANT, we analyze the initial uncertainties, the problematization processes that position the PA as a solution, the search for synergies and the challenges of the material and relational inscription of this new analytical reality.

We have seen how agency is distributed between human and non-human actors and how this new order's stabilization (or contestation) requires the reordering of cognitive, cultural, and functional structures in contemporary HRM. At this point, we present an initial practical contribution to the field of HRM by detailing, based on empirical data, how this translation of reality occurs in practice, revealing the mechanisms by which the PA begins to shape the knowledge and practices of professionals in the area, proposing that the new HRM be a calculation center (Latour, 2003).

A perspective that enables the new HRM to constitute a cycle of accumulation of knowledge based on the data of the people in the organization. The management of this cycle of accumulation tends to reduce distances – internal and external – of HRM relationships, based on the integrative potential of data – a unanimous proposition among the interviewees. This analysis, however, represents a part of the general panorama proposed by this research.

As we move on to Theme 2, Objectivity Required, we shift the focus of the analysis to the consequences of this translation and the search for the consolidation of the PA as an inherent and reliable element in new HRM. This chapter will deepen the understanding of how knowledge generated by PA and AI transcends the translation phase to achieve a status of objectivity. We will investigate the processes by which the new HRM, driven by data and analysis, traces a path to institutionalization within organizational structures and is legitimized by its actors.

3.2 Objectivity Required by PA

Reality Translation driven by the PA is not completed only with the registration of technology; it creates a demand for a new form of Required Objectivity. This objectivity is not inherent to data but rather an attribute that needs to be socially and technically constructed and validated. This theme explores how organizations in the BPI seek to establish and consolidate this objectivity through institutionalizing analytical practices and the constitution of legitimacy for the knowledge and decisions derived from them. This process is fundamental to respond to the SO02 (changes in and of processes) and already introducing the SO03 (human/non-human protagonism).

3.2.1 Institutionalizing: The New Analytic Order in the Making

The institutionalization of PA in HRM represents a transformative shift towards a data-driven approach, enhancing strategic decision-making and aligning HR practices with broader organizational goals, strengthening synergies, and creating new values and beliefs(Ferrar & Green, 2021; Gittell & Ali, 2021). This process involves several stages, starting with the adoption of PA, which is often driven by competitive pressures, institutional mechanisms, and the need for human resource management to fit into the broader and more complex business context. Adopting PA involves integrating new technologies – digitalization- and reshaping organizational culture to embrace evidence-based decision-making, leading to improved employee performance and organizational resilience – datafication (Thorpe & Sellar, 2023). Implementing PA requires addressing challenges such as data integrity, ethical considerations, and the rapid pace of technological change (Holz et al., 2024b).

The integration of PA into HRM facilitates intra- and inter-area strategic partnership (vertical and horizontal dimensions of HRM action), enhancing the alignment of HR functions with organizational strategies and demonstrating causal links between HR practices and business performance(Suri & Lakhanpal, 2024). This strategic integration is decisive to transform HR into a profit or calculation center (Latour 2001) and ensure its contribution to the business's success (Rehman, 2023).

However, the process of institutionalization is complex and involves overcoming barriers such as lack of analytical skills, data use reservations, and legal and legal uncertainties(Hammermann & Thiele, 2019). The process also requires a nuanced understanding of how ideals are translated into organizational discourses and control

techniques, as highlighted by the theoretical framework of institutionalization. By fostering a continuous improvement and innovation culture, PA enables organizations to navigate the dynamic business landscape effectively, provided that ethical and practical considerations are critically addressed and considered(Rehman, 2023; Westover, 2024). Ultimately, the successful institutionalization of PA in HRM depends on HRM's ability to balance the opportunities and challenges it presents, ensuring that data-driven practices enhance, rather than undermine, organizational values and employee trust (Rehman, 2023)

3.2.1.1 New Role of HR

Institutionalization requires that the role of HR as an analytical and strategic partner be not only declared but was experienced and appropriate to the practice, as well as to the specificities of each organization. This is a relevant aspect, as it relativizes a common practice of innovation absorption: benchmarking (Rasmussen et al., 2024). This depth is meant for the interviewees as: going beyond taking data (E01) but effectively participating in business decisions based on this data. E02 describes this performance: “HR as a business partner [...] we are giving the moment, for the initial discussion, so HR has to be present”. In the conception of interviewee E06, with her previous experience in production and R&D, she exemplifies a profile that has already internalized this need: “[…]When I sit down to talk, I do not just sit down to talk, I discuss it with the manager”. However, there is difficulty in getting out of the operational (E10), or the lack of familiarity with numbers (E07), which are still barriers to the full institutionalization of this new role for many.

HRM is increasingly seen as a critical enabler in harmonizing AI's technological capabilities with human-centered needs and ensuring that AI implementation aligns with organizational goals and human values (Som et al., 2011). This assertion is implicit in the E04 interview, which states: “[...] But we forget that people continue to work. In a more previous mentality, you know? So I think the structure can be a..., a thud...., but I think it's more about change management." From a macro perspective, interviewee E05 ratifies that: "[...] ... so, how are we going to deal with them [people and AI tools], we are still making these definitions, I don't think even Global itself has 100% of these definitions, yet [...]”.

For E07, there is a pressing need to understand that this new HRM must happen and that this depends on each HR professional, according to his report: “[...] They [HR professionals] need to understand that what brought them here doesn't take them where they need to go in terms of the future, right? Because the world has demanded different ways of

doing things. At all times, we have to unlearn and learn again, right?" A proposition corroborated by Marler & Boudreau (2017) and Sharma (2020).

From the perspective of Berger and Luckmann (1966), institutionalization can be conceptualized as a comprehensive and multifaceted process through which various social practices transform, becoming codified and routinized to such an extent that they are perfectly integrated into a system of significance and collectively understood regulatory structures. The main factor determining the breadth and depth of this phenomenon is to what extent are structures of relevance commonly recognized and accepted among the various members of a given society, which in this case is the organizational environment.

In cases where a society – or the organization – demonstrates a remarkably high degree of shared relevance, the process of institutionalization is likely to be extensive and comprehensive, encompassing a wide range of actions, interactions, and behaviors that reflect this collective understanding. This relevance has been required, but still in a fragmented way, according to the interviewees cited above. However, such fragmentation somehow contributes to construction in flux: from the micro to the macro, from the individual to the group, from the sectoral to the organizational, when and if signified or resignified without generating crisis (Berger & Luckmann, 1995).

On the other hand, in situations where shared relevance is confined to a limited subset of social practices, the institutional scope tends to remain quite narrow, resulting in a scenario where multiple practices can become highly segmented, fragmented, or disjointed from one another (Forsten-Astikainen et al., 2017). Thus, for Riitta Forsten-Astikainen and colleagues (2017), the problem of silo syndrome is widespread and affects a wide range of organizations, highlighting that no modern company, public institution, or government agency is immune to the harmful effects caused by an insular and compartmentalized work environment. Despite technological and structural advances aimed at promoting the comprehensive dissemination of information and developing shared competencies, strengthening internal borders still impedes cross-functional integration.

In this case, Forsten-Astikainen et al. (2017) argue that communities of practice (PCs) have the unique ability to facilitate the dissemination of knowledge and skills across entrenched boundaries, potentially remedying the negative impacts of silo mentality in a reconstruction process, as reported in interviewee E04's PA project implementation report. Thus, the dynamics of institutionalization are intrinsically linked to collective consciousness

and mutual recognition among individuals within the social structure, shaping how social norms and practices evolve over time.

3.2.1.2 Change Management

Successful implementation of PA requires a structured approach to change management that addresses the broader implications of change, particularly how individuals and teams adapt to new processes and systems (Huschbeck, 2021; Tursunbayeva et al., 2021). To do this, it is necessary to understand how to move the organization towards change. Therefore, obtaining knowledge of the "engines" that constitute change tends to corroborate so that this process is managed efficiently, effectively, and with agility (Poole & Van De Ven, 2021). This is essential as PA transforms HR into a strategic partner, enabling data-driven decision-making, which aligns HR strategies with organizational goals and fosters a culture of continuous improvement and innovation with different conceptual approaches

Change management strategies should, therefore, focus on building organizational commitment and encouraging innovative behavior among employees, as demonstrated by sustainable HRM practices that emphasize environmental and social responsibility, defended by Çiçek (2024). Also, the adoption of PA is facilitated by advances in technology and information systems, which make data analysis more accessible, thus supporting the necessary transformation of HR into a more efficient and strategic function. Effective change management ensures that PA's potential is fully realized, leading to better employee recruitment, retention, and engagement while addressing the ethical and practical challenges associated with its implementation (Westover, 2024).

Institutionalization requires continuous and profound change management. It is necessary to deal with cultural resistance and preexisting habits. For interviewee E08, the need to: "[…]Deconstruct this prejudice [refers to the adoption of AI in HR] and that technology drives people away should be highlighted". E07 corroborates this thought and emphasizes the importance of training and engagement to; "[…]break the resistance of the other". There is a silent mention of the need for greater internal interaction and cooperation. The reflective posture of Braun and Clarke (2020) is essential here, as both the researcher and the HR professional must be aware of how their dispositions influence the perception and management of this change. Still, under this lens, a difference in the attitude of each interviewee is inferred when considering their professional historicity.

In this respect, respondents who are graduates of other organizational departments (not native to HRM) tend to have a different way of using their critical reflective power and putting it into practice (E01, E05, E06, E07, and E10) when compared to professionals (HRM natives), less pragmatic or with a less acute business mentality. Such differences lie in identifying indicators of resistance to the changes necessary for the success of such processes and acting promptly and punctually on these challenges because, according to E01, it states "[...]we humans tend to zero". An important reference on how each one deals with the challenge and the force(s) of the status quo of everyday life, which will be covered in more depth in section 3.3.2.

Nevertheless, from the perspective of Berger & Luckmann (1995), in their work, *Modernity Pluralism and the Crisis of Meaning: The orientation of modern man*; The institutions, play a fundamental role in controlling the distribution of meaning in society, acting as creators, repositories and transmitters of socially constructed knowledge, therefore, responsible for managing the possible changes demanded in the dimensions: time/temporality and depth. These meaning-making movements, which emerge in institutionalization, are designed to relieve individuals of the constant need to reinterpret the world by providing structured "programs" for social interaction, intercourse, and life conduct. These programs offer patterns of behavior, already tested, that individuals can follow – or not – , aligning their actions with society's expectations and roles, such as those of husband, father, or employee.

Social relations are inherently meaningful because they are embedded in an intersubjective framework in which individuals act and live (Berger & Luckmann, 1966, 1995). This intersubjective framework assigns meaning, as it provides a framework within which individuals can understand their roles and actions about others – to live and relate positively and productively socially. In relationships, meanings – constituted and shared – are not static and are constantly negotiated and redefined through social interactions and will depend directly on the intensity and frequency of these interactions. Therefore, when the social structure fails to provide sufficient coincidence in intersubjective reflections, the basis for a personal identity with constant meaning is undermined. This can lead to subjective crises of meaning, in which individuals struggle to find coherence and purpose in their social interactions representing their identity (Berger & Luckmann, 1995).

In organizational practice, these conflicts are inherent to everyday life, for example, in the statement of interviewee E07, who reports: "[...] my CEO does not agree with any

automation, because he thinks that such a process will disintegrate his HR team and that such a posture does not correspond to HRM, which he wants". This important statement can be "translated" as: First, the CEO owns the company, so a company is family-owned, which gives him or her a "total and unrestricted decision-making power" that is generally greater than another type of organization, which depends on other decision-making levels. Secondly, it is in their concern – which is valid – with people, the provision of employment, and human relations. Third, this CEO takes a position of discredit in the evolution of contemporary HR operating systems. Such a position may, shortly, decrease organizational competitiveness and agility due to inertia or omission in technological innovation and its connection and relationship with the people of the organization.

In a less forceful way, the rejection of change can be externalized in apparently ordinary phrases as characterized by the E07 report:

> "[...] I perceive it within my area a lot like this, ah, but it has always been like this, ah, but he likes it like this, ah, but it has to be like this. Why? Because it's challenging to have to argue, set up a strategy, try to sell ideas, you know? So, I've been trying to go along this line, ah, I see, but what path could we look at and do differently that will improve for us as well as for everyone else, so... [...]I perceive a certain air of accommodation, it doesn't touch mine, no one bothers anyone, no one exposes anyone, each one in their own, everyone has their salary, their position and that's okay[...]"(E07).

In an opposite situation to the previous case – CEO who rejects HR innovation – in E01's interview, the implementation process and changes necessary for the implementation of PULSE are portrayed, which redefines the conception of quantitative analysis of data on HRM engagement. Also, according to E01, the first significant change was in himself as HR Director. A change that was demanded by his perception of the need to expand his understanding of an important topic for the BPI: engagement. Given the functional configuration of the sales force, this topic is of special concern for this type of industry – they are all positions of trust. Therefore, maintaining an engaged team is fundamental and critical for organizational success. From the organizational perspective, E01 reports that the process of change in the perspective of analysis was inevitable. Taking responsibility for implementing a new process required overcoming the challenge of defining which tool would be the most

appropriate and how to carry out the proper integrations and “signify” such a change in a general broadway in the organization.

However, to signify the process for the “mobilization of actors” outside its departmental limit influence, and priority – the different directorates (director of sales force, training, marketing, finance, accounting, and legal). To their surprise they generated a positive change in the posture and acceptance of these directorates in the face of the proposal presented. This change in posture – positive – consists of balancing and adapting the information/communication that the organization had analytical gaps, that a project with the configurations presented would eliminate this gap; a new social reality being built, considering the combination of different organizational actors (Callon, 1986; Latour, 2005).

Such a PA project (E01, PULSE) is capillary. It achieves a new level of organizational integration, which breaks the internal divides of the organization, which arrives to offer a direct link between departments, the sales force and the organization's president, which in this case would be with weekly communications – breakfast with the CEO. It leads us to another dimension of analysis of the institutionalization process: the power of communication as a mediator of meaning, construction of meaning, as well as the assumption of risk in transforming it to a crisis (Berger & Luckmann, 1995).

In this specific case (E01 – PULSE) it was positively strengthened, it expands and aligns itself in solid relationships between humans and non-humans; being the humans – the interviewee, the directors of the various departments, the president, the FDV, the CEO of the company that provided the program that generated PULSE, the technicians of the supplier and the interviewee; non-humans – the congress space in which the interviewee got to know the service provider, the service provider's publicity material, the service provider's system, the operating system adapted to the interviewee's company, the data collection and storage systems, the internal communication systems, which provide the opportunity to disseminate results to stakeholders, above all, a resignification of the intersubjectivity of organizational actors, a solid social construction for a common goal – a new level of organizational knowledge.

Berger and Luckmann's theory emphasize the dialectical relationship between individual actions and social structures, where social reality is constructed by individuals and imposed on them as an objective reality but is complemented by using multiple organizational actors in ANT(Latour 2005, Callon 1984; Mustafai, 2019). E01's reports portray a complex process that demands constant negotiation. Therefore, adopting an PA project requires precise

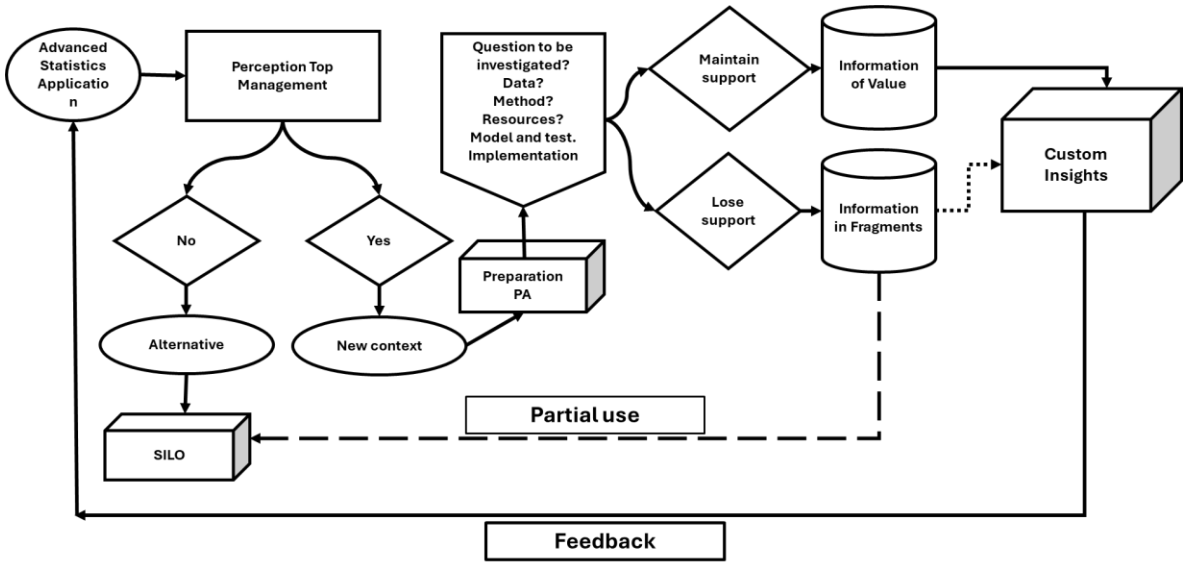
management and effective alignment of internal and external networks crucial for managing and implementing change, as postulated by Poole and Van De Ven (2021). In this aspect, the organizational or department size is not relevant because, in both small and large organizations, resources are generally limited – keeping the due proportions – and must satisfy the objectives of all stakeholders, as well as manage the decision-making forces – even contradictory – of the area that proposes the project, a fact that is related to the need for alignment with the organization's top management. For a change to be successful in an PA project, a proactive network with internal and external actors is still required while paying attention to the influence of non-human actors who can alter such flow to meaning or the crisis of meaning (Grabowski & Mathiassen, 2017).

3.2.1.3 Communication

Communication emerges as a critical element in the process of institutionalizing the objectivity required by an PA project. As the primary vehicle for the transmission of meanings and the negotiation of realities (Berger & Luckmann, 1966), communication is fundamental for the new practices and knowledge derived from the PA to be objectified, legitimized and internalized in the organization. PA tools, as discussed, act as non-human mediators (Latour, 2001) that process data and generate insights, but the transformation of these insights into action and their integration into social reality depend intrinsically on human communication and the strategies adopted to overcome the barriers inherent to organizational structures, notably knowledge silos, which can be enhanced by translating the strategic needs of HRM and organizational, are not clearly aligned with the analytical construction of the AP project.

The flowcharts proposed in (Figure 17) illustrates how failures in the perception or support of senior management – often linked to communication deficits – can lead to fragmentation of information or silo isolation, preventing the realization of the potential value of the PA.

Figure 17: PA Flow – Analytical Frame



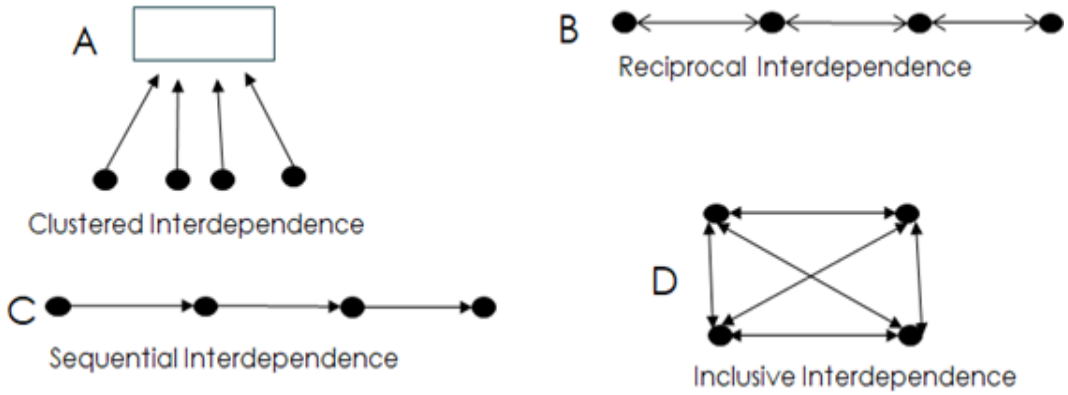
Source: Research (2025).

Analyzing the implementation of PA as an innovation process, the perspective of Utterback & Abernathy (1975) serves as a referential basis for the flowchart of Figure 17, which reinforces the premise of the centrality of communication in the process of adoption and implementation of PA. The model describes distinct phases in the evolution of an innovation, from an early fluid stage to a specific, standardized stage. Communication plays different but integrated roles at each stage. Specifically in its initial phase, of greater uncertainty (corresponding to the Upper Management Perception and the Yes/No bifurcation in the flowchart), communication is vital to explore needs, define the problem (problematization - Callon, 1986) and build initial support (*interestment*). Faulty communication here can lead directly to "No" and the reinforcement of the "SILO". If innovation advances ("Yes" → "New context"), communication in the preparation and implementation phase (the "question box" of the flowchart) becomes a guide to define methods, mobilize resources, test models, and manage change, ensuring the necessary support. Loss of support, as indicated in the flowchart, leads to fragmentation, suggesting a failure in ongoing communication about the value and progress of innovation.

The need for effective communication becomes particularly acute when we consider organizational interdependence, as theorized by James D. Thompson (1976). Many of the issues that the PA seeks to address in HRM – such as talent management, workforce planning, engagement analysis, or cross-cutting process optimization – involve high reciprocal interdependence between different units or functions (HR, finance, operations, business areas). Thompson (1976) argues that reciprocal interdependence requires coordination

mechanisms based on mutual adjustment, which requires intensive, frequent and information-rich communication. Such interactions can be represented under the different levels of interactions as shown in Figure 18 that will be detailed below.

Figure 18: Types of Interdependencies



Source: Adapted from Thompson (1967).

The emphasis and effectiveness of organizational communication becomes structures elements to implementing a PA project in any organization, which is profoundly influenced by the interdependencies between units, teams, and/or individuals. Based on James D. Thompson's (1967) concepts of interdependence, we can distinguish different interconnections directly impacting communication needs. Sequential interdependence (Figure 18 – Item C) occurs when the product of one unit becomes the input for the next, requiring clear and coordinated communication flows to ensure efficient transitions and the correct sequencing of activities. Reciprocal interdependence (Figure 18 – Item B), being the most complex and dynamic, involves mutual and interdependent exchanges of inputs and outputs between different units or individuals. This type demands intense, frequent, and multidirectional communication to facilitate mutual adjustment and continuous collaboration. Inclusive interdependence (Figure 18 – Item D), often related to Thompson's concept of pooled interdependence, describes situations where units.

However, they may operate with a degree of independence in their daily tasks and are interdependent on the organization's overall performance and sharing common resources, requiring communication that reinforces alignment with strategic objectives and utilization of shared resources. Finally, grouped interdependence (Figure 18 – Item A) can be understood as organizational arrangements where subgroups of units or teams exhibit high internal interdependence (often reciprocal or intensive sequential within the grouping) while grouping as a whole connects with other parts of the organization; This requires communication

strategies that ensure both the internal cohesion and coordination of the cluster and the effective interface with the rest of the organizational system. Understanding these interdependencies is essential to designing communication channels and processes that support the collection, analysis, and effective use of PA data.

Therefore, organizational silos function precisely as barriers to this type of communication, limiting the flow of information and the ability to coordinate adaptively. E02's report on integrating data from different areas and the perception of different interests exemplifies this challenge. In this context, PA is often introduced with the promise of overcoming silos – an integrative pathway – by providing a common, objective, accessible, and intelligible database. Tools such as shared dashboards (E01, E03) or integrated systems (ideally, or at least close to) work as engines to obtain registrations, which seek to facilitate communication and mutual adjustment (Latour, 1988).

However, as the interviews show (E01, E05, E09), the mere availability of the tool does not guarantee effective communication. An active communicational effort is needed to interpret the data together, negotiate their meanings and coordinate actions, overcome the boundaries imposed by low interdependence and structural silos. Thus, Communication becomes essential to activate the integrative potential that the objectivity of data promises (Lee et al., 2023; McKinney & Shaffer, 2023; Prasetyaningtyas et al., 2022).

Interviewee E10's experience with gradual implementation and pilots tests (hypothesis tests), or E07's strategy of building a case based mainly on the perception of ROI, are examples of communication strategies that seek to ensure the success of the innovation process represented by PA. These strategies align with the attributes that Rogers (2003) identifies as stages to be overcome for dissemination: Relative Advantage and Trialability. Effective communication is, therefore, the lubricant that allows the PA innovation process to move toward value creation rather than stagnating in fragments or silos.

No less important, the perspective of Erving Goffman's (1974/1986) *Frame Analysis* offers a powerful insight into how communication, mediated by PA, attempts to redefine the very organizational reality that it tends to alter in its analytical and relational propositions in the organization. Goffman suggests that individuals interpret the world through “frames”—interpretive schemes that define what is going on here. Organizational silos can be seen as generating such distinct and often conflicting frameworks about the reality of HRM, as an example: The financial frame focuses on costs, the HR frame focuses on engagement, the sales frame focuses on goals and KPIs, and so on and recursively in the complex reality of each

organization. Therefore, communication between silos is often flawed because actors operate within incompatible frames.

A PA project, by presenting allegedly objective data – because I emphasize again that this will depend on the quality of the inputs and training in their cross-sectional analyses – tends to function as an attempt to introduce, institutionalize, and legitimize a new dominant framework – the analytical framework (Berger & Luckmann, 1966; Goffman,1986),. The communication of PA results (in reports, presentations, and dashboards) is the vehicle through which this new framework is proposed to the organization. As E02's example of turnover analysis demonstrates, the data was used to reframe the board's perception of the problem – "that it was not a loss of talent to a specific competitor, but rather a process of normal oxygenation of the organization".

Similarly, E10's insight into his lack of interaction with Sales, generated by Viva Insights Microsoft®– a simple but compelling example – allowed him to reframe his communication priorities. The success of this attempt at reframing depends on the ability of communication to make the analytical frame not only intelligible but also plausible and relevant to the different actors, overcoming the inertia of the frames established by silos and contributing to the construction of a more integrated and objectively shared organizational reality – even if this objectivity is itself a social construction mediated by the socio-technical network of the PA (Gergen, 2023; Heiland, 2023; Tenakwah & Watson, 2024).

To fully understand how this new objective reality is mediated by the socio-technical network, as mentioned, it is essential to recognize the agency of its technological components, as proposed by the ANT (Latour, 2005). The tools contained in the concept of an PA project – such as software, hardware with processing power, and complex algorithms – are not merely passive channels but act as central non-human actors in this organizational network (Latour, 2003).

Through their ability to process and analyze data, they effectively mediate the construction of the new analytical frame, providing insights that shape organizational decisions and communication. For example, PA tools can identify key performance indicators and predict future trends, influencing how employees and managers interact and make decisions (Lee & Jacobs, 2023; Rehman, 2023). Integrating PA tools into organizational networks highlights their role as agents, shaping the social construction of reality. By providing data-driven insights, PA tools influence how employees perceive their work

environment and interact with each other, thus contributing to social reality within the organization (Plesner, 2009).

People Analytics, by analyzing communication and interaction data, can identify key influencers and leaders within an organization, whether these leaders are formal or informal. This information can be used to (re)shape social dynamics, influenced by how employees construct their reality. For example, by identifying bottlenecks or key individuals in the communication process, organizations can intervene to improve the flow of information, thus changing the social construction of reality within the organization, strengthening the interdependence between areas and/or departments, through these “organizational lawyers” (Dahlbom et al., 2019; Ijanperä & Vuori, 2021).

PA can reveal patterns of collaboration and conflict, which are essential to understanding how social reality is constructed. By addressing these issues, organizations can influence the process of social construction, fostering a more collaborative and productive work environment (Korichi et al., 2023). Communication is not just about introducing the PA but also about integrating its results into organizational conversation. In the interview with E10, he mentions the need for “frequency and sequence” in communication, similar to advertising (narratives repeated continuously to fix concepts and differentials of the products they propagate), to fix the importance of data.

Again, while implementing PULSE (E01), the interviewee admits that some failures in communicating the results effectively may have prevented the tool from becoming part of the shared knowledge and decision-making routine, failing to institutionalize it more quickly and deeply. Interviewee E01 refers that each board used the results presented by the survey – when updated – and "closed" itself in analyses that corresponded to its sector of responsibility – nothing strange in this – however, the sharing of the insights generated was maintained in the individual knowledge of each business area, strengthening the knowledge silos, as in the statements:

> “[...] I think this is natural, as it happened with "sales" too, I think that the company that bought it did not exist a person for a long time who had general knowledge, so there was only financial, spoke only with financial, HR only with HR and everything, so there should be "one out" [a person] eventually that you put everyone together, but what is not simple either, you put everyone in a room that has different

> interests, regulatory has one interest, financial has another, HR has another, has a series of information (E02)".
>
> “[...] there was no indicator panel here, each one who had an experience that came from a different company, each place did it in a different way (E06)".
>
> “[...] The demand area, we have a commercial intelligence area, which manages data to supply demand, marketing, commercial strategy. So, I think it is an area that generates and unites areas in favor of a common goal. I see this much more matured in the area of sales force strategy. I think it is the most evolved area here in our context (E08)".

As one of its strengths, PA can collaborate to reduce this fragmentation. Another example of how managing the flow of interactions and information exchange can generate important insights is in the report of interviewee E10. On his initiative, during the interview, he showed me how his Copilot®, which has a task “manager” named VIVA,® can practically instantly provide a reasonable analysis of your interactions and time management. When he introduced me to his internal connections, he discovered that he, as HR director, was not maintaining frequent enough communication with the Sales Force Director (FdV). This specific director did not appear as a 1st level relationship with E10. As simple as it may seem, this insight generated a proactive action on the part of E10 about its primary internal customer, the FdV director (Korichi et al., 2023).

This event – the relationship between E10 and the director of FdV – under the lens of the social construction of Berger and Luckmann (1966), represents the capacity that PA has to externalize patterns, whatever they may be, indicating in this case, the lack or at least the little daily interaction with the FdV by E10. Also, PA institutionalizes a socialization practice when: First, it makes it habitual for E10 to verify their system of interactions through AI recurrently. Secondly, PA typifies these relationships visually and hierarchically, facilitating the recognition of the possible relational “gaps” of E10 and their opposite, when the relationship is overestimated, in a state of “overrated”. From Latour's perspective, the “agency” of the non-human, in this case, becomes manifest when Latour (2001, pg. 203); contrasts this perspective with that of M. Heidegger (1977) in The question concerning technology, who, although he recognizes that technology is not a mere instrument, ends up placing us as instruments of instrumentality itself because this perspective does not capture the reciprocity and mutual

transformation that occur in technical mediation and the construction of new or complementary processes in both actors.

Still using the experience of interviewee E10, Copilot® VIVA, in the case of interactions, communicates them under a fundamental dimension, that of technical mediation that generates the "composition" of the action. Latour (2001) argues that action is not an exclusive property of humans, but rather of an association of actors. And who practices the action, E10 or Copilot® VIVA? Does E10 respond to a stimulus or Copilot what is that stimulus? In order to clarify this complex connection between the actants, Latour (2001) uses the example of the action of flying, which is not only of the individual, but of the entire association that includes airports, airplanes and procedures, means of locomotion used to arrive at the airport, ticket purchase platforms, among others. This understanding challenges the traditional view of *Homo faber*, suggesting a *Homo faber fabricatus*, where we are molded by the tools we create, in a dialectical process of mutual constitution.

3.2.1.4 Integrated and automated processes

The adoption of PA in HRM is an integration process. Initially, the idea of automating HR processes may arise as an innovation or response to specific needs in efficiency and standardization, among others, or a simple digitalization. Over time, through the habituation of actions by types of actors – HR professionals, managers, employees, or users of these systems – these automated processes become established standards, as previously portrayed with interviewee E10. From a macro perspective, the use of software for recruitment and selection, initially a choice, can become the "way things are done" – an institution – in that organization (Berger & Luckmann, 1966, pg. 70-71).

As previously highlighted, institutions arise from the reciprocal typification of habitual actions. In HRM, this would mean that repeated interactions with automated systems, such as registering vacations, evaluating performance through a platform, and performing payroll, lead to typifications on how these tasks should be performed and who are the actors involved, for example, the employee who requests, the system that processes, the manager who approves, evidenced by the relationships. Passing on these institutionalized practices to new organization members (new HR employees or the company at large) solidifies the objectivity of these practices performed by algorithms (AI). For the new generations of employees, PA(IA) already exists as an external and coercive reality (Berger & Luckmann, 1966, pg. 74-77).

In a complementary way, Latour (2001) explicitly postulates that technical mediation offers a different perspective, focusing on how humans and non-humans – HR and PA Directors – interact and mutually transform each other. Adopting PA projects in HRM transforms all actors into actionable mediators who produce results. They have an "action program" or "script" that influences the behavior of users. For example, an automated performance appraisal system with pre-defined fields and scales not only collects data but shapes what is considered a valid appraisal – with minimum and maximum thresholds – which competencies are valued – for each specific role or area – and how feedback should be structured and applied, or again the example of the PULSE system, which aims to increase the engagement of FdV (E01).

Therefore, from Latour's perspective, the HRM professional with the automated system is a different actor from the HRM professional without it. The introduction of the PA creates a new hybrid – a theme addressed in section 3.3 more broadly – a new association of actors (human and non-human) that redefines the objectives, objectification and contemporary competencies of leadership (Boyatzis, 2008).

For respondents, proper institutionalization happens when HRM processes are redesigned around data analysis, ideally in an integrated and automated way. Examples such as digital admission (E02, E08) or the vacancy flow (E09) are steps in this direction. However, the persistence of manual processes, "on the nail" as emphatically mentioned by E07 and E09, or the need to cross-reference data from multiple systems (E1, E2, E5) indicate that the institutionalization of PA is still partial.

3.2.1 Algorithms: an institution to be legitimized

The search for Required Objectivity in HRM, catalyzed by the introduction of PA, can be enriched by the perspective of Algorithmic Institutionalism. As detailed by Mendonça, Filgueiras, and Almeida (2023), this approach proposes that algorithms transcend their function as mere technical tools, operating as institutions. They are considered institutions because algorithms establish sets of formal and informal rules that shape interaction contexts, define development paths, and restrict or induce specific behaviors of human actors and the (non-human) systems themselves, and generate significant collective consequences for the organization.

For HRM, this means that PA algorithms not only process data in search of objectivity and/or standardized rationality but actively (re)construct the rules of the HRM game, that is,

the interaction between HR practices and policies with organizational strategy, influencing everything from the definition of what a “talent is, for each organization” to the evaluation and decision-making processes. Such changes generate and impute effective changes in the interviewees' values, rules, habits, and daily routines (Nawaz et al., 2024; Parsons, 1991).

Algorithmic Institutionalism allows us to analyze how PA systems – when and if implemented – become part of the institutional fabric of the organization through HRM intervention. New norms and routines crystallize through the habitualization, and reciprocal typification of actions mediated by algorithms – such as using a dashboard to evaluate performance or a system to filter candidates. These routines, inscribed in the code and the interfaces of the systems, acquire a facticity, an “objective reality” that tends to confront HR professionals and managers in their deepest conceptions. Mendonça et al. (2023) argue that these algorithms function as “rulebooks” that define meanings and organize action. Therefore, the “objectivity” through the process of algorithmization that an PA project is not translated into neutral data extracted from reality but as a result of a specific institutional process, mediated by the rules and logics embedded in the algorithms themselves that are sought to implement specific strategic propositions (Mendonça et al., 2023; Weiskopf & Hansen, 2023).

Adopting the lens of Algorithmic Institutionalism has important implications for the analysis of HRM at the BPI. First, it highlights the power relations inherent to algorithms and the hierarchies instituted in their activity and action: they are not neutral since they incorporate values, biases, and interests of their creators and the contexts in which they operate, distributing resources (such as attention, visibility, agility, promoting accessibility, with various roles), and defining hierarchies of knowledge, relations, and standards (Burrell & Fourcade, 2021).

Second, it complicates the legitimation process, as algorithmic objectivity needs to be justified and validated not only for its technical effectiveness but also for its adequacy to organizational norms and values, corresponding to the moral and ethical values of the environment in which it is inserted (K. Dasari & Sujatha, 2023; Di Lauro et al., 2024). Third, it challenges traditional notions of agency since, as argued in the ANT, algorithms act as non-human mediators with their own “scripts”, forming hybrids with human actors and transforming the nature of decision-making and the reality of HRM. Thus, understanding how these algorithms effectively become organizational institutions, redefining rules, and practices, becomes a central point for the analysis (Latour, 2001;Wei, 2024).

The emergence and growing ubiquity of algorithms in organizational processes invite an in-depth analysis of their role and impact. More specifically, the ubiquity of algorithms in HRM has transformed traditional practices and introduced opportunities and challenges (Olhede & Wolfe, 2018). This shift towards algorithmic HRM is met with a duality of perceptions among HR professionals, who experience both aversion due to uncertainties and appreciation for the efficiency it can bring (Shore et al., 2024). Such uncertainties can be described as:

> "[...] I have to do it, now, after here [sic], I have to take a PA course, to try to understand this PA a little more, too. Then, the question comes back to who asks the question. Whoever asks the question needs to ask correctly [sic]. What do I want? Oh, I want to know why my HR is talking too much with the legal department, so what are my problems that can impact me legally?(E01)"

Integrating AI into HRM processes, such as recruitment, performance appraisal, and talent management, offers the potential for streamlined—but not simplistic—operations and strategic workforce planning (Bock, 2015; Madanchian & Taherdoost, 2024). Some consensus is that automation tends to reduce the administrative burden on HR professionals, allowing them to focus on key business objectives and adapt to the evolving analytical and strategic landscape(Bahuguna et al., 2024; Shore et al., 2024). Despite the benefits, there is a pressing need for conceptual clarity and a multidisciplinary approach to fully harness the potential of algorithmic HRM, as current research often lacks an emphasis on socially integrative strategic contexts of HRM (Weiskopf & Hansen, 2023a).

Assertions that are of fundamental importance – for this research and the practice of management – because the use of algorithmic systems affects the motivation and behavior of employees, with clear inferences in managerial control, leading to extrinsic motivation, or its opposite when internalized, and to the provision of feedback processes, on each respective metric, more objective, personalized, frequent and data-driven (Ateeq et al., 2024; Edwards et al., 2024; Huang et al., 2023). As organizations grapple with the complexities of algorithmic HRM, become focus themes; data privacy, ethical considerations, and the potential for bias produced, ensuring that AI enhances, rather than supplants, the human element in HRM practices. The evolving landscape of algorithmic management, particularly in platform work and the gig economy, further complicates traditional HRM frameworks, requiring a

reassessment of the boundaries of work and the development of more nuanced approaches to understanding and managing its dynamics (Keegan & Meijerink, 2025).

Overall, integrating algorithms into HRM processes presents a transformative opportunity for organizations, as long as they (organizations) accept and address the challenges associated with such an endeavor and maintain a balance between technological advancement and human interaction. In this aspect, the interviews generally portray this concern, which at the BPI, is a priority for managers. Parsimony, as well as the search for a greater understanding of how to carry out such functional and strategic changes in and at work, tends not only to protect the employee but to avoid possible disruptions of management systems that are already institutionalized and legitimized (E06, E07, E02, E05).

Furthermore, the introduction of an algorithm in an organization, under the constructionist lens of Berger and Luckmann (1966), is an act of externalization, where a solution is proposed to a problem or a process that needs to be (re)formulated or (re)inserted in a "new organizational context". Externalizations can emerge as: "[…]Need for greater engagement" (E01), "[…]improve functional mobility" (E04), "]…]understand turnover" (E09), or simply a "dashboard" that speeds up data visualization (E07).

With repeated use, interactions mediated by the algorithm begin to be habituated by the organization's members (Berger & Luckmann, 1966, p. 70). These habitualized actions are then typified and/or signified (Berger & Luckmann, 1995); they are recognized as expected behavior patterns in certain situations mediated by the algorithm. For example, the algorithm of a customer relationship management (CRM) system not only dictates the steps to record an interaction but also typifies the "good" and "bad" customer service according to the parameters embedded in it, which can be analyzed, as well as assigned actions if necessary, therefore, this reciprocity of habitual actions, by analysts and customers, through the system, are shared and taken for granted by the members of the organization, formatting new relationships and interactions of the actors (Berger & Luckmann, 1966, p. 72).

The algorithm considered an organizational actor, is no longer just a piece of software but an institution that defines "how things are done here". This institutionalization gains historicity and objectivity, mainly when transmitted to new members of the organization, who find it a pre-existing and coercive reality (Berger & Luckmann, 1966, pp. 76-77). The algorithm, as an institution, comes to be perceived as a "thing" or an "entity", an objective social fact that does not depend on the individual will of this new member (Berger & Luckmann, 1967). The maintenance of this objectified algorithmic reality must be legitimized.

Berger and Luckmann (1966, p. 110) describe legitimation as an objectification of “second-order” meaning that explains and justifies the institutional order. In relation to organizational algorithms, such legitimator processes can be identified at the following levels: (A) Incipient level: They are represented in the interviewees' statements with their language, such as: “[…]our approval system” and “[…]the suggestion algorithm”; “[…]I think it would be amazing such an evolution” and “[…]someone doing this for me [laughs] or at least an algorithm doing it” are assertions that already carry a fundamental legitimacy (E01,E03,E05,E10).

When the process advances to the third level of legitimation, the recognition of the need for explicit theories being used by experts – whether internal or external – that can be developed and increasingly elaborated and complex on: The efficiency, fairness, or innovation provided by the algorithm, creating a differentiated body of knowledge that supports its use. On this aspect, the algorithm used in the projects of E01, E04, E09 and E10, each of which was externalized with a clear and shared purpose, as well as being hypothesized and disseminated as an agent with transformative power in its premises of engagement, mobility, turnover, and interdepartmental relations, respectively, but it is still not a standard level among the interviewees.

At a level of legitimacy that encompasses the symbolic universes – fourth level – the interviews reveal that the use of algorithms can be integrated into broader values of the organization, such as: “[…]the need for digital transformation”, “[…]the construction of a data-driven culture”, “[…]in search of operational excellence”, “[…]providing agility” by giving them a meaning that transcends pragmatic application and anchors them in the identity and mission of the organization, which is still in the dimension of a desire somewhat distant from the reality of most of the interviewees (Berger & Luckmann, 1966). On the other hand, the legitimacy obtained by the symbolic references of this new actor, in the cases of application of PA projects (in the cases of E01 and E04 mainly), does not subjugate or co-opt human protagonism.

On the contrary, it is found that the interviewees have different levels of internalization of this new algorithmic reality in their intersubjectivity. As an example of this symbolism, there is, in the construction of new discussion groups, in the perception of greater interaction and integration about a problem, in the acceptance of suggestions that transcend the limits of their department or portfolio, a process of secondary socialization that occurs when entering or assuming new functions in the organization. They still learn and apprehend the "*stratum*"

defined by the algorithms, including specific vocabularies and routines, which become part of their reality at work, but clearly, they do not deify this process, reiterating the human protagonism of this evolutionary process as follows:

> “[...]the tool that will segment, but someone behind it has to bring all the intelligence, logically, to create the resume (E01)”;
>
> “[...]I'm going to say it like this, the analysis of the human being, of the human factor, it will continue, but our confirmation, our process analysis, it will continue to be done(E02)”;
>
> “[...]I say, the chat GPT will no replace this here.... I joke with my colleague from the legal team, I make the contract in chat GPT, but now no one will discuss performance with you (laughs)(E03)”
>
> “[...]So, I think that if we know how to really use this to our advantage....., and then using our filter, you know [sic]. Helping the organization to understand that not everything we can count on artificial intelligence or technology in general, can be good, you understand...(E04)”
>
> ”[...] Now, when I look at human resources, I think it depends much more on humans than on artificial intelligence. At least in the industry where I work, and I don't go to any other industry that's different.(E05)”.

The quotes reinforce the conduct of the BPI 's HRM as an unrestricted focus on valuing its human capital – beyond rhetoric. This conviction is increasingly strengthened and aligned with the profile of the required and developed FDV professionals at the BPI, who, due to the autonomy granted, need a high degree of intrapreneurship (van Wetten et al., 2020). Therefore, it is the responsibility of HRM to use all possible apparatus – conceptual, technological, cognitive – to foster, develop, and maintain the valorization of human capital. Understanding integrated and automated processes in HRM or any other organizational domain requires, therefore, not only analyzing algorithms as technical tools but as complex institutions that emerge, and feedback, social and sociotechnical processes, need legitimation. This dual perspective – institutionalization/legitimation and non-human action of the algorithm – allows for a richer understanding of how algorithms are changing the rules of the game in organizational life (Berger & Luckmann, 1966, 1995; Latour, 2001, 2011).

3.2.2.1 Algorithms as Mediators and Actors in Sociotechnical Networks

Algorithms are not passive structures, but mediators or non-human actors who actively participate in the reconfiguration of organizational practices. In "*Pandora's Hope*", Latour (2001) argues that techniques mediate action in complex ways, transforming the actors and their objectives; algorithms, from this point of view, are not mere intermediaries that transmit human intentions without changing them; as mediators, they have their own "scripts" or "programs of action" , they "do" things, influencing user behavior and process outcomes (Latour, 2001, p. 205).

A recruitment algorithm, for example, not only organizes resumes but actively filters, sorts, and even decides which candidates are most promising, transforming the hiring process and the role of human recruiters. The HR professional with the algorithm is a different actor, with different capabilities and limitations, than he would be without it (Bock, 2015). This statement is recognized in the interviews, and how much this change is self-portrayed: "[…]Today my emails are much more objective and assertive (E02)"; "[…]My communication with all the departments is in order, thanks to this system (E10)"; "[…]I do not spend days going through endless Excel tables anymore(E07)".

The introduction of an algorithm in an organization leads to a process of translation of interests and objectives (Latour, 2001). Different actors (humans such as developers, managers, users, and non-humans such as the algorithms themselves, databases, and interfaces) have their interests aligned, diverted, or transformed according to organizational contingencies and/or individual specificities. An algorithm implemented to increase efficiency can, for example, translate the objective to the standardization of processes – as portrayed by E07, or even create new unexpected objectives, such as generating new types of data and insights (Ferrar & Green, 2021). This theoretical conception was described by E01 when he states: "[…]I witnessed (with the adoption of PULSE) a new conception of analysis being built and disseminated in the organization."

This dynamic creates complex socio-technical networks in which humans and algorithms are inextricably linked. The stability of these networks depends on the continuous work of alignment and the performance of their components. An algorithm does not work in isolation; It depends on infrastructure, quality data, maintenance, and human users' cooperation (or resistance). Thus, algorithmic failures often reveal the complexity and fragility of these connection networks, and these failures can occur anywhere in that network(Yao, 2023). At BPI, this integration and complexity can be described in the fieldwork of each employee. In

this network, there is a large relational apparatus between humans and non-humans, technological and non-technological, which is made available and integrated – by the industry – to enable a specific objective – the medical visit: the cell phone, the tablet, the car, the cards (credit, food, fuel, health), the fleet dealership network, the drug distribution network, the pharmacies and their visit logistics (today managed by Waze®), the internal operating system that records your visits, as well as the results of it, the telephone network that makes your connections available in real-time. As well as the human-human interactions routinely performed, employee-secretary, employee-doctor (client), pharmaceutical employee, employee-employee (inter and intra organizational), and immediate employee-superior, among many other connections that can somehow influence some "algorithmic failure".

Therefore, Mendonça et al. (2023) argue that, as institutions, algorithms establish interaction contexts, create development paths, and induce or restrict certain behaviors with collective consequences, such as online development and training plans. This view is aligned with that of Berger and Luckmann, as well as Latour, where algorithms, by mediating action, help to build and stabilize – or destabilize – the organizational social order. Thus, when considering algorithms as organizational institutions, we realize that they are fundamentally hybrid: they are social constructions that objectify themselves and gain normative power, and simultaneously, they are non-human actors that mediate and transform interactions and collectives.

3.2.2.2 Validation and Algorithmic Value: Shaping Everyday Processes

As argued, the algorithms that underpin PA projects can be understood not only as technical instruments but as new organizational institutions in formation (Mendonça et al., 2023; Berger & Luckmann, 1966) and as active mediators in the sociotechnical network (Latour, 2005). For this algorithmic mediation to be consolidated and effectively change the daily processes of HRM at the BPI – and with a focus on the declared Specific Objective 02 (OE02) – it needs to go through rigorous (or not so rigorous) validation processes and have its value recognized by the actors involved.

The validation of an algorithmic innovation in HRM can be understood as a translation process (Callon, 1986), specifically at the moments of enrollment and mobilization. It is necessary to convince the actors (humans and, sometimes, other non-human systems) that the algorithm works, delivers what it promises, and consequently, that its results are reliable(Grodal et al., 2024). The interviews reveal different strategies for this validation. A common approach is to carry out pilots and gradual tests, as described by E10 when recalling

the implementation of SAP: "[...] There was a lot of optimizations. There was a lot of adjustment; they made several pilots this story, right, but you know that one thing that the pharmacists still taught us [...] was the issue of the pilot, right? [...] Here we had a very gradual process".

This incremental approach allows not only to test the technical functionality but also facilitates the internalization of the new logic by users, managing resistances and allowing adjustments in search of legitimation (Berger & Luckmann, 1966). Interviewee E4 also mentions this practice when comparing it to Stefanini[2]: "[...] they make pilots, they do sprints [...] and then a process gains strength in there, and people even talk about it and, you know, the value comes, now we move on to the next step". These pilots function as tests of strength where the algorithmic network demonstrates its ability to perform and resist controversies, as well as to create new meanings by mitigating crises (Berger & Luckmann, 1995; Latour, 1988),

The validation and value perception dynamics of PA tools can also be understood through Everett Rogers' (2003) theory of Diffusion of Innovations. Rogers argues that the adoption rate of innovation (in this case, the PA practices and algorithms) is significantly influenced by attributes perceived by potential adopters within a social system (the organization). Demonstrating relative advantage – the degree to which PA is perceived as a better source of analysis of past practices, as evidenced by the pursuit of ROI (E07) or efficiency (E03) – is crucial. Compatibility with existing organizational values, needs, and experiences is also vital; the cultural resistance mentioned by (E08) or the difficulty of HR in dealing with numbers (E06) points to compatibility challenges with existing data. The perceived complexity of the tools (E05) can slow adoption, while the possibility of experimentation (Trialability), as in the pilots mentioned by E10 and E04, can facilitate it. Finally, the observability of positive results and seeing the value that the tool brings accelerates the process of spreading innovation. Therefore, the AP's ability to effectively alter the day-to-day processes of HRM is directly linked to how these perceptions about their attributes are constructed and communicated within the organization, influencing the individual and collective decision to adopt (or reject) algorithmic innovation(Rogers, 2003). The existence of different adopter profiles (innovators, early majority, etc.) within HRM itself

[2] Technology company with more than 32,000 employees globally

and the organization also explains the uneven speed with which these procedural changes occur.

Once PA is validated (even partially) and perceived as a value generator (either by its relative advantage, perceived compatibility, or observable results), the PA algorithm and tool begin to impact and change the day-to-day processes of HRM. Perceived value may lie in operational efficiency, agility, quality of information, or the ability to generate new insights. Thus, the urgent need for strategic alignment of HRM with the organization's central objectives is reiterated (Piasna, 2024). Efficiency is a value often associated with automating repetitive or labor-intensive processes. E02's experience with digital admission is clear: “[...] We have already gained [time]. If each admission I spent two hours to make the input, now I spend 30 minutes to review”, a fact portrayed in detail by Laszlo Bock (2015), in his book Work Rules!, portraying the pressure for agility and the need to scale Google's® recruitment and selection processes at a time of exponential growth of the organization, at the same time, the need for hiring to follow a standard of high professional qualification. Innovation that is benchmarking to this day.

Similarly, E03 values the HR Dashboard because it frees it from operational tasks: “[...] It will free me up time to do more strategic things. Because there's nothing strategic about me generating many reports [...] The result is strategic, but the process itself is quite operational”. In this sense, the algorithm acts as an actant that takes over tasks previously performed by humans, changing the workflow and, potentially, the very content of the HR professional's work, who can dedicate himself to activities of greater complexity or added value. This process leads to the habitualization of new routines that incorporate the tool and foster the analytical mindset, mitigating the functionalist bias of HRM's daily life (Latour, 2001,2005; Berger & Luckmann, 1966, 1995).

The agility and quality of information for decision-making are other value vectors that modify processes (Falletta & Combs, 2021; Levenson, 2018). The ability to quickly access HR KPIs (E03), obtain turnover analyses for strategic conversations (E02, E06), or use generative AI tools to write communications (E03, E08) fundamentally changes the dynamics of daily task management. The information “objectified” by the algorithm starts to circulate more quickly, allowing for more agile responses and informed decisions. As interviewee E02 argues, having data to present to the board prevents the discussion from being based only on perceptions: “[...] We needed to take away a little of the perception that we were losing [talent] [...] the reality when they saw it is, okay, there were exits, but they are exits that happen [...]

there is no target company". This legitimization of data-based decisions not only validates the Vizier® tool but changes the board's decision-making process, which starts to require and value quantitative HR analysis before endorsing actions, modifying the preparation routine and the format of strategic meetings. The algorithm, here, mediates (increasing knowledge and the value of information) in the relationship between raw data and managerial decision-making, transforming the data into validated information that reorients action (Hassenstein & Vanella, 2022; Latour, 2001; Strong et al., 1997).

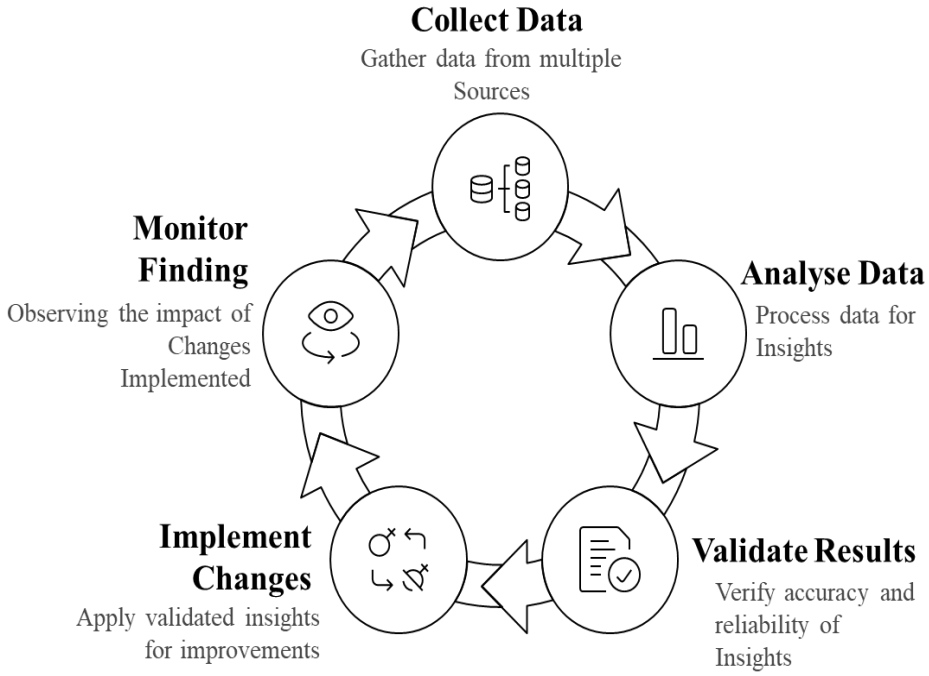
The algorithmic ability to generate new insights and support the resolution of complex problems is perhaps the most strategic value to be sought – although still incipient in most reports at a macro level – and has been consistently "experienced" at the micro level. Such individual experiments are portrayed in the wide use of Chatbots – primarily Copilot® due to the organizational regulation of Windows® systems – a fact that does not limit the use of other options, such as Open AI'® s ChatGPT, which more intuitively and visually, attributes considerable value concerning time management and relationships exercised and managed by one of Copilot's® algorithms (E10), that, in his opinion, these visual insights provide him with greater assertiveness and efficiency. Algorithmic insights can evolve into macro-level projects such as E10 itself, which aims to correlate hiring profile and their result, that is, use algorithmic analysis to validate or refute hypotheses about the success profile and, with this, improve the recruitment and selection process already based on an improved profile and more aligned with the organization's talents (Sharma, 2023).

The combined analysis of quantitative and qualitative data for a better understanding of the data from a climate survey. For example, the causes of the low score about the benefits offered by the organization can be systematically crossed with other variables in a practical, fast, and consistent way and produce insights that go beyond the superficial description, as portrayed by E01. When the PA can offer this level of insight, it changes not only the decision-making process but the very understanding that the organization has about its employees and its internal dynamics, leading to potential changes in policies, people management strategies, and management or leadership style (Zieglmeier et al., 2022).

However, these complex processes are not without challenges and contingencies, limiting changes and understanding in everyday processes. The systems' fragmentation (E01, E02, E05) makes it difficult to obtain reliable and integrated data, undermining the validation of algorithmic outputs. Low analytical maturity, or the habit of collecting and analyzing quality data (E08, E01), can also lead to distrust or misinterpretation of the expected results,

preventing them from translating into factual and/or applicable changes. Lack of resources or priority (E01, E07) can prevent acquiring or developing more sophisticated tools. And, the need for human oversight and interpretation remains. As E06 warns, one cannot simply "leave the process in the hands of AI"; it is necessary to "analyze on top of AI." Interviewee E9 compliments, pointing out the risks of misinformation generated by AI, especially in legislation, reinforcing that human validation ("feeling" of the context) is still essential. Figure 19 summarizes the data validation cycle.

Figure 19: Data Validation Cycle



Fonte: Pesquisa (2025)

In short, the analysis of the daily changes in HRM processes reveals that the application of PA tools is mediated by the way they are validated, the value and meaning attributed to them, and the solidity in the co-construction process. Therefore, algorithms, as emerging institutions and mediators in the socio-technical network – as an organizational actor – only effectively transform routines, strategies, and decisions in HRM when their performance is considered reliable and value-generating (whether in efficiency, agility, or insight). This process of validation and valorization is itself a social construct and new meanings (Berger & Luckmann, 1995); it depends on hypothesis, prototyping, demonstrations that produce measurable results (ROI), communication, and the development of analytical maturity while reflecting the capacity of the actor-network (Latour, 2001) to stabilize itself and perform convincing and sustained results. The changes in everyday processes, therefore, are not a direct result of technology but of the complex interaction between the agency of

algorithms and the continuous negotiation of their legitimacy and value in the organizational context of the BPI, which is primarily in human capital.

3.2.3. Algorithmic Mediation in HRM – Co-constructed legitimation.

The argument developed so far, both in the Reality Translation (section 3.1) and in the search for the Required Objectivity (section 3.2), consistently points to an understanding that PA and AI (which in this case are two sides of a single coin), transcend their nature as mere passive tools or instruments, to non-human constructions with meaning and agency for the social transformation of reality. As already addressed and referenced, PA projects operate as organizational actors that actively shape and are shaped by the social reality of HRM.

To deepen this understanding and argumentative consistency, it is essential to explore the concept of mediation as proposed by Bruno Latour (2001; 2005), which offers a powerful lens to analyze how the association between humans and non-humans (in this case, HR professionals and AP algorithms/systems), which transforms the action and the very constitution of the actors involved, and, consequently, the organizational social reality. This analysis of algorithmic mediation will serve as a bridge for the subsequent discussion on the nature and implications of AP, as a support for a new era of hybrid relationships – human-machine, as an organizational strategy, beyond industry 4.0 (Bahuguna et al., 2024; Dreković et al., 2023; Mahdia, 2024).

Latour (2001, pp. 207-249), in "*Pandora's Hope*", uses the metaphor of the relationship between a citizen and a weapon to illustrate that non-humans are not simple intermediaries who carry human intention without altering it, but rather mediators who actively participate in the definition and execution of the action. He distinguishes four ways in which this mediation occurs, modifying the original "program of action" of each entity. Applying this typology to contemporary HRM, I argue how PA algorithms mediate people management, theorized in this section, seeking to detail the four ways of mediation: (1)Translation – Deviation from goals; (2)Composition – Construction of new goals; (3)Reversible Delegation – Distribution of Competences; finally the (4)Modification of attributes/Crossing – transformation of the actors.

3.2.3.1 Mediation 1 - Deviation from goals

In the first way, Translation – Deviation from goals, the non-human actor – specifically, the PA algorithm – translates the objectives or the initial “action program” of the human actor (the HR professional, the manager) into something different, or at least, more

specific and operationalized within the logic of the algorithm itself. In the daily life of HRM, the Vizier® system, for example, interprets its user's requests (human goal) according to its own parameters. Latour (2001) describes this as a "deviation" or a "derivation" from the original goal: "[…]You define the goal, but I [the non-human] translate it[…]" (Latour 2001, p. 216). This deviation or derivation of the goal does not necessarily imply a betrayal of the goal but an inevitable reconfiguration when the action is delegated or shared with a non-human. An HR professional may have the broad goal of "[…]improving team engagement (E01)". By using an PA platform that analyzes data from a new survey collected from this perspective, in this case the PULSE project is translated into identifying patterns or monitoring the frequency and relationships between variables, providing feedback in the form of actionable insights. The intention to "*improve engagement*" is thus converted into a series of actions and metrics that the platform can process and present, objectively, so that they can be discussed, analyzed, and implemented (Tyagi & Pandita, 2022).

Recent literature on the implementation of AI in organizations corroborates this notion of translation. (Kellogg, Valentine and Christin (2020), for example, when studying the introduction of AI in professional contexts, observe how practitioners often need to adjust their own goals and workflows to align with the capabilities and outputs of algorithmic systems, a process they call "*mutual adjustment*". This adjustment is not just a human adaptation to the machine but a translation of the original intention into a format that the human algorithm network can perform. The report of interviewee E06 about the turnover analysis illustrates this translation well the initial concern with the "departure of people" is translated by the need to "understand the cause"; "[…] through indicators to be able to guide you, which in my case when I joined here, there was no guide for this evaluation".

The algorithm (or the analytical system) does not directly answer the broad concern. However, it translates the problem in terms of the variables, correlations, and indicators it is programmed to process. As E01 points out, when using data from Great Place to Work (GPTW) to understand the perception of benefits, the initial data: "[…]we have the worst GPTW indicator of benefits[…]" (E01), needs to be translated into deeper investigation actions; "[…]we do focus groups to go to the qualitative and try to understand why? (E01)". Here, the PA tool (GPTW) translates the subjective experience about benefits into a quantitative indicator, diverting the manager's action to a qualitative investigation focused on this specific indicator. The action program "improving benefits" is translated into "investigating the low score on indicator X or Y."

Another example that we can use in this process of translation and derivation that the non-human produces from the original goal is represented by interviewee E02 when he mentions the concern of senior management about the; “[…]we loss of talents[…]” of the organization to a specific competitor. In reality, the core of this statement is – Are we developing talents for our competitors because we are not managing to keep them? Are we, consequently, about to compete with our strategies – carried out – by these dissidents? This apparent “deviation” allows E02 to interact with the non-human (Vizier®), deepen the knowledge of this process – supported by historicity – and clarify to the board that such propositions have been refuted.

This translation of goals – a process of systematic and quasi-scientific research – can have significant consequences. It can lead to greater specificity and operationalization of previously vague objectives, as described in E01, E02, and E06. On the other hand, as Ziewitz (2015) warn in their analyses of algorithmic governance and “algorithmic morality”, there is a risk that the nuances and unquantifiable aspects of the original goal will be lost or distorted in the process of translating into what is algorithmically measurable. E09's concern about AI “[…]not having 'feeling,' or context[…]” when analyzing resumes reflects this potential deviation: the goal of “finding the best candidate” can be translated by the algorithm into “finding the candidate with the most matching keywords”, potentially ignoring soft skills or valuable atypical experiences. Therefore, it is necessary to reaffirm that the “translation” performed by the PA algorithm is not a neutral process; it actively rewrites the program of action, and HRM professionals need to be aware of this mediation to ensure that the translated goals still serve the broader organizational and ethical purposes, maintaining, as Latour (2001) suggests, the possibility of “adding a second translation to the first” (p. 217); through critical interpretation and human complementation.

3.2.3.2 Mediation 2 – Composition

The second way of mediation described by Latour (2001) is composition, where the association between the human and the non-human results in creating a new goal, a new goal that did not previously exist either for the isolated human or for the non-human itself. “[…]You aim at a goal through me [the non-human], but I end up aiming at another goal, which you endorse” (Latour, 2001, p. 217). This pathway is particularly relevant to understanding how PA can be more than an optimization tool and become a driver of strategic innovation in HRM. Interacting with algorithms' analytical and predictive capabilities can lead

HR professionals to formulate and pursue goals that would previously have been unimaginable or considered outside their traditional scope.

From the perspective of HRM at the BPI, the ability of PA systems to process large volumes of data and identify subtle patterns can lead to the composition of new strategic goals for talent management, employee well-being or organizational effectiveness, internal mobility, personalization of HRM policies and practices in a broad and unrestricted way. For example, the simple goal of "reducing turnover" can be transformed, through interaction with a predictive algorithm, into the more complex and proactive goal of identifying and intervening in micro-segments of employees at high risk of leaving before they express an intention to leave, using personalized development or engagement recommendations(Zander & Zieglmeier, 2023).

This last goal only becomes feasible and attainable due to the composition of HR's intentions (retaining talent) with the algorithm's capabilities (predictive analytics and segmentation). E01's experience with the PULSE tool, which could: Indicate "[...] risks of losing talent, we could have a conversation, something like that[…]", he points to this composition: the tool not only has the power to mediate the analysis of the level of engagement (translation) but also enables the creation of a new proactive goal of early intervention. The literature on algorithmic management and the future of work echoes this idea of compounding new goals. When discussing working with AI, authors such as Faraj, Pachidi, and Sayegh (2018) suggest that human-AI collaboration can lead to new forms of value creation and the redefinition of tasks and professional goals. AI not only performs tasks but can, through its "discoveries" in data, suggest new directions or problems (Faraj et al., 2018). E10 uses Viva insights to analyze their interaction patterns – "[...]He shows here my network that I'm [sic] and the least frequent [...] Habits and meeting how cool right[...]? This is artificial intelligence helps you a lot" – illustrates how the tool can lead to the composition of personal goals for time management and collaboration that might not have been prioritized or even identified without the mediation of the system. The tool composes, with Irineu, the goal of "optimizing meeting habits and expanding the network of less frequent contacts, "as well as helping him to analyze his efficiency and focus visually and in real-time" (Begkos et al., 2024).

However, this composition of new targets is not an automatic or tension-free process. It requires HR professionals to be open to internalizing the possibilities offered by algorithms and to integrate these new goals into their professional symbolic universe (Berger & Luckmann, 1966). As E04 points out, when discussing the potential of the PA to generate

insights, HR needs to use its “filter” and help the organization understand the new paths proposed. E08's concern with ethics in the use of AI and the need for humans not to “copy and paste” what AI suggests also indicates that the composition of new goals is a negotiated process, where human agency defines whether the new goals “composed” with the algorithm are desirable and aligned with organizational values. Thus, mediation by composition not only expands the repertoire of HRM goals but also imposes new challenges of governance and ethical discernment in defining the “new reality” of HRM. (Tursunbayeva et al., 2018, 2021).

3.2.3.3 Mediation 3 – Reversible Delegation

The third way of mediation identified by Latour (2001) is reversible delegation, where humans delegate specific actions, competencies, or “action programs” to non-humans who execute them. However, this delegation is not a passive and definitive transfer; it is "reversible" or, more precisely, constantly negotiable and implies a transformation in how competence is exercised. “[…]I transmit to you the program of action A”, says the human to the non-human, “[…]but I transform it into program B, which you can then ignore or follow” (Latour, 2001, p. 218). As for the integration of PA into HRM, this is manifested when tasks previously performed by HR professionals are delegated to algorithms and systems, which then perform them in specific ways, distributing and reconfiguring competencies in the network.

A clear example of this delegation is the initial screening of resumes by Applicant Tracking Systems (ATS) systems or AI algorithms, as mentioned by E02 and E05. The competence to “identify qualified candidates” is partially delegated to the algorithm (Bock, 2015). The system performs this task based on pre-defined criteria (keywords, experiences, etc.), changing the action program that would previously involve the manual reading of hundreds of resumes (E07). HR delegates this process phase, hoping the algorithm will select efficiently. This delegation is also visible in the automation of transactional processes, such as the digital admission described by E02 and E08, where the system takes over the collection and processing of documents, or in the use of a chatbot to answer frequently asked questions from employees, delegating part of the communication and HR support.

The literature on automation and the future of work has extensively explored this delegation of competencies to AI systems (Zirar et al., 2023). However, the Latourian perspective reminds us that this delegation is not a simple substitution but a transformation of competence and of action itself. The algorithm does not “do the same” as the human but does it differently, with different biases, criteria, and capabilities. As E04 points out, even with

automation, it is necessary: “[…]using our filter, right? Helping the organization to understand that not everything we can count on artificial intelligence[…]”. This points to “reversibility” or, instead, the ongoing negotiation of delegation, so HR professionals need to constantly monitor, adjust, and sometimes intervene or “reverse” delegation when algorithmic results are unsatisfactory or ethically questionable (Mittelstadt et al., 2016; Tursunbayeva et al., 2021).

In addition, the delegation of tasks to algorithms often requires the development of new skills on the part of humans, such as the ability to formulate the right questions – hypothecation – for systems, interpret algorithmic outputs, identify biases, and manage human-machine interaction (Devi, 2023). E01's perception that it is necessary: “[...] think about a different person's profile to be able to dream [with PA]” or the emphasis of E06 on the need for HR to “[...]learning from the business” and “[…]knowing how to ask questions” reflect this transformation of skills that accompanies delegation. Thus, reversible delegation is not just a workload shift but a dynamic process that reconfigures the division of cognitive and practical labor between humans and algorithms in HRM, leading to the institutionalization of new forms of socio-technical collaboration. The objectivity that emerges from this delegation is, therefore, a negotiated objectivity, where the possibility of human intervention and reconfiguration always conditions the apparent autonomy of the algorithm. (Suchman, 2007).

3.2.3.4 Mediation 4 – Modification of Attributes

The fourth and perhaps most profound path of mediation proposed by Latour (2001) is the modification of attributes or the crossing (pliage, foldable), where the association between humans and non-humans results in the mutual transformation of the actors involved. “[...]You are different because you hold the gun; the gun is different because you hold it” (Latour, 2001, p. 219). From this perspective, neither the human nor the non-human – in this case, the PA algorithm, as non-human – remain the same after the interaction; both acquire new properties, roles, competencies, and even identities. This mutual transformation is fundamental to understanding how the introduction of PA changes tasks and resignifies the HRM actors themselves and the nature of people management, directly impacting everyday processes.

This daily routine of the HRM professional at the BPI, who interacts with PA systems, is no longer the same. This professional now becomes an HR professional with analytical skills, regardless of their degree of development (E06, E08, E09), data-driven HR professional (E01, E04, E06) or, as some interviewees want, a strategic business partner based on evidence,

as stated by E01: “[...] But 43% is already representative, because what we needed, we could already have a small ‘pulse’ (a term used in allusion to the PULSE project) then, we wanted a little more[…]”. This transformation is evidenced by the search for new competencies, such as the analytical mentality mentioned by E01 and E09, or the need to know how to “ask the right questions” for systems and critically interpret their outputs, as highlighted by E02: “[...]I need to stop and think, first, what data is important? What data do I want to show?”. The professional experience of E06, who migrated from technical areas to HR and implemented a panel of indicators, demonstrates a professional whose profile was modified by the internalization of analytical logic – her experience as a production manager, enabling her: “[...]discuss with the manager, on an equal footing about the data”. The very act of using tools such as Copilot® to summarize texts (E03, E08) or analyze interaction patterns (E10) modifies the manager's attributes, who gain new information processing and relational self-knowledge skills (Huang et al., 2016).

At the same time, the PA is also transformed. When implemented in a specific BPI organization, a generic algorithm, fed with its particular data and adjusted to its needs and culture, it becomes a “PA algorithm of company-X or Y”, as recommended (Rasmussen et al., 2024). It acquires a history a content of use, and its outputs start to carry specific meanings within that organizational reality. Interviewee E06's concern with the management of the General Data Protection Law (GDPL) and the need to build a proper chatbot to avoid the use of open networks with confidential data illustrates how the non-human – the algorithm – is modified and resocialized to fit the legal and ethical context of the organization. The underutilization of robust systems, such as Workday®, Vizier®, and even SuccessFactorsSAP®, tends to be a limiting factor to growth and greater analytical knowledge, as described by E01: “[...]We use 10, 15, 20% of what it can deliver”, E05 reinforces this premise. Therefore, we can infer that a non-human has attributes and potential limited or redefined by the human network that interacts with them – or ceases to interact fully. (McCarthy et al., 2022).

This mutual transformation of attributes is central to understanding the everyday changes in HRM processes (Belizón & Kieran, 2022). Processes previously based on purely human interactions or manual analysis are now co-performed by a human-algorithm association, which has new capabilities and, consequently, allows (or requires) new routines(Belizón & Kieran, 2022). The literature on sociomateriality in organizations emphasizes precisely this constitutive intertwining between the social and the material, where

actors and artifacts co-produce each other (Orlikowski, 2007; Scott & Orlikowski, 2025). The adoption of PA is not a simple application, adding another technology to preexisting processes; it is the emergence of new hybrid processes(Tenakwah & Watson, 2024). For example, the performance appraisal process, previously a subjective annual ritual, can be transformed into a continuous flow of feedback and data analysis – as desired by some, although not yet fully realized in the interviews (E05), where the manager, the employee, and the algorithmic system change and act together. The manager gains the ability to have a more "objective" (or at least different) view of performance, the employee receives more frequent inputs, and the algorithm "learns" from the data generated.

This crossing of attributes, however, also generates tensions and ambiguities. The concern with dehumanization or objectification (E07, E10) reflects the fear that, in this mutual transformation, valued human characteristics – such as empathy contextual judgment – will be suppressed or devalued in favor of purely algorithmic logic, a concern evident in the literature on datafication, which is a term coined by Viktor Mayer-Schönberger and Kenneth Cukier in 2013 and can be defined as the transformation of social action or real-life aspects into quantified data, allowing the analysis and use of this data to inform, characterize, and streamline decision-making processes (Thorpe & Sellar, 2023). Datafication is a growing and continuous process of transformation, and it has affected organizations, governments, individuals, and society (Ayokanmbi, 2021; Diab, 2017; Lavazza & Farina, 2023; Posada et al., 2021). The constant need for human validation (feeling, context), as highlighted by (E09), or to "[…]analyze on top of AI[…]" (E06), indicates that the transformation of attributes does not eliminate human agency, but repositions it, re-signifies it, as essential for the interpretation and ethical direction of algorithmic mediation(Bar-Gil et al., 2024; Bist et al., 2024). Therefore, the new reality of HRM is a reality co-constructed by the performance of this network of mutually transformed humans and non-humans, where every day processes become the stage for this continuous reconfiguration.

By mediating action in and from HRM through these four pathways, PA algorithms are not mere neutral instruments. They actively participate in problem definition, goal creation, distribution, and competence generation – data literacy and the actors' transformation (Khan, 2024; McKinney & Shaffer, 2023). This algorithmic mediation is essential to understanding why the PA can be considered a strategic organizational actor today because it optimizes existing processes and reconfigures the very field of possibilities of people management, introducing new ways of seeing, knowing, and acting.

The Objectivity and Rationality that the PA promises – addressed in theme 2 – is, in itself, a product of this mediation, a reality objectified through the interaction between the analytical capacities of the algorithm and the interpretations and validations of human actors (Berger & Luckmann, 1966, 1995; Latour, 2001, 2005). This understanding of algorithmic mediation prepares us to explore, in the following theme (3.3), the broader implications of considering the PA as a full-fledged organizational actor with its agency and impact on the construction of the social reality of contemporary HRM.

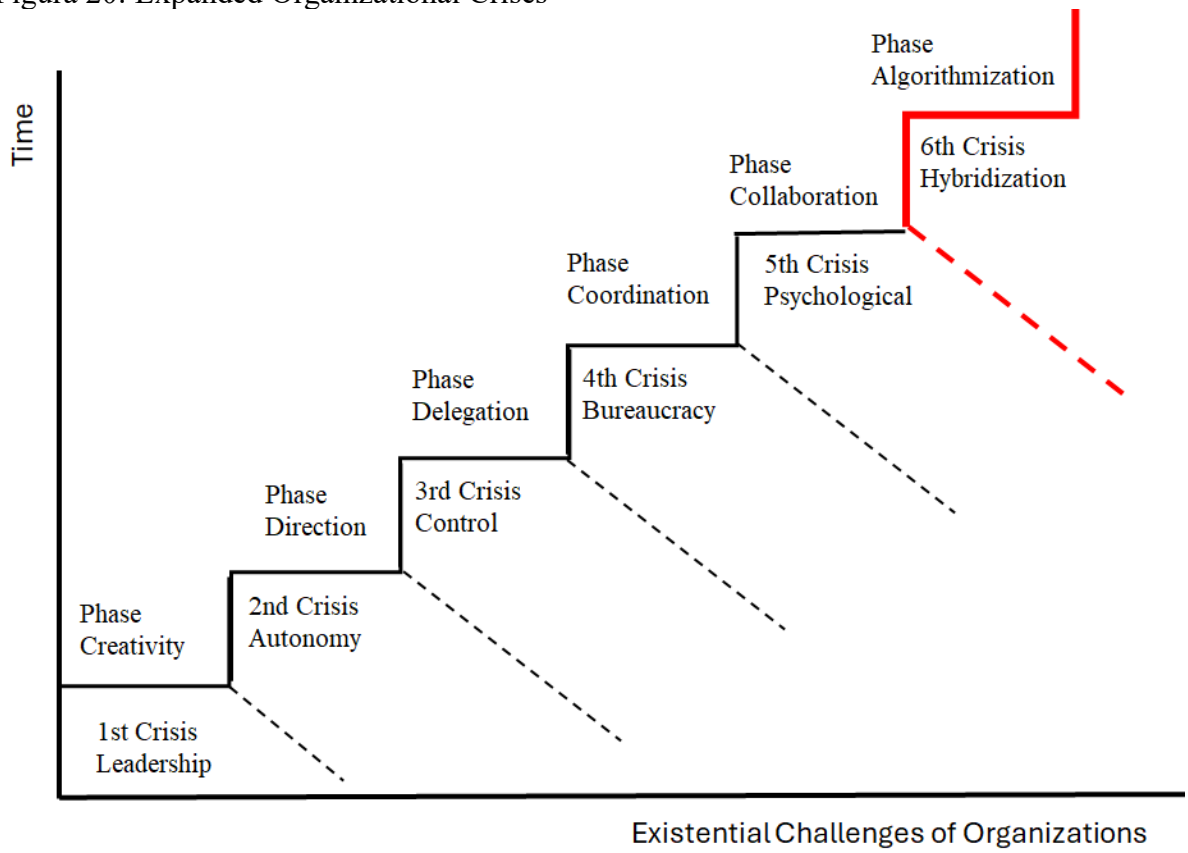
3.3 Hybridization as a Management Strategy in Everyday Relationships

The immersion of organizations in the digital age culminates in a scenario where Hybridization – the intricate coexistence and interdependence between human and non-human actors (algorithms, AI) – ceases to be a future possibility to become an everyday reality and an inescapable management strategy. To understand the depth and implications of this new configuration, it is instructive to turn to organizational development models, such as the one proposed by Larry Greiner (1998), which maps growth through phases of evolution punctuated by revolutionary crises. However, the magnitude of the current algorithmic transformation raises questions about the continued applicability of Greiner's classic model to explain contemporary organizational development. Against this backdrop, this study explores the plausibility of a theoretical extension to his model. We therefore propose the emergence of a sixth evolutionary phase, which we term Algorithmization. This phase is characterized by the infusion of AI and Machine Learning into core organizational processes and the reconfiguration of management logic around data analysis and automated decision-making (Davenport & Mittal, 2023).

Following Greiner's dialectical logic, wherein each evolutionary period gradually shows its specific revolutionary crisis, the Algorithmization phase may be fostering a new set of systemic tensions. As an analytical construct for this research, the apex of these tensions can be understood as the sixth major organizational crisis: the Hybridization Crisis. The contours of this possible crisis would be defined not merely by the realignment of work models (remote, in-person, or hybrid) but by deeper, more fundamental challenges. These include the negotiation of boundaries between human cognition and machine execution, the growing challenges to autonomy, purpose, and subjectivity in the workplace, and the pressing need to develop new forms of algorithmic and ethical governance to mediate these interactions (Manuti & de Palma, 2023).

Figure 20 illustrates this extended conceptual model. It is uppercase to emphasize that it is presented here not as a declaration of a new universal stage but as a theoretical lens we have adopted to guide our investigation, comprehension and deepen the analysis of the complex transformations that modern HRM is embracing.

Figura 20: Expanded Organizational Crises



Fonte: Elaborated and expanded by Greiner (1998).

Thus, the introduction and progressive consolidation of PA projects – even if still in early stages – as well as the use of AI tools in HRM – specifically in the BPI, as analyzed in the previous themes Reality Translation and Required Objectivity, of this analysis, reinforces the thesis that PA and AI are new organizational actors; therefore, they transcend the conception of a mere implementation of new technological tools (Nawaz et al., 2024). They catalyze the emergence of a fundamentally hybrid work and management environment, while the boundaries between human agency and algorithmic agency become porous and interdependent (Ijanperä & Vuori, 2021; Thompson, 1967, 1976). The previous observation that the PA operates as an active organizational actor, mainly using its algorithmic mediation capacity (Latour, 2001), establishes the basis for discussing, in this section, the (im)possibilities of its adoption and the protagonism of human and non-human actors in this new era of algorithmic inevitability and increasing hybridization of management (SO03).

This hybrid environment, introduced by PA, is not only a space where humans use machines but takes shape from an ecosystem, an organizational fabric, in which human capabilities and algorithmic performances are intertwined, generating new forms of interaction and iteration, new challenges to the status quo, new structural demands and, new limits, risks and ethical dilemmas(Mahdia, 2024; Simbeck, 2019).

The analysis of the SCRT will help us to understand how this hybrid reality is socially constructed and negotiated by the actors, while the ANT will illuminate the dynamics of the socio-technical network that emerges. Additionally, the concepts of human-machine co-intelligence addressed by Boucher et al. (2023), Mollick (2024) and Yao (2023), and Wallace's (2021) recognition of the indissoluble nexus between human and artificial intelligence – which I call an era of algorithmic inevitability – will provide a contemporary vocabulary to discuss shared protagonism and the (im)possibilities inherent in this hybridization in HRM.

3.3.1 Interactions between Human and Non-Human Actors in Hybrid Daily Life

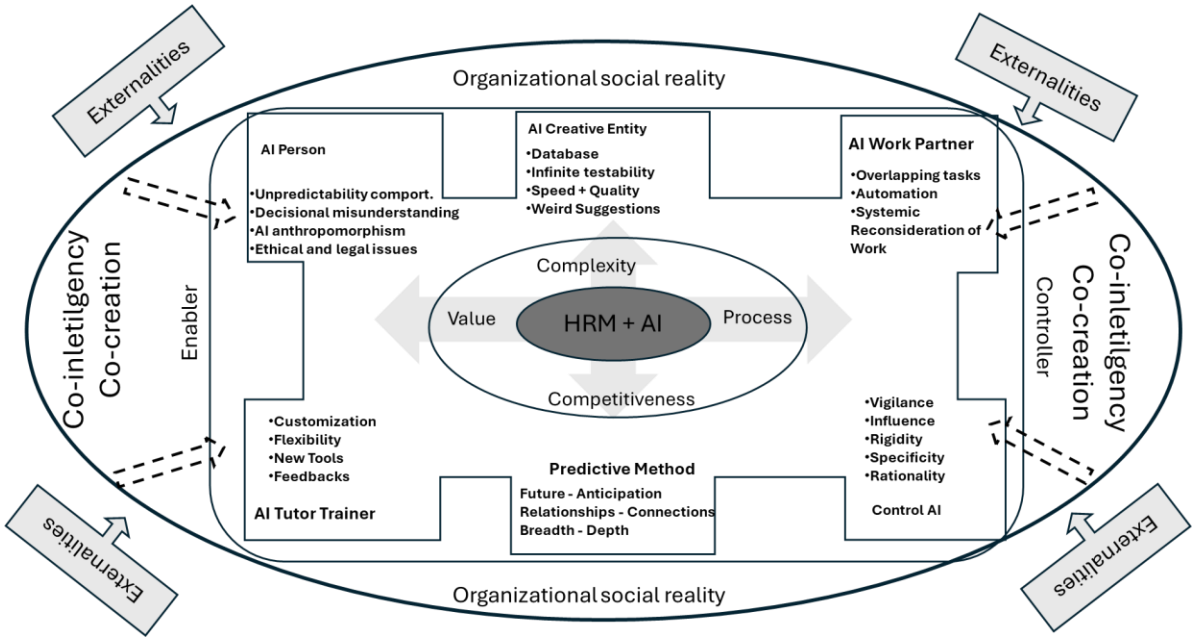
The insertion of the PA as an active actor reconfigures the daily interactions in HRM. A context that challenges individualistic patterns, requiring new forms of interactions and construction with co-responsibility and, at times, exacerbating or mitigating existing knowledge silos and their consequences (Hüllmann et al., 2021; Shrivastava et al., 2018).

The title of this section invokes the term hybrid, which can be understood – in this reflective lens as being: When analyzed through the lens of agency, action is distributed and shared; Under the lens of processes, they are increasingly co-performed. As for the interaction with human and machine intelligence, they are the result of a symbiosis between human capabilities and algorithms. Thus, the hybrid reality is co-constructed and mediated by other actors in a new vision. However, this process does not discard their dualities since, as Mollick (2024) points out, it depends intrinsically on them, which will be highlighted below. The complexity of these interactive dynamics and the dualities of hybridization are better elucidated when we consider the multiple roles that AI and, by extension, PA assume in HRM as an organizational actor. Such roles, which emerge both from the literature and from the experiences reported by the interviewees, range from an “AI work partner”, which aims at the automation and optimization of tasks – reflected in E02's search for efficiency in digital admission or in E01's desire for a system that solves operational issues – to an “AI as a tutor and coach”, able to offer personalization and new development tools, as targeted by E01, to segment training or experienced by E10, by using Copilot® Viva Insights to improve their

interaction management. Nevertheless, “AI as a person” is also manifested, with its perceived unpredictability and inherent ethical issues, raising concerns such as those of E06 about data security and the need for human curation or E09's caution about the accuracy of AI in legal contexts.

To visualize this multifaceted algorithmic performance and its implications for the new construction of organizational social reality, the framework represented by Figure 21, adapted from Mollick (2024) himself, is proposed. This schema not only categorizes these different roles of AI (AI Person, AI Ent. Creative, AI Work Partner, AI Tutor Trainer, AI Control) but also situates them in relation to the challenges of managing complexity, increasing competitiveness, generating value and optimizing broadly in the HRM process, and considers externalities and co-intelligence cycles Co-creation. The subsequent analysis of the dynamics of interaction in the hybrid daily life, therefore, will use this framework to deepen how PA, in its various manifestations and constant dialogue with human actors, influences collaboration, co-responsibility, and time management, thus responding to the (im)possibilities of its adoption (SO03).

Figure 21: HRM in an Algorithmic Context



Fonte: Adapted and expanded from Mollick (2024)

The model presented in Figure 21 highlights how different manifestations of AI – from “Person AI,” with its potential unpredictability and ethical-legal issues, to AI as a creative

entity and its sometimes weird suggestions, “AI as a work partner”, with its capacity for automation and overlapping of tasks, “AI as a tutor or trainer”, promoting personalization and new tools, even control AI, with its emphasis on vigilance and rationality – interact with the imperatives of suppressing complexity, increasing organizational capacity for competitiveness, generating real value, and optimizing the processes inherent to the HRM relationship through AI.

Surrounding this central interaction, the framework also points to the “Externalities”. Such externalities may be market requirements and/or opportunities and legal and regulatory limitations, which are pretty standard for the BPI. Also, the “Co-intelligence Co-creation” cycles can act as an enabler or controller (limiter), influencing the general dynamics of the new HRM. The subsequent analysis of the (im)possibilities of adopting PA in HRM and the protagonism of human and non-human actors (OE03) will benefit from this scheme to unfold how these different facets of PA/AI effectively reconfigure everyday relationships in the hybrid work environment. From this perspective, we examine how individualism and collectivism manifest themselves in the co-responsibility of data-mediated management.

3.3.1.1 Individualism vs Collectivism: Data-Mediated Management

The emergence of a hybrid organizational routine, increasingly mediated by PA/AI, puts traditional work dynamics in check, notably the tension between individualistic practices and the desired construction of an environment of collectivism and co-responsibility in management (Hüllmann et al., 2021). The different facets of AI, as outlined in the framework of Figure 21, act in different ways in this clash. On the one hand, AI can be perceived as a working partner, with its capacity for automation and optimization of processes, such as digital admission that frees up time for managers (E02; E08), could foster a greater focus on collaborative and strategic activities.

However, this same AI can reinforce individualism if the perception is that “everyone has their cockpit” of data without transparent governance that promotes sharing, as E01 points out when describing the underutilization of Workday and the creation of isolated dashboards. The statement of (E1), “[...] The day to day kills people [...] no one has '[sic],' everyone wants things ready here”, evidences a culture where individual overload makes it difficult to internalize a co-responsibility mediated by data, even if tools such as Great Place to Work (E01) or Workday (E01, E03, E05) aim to democratize access to information.

AI as a tutor or trainer, in turn, by offering personalization (E01), can both empower individuals for greater autonomy and, if there are no mechanisms to share the acquired knowledge, deepen silos of expertise. The promise that technology "[…]democratizes access to information and decision-making a little[…]" (E01), transferring responsibilities to managers, encounters barriers when "[…]HR does not have time to manage people" (E10) or when the culture persists that HR is the sole guardian of information (E01; E06). Thus, the transition to effectively co-responsible management, mediated by data, requires not only the availability of tools but in a translation of a "new capital"; economic, cultural, social and symbolic (Bourdieu, 2007), where the value of collective intelligence supplants the convenience of individual delegation of analytical tasks (Weng et al., 2018).

Based on the empirical content analyzed here, the organizational culture still reflects a confident individualism in management, represented by the hierarchical structure of the companies investigated (Hofstede, 2011). However, there is a concept – or desire in a certain amount – of building an environment conducive to co-responsibility or co-construction. The introduction of PA projects, which by nature tend to centralize and make information more accessible, creates a tension with this logic – in a movement of retraction and expansion(Schumpeter, 1934). Therefore, the promise of a more objective, horizontal, and co-constructed and data-based management tends to share responsibilities and power over each frame of the process that is in flux. To corroborate the previous assertions, I use the words of E01 when discussing the process of delegating small parts of the process, using tools such as Workday®

> "[...]Because sometimes I'd rather pick up the phone, or send an email to my business partner here next to me, 'Hey so-and-so, can you pull this thing for me?', than go there and extract the data myself. [...] The day to day kills people [...] no one has '[sic]', everyone wants things ready here (E01)".

This statement illustrates, in the first instance, that this Director portrays his work overload, represented by the delegation of a functional task, which, in his view, would only result in a loss of time. As his time is analyzed as very valuable, the delegation of this bureaucratic task to a person on the team – for him – is more consistent. In a second analysis, it can be inferred that – if, on the one hand, the Director – has an adequate staff in terms of

the amount of human capital, this human is being overloaded by the routine of both – manager and collaborator – while the non-human is being underutilized.

The second reflection on this statement is in the evident overlapping of tasks, in the exhaustion that daily life generates, and in a functionalist emphasis that commands human actions. This behavior is a feeling of wasting time in delegating repetitive tasks so that this habit, or agency, is not delegated. A paradox is created, where there is a complaint about: how busy and overloaded I am, in addition, I prefer to do it my way and speed, which culminates in pressure on the results to be obtained, which is my entire responsibility as a manager, and finally the need for the processes to work efficiently and effectively, however, the jumps or shortcuts described are contingency actions that mitigate the fluidity of the process as a whole, even when PA tools offer paths to a more autonomous and potentially shared analysis.

The effectiveness of co-responsibility depends, therefore, on the institutionalization of new practices, as well as a greater knowledge of the potentialities of their partners or peers – human or non-human – and that these relationships and interventions are carried out systematically, where data is seen as a common good and a resource for collective intelligence, and not as the property or responsibility of an individual or department (Arcidiacono & Sartori, 2024). The concept of "user-perceived value of collective intelligence" (VPUIC) by Weng et al. (2018) becomes relevant here if actors do not realize the social, functional, or problem-solving value of sharing and using data collectively with the use of an PA project, individualism will prevail. Inferences that are corroborated in the comment of interviewee E07 are as follows:

> “[...]In a way, it's a thrust for a movement. Yes, but I think that if the company doesn't move towards this [co-responsibility and co-creation], it will be far behind. And then we will always depend on what others are doing [maintenance of individualism]. Then I'll have to do it because I'm doing it. I'm going to have to change because I... And no one guarantees that we will be able to get where [sic] we need to go. We need to be more open to this.”

The PA platform, as an actor (ANT), tries to inscribe this co-responsibility, for example, through shared dashboards or interdepartmental goals based on standard indicators. However, its success depends on the enrollment of human actors in this new logic. Interviewee E06 mentions the attempt to implement shared goals in her organization but acknowledges: “[...] we are not yet at that degree of maturity here [...] but everything is fine”, indicating that

the transition to effective co-responsibility and a more prominent analytical mindset, is a gradual process of social construction.

3.3.1.2 Day-to-Day Management vs Strategic Analysis Tension

Day-to-day management, with its urgencies and operational demands, often consumes the time and energy that could be dedicated to a more strategic use of the PA or various analytical instruments. As E10 laments: "[...] HR doesn't have time. HR doesn't have time to manage people [...] we know the strategy, but the day-to-day consumes us, so we can't get much out of the operational". Argument corroborated by E02: "[...]so today I take care of the operations part, I take care of the compensation and benefits part, I am a business partner for the functional areas, I take care of IHS and fleets, so this is my reality, these five areas basically, so I say, it's a little monster [...]"; Or even, "[...] My day is without routine. If I'm in the office, like today, for example, it's quite common for the person to say this: Don't you have a minute? Can we have a coffee? So, as much as I am, I can't do it now because it's not even crazy today (E03)". In general, all interviewees have the same feeling that they are not enough to meet the demands received in their daily lives.

This tension between the operational and the strategic is a central challenge to the PA's protagonism. If HR professionals and managers are immersed, in putting out fires, and primarily contingent actions, the PA's ability to generate insights for long-term decisions is underutilized (Ferrar & Green, 2021; Greene, 2018). At this point, the concept of mediation (Latour, 2001) can aggravate and alleviate the problem since each of the four mediation processes – already discussed above- will exert a specific pressure, which will create or eliminate a greater volume of interactions and amplify their complexity. Tools that automate reporting or transactional processes, such as Vizier®. Workday®, among others, should be fully qualified to mitigate this daily pressure (reduce functional effort and expand strategic exercise), such as digital admission (E02, E08), can "free up time", according to E03's expectation: "[...]It will free me up time to do more strategic things".

However, suppose the interaction with the tool is complex. In that case, if the data requires a lot of manual preparation, as mentioned by E05, or if the outputs are not immediately actionable, the PA can be perceived as another burden in day-to-day management; again, I refer our reflection to the research carried out by E01 in the PULSE project, which did not obtain the expected response rates of the respondents, for being perceived as "one more thing in my daily life". Yao (2023) and Mollick (2024), in this sense, rely on a division of labor and or empower human and AI common role significantly

effectively, where the machine takes over the routinize work so that humans can focus on the strategic and interpretation (Yao, 2023; Mollick, 2024).

Thus, the conception and strengthening of integration and interdependence are objectives of legitimation. This ensures that the entire system of institutions makes sense to everyone involved. This is important because people participate in different functions and processes in a "world" with purposes and ways of being that must be aligned due to their multiplicity and complexity. For example, a person may be a father at home and a military commander at work. Legitimation helps them (individuals) and others (their relational) to see how these roles fit together, creating a coherent, productive, and long-lasting social order. This is called "horizontal" integration, in which different functions and processes are connected between individuals and situations. (Berger & Luckmann, 1966, 1995; Fraser & Turcan, 2025).

3.3.1.3 Silo and its consequences for hybridization

Organizational silos – whether departmental, systems, or knowledge containment – are significant barriers to realizing the potential of the PA and building a cohesive hybrid reality. As previously discussed, departmentalization or Organizational Frames was disseminated to reduce organizational complexity (Goffman, 2011). As a consequence of this process, there is a considerable effort to make connections, or greater interdependence between these departments, so that the organization works as a whole and at an acceptable pace.

In this sense, the effort to transcend the barriers inherent to departmentalization mentioned above is echoed in the propositions of Richard Daft (2021). In his analysis of organizational theory and design, Daft (2021) underlines the criticality of horizontal linkages as antidotes to the excessive differentiation that fosters silos.

> “[…]Organizations may have to experiment to find the correct degree of centralization or decentralization to meet their needs. For example, a study by William Ouchi found that three large school districts that shifted to a more flexible, decentralized structure, giving school principals more autonomy, responsibility, and control over resources, performed better and more efficiently than large districts that were highly centralized. Walmart is experimenting with greater decentralization in some stores by having fewer managers and giving

> more decision-making authority to people on the sales floor. The company is testing several versions of the concept, called “Great Workplace,” in around 100 Walmart Neighborhood Markets and smaller supercenters. Even Japanese companies such as Toyota, which have a strong tradition of centralization, are seeing the power of decentralization for promoting a sense of ownership, as described in the following In Practice example.” (Daft, 2021, p. 98)

This fragmentation becomes particularly dysfunctional in the emerging era of work hybridization, where effective collaboration between distributed teams and synergistic integration between human and non-human intelligence – such as People Analytics (PA) systems – are vital. The absence of such mechanisms, or their ineffectiveness, compromises the organization's ability to process information in an agile way and make informed decisions, which, according to), hinders relational coordination and the construction of cohesive organizational knowledge. This scenario of structural and informational disconnection, as we will see below, manifests itself concretely in the fragmentation of systems, a direct obstacle to fully realizing the analytical and strategic potential of People Analytics.

The fragmentation of systems, where “[…]each area has its system” (E05) or where there are “[…]multiple systems that do not talk to each other” (E01, E02, E06), prevents the integrated vision that an PA project promises and hinders the translation of data into shared strategic knowledge (Latour, 2005). Interviewee E02 is explicit about the negative impact of fragmentation: “[...]So if I have two distinct bases and I want to do some analysis, I'll have to do some manual work. And then we lose agility; we lose confidence in the data because manual data we know that there may be something wrong there along the way”. These informational silos, not only technical but also social and cognitive, represent different provinces of meaning as promulgated by Schultz (1973), hinder communication and collaboration. Both the process of departmentalization – a concept widely and solidly disseminated and applied in organizational management (Mintzberg, 1979), as well as the concept of PA, which in contemporary times, trying to impose a unified analytical framework (Goffman, 1986), confronts these departmental fragmentations, in search of integration and interdepartmental of the “new HRM”

Communication, discussed in section (3.2.1.3), becomes pivot for negotiating these different frames and constructing an intersubjectivity around the data (Goffman, 1986; Berger & Luckmann, 1966). However, as the flowchart (“PA Flow: Analytical Frame”) points out, if

communication fails or the perception of top management is negative, the result can be the reinforcement of silos. The PA can, paradoxically, create new silos of data specialists if there is no conscious effort to disseminate the analytical culture and tools in a broad and integrated way, promoting what Wallace (2021) calls the indissoluble nexus between human and artificial intelligence throughout the organization, and not just in isolated pockets. Overcoming silos requires, therefore, not only technological integration but reconfiguration of power relations and knowledge construction processes in the organization.

3.3.2 The Challenge to the Status Quo: Confronting Everyday Life

The introduction of PA/AI as an actor that seeks "objectivity/rationality" inevitably challenges the *status quo* of HRM, marked by an often reactive daily life, a predominantly functionalist focus, and the limits inherent in human subjectivity in decisions and evaluations.

3.3.2.1 Daily Life as a Challenge to Analytical Objectivity

The daily life of HRM, with its complexities, ambiguities, and the need to deal with the human and relational dimensions, presents a constant challenge to the linear application of algorithmic logic. The "first-hand reality" of lived experience (Berger & Luckmann, 1966) does not always translate easily into quantifiable data or patterns identifiable by algorithms. Interviewee E07 makes this premise clearer when reporting the difficulty of justifying the need for cutting-edge analytical tools for a board focused on the commercial and financial, says: "[…]They do not have this experience, they cannot understand that if I do not have a good tool, it will take me much longer to deliver a selection process".

This difficulty in translating 'first-hand reality' into algorithmic logic is where different manifestations of AI come into play. The attempt to impose an AI of control, which presupposes rationality and structured data, can come up against the fluid nature of everyday life. An AI as a working partner that helps organize these complexities or an AI as a creative entity that suggests new ways of capturing or interpreting these experiences could offer alternative paths. However, as evidenced by E07's speech, if the perception of the value of these tools is not built and if the "Enabler" of "Co-intelligence Co-creation" is not activated to overcome the externalities of interdepartmental misunderstanding, the potential of AI to assist in the management of daily life remains underutilized. Furthermore, as Horgan and Dourish (2018) and McKinney and Shaffer (2023) point out, it is substantial to recognize that the data themselves, the raw material of PA, carry a significant degree of ambiguity.

This proposition attempts to show how the daily life of HR, with its specific processes and challenges, may not be immediately understood or valued by other areas that operate under different logic. Consequently, it is necessary and urgent to understand, in a broad way, the value that this new organizational actor, a PA project, has in integrating and generating a less stressful, more productive, and strategically aligned daily life (Ellmer & Reichel, 2021). This inherent ambiguity, added to the complexity of the organizational routine, imposes that the "objectivity required" by analytical tools is not passively accepted but actively constructed and negotiated in practice. Human interpretation, the "[…]ending of people's individuality" (E06), and contextualization with the use of mechanisms such as focus groups (E01) become, therefore, not mere complements but essential components to decode algorithmic outputs and mitigate the risks of oversimplification or decisions disconnected from lived reality.

In this scenario, explainable human-AI interaction is vital to turning algorithmic results into actionable, legitimate knowledge (Ijanperä et al., 2022). However, overcoming these challenges in the day-to-day application of PA/AI does not depend solely on the quality of human-machine interaction or the ability to deal with data ambiguity; it is also profoundly influenced by the orientation itself and the historically constructed role of HRM within organizations. This orientation has often been marked by a functionalist focus on HRM and how the PA can broaden – or, paradoxically, only reinforce – this perspective will be the subject of the subsequent analysis.

3.3.2.2. HRM's functionalist focus transformed by PA

Historically, HRM has often operated under a predominantly functionalist and administrative focus, focusing on executing processes (payroll, recruitment, training) efficiently, but not always with a clear connection to the overall strategy of the business. This focus represents an institution, an established standard of "how things are done" (Berger & Luckmann, 1966, 1995). E10 describes this reality: "[...]So today, when we talk about HR, you do not have a strategic HR. You don't have it, you have a functional HR, which supports the areas in their operational needs."

PA/IA is often introduced with the promise of breaking with this merely functionalist focus, elevating HRM to a strategic level. However, this perspective must be analyzed critically and deeply (Noponen et al., 2023). If the promise is a more strategic HRM, the analysis by Noponen et al. (2023) warns us of the risk that PA/AI may actually institute a "Digital Taylorism" or an "Anabolic Taylorism" or Taylorism on steroids. This would occur if PA/AI tools were primarily employed to intensify the standardization of tasks, the

decomposition of work into measurable components, continuous digital surveillance, and the exhaustive measurement of the performance of employees or even HR professionals themselves (Noponen et al., 2023). In this scenario, instead of raising HRM, the PA could reduce it to a more efficient executor of rationalized control and, paradoxically, less strategic in the broad sense of human and organizational development. The "uberization" of work, a consequence of a logic that individualizes risks and responsibilities, finds in the ER a sophisticated management apparatus, further distancing HRM from a genuinely strategic and people-centered role (Fleming, 2017).

3.3.2.3 Limits of Human Subjectivity and the Algorithmic Promise

The subjectivity inherent in human evaluations and decisions in HRM (biases in recruitment, performance evaluations, promotions) is often cited as a justification for the introduction of an PA project, which promises a more “objective,” “rational”, and “fair” basis. Interviewee E04 expresses this hope: “[...]To improve our perception a little that, sometimes, we are not a 100% fair area, because everyone has bias [...] I think we can make better decisions [with tools/data]”. In this context, the algorithmic tool is seen as a non-human actant capable of "correcting" or mitigating the limits of human subjectivity (Latour, 2005).

However, algorithmic objectivity is not absolute. Algorithms are social constructions produced and produced, reflecting the data with which they are trained and the assumptions of their programmers, and can perpetuate or even amplify existing biases (Noble, 2018; O'Neil, 2016). The algorithmic “black box” can make these biases even more difficult to identify and challenge. Therefore, the challenge to the status quo of human subjectivity is not simply to replace it with machine objectivity but to build a hybrid environment where human subjectivity and algorithmic analysis complement and correct each other. As Wallace (2021) argues, this requires recognizing the indissoluble nexus between human and artificial intelligence and developing processes that allow for the explainability and contestability of algorithmic decisions(Sreedharan et al., 2022). The objectivity required by the PA only becomes legitimate when it is perceived as part of a transparent and ethically accountable process. It does not ignore but qualifies and enriches human judgment.

3.3.3 Implementation and the Role of Data Governance

Existing organizational and technological structures profoundly influence the adoption and effectiveness of the PA. System fragmentation, departmentalization, and the need for

support from senior management emerge as elements that define the (im)possibilities of building an effectively hybrid and data-driven HRM.

In analyzing the structures and implementation challenges, the framework in Figure 21 highlights the externalities. Externalities can be considered as internal and/or external influences on the process of adoption or implementation of an AP project, such as the pre-existing organizational culture, technological maturity, market pressures, the complex ecosystem of organizational partners, and the internal power dynamics themselves. Therefore, the ability to foster the construction of processes that use co-intelligence and co-creation, or at least a combination of forces that act as a powerful enabler, thus become central to overcoming these structural obstacles.

Recognition of these externalities is necessary because, as Ojanperä and Vuori (2021) argue, when discussing platform strategies, success depends not only on the technology itself but on the organization's ability to navigate and shape its ecosystem, transforming potential "fears" (organizational, and partners) into "energy" for change. Systems fragmentation, for example, can be seen as an internal externality – for PA – that limits the ability of AI as a working partner to deliver expected value, a point that resonates with E02's difficulty in consolidating data between ADP and Workday or E05's frustration with the inconsistencies between CLIC and Workday. Similarly, the culture of departmentalization, an organizational externality, can prevent the insights of an AI as a creative entity from being disseminated or an AI as a tutor or trainer from effectively empowering the organization across the board, as implied by E10's perception of the need for greater interaction between HR and Sales.

According to Yao (2023), the perspective of human-machine co-intelligence suggests that these externalities are not mere noise but constitutive elements of the SMV Symbols-Meaning-Value space shared (or not) between humans and machines. The difficulty in establishing an effective "compute center" for PA, as lamented by E01 due to the lack of a "clear owner of the tools," can be interpreted as a failure in building a cohesive SMV space, where the externalities of the lack of governance undermine the ability to translate data (Symbols) into meaning and subsequently in value for HRM. Additionally, the need for explainability in human-AI interactions, highlighted in Sreedharan et al. (2022), can be considered an externality that arises from the complexity of "Person AI" or "Control AI" (Figure 21). If users' mental models of how PA works are not adequately accounted for – a cognitive and social externality – trust and adoption can be compromised, as E06's fears about data security and human curation warn. Therefore, overcoming the implementation challenges

discussed below necessarily involves the recognition and proactive management of these multiple externalities, aiming to transform them from potential obstacles into catalysts for an effective hybridization of HRM..

3.3.3.1 *Fragmentation of HR Systems*

Most pharmaceutical organizations surveyed operate with many HR systems that are often not integrated, a significant technical and conceptual obstacle. As experienced by E02 – "[...] If I have two distinct bases and I want to do some analysis, I'll have to do some manual work. And then we lose agility, we lose confidence in the data" – and E05 – "[...] You can't beat the information 100%." Therefore, under the perception of these two interviewees, such fragmentation acts as an externality (Figure 21) that prevents the PA from performing effectively in roles such as AI as a work partner or AI as a creative entity. This multiplicity of systems and the consequent fragmentation of data can still be understood, from the perspective of Latour (2005), as a reflection of the first source of uncertainty: 'There is no group, there is the formation of groups' (p. 29). Each HR system (ADP, Workday, CLIC, among others mentioned by the interviewees) with specific data, operational logic, and user communities represents a continuous effort to group around certain functionalities or departmental needs..

The dispersion and inconsistency of data compromise the integrative process (Figure 21) of value generation, undermining the validation of algorithmic outputs and their legitimacy. The "objectivity" promised by the PA thus loses plausibility when the input data is perceived as flawed. Overcoming this dilemma, for HRM to establish itself as an effective "calculation center" (Latour, 2003), requires not only integration technologies, but robust data governance (Henninger, 2024; Kalpokas, 2019). This governance is a central point to transform Co-intelligence Co-creation (Figure 21) into an enabling role, ensuring the quality and fluidity of the data that feed the organizational social reality resignified and mediated by the PA.

3.3.3.2 Organizational Departmentalization and the Necessary Cross-Cutting View

Departmentalization, although designed to simplify organizational complexity and promote specialization (Mintzeberg, 1979), often results in the creation of silos that become barriers to collaboration and the flow of information, as already discussed. This internal fragmentation, an apparent organizational externality (Figure 21), prevents the PA from reaching its strategic potential. The effectiveness of the PA, especially in the manifestations and assumption of its various roles in cross-cutting projects, depends intrinsically on the

ability to access and integrate data from multiple departmental sources – a recurring challenge. E01's statement, when mentioning in his organization, "[...]is born thinking about data, but there is still much evolution in connecting the areas", or E02's perception about the difficulty of obtaining a vision of "[...]where is it going" the company without integrated data, underline the urgency of a more transversal organizational vision, which transcends functional boundaries.

In this context, Organizational Network Analysis (ONA) emerges as a powerful diagnostic and strategic tool (Cross & Parker, 2004; Tichy et al., 2010). ONA allows us to visualize the organization through its formal departmentalized structure and as a network of interactions and formal and informal knowledge flows, revealing the true "centers of influence" and the "communication bottlenecks." By mapping how actors (humans and, potentially, PA systems as actants) connect and share information across departmental boundaries, ONA can expose the disconnect between formal structure and operational reality. The example of E04, which aims to "identify" employees' competencies at a global level to provide analysis on functional mobility, tends to use the analysis of these networks to build this new integrative and hybrid reality of the organization. ONA's ability to leverage insights by unveiling connections and/or optimizing collaborative workflows would be immense.

Integrating the PA with the insights from the ONA could be a promising way to foster the necessary transversal vision. PA, fed by more holistic data (E04, for example, in a global project) that reflects real collaboration networks, could generate richer and contextually relevant analytics to inform more strategic decisions. However, the mere visualization of these networks does not guarantee overcoming silos, as highlighted in 3.3.1.3. As E09's experience on the need to "sell" internally the importance of data integration beyond HR boundaries, or E01's difficulty in engaging other areas, points out: "[...]we as HR still need to have the voice to say, 'I need this information,' or E07 that the CEO does not accept such approaches, demand a culture of sharing and the perception of mutual value are indispensables . ONA can identify the "champions" or "connectors" (Rob Cross) who can act as enablers or mediators (Latour 2001) of this interdepartmental collaboration. However, data governance and support from senior management (discussed earlier) are key to institutionalizing these cross-cutting flows.

Therefore, there is a possibility for the organization – when using a PA project, adding value to its analysis using the ONA concept can provide valuable subsidies for the design of data governance that actively promotes the transversal view and Co-intelligence Co-creation between humans and AI (Boucher et al., 2023). By understanding how the data and knowledge

reveals flows (or does not flow) between departmental "groups," organizations can design new policies and/or practices (Figure 21) and incentives that reward collaboration and data sharing. ONA can help identify where the intervention of an AI as a working partner would be most effective in automating the exchange of information between departments or where control AI – used ethically and transparently – could monitor the effectiveness of cross-departmental workflows. Overcoming limiting departmentalization, therefore, is not only a matter of technology but of a deep understanding of the social and information networks that constitute the organization and of a strategic will to foster a more integrated and collaborative resignification of the organizational social reality.

3.3.3.3 Data Governance Challenge

The multiplicity of systems (ERPs, payroll systems, ATS, e-learning platforms, etc.) is not only a problem of fragmentation but also of data decentralization and governance. Who is the “owner” of the dice? Who guarantees its quality, consistency, and up-to-dateness? E01, when talking about the underuse of Workday and other tools, points to the lack of a clear owner of the tools or global processes that ensure their correct use and updating. This lack of governance leads to the production of "information in fragments" and distrust in data, a direct obstacle to the legitimization of PA projects.

Data Governance is not situated in the mere technical management of databases, but instead establishing itself as a strategic pillar for organizations that seek to maximize the value of their informational assets and mitigate associated risks. It is a system of decision rights and responsibilities for information-related processes, executed according to agreed models that describe who can take what actions, with what information, when, under what circumstances, and using which methods(Weill & Ross, 2004, 2011). This structured approach ensures data quality, consistency, usability, and security and aligns information management with the organization's business objectives. By defining clear roles, standardized processes, and comprehensive policies, data governance empowers companies to make more informed decisions, optimize operations, comply with regulations, and transform raw data into actionable intelligence, fostering a data-driven culture at all hierarchical levels.

Effective data governance is, therefore, a *sine qua non* for the “required objectivity”. It involves defining roles, responsibilities, quality standards, and access protocols and creating a trusted infrastructure for the PA network operation. Without this infrastructure, the multiplicity of systems becomes a source of informational entropy rather than an asset for organizational intelligence. The complexity of data management and the need for a cohesive

strategy for people analytics and AI have led contemporary organizations to consider and often institute dedicated executive roles, such as the Chief Data Officer (CDO) or, with more specific focuses, the Chief Analytics Officer (CAO)(Davenport & Harris, 2007) and the Chief AI Officer (CAIO). These roles emerge as a response to the need for robust governance to transform data into strategic assets and manage the multiple externalities (Figure 21) associated with its use (Laney, 2017).

As highlighted, a new or functional position CDO, CAIO, CAO, is insufficient to manage the complexity represented in Figure 21. Senior management support and lack of it (E07) consistently emerges in interviews as a critical factor for PA adoption and success. As E01 points out, he needed the president's approval to implement PULSE. With no response, E07 sought the support of the VP and the company's owner for his initiatives. E10 highlights that HR's strategic planning, including the "data-driven" pillar, needs the leadership's approval. This support is not just about approving budgets but about legitimizing the analytical approach as a strategic priority and mobilizing the organization around it (Berger & Luckmann, 1966; Latour, 2005).

Therefore, top management also acts as a potent mediator (Latour, 2005), whose endorsement can facilitate the overcoming of departmental resistances, regiment resources for systems integration, and the development of analytical skills in the short and long term, and consequently, impact the life cycle of each PA project. Without this support, PA initiatives risk remaining isolated, experimental, or underutilized, failing to institutionalize as part of the everyday reality of HRM. The indissoluble nexus between human intelligence (in this case, the vision and power of leadership) and artificial intelligence (the capabilities of PA/AI) is particularly evident here: one cannot thrive without the other (Wallace, 2021).

3.3.4 Algorithmic Inevitability Era

The growing hybridization of HRM, driven by algorithmic inevitability, brings with it not only opportunities but also significant limits, risks, and challenges that shape the (im)possibilities of its adoption and the protagonism of its actors. These challenges reflect the complexity of human-AI interaction and the inevitable need in absorbing innovation and its application in a parsimonious way. This reflection is necessary because there is a pressing risk of disruption if an PA project, for example, is implemented inappropriately (Minbaeva, 2021; Sivathanu & Pillai, 2018; Tursunbayeva, 2019).

This algorithmic inevitability does not appear as a decree but as a logical consequence of the dynamics of the exponential age (Espindola & Wright, 2021, p.17-19). The accelerated convergence of technologies, such as artificial intelligence, Big Data, and ubiquitous connectivity, tends to create an ecosystem where the adoption of algorithmic systems ceases to be a strategic option to becomes an evolutionary imperative (Espindola & Wright, 2021, pp. 27-29, 68). For the “new HRM”, this transition is particularly pressing, as the ability to analyze large volumes of data, predict trends, and customize interventions on a scale never seen before becomes vital for the effective management of human capital in complex and volatile environments (Bock, 2015; Tenakwah & Watson, 2024).

Only by mastering this algorithmic inevitability can expanding organizations optimize their time while being efficient in the war for hiring the best talent. Bock (2015, p. 69) illustrates the size and complexity of this challenge, that occurred in 2013:

> “[…]In fact, we’ve grown by about five thousand people almost every year. To get there, we start with the 1,000,000 to 3,000,000 people who apply for jobs each year, which means we hire about 0.25 percent of the people we consider. As a point of comparison, Harvard University in 2012 extended offers to 6.1 percent of its applicants (2,076 admitted out of 34,303 applicants). It’s a very hard place to get into, but almost twenty-five times easier than getting hired by Google.”

This small example has enough impact to affirm that it is not just knowing what IA is but also using this definition as a guide. So, the definition of AI as the search for machines capable of acting effectively in new situations signals a trajectory of increasing integration in organizational practices. Thus, the question is not “if” but “how” organizations and individuals will adapt and co-evolve with this algorithmic reality (Russell & Norvig, 2021).

From the perspective of the Social Construction of Reality (Berger & Luckmann, 1966), this era of algorithmic inevitability represents the emergence of a new symbolic universe – about the way we communicate, relate, solve problems, and move around – which is progressively institutionalized and legitimized. The analytical reality promoted by algorithms, initially an externalization promoted through the "new perceptions" of experts and technologists, begins to objectify itself in new practices, tools, discourses, and values in HRM (Burrell & Fourcade, 2021, p. 215). In this context, individual evolution implies the internalization of these new logics and the development of new skills, such as data literacy

and the ability to interact critically with AI systems, transforming social action (Tenakwah & Watson, 2024).

This proposition of deeper interaction between man-machine-data emerges in the conceptions of the interviewees – even if they are still early – as in the reports of E01 and E02 about the need to: “[...]thinking that data is important”, as being a path of no return, inevitable. From another perspective, the reports on the meaning of the data for each interviewee sound like a unison cry: “Data is a means of organizational integration." Therefore, organizationally, the evolution involves the reconfiguration of roles, processes, and structures to accommodate and capitalize on the distributed agency that algorithms introduce, as discussed in section 3.2.3 on algorithmic mediation through data (Horgan & Dourish, 2018; Kels & Vormbusch, 2020; Latour, 2005).

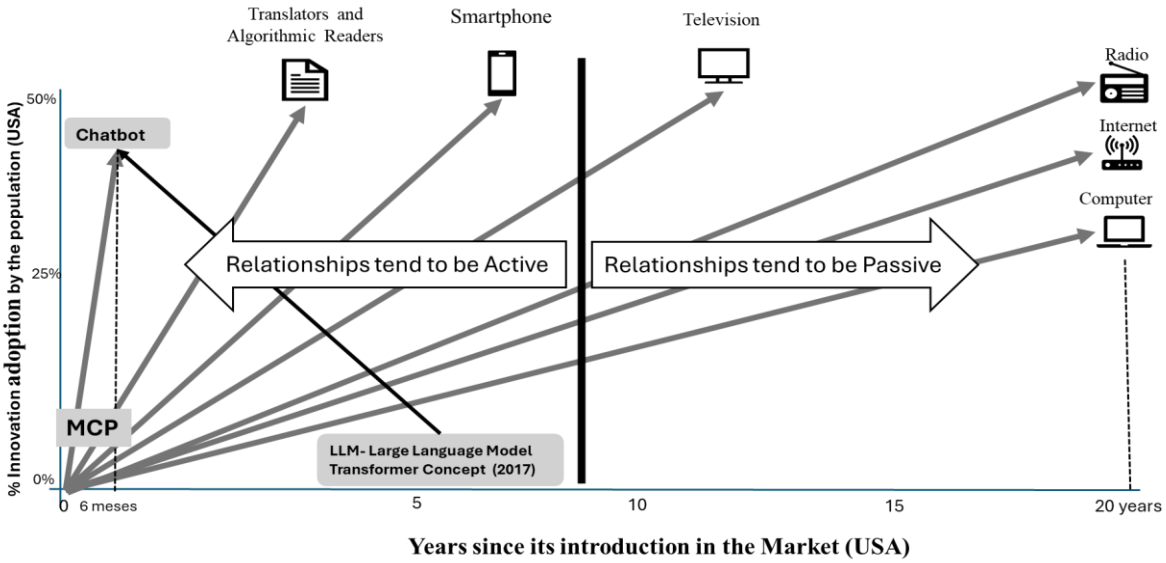
This proposition of algorithmic inevitability becomes increasingly complex due to the issue of technological volatility and the consequent challenge of continuous innovation. PA or AI technology, which is constantly and rapidly evolving, uses tools that are cutting-edge today and can become obsolete in a short time. This technological volatility challenges organizations regarding investment, updating systems, and continuous skills development. Interviewee E04 mentions how quickly generative AI tools, such as ChatGPT, have evolved and been adopted by her team, showing both the opportunity and the pressure to adapt: “[...] Today we already use a lot [...] a speed of things that we are using [...] ChatGPT for everything” (Russell & Norvig, 2021).

The challenge of innovation is not only to keep up with new technologies but to integrate them in a meaningful way into organizational processes and culture, avoiding superficial adoption or fads or yet, deifying them. Such critical capacity requires organizational learning and a learning organization (Senge & Sterman, 1992). This culture values experimentation and adaptation, a principle that resonates with the work of Rogers (2003) on the diffusion of innovations. Specifically, Rogers identifies “trialability” – the degree to which an innovation may be experimented with on a limited basis – and “compatibility” with existing values and practices as critical attributes that determine the rate and success of its adoption.

The institutionalization of analytical innovation as an ongoing process, rather than as a one-off project, is crucial to navigating this volatility and hybrid change processes, which should be understood using much flexibility (Poole & Van De Ven, 2021). This dynamic of constant transformation, intrinsic to the era of algorithmic inevitability, can be visualized as a

cycle of continuous evolution, where algorithmization acts as the central driving axis. Figure 22 illustrates this conception, representing hybridization not as a final state but as a perennial process of organizational development, and that in contemporary times, human and non-human interactions are inevitably increasingly co-creation (Mollick, 2024).

Figure 22: Algorithmic Inevitability



Source: Research (2025), Adapted from Anand (2025); Bock (2015); Hou et al. (2025); Vaswani et al. (2023)

The framework presented in Figure 22 encapsulates the essence of organizational adaptation in the algorithmic era: innovation, often driven by new technological capabilities, generates disruptions that require continuous adaptation, culminating in an evolution that feeds back into the innovation cycle. Algorithmization permeates each phase of this process, consolidating hybridization as an amalgam that imputes new norms for social relations and management strategies. However, this evolutionary journey has complexities and risks, which demand critical attention, especially about ethical, legal, and safety aspects.

3.3.4.1 Building Trust in the Hybrid Network: Security, Ethical and Legal Issues

The transition to an increasingly hybridized HRM, where algorithmic agency and human intelligence are intrinsically intertwined, is inseparable from an in-depth reflection on the pillars that sustain trust in this new and complex ecosystem: information security, ethics in data processing, and strict compliance with the current legal framework (Ilori et al., 2024). As previously mentioned, the use of Big Data by employees and the increasing autonomy of algorithms raise serious questions of information security, ethics in the use of data, and legal compliance, especially for the BPI (Gal et al., 2020; Hamilton & Davison, 2022; Simbeck, 2019). Far from being peripheral or secondary concerns, these elements are, in fact,

constitutive of legitimacy itself, institutionalized by the operational and strategic viability of algorithmic HRM in contemporary times.

With regard to information security in the PA context, the scope of action transcends the traditional protection of data against unauthorized access or leaks (Tursunbayeva, 2019; Tursunbayeva et al., 2021). It mainly maintenance da encompasses the integrity of the data that nourishes and trains the algorithms, the confidentiality of employees' personal and often sensitive information, and the continuous availability of systems that are becoming increasingly critical to HRM's fluid and strategic operation. Recently, the ethical implications of human resource management have intensified the focus on the perspectives of caring for people in organizations' management studies. Calls have also been made for the concept of organizational care to be grounded in philosophies of care, rather than business theories. Perspectives of care see individuals, especially women, as primarily relational and see work as a means by which people can increase their self-esteem, self-develop and feel fulfilled. The ethics of care has received attention in feminist ethics and is often interpreted socially as a feminine ethic (Elley-Brown & Pringle, 2021).

The statement of interviewee E06, when expressing her fear in: “[...]using open networks, because it is collaborators' data [...] I cannot guarantee security”, encapsulates a central, valid and widely shared concern by professionals and researchers in the field. In an organizational environment where a myriad of data – from performance indicators, behavioral patterns, records of social and professional interactions, to biometric information – can be systematically collected, stored, and processed, the inherent vulnerabilities multiply exponentially (Lupton, 2020). The very sophistication of PA algorithms, which often require access to vast, heterogeneous, and longitudinal data sets to reach their analytical potential, concomitantly increases the attack surface for security incidents and the complexity of permissions management, access control, and risk minimization with the use of the blockchain concept (Handayani et al., 2023; Vidhya K & Sangeetha P, 2024).

Techniques such as data anonymization and pseudonymization, although important tools in the privacy arsenal, may prove insufficient to ensure robust protection, while advanced algorithms demonstrate the ability to re-identify individuals by combining and cross-referencing multiple apparently unrelated sources of information (Ohm, 2010). In addition, the underlying technological infrastructure that supports the PA platforms – whether deployed on-premise or using cloud services such as Visier®; SuccsessFactor® and others – introduces new layers of security considerations that need to be proactively managed. The security of the

algorithms themselves, protecting them against various forms of manipulation, such as adversarial inference or data poisoning, which could distort their learning and lead to biased or harmful decisions, also emerges as a new and significant challenge for corporate cybersecurity (Barreno et al., 2010; Zuboff, 2021). Building lasting trust, therefore, is based on the unequivocal and continuous demonstration that the organization adopts the best practices, policies and technologies available to safeguard this critical informational asset. In ANT this translates into the stabilization of a socio-technical network where security protocols, firewalls, advanced encryption systems, and strict access policies act as effective and resilient actors in the integral protection of the system (Latour, 2005; Milan & Beraldo, 2024).

In parallel with safety, ethical issues permeate all phases of the PA life cycle in HRM, from the conception and design of systems through data collection and processing to the application and monitoring of algorithmic decisions. The primary and most debated ethical concern lies in the intrinsic potential for discrimination and the perpetuation of systemic biases (Hamilton & Davison, 2022; Misut & Jurik, 2021). It is well known that algorithms learn from historical data, and if that data reflects past social inequalities and biases – whether in terms of gender, race, age, socioeconomic background, or other (un)protected characteristics, AI models can inadvertently or explicitly replicate and even amplify these biases in key processes such as recruitment and selection, as well as, in the talent identification, succession planning, performance evaluation, and compensation definition (Ferrer-Conill et al., 2023; Gobble, 2017; Kakulapati, 2021; Noble, 2018).

The pertinent observation of interviewee E09, who warns of the risks of misinformation generated by AI, can be extended to the risk of decisions that, although "correct" according to the internal logic of the algorithm, are deeply flawed from an ethical point of view or result in discrimination, injustice, or dehumanization for certain groups of employees (M. Schultz et al., 2022). Another fundamental ethical dimension is algorithmic transparency and explainability (Galhotra et al., 2021). If decisions that directly affect employees' career paths, development opportunities, and ultimately lives are made or significantly influenced by algorithmic “black boxes” whose inner workings remain opaque and inscrutable to affected individuals and sometimes even to HR managers themselves, the perception of fairness, equity, and due process is irreparably compromised (Burrell, 2016; Burrell & Fourcade, 2021; Diakopoulos, 2016; Žlahtič et al., 2024).

The ability of an employee to understand the rationale of an algorithmic decision that affects them and, if necessary, to have mechanisms to challenge it or request a human review is a fundamental aspect of dignity at work and respect for their rights (Vayena et al., 2018). Therefore, an PA project, is the construction of a genuinely ethical co-intelligence ecosystem that requires a minimum of transparency in the algorithms added to the explainability in their decisions (Blattner et al., 2024; Lavazza & Farina, 2023). This does not necessarily imply the full disclosure of the source code of the algorithms, which could expose trade secrets or facilitate manipulations, but rather the ability to provide clear, understandable and contextual explanations about the main factors that influence algorithmic decisions and their possible impacts on individuals and the organization.

The search for this clarity and greater transparency can minimize the perception of a surveillance process and the continuous monitoring of employees – practices exponentiated by the capabilities of PA technologies – which can analyze everything from the content and frequency of emails to patterns of physical movement and social interactions also raise serious and complex ethical questions about the right to privacy. Individual autonomy and the balance of power in labor relations (Noble, 2018; Zuboff, 2019, 2021) The boundary between legitimate process optimization and the potential invasion of employees' personal and private spheres becomes increasingly tenuous and contested, which generates relational stress, requiring a robust and continuous ethical debate within organizations about what data is ethically permissible to collect, for what specific purpose, with what level of truly informed consent, and with what degree of knowledge on the part of employees (Yener et al., 2021).

The concern expressed by E10 that HR will become “Too many people and too few enabled” reflects precisely this fear that the human and relational dimension of work will be subsumed and devalued by the cold and sometimes merciless logic of datafication (Schweitzer & De Cremer, 2023). Ethics in algorithmic HRM, therefore, is not a mere compliment or an afterthought, but a continuous and proactive commitment to the principles of justice, equity, dignity, non-maleficence, and respect for individual autonomy. These principles must be carefully embedded in the design, implementation, and governance of PA systems (Floridi, 2018).

Additionally, the use of PA in HRM must operate in strict accordance with an increasingly detailed, complex, and constantly evolving legal framework. Personal data protection laws, such as the General Data Protection Regulation (GDPR) in the European Union and the General Data Protection Law (GDPL) in Brazil, impose a significant set of

obligations on organizations regarding the collection, processing, secure storage, sharing, and eventual deletion of personal data (Voigt & von dem Bussche, 2017) and Law Nº 13,709, 2018. These legislations, in general, emphasize fundamental principles such as the free, specific, informed, and unambiguous consent of the data subject, the minimization of data collection (collecting only what is strictly necessary for the declared purpose), the legitimate and explicit purpose for the processing, transparency towards the data subjects, and respect for the rights of these subjects, which include the right to access your data, to request its rectification, to object to certain types of processing and, in certain circumstances, to carry out the portability or deletion of your information.

Apprehension, reported by E06 amid its concern about the security of employee data, demonstrates the growing awareness and internalization of these legal responsibilities by its HR professionals. In addition to general data protection laws, there is the emergence of specific regulations or sectoral guidelines aimed at the ethical and responsible use of AI, as is the case of the AI Act proposed by the European Union, which adopts a risk-based approach, classifying AI systems according to their potential impact on fundamental rights and imposing progressively stricter requirements for applications considered high-risk (Gal et al., 2020). This category includes a variety of applications used in the context of employment, recruitment, and worker management (Commission, 2022).

Legal compliance, in this dynamic scenario, transcends the mere issue of avoiding financial sanctions or reputational damage – even if these are of significant impact; it is configured as an essential component of the "social license to operate" with PA systems and the management of your agency, in this conception of algorithmic inevitability (Vantrepotte et al., 2022). It requires HRM, in close collaboration with the legal, compliance, and information technology departments, to develop and maintain a deep understanding of applicable legal obligations and to implement robust and auditable organizational processes to ensure ongoing compliance. Such processes include, among others, the systematic conduct of Data Protection Impact Assessments (DPIAs) for new PA projects, as well as the possible appointment of a Data Protection Officer (DPO) or by the aforementioned CAIO (Chief AI Officer), or Chief Data Officer (CDO), when legally required, and the creation of effective and accessible mechanisms to deal with requests, Questions and complaints from data subjects (Malgieri & Comandé, 2017). The topic's importance is reiterated in E09's statement on the risks of misinformation provided by AI in sensitive contexts such as labor legislation. This statement underlines the critical importance of ensuring that AI systems not only respect

data privacy regulations but also do not induce, directly or indirectly, violations of other labor or human rights legislation.

The BPI's interdependent dimensions of information security, applied ethics, and legal compliance and regulations ultimately converge on a central and inescapable imperative: building and maintaining a solid trust in the hybrid network that is formed so that it supports the decisions of the new HRM with the necessary efficiency and effectiveness (Babu & Govindappa, 2022; Boudreau, 2017; Bucher et al., 2021). Because no matter how automated and/or automaton HRM's processes are, there is above all the work of constituting trust in and in employees, as well as in society, in the way their data is collected, processed and used, as an intangible asset without which the required objectivity becomes suspect. Under the lens of SCRT (Berger & Luckmann, 1966), the legitimacy of the PA as a valid and accepted institution in HRM depends fundamentally on this trust (Kolb et al., 2020). Suppose employees and other stakeholders perceive algorithmic systems as being operated in a way that is fair, secure, transparent, and respectful of their rights and dignity. In that case, they will be considerably more likely to internalize this new analytical reality and cooperate constructively. Otherwise, resistance—whether passive or active—widespread distrust and systematic contestation will undermine its effectiveness and long-term sustainability (Lagios et al., 2023).

Building this indispensable trust, in turn, requires transparent governance and accountability mechanisms that cannot be merely symbolic or protocolary (Yao, 2023). This translates into the effective implementation of data and AI governance frameworks that define clear and accessible policies, establish well-defined roles and responsibilities, promote the necessary human oversight over critical algorithmic processes, ensure that regular and independent audits of algorithms are carried out – to assess bias, performance, and ethical compliance – , and create open and safe channels for dialogue, feedback and resolution of employee complaints (Baumgart et al., 2023; Bucher et al., 2021; Milan & Beraldo, 2024; Mittelstadt et al., 2016). I again make use of the ANT, which warns us that the stability of the complex socio-technical network – which in this case constitutes the PA – depends firstly on the alignment and reliable performance of all the actors involved. In this perspective, laws, codes of ethics, security protocols, and governance structures are non-human actants that inscribe and materialize trust in the system. If these normative and procedural actants fail in their function or are perceived as ineffective, the network as a whole is destabilized, and symmetry is lost (Latour, 2001).

It is imperative to recognize that organizational culture plays a vital and sometimes decisive role in this process. An organizational culture that genuinely values radical transparency, open and constructive dialogue about the complex ethical challenges posed by AI, continuous and adaptive learning, and fundamental and non-negotiable respect for individuals and their rights will be intrinsically more conducive to building an algorithmic HRM that is both effective and reliable (Ferrar & Green, 2021).

The desired digital ecosystem of co-intelligence that PA can create is, therefore, not just a matter of technical implementation or regulatory compliance but is, deeply and essentially, a cultural artifact that demands specificity, that it is a unique project (Boucher et al., 2023; Ojanperä et al., 2022; Rasmussen et al., 2024). It is nourished by a leadership that does not limit itself to implementing new technologies but actively models, promotes, and demands an ethical, responsible, and human-centered use of AI. Such leadership ensures that the "algorithmic inevitability" that marks our era does not translate, by omission or carelessness, into an undesirable and avoidable erosion of fundamental human values in the workplace (Mendonça et al., 2023).

A HRM, strategically positioned at the critical intersection between technology, people, and broader organizational strategy, holds both the responsibility and the singular opportunity to lead the way in building this essential level of trust, ensuring that the increasing hybridization of day-to-day work relationships is, in fact, a path to management that is not only more efficient and competitive, but also intrinsically more just, equitable, and humanly enriching for all involved (Gergen, 2023).

In this trajectory of increasing hybridization, permeated both by a transformative potential and by the security, ethical, and legal challenges discussed, a fundamental reconfiguration emerges in understanding agency and the very constitution of organizational reality. Table 12 aims precisely to synthesize this evolutionary transition, illustrating how the conception of a purely human agency gives way to a hybrid model or, in the hybridization process, where the complementarity between different actors becomes central.

The theoretical perspectives that underlie this thesis – the SCRT, which illuminates the dimension of meaning and social legitimation, and the ANT, which unveils the constitutive role of non-human elements and socio-material networks – converge to explain this flow of development. This is a continuous process, exponentiated by mutual interactions and fostered by the complexity of contemporary HRM, providing a rich analysis of the phenomenon of algorithmic inevitability, represented here by AP projects, as an organizational actor. The

hybrid conception, by contemplating the agency and the influence of non-human actors in the organizational fabric, not only enriches the readings of the SCRT and the ANT but also responds directly to the analysis of everyday changes (OE02) and to the multifaceted protagonism that defines this new era (Mendonça et al., 2023; Mollick, 2024).

Tabela 12: A Process of Hybridization

Analysis Dimension	Constructionist Perspective	Sociomateriality Perspective
Primary Construction Agent(s)	Human beings in intersubjectivity; emphasis on collective human agency.	Heterogeneous socio-technical network, composed of human actors (HR professionals, managers) and non-human actors (algorithms, PA/AI systems, data).
Nature of Organizational Reality	An objectified social reality, a product of human externalization, which becomes a "social fact" perceived as external and coercive.	A performed and mediated reality, co-constructed through the continuous interaction between humans and non-humans; less stable and more fluid.
Role of Technology (PA)	Tools or products of human externalization, which, once objectified, can influence reality, but their agency is derived from the human.	Active mediators and actants with their own agency, who translate, compose and transform human action programs, not being mere instruments.
Institutionalization Process	Ocorre através da habitualização de ações, tipificação recíproca de atores e comportamentos, e legitimação por universos simbólicos humanos.	Occurs through the habitualization of actions, reciprocal typification of actors and behaviors, and legitimation by human symbolic universes.
Formation of Institutions	Institutions as common sense knowledge sedimented, transmitted and maintained through socialization and human social control.	New institutions (e.g., "data-driven decisioning") emerge from network performance, where algorithms act as "new organizational institutions" (Mollick, 2024).
Nature of "Objectivity"	A quality intersubjectively constructed and shared by humans, which confers facticity to the social world.	A quality performed and mediated by the socio-technical network; "Data objectivity" is an effect of algorithmic inscription and translation, negotiated in the interaction.
Change in Everyday Processes (HRM)	Processes change as new typifications and legitimations are internalized by human actors, altering the "way of doing things".	Processes change as algorithmic mediation reconfigures workflows, decision criteria, and actors' capabilities, imposing new routines and logics.
Protagonism	Primarily human; Humans are the "architects" of social reality and its institutions.	Distributed and hybrid; Protagonism is shared between humans and algorithms, which mutually transform each other in interaction.
Analysis Focus	Emphasis on consciousness, subjective meanings, intersubjectivity and the dialectic between individual and society.	Emphasis on associations, mediations, performance, materiality, and agency of non-humans in conjunction with humans.

Source: Research (2025).

3.3.4.2 Hybridization of HRM through the Lens of Sociomateriality

The perspective of sociomateriality, as articulated by authors such as Orlikowski (2007) and Orlikowski and Scott (2008), challenges the analytical separation between the social and the material, including technology, here considered as a condition of hybridization. Therefore, rather than seeing technology, labor, and organization as distinct entities that interact, sociomateriality postulates an inherent inseparability or constitutive entanglement. This means that "[...]there is no social that is not also material, and no material that is not also social" (Orlikowski, 2007, p. 1437). This approach strengthens the ANT view, which identifies humans and non-humans as actors in a network by suggesting that they are interdependent and co-constituted in recurrent practices.

In the emergence of HRM and PA, the lens of sociomateriality allows us to deepen the conception of the PA as an "organizational actor" that influences or is influenced by HR professionals. Instead, HRM practices and PA technologies (algorithms, data, platforms) are seen as an emergent, indivisible socio-material practice or nexus (Wallace, 2021; Yao, 2023). Data analysis, the insights generated, and the decisions made are not only products of technology or only of humans but of the performance of this socio-material amalgam. As Schatzki (2025) points out, the agency itself can be understood as a particular form of causality, where multiple entities and their doings combine to produce effects, and sociomateriality helps us to see this causality distributed and intertwined.

The interviews conducted with the BPI managers offer glimpses of these socio-material dynamics. When interviewee E02 states that the IA helps to "[…]remove this perception, ah, I think it's coming out [...] Here are facts [search for talent loss]", we observe how data (material) and human (social) interpretation/presentation intertwine to constitute a new "reality" about turnover, which then guides strategic decisions. Similarly, E01's statement about the need to "[…] think about a different person's profile in order to be able to dream [about PA]" can be interpreted not only as a need for new skills but as a reflection of how the socio-material practices emerging with PA are actively (re)shaping professional identities and ways of "being" an HR professional. The "protagonism of data" or the perception that "data integrates", the perception of "PA as a solution and protagonist," and how algorithms act as mediators that modify the attributes of both humans and systems – all these points discussed earlier – gain additional depth. It is not just a tool gaining prominence; HRM is becoming a fundamentally socio-material practice. Sandberg, Dall'Alba, and Stephens (2025) argue that "things" (such as algorithms and tools) are constitutive of performance at work, belonging to

a nexus of useful things and co-constituting concert performance with practitioners (Sandberg et al., 2025).

The difficulties reported by the interviewees, such as the fragmentation of systems (E01, E02, E05) or the absence of a clear sponsor for the PA tools (E01, E03), can be seen as challenges in the stabilization and governance of these new socio-material configurations. In this perspective, the search for integrated and automated processes is an effort to give shape and coherence to these emerging hybrid practices. Leonardi (2013), when discussing the theoretical foundations of sociomateriality, points to the importance of understanding how the social and the material become entangled to produce practices that seem indivisible, which resonates with the search for integration reported. The discussion of algorithmic inevitability is also enriched by this perspective. In that case, the growing penetration of algorithms in organizational practices represents a greater use of tools and a fundamental transformation in the very nature of organizational reality and, therefore, of HRM.

Scott and Orlikowski (2025) treat AI not as a thing but as a phenomena-in-construction whose performativity and accountability need to be examined using a genealogical socio-material approach, highlighting how the boundaries between humans and technology are continuously (re)configured. The various manifestations of AI – such as AI Person, AI Work Partner, AI Tutor Trainer, or AI Control – as outlined in Figure 21 – illustrate the depth of this intertwining, where algorithmic capabilities and human practices co-construct each other dynamically.

Costa Júnior and Rivera (2024) reinforce this view by discussing how artifacts, especially technological ones, influence the performative scheme of human routine and agency, emerging from a heterogeneous network of technical and social elements. They argue that the role of artifacts evolves from the knowledge and know-how embedded in routines and their actants. In the context of PA, this means that algorithms and platforms are not static; their role in HRM is continuously negotiated and (re)constructed in practice as HR professionals learn to interact with them and as the systems themselves learn from the data and interactions, becoming more and more an organizational actor – as an example the relief of not analyzing another hundred tables with data (E07), or, my emails are much more efficient and transparent with the help of AI (E08).

Wherefore, sociomateriality offers a conceptual framework that allows us to understand the hybridization of HRM, driven by PA and AI, not as a simple addition of technology to existing practices but as the emergence of new sociomaterial practices. In these

practices, agency is distributed, realities are performed, and the boundaries between humans and technology become increasingly porous and co-constituted. The socio-material perspective, for the analysis of the new actants of HRM at the BPI, allows a deeper and more contemporary understanding of the ongoing transformations – summarized in Table 12, p. 223 in this work – aligning the analysis of organizational and technological studies and responding to the challenge of understanding the multifaceted protagonism of the various actants in the era of algorithmic inevitability. The consideration of socio-materiality not only enriches the PA's analysis of Reality Translation and Required Objectivity and Hybridization as a organizational strategy, but also underlines the critical importance of issues of security, ethics, and legality as constitutive of the very trust and legitimacy of these new hybrid configurations.

CHAPTER IV

4 FINAL REMARKS

We have arrived at the axial moment of framing the final remarks of this extensive and deep investigative journey. This journey, which is almost complete, represented a dense and meticulous dive into the intricate and emerging reality of HRM at BPI, a field in vibrant transformation due to the introduction and growing influence of PA and AI in its daily life. The weight of this trajectory lies not only in the complexity of the phenomenon itself but also in the effort undertaken to unveil its multiple layers of meaning and its practical consequences. The central purpose that guided and sustained each stage of this thesis is: To evaluate the potential of PA as a new organizational actor that transforms and resignifies the daily reality of HRM at BPI. To achieve this aspiration and to give due importance to this straight of arrival where the threads of investigation intertwine, we embarked on a qualitative, interpretative, and reflective exploration, anchored in a critical and revealing dialogue between the SCRT of Berger and Luckmann (1966) and the ANT of Latour (2005) and Callon (1986).

This section, therefore, assumes the responsibility of being more than an epilogue; it is configured as a space for in-depth and critical reflection on the findings of this research, its multifaceted theoretical implications, and its practical resonances. The importance of this project as a whole lies in its attempt to shed light on a fundamental transformation underway in HRM, a vital field for contemporary organizational dynamics. By outlining the contributions that emerge from this analysis, we aim to echo and effectively reverberate in the academic field, fostering new debates and investigations and, equally, in the practice of HRM, offering insights and paths for professionals who navigate this new hybrid territory. To this

end, it was sought, with rigor and constancy, to maintain fidelity to the data collected and to the theoretical framework built and amalgamated in order to present a robust narrative and, perhaps, that the considerations outlined here can be applied, contributing, to some degree, to the continuous evolution and organizational innovation, especially about the increasingly symbiotic relationship between human and artificial intelligence in people management.

To fulfill its greater purpose, the present investigation unfolded into specific objectives that guided our immersion in the field. First, (SO01), we sought to understand how PA is experienced and signified in the complex context of HRM at BPI. Next, (SO02), we analyze the daily changes in HRM processes that result from the increasing application of AI tools and/or specific PA projects. Finally (SO03), the research proposed to explore the (im)possibilities of adopting the PA in the HRM of the BPI, considering the multifaceted protagonism of the human and non-human actants that make up this new reality. The answer to these objectives, which form the core of this thesis, was meticulously woven throughout the data analysis, a process made possible by the application of a qualitative methodological approach, of an interpretative and reflective nature, and by the reflexive thematic analysis of the collected narratives, always in the light of the dialogue between the Theory of the Social Construction of Reality and the Actor-Network Theory. We now revisit and consolidate the main lines of force that emerged from this analysis.

4.1 The Emergence of People Analytics: Between the Translation of Reality

The first and fundamental line of strength that this thesis unveiled, and which we consolidate here, lies in the deep understanding of the process by which the introduction of PA and AI in the HRM of the BPI constitutes a complex Reality Translation. This central finding, detailed in section 3.1 of the present study, emerged from the detailed analysis of the experiences and meanings attributed to by the interviewed HR Directors and Business Partners, whose speeches were the empirical foundation for this finding. Section 3.1, in its entirety, demonstrated that the emergence of the PA transcends a simple technological adoption, configuring itself as a socio-material and symbolic reconfiguration of HRM itself. In HRM, uncertainties are paralyzing substances.

The subsection Uncertainties and Problematization of the PA, for example, revealed, based on reports such as those of E01, E04, and E07, the initial ambiguities, the doubts about the return on investment (ROI) and the prioritization of the PA over other demands, and the crucial confrontation with the traditional *habitus* of HRM, less accustomed to analytical

cognition, as pointed out by E02 and E06. The literature on the ANT, especially the concepts of problematization and interessement of Callon (1986), allowed us to interpret how the need for the PA is socially constructed and how credibility (E04) needs to be laboriously established, often facing challenges in the formation of groups of notables and in the management of mediators and intermediaries that can both facilitate and obstruct this initial process (Latour, 2005).

In an initial contribution, this thesis empirically demonstrates that introducing PA and AI in HRM at BPI constitutes a complex Reality Translation. This finding echoes the perception that the relationship between technology and organizations is reciprocal and that AI can disrupt and reshape HRM, going far beyond a mere technological adoption and configuring itself as a socio-material and symbolic reconfiguration. The starting point of this process is the uncertainties and the capacity for problematization offered by an PA project, which is marked by ambiguities. These ambiguities resonate with the four ambiguities for AI-driven HRM identified by Böhmer & Schinnenburg (2023) – work design, transparency, performance, and data – and with the barriers and lack of knowledge about PA pointed out by Di Lauro et al. (2024). Reports from interviewees such as E01, E04, and E07 about the return on investment (ROI) and prioritization of PA, and from E02 and E06 about the traditional routine of HRM, less accustomed to analytical cognition, vividly illustrate this initial field of paralyzing uncertainties in a pressing context for translation and action.

This difficulty with the analytical mindset in HR is corroborated by Böhmer and Schinnenburg (2023) observation that HR professionals often lack an understanding of analytical approaches and by Di Lauro et al.'s (2024) finding about the lack of a data-driven culture in Italian HR departments, propositions corroborated in this research.

The articulation of these experiences with the concepts of problematization, interessement, and the management of mediators of the ANT (Callon, 1986; Latour, 2005), allowed us to interpret how the need and credibility of the PA (E04) are laboriously constructed socially, a process that Böhmer and Schinnenburg (2023), approach from the perspective of the Resource-Based View as an organizational learning journey.

Thus, this work elucidates the indispensable process of negotiation and signification for the PA to begin its trajectory as a potential transforming actor, highlighting the intrinsically socio-material and symbolic nature of this emergence since its conception. However, if PA provides this translation of reality, he can be considered a protagonist, regardless of the degree of autonomy he possesses as a non-human actor.

In addition, the PA as a solution and protagonist, the translation of interests is included in this complex environment, demonstrating how, once the uncertainties are managed, the PA is actively positioned as a strategic actor. The narratives of E04 about the PA giving a "seat at the table" to HRM, or of E01 and E06 about using data for a more robust dialogue with areas such as finance, illustrate this construction. Here, the five moments of the translation of Latour (2011) and Callon (1986) were used to understanding the dynamics of alignment of interests and the constitution of the PA as a Mandatory Waypoint (PPO) for an HRM that aims for strategic relevance and an integrated vision, despite the challenges imposed by the lack of a clear sponsor or effective data governance, as lamented by E01 and E02. The perception of data as a strategic instrument and means of integration was almost unanimous (E01-E10), and technology was recognized as an essential enabler (E02, E06, E10) in a process that SCRT helps us to see the externalization and objectification of new categories and analytical logics.

If the overcoming of uncertainties – as a paralyzing substance – with the alignment of interests being aligned, it is necessary to strengthen this new collaborative network – human and non-human – to seek synergies and inscriptions in a socio-technical network fostered by PA. The search for synergies, such as the democratization of access to data (E01) and the challenges of system fragmentation (E02, E05), were contrasted with the materiality of the AP's enrollment in several tools (Workday, SAP, Vizier, ChatGPT, among others, cited by E01, E02, E03, E04, E05, E07, E09, E10).

The agency of the non-human (Latour, 2005), where the system indicates or filters (E01, E02), and the attribution of value in technological adoption (E01, E05, E06, E07, E08, E10), were central aspects. The literature on mimetic isomorphism (Serra et al., 2021); and the need for unique and contextualized projects urged by Rasmussen et al. (2024) enriched the analysis, while enrollment and internalization helped to unveil how this weaving of the analytical network and the overcoming of silos are contingent processes full of challenges.

Changes that generate a new order require mobilization and reordering. An evolutionary and summative stage in the Translation of Reality for the new HRM, addressing the complex phase of stabilization (or contestation) of the new analytic order. At this point, the continuous mobilization of actors (human and non-human) and the indispensable reordering of organizational, cognitive, and cultural structures – analyzed through the concrete PA projects reported by E01 (PULSE), E04 (Global Mobility) and E09 (Turnover) – reveal the effort to build the pillars of the necessary foundation, such as governance and analytical

culture, and optimizing the resources demanded, such as data, technology, and skills, elements to achieve the desired analytical excellence. It is from this strengthened base and the subsequent demonstration of value that PA can generate – the third pillar – PA begins to reach the level of Translation Five – and become indispensable (Latour 1988, 2011).

The new HRM transformation into an effective calculation center depends intrinsically on this ability to consolidate its network and generate perceptible and strategic value. Only then does the mediation of the PA become active element for decision-making and the performance of analytical HRM, even though the persistent structural challenges and fragmentation of systems require continuous and adaptive management.

Based on the evolutionary and cumulative analysis we have covered so far, we can now consolidate the central contribution of this section. What this thesis unveils in principle, in its essence, is that the rise of PA in the reality of HRM is not a linear adoption of technology or tools for managing metrics and/or data dispersed throughout the organization but rather an arduous and contingent journey of translating reality. It is a journey that begins with overcoming paralyzing uncertainties, such as: What to do? How to do it? Where to apply changes? Moreover, when to start and or stop? All of this advances through the complex articulation of interests to position the PA as a protagonist actor – who needs and expects responsibility and autonomy – which materializes in the intricate fabric of a socio-technical network and aspires to culminate in the consolidation of a new order, where HRM seeks to reinvent itself as a calculation center indispensable to the organization located in HR.

This study's contribution, therefore, is to map and make sense of this trajectory. We demonstrate that, before any data generates value, there is an intense and sometimes conflicting work of negotiation and signification between humans and non-humans in a reality that is in a continuous process of creation and destruction. Ultimately, true transformation is achieved not through algorithmic sophistication but through arduous forging alliances, dismantling cultural and structural resistance, and translating a technological promise into a shared, meaningful, and trustworthy organizational reality. This understanding paves the way for the subsequent analysis of the new forms of objectivity that translated reality demands and the dynamics of hybridization that it engenders.

4.2 Institutionalization and Algorithmic Mediation

Once the complex Reality Translation is initiated, as unveiled in the previous section, the trajectory of PA as a transformative organizational actor enters a new, equally challenging

stage: the search for the objectivity that PA requires. Is not a return to a naive positivist view, where data would be a faithful mirror of a pre-existing reality. On the contrary, in one of its contributions, this thesis demonstrates that objectivity is a social and socio-material reality that needs to be laboriously constructed, negotiated, and validated. In order for the insights derived from the PA to transcend the field of mere opinion or point analysis and become the basis for legitimate and effective decisions – so that they become organizational "facts" – a double movement is imperative: the institutionalization of new practices and roles and the understanding of the algorithmic mediation that is at the heart of this new way of knowing and managing.

4.2.1 Institutionalizing the New Analytical Order: The Building of Reality

The first battlefront in constructing this objectivity is institutionalizing a new analytical order in HRM. Under the lens of SCRT, institutionalization is the process by which human actions become habitual, are typified, and solidify into an objectified reality that governs social conduct. In the context of HRM at BPI, this translates into an attempt to transform the use of data from a sporadic event into a routine and expected practice.

The narratives of the interviewees illuminate the multiple facets of this process. The consolidation of a new role for HR is the first step. As reported by E01, it is necessary to go beyond taking data, and, as E02 describes, HR must be present as a business partner “in the initial discussion”. This transformation, however, requires deep change management to address cultural and cognitive resistances. The need to deconstruct this prejudice that technology distances people (E08) or to break the resistance of the other (E07) evidences the confrontation with preexisting institutions and *habitus, the field and the capital,* made up of long data. The implementation of PA, therefore, can be seen as a unfreeze-change-refreeze process where the PA offers the data to justify the change and monitor its effectiveness in an analytical order (Weick & Quinn, 1999).

Communication emerges as the primary vehicle for this institutionalization. As argued by Goffman (1986), organizational silos can be understood as generators of distinct interpretative frames, solidifying the changes into small portions. People Analytics seeks to establish a new, unifying analytical framework by presenting data as objective. E02's experience using data to reframe the discussion on turnover with the board is a practical example of this attempt to reframe reality. The need for frequency and sequence in communication, as mentioned by E10, to establish the importance of data reinforces the notion

that the legitimation of a new frame requires a continuous and strategic communicational effort.

Lastly, institutionalization materializes in the creation of integrated and automated processes. Digital admission (E02, E08) or the vacancy flow (E09) are examples of how new logics are inscribed in material routines, becoming "the way things are done". The persistence of manual processes "in the fingernail" (E07, E09) indicates, however, that this institutionalization is still partial and that the fragmentation of systems (E01, E02, E05) remains a significant challenge for the complete objectification of the new analytic order.

4.2.2 Algorithms as Institutions: The Validation of Objectivity

The search for objectivity in contemporary HRM forces us to go beyond the institutionalization of human practices and recognize algorithms as institutions. Algorithms are not just tools but sets of formal and informal rules that shape interaction contexts and define development paths. They are the rulebooks of the new organizational reality, with a special form of communication and special language, combining the objectification of meaning with the agency of the non-human.

For an algorithm to become a legitimate institution, it needs validation and the perception of value. Validation occurs when processes such as piloting and gradual tests are activated, as described by E10 and E04. These strength tests are constituents for the enrollment the actors in the algorithmic network (Latour, 1988). The perception of value, in turn, is linked to the attributes of innovation, such as its relative advantage and observability (Rogers, 2003). The value is perceived in efficiency – "[…]now I spend 30 minutes to review" (E02) –, in the agility and quality of information – "[...] here are facts" (E02) – and in the ability to generate new insights, such as when E10 uses Viva Insights to analyze its interactions or when E01 uses GPTW data to deepen its understanding of perceived benefits.

The objectivity produced by these institutionalized algorithms is not an absolute truth but the result of a specific process mediated by the rules and logic embedded in the systems themselves, which need to be continuously validated and perceived as valuable to sustain their legitimacy.

4.2.3 The Mechanism in Action: The Four Ways of Algorithmic Mediation

If algorithms operate as institutions, mediation is the mechanism by which they exercise their agency and construct this objectivity. Inspired by Latour (2001), we can analyze

how PA algorithms mediate people management through four transformative paths, which alter daily processes (SO02) and redefine the role of actors (SO03). Translation (Deviation from Goals): The algorithm translates the human goal into something operationalized in its logic. The PULSE system translates the broad goal of E01 to improve engagement into identifying patterns and monitoring the frequency of certain variables, which have been negotiated and inserted into a system so that the translation of reality is understood and, if possible, changes are made into continuous and institutionalized action. E09's concern that AI might ignore feeling or context when analyzing resumes reflects the fear of a deviation that misses important nuances. Objectivity here is produced by a translation that specifies but potentially simplifies the original goal.

PA can mediate from the perspective of composition or construction of new goals because, often, the human-algorithm association creates new goals that were previously non-existent. By indicating risks of talent attrition (E01), the PULSE system transcends simple engagement analysis, enabling managers to formulate the new strategic goal of proactive intervention. E10's use of Viva Insights to analyze their interaction patterns leads to the composition of the new goal of optimizing meeting habits and expanding the network of contacts. Here, the PA optimizes and innovates, expanding the strategic scope of HRM. Human skills are delegated to algorithms, which perform them differently. Screening of CVs (E02, E05, E07) or digital admission (E02, E08) are clear examples. This delegation, however, is negotiable. The need for a human filter (E04) or to analyze on top of AI (E06) shows that HR professionals monitor, adjust, and, if necessary, reverse the delegation, maintaining control over the built objectivity.

PA can perform mediation by modifying attributes. PA is the deepest mediation, where humans and algorithms are mutually transformed. The HR professional who interacts with PA becomes an HR professional with better analytical capacity when compared to his previous state (E06, E08, E09), with new skills and a new identity. The experience of E06, who migrated from technical areas, demonstrates a professional whose profile has been modified, enabling her to discuss with the manager, on an equal footing, about the data. Simultaneously, when implemented and fed with local data, a genetic algorithm becomes a specific system of that organization, modified by its context of use, as in E06's concern with building its own secure chatbot. This mutual constitution is the core of sociomateriality, where new hybrid processes emerge, reconfiguring the everyday reality of HRM (Orlikowski, 2007).

The detailed analysis of section 3.2 of the work, deepened here, offers a contribution: it demystifies objectivity in the age of PA. This study demonstrates that the "Objectivity Required" by the PA is not an inherent property of the data or a return to positivism but rather a complex social and socio-material reality, co-constructed and continuously negotiated. It emerges from the difficult task of institutionalizing a new analytical order for the new HRM, redefining roles, managing change, and communicating new frames of everyday reality. Moreover, fundamentally, this objectivity is the performative result of algorithmic mediation, where humans and non-humans associate, translate goals, compose new objectives, delegate competencies and mutually transform each other. Understanding this negotiated objectivity is an indispensable step in analyzing the emergence of a truly hybrid HRM. It is the understanding of how this new basis of everyday reality is built that allows us, subsequently, to analyze the dynamics, tensions, and crises of this new configuration, paving the way for the discussion on the Hybridization Crisis and the proposition of an extension to Greiner's (1998) model, another contribution of this thesis.

4.3 The Hybridization: A New Organizational Crisis and Its Implications

The preceding analyses of Reality Translation and Required Objectivity converge on an ineluctable revelation, the central finding of this investigation: we are witnessing not just a change, but a profound ontological reconfiguration of HRM, with the emergence of a fundamentally hybrid HRM. Driven by the relentless force I term algorithmic inevitability, this hybridization transcends the superficial notion of merely adopting new technologies. It is, in fact, the creation of a new and complex organizational ecosystem in which the classic distinction between the subject – the human – and the object – the tool – dissolves.

In this new conception, the boundaries between human agency and algorithmic performance not only meet but intertwine in an indissoluble symbiosis, becoming porous, interdependent, and, more and more co-constitutive. It is a scenario that can only be fully understood through the lens of sociomateriality, which compels us to see not the interaction between distinct entities, but the constitutive entanglement of practices, where the social and the material are inseparable. In this new organizational fabric, people management ceases to be an exclusively human prerogative and becomes a complex choreography, performed by a multiplicity of human and non-human actors, whose dynamics, challenges, and potentialities we will now explore.

To unravel the complexity of this phenomenon, the dual analytical architecture that underpins this thesis – the dialogue between the SCRT and ANT – proved to be an epistemological necessity. SCRT was fundamental to understanding how PA is socially signified and how its legitimacy is constructed. However, an analysis through this lens alone would risk treating technology as a "black box". It is by introducing ANT that we open this box, revealing the distributed agency in the socio-material network and the protagonism of algorithms not as mere instruments but as actants that actively reconfigure the materiality of HRM practices. It is at the intersection of these theories that the emerging hybrid reality reveals itself, not as a state of technological harmony, but as a field of dialectical tensions.

The contradictions observed in the interviewees' speeches – the dispute between individualism and the desired co-responsibility, the clash between the urgencies of daily life and strategic analysis, and the persistence of organizational silos – are not mere obstacles, but the very manifestations of the complexity of this new condition. At the epicenter of these tensions, the PA/AI actor positions itself as an ambivalent and paradoxical catalyst. The same tool that promises to break down silos with transparent dashboards can, paradoxically, deepen individualism by focusing on isolated performance metrics. The same algorithm that could free up time for strategic reflection can, if poorly implemented, intensify the focus on operational micro-tasks, reinforcing "Digital Taylorism" (Noponen et al., 2023).

It is precisely at this inflection point that the present thesis advances by suggesting a new lens. Inspired by Larry Greiner's (1998) evolutionary architecture, we argue that the magnitude of the current transformation requires an extension of his model. Therefore, is propose the existence of a sixth evolutionary phase: Algorithmization. This phase is defined by the massive and deep infusion of a non-human agency endowed with learning capabilities into the very core of decision-making processes. This new stage, by its transformative nature, inevitably triggers a crisis of unprecedented proportions, which we call the sixth great organizational crisis: the Hybridization Crisis. This is not simply a crisis of technology; it is a fundamental crisis of coordination between human and non-human intelligence, a crisis of governance that demands new ethical frameworks, and a crisis of identity that destabilizes the roles and meanings of what it means to be a manager and an employee (Eubanks, 2022). Following Greiner's dialectical logic, wherein each evolutionary period gradually sows the seeds of its revolutionary crisis, the Algorithmization phase may be fostering a new set of systemic tensions. As an analytical construct for this research, the apex of these tensions can be understood as the sixth organizational crisis: the Hybridization Crisis.

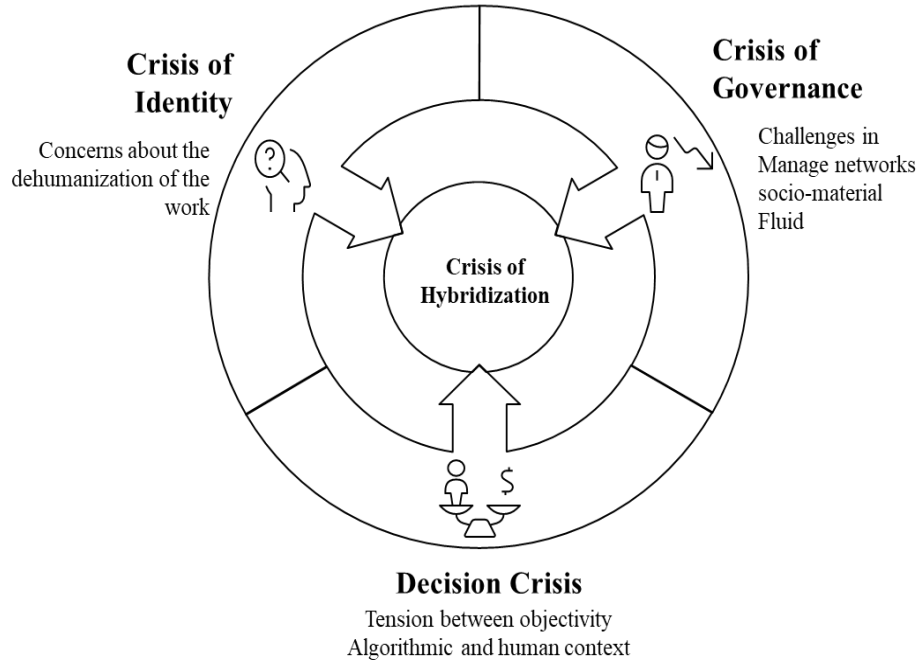
The Crisis of Hybridization, as conceived in this thesis, represents the epicenter of the tensions that unfold in multiple interdependent domains: a crisis of governance and structure, manifested in the difficulty of managing the new and fluid socio-material networks; a crisis of decision and subjectivity, evident in the tension between algorithmic objectivity and the need for human "feeling"; and, a crisis of identity and meaning, as summarized in Figure 23. In view of this new conception of hybrid management, the study suggests that an apparent state of "stupor" emerges from the interviewees' statements. In classical neurology, "the state of stupor" is a precise technical concept that describes a severe lowering of the level of consciousness. An individual in a state of stupor appears to be asleep and can only be awakened or brought to a state of alertness by means of vigorous, repeated, and sometimes painful stimuli (Plum & Posner, 1982). Even when aroused, the person in a stupor state demonstrates markedly reduced cognitive function, with slow, incoherent, or inappropriate responses, and tends to return to the previous state – lethargic or inertial – as soon as the stimulus ceases. Stupor is situated on the continuum of consciousness between lethargy – in which the patient is easily awakened, and coma – in which the patient cannot be awakened at all (Posner et al., 2007).

Here I refer to the stupor state, not as a paralysis due to incompetence, but a momentary astonishment, or a state of bewilderment, at the speed and complexity of contemporary algorithmic inevitability. This stupor often seems to translate into a certain lethargy, which implies resistance in the process of organizational change (Weick & Quinn, 1999). Many professionals report feeling overwhelmed by the day-to-day demands that consume them, and they lack the energy or strategic space to fully embrace the new analytical demands. This lethargy, in turn, is reinforced by a powerful inertial tendency, anchored both in cultural resistances – the recurrent, "but it has always been so" – and in structural barriers, such as the persistent fragmentation of systems that makes integrated action a constant challenge (Minbaeva, 2018; Shore et al., 2024). Therefore, the cycle of stupor-lethargy-inertia can therefore be understood as a central symptom of the Hybridization Crisis, reflecting the profound difficulty of actors in navigating a reality where old certainties have dissolved, but the new rules of engagement have not yet been fully institutionalized.

Transposing this concept as an analytical metaphor to the organizational context, the stupor state that we suggest can be understood as a state of responsive inertia of management. In this state, the organizational actors, although not totally paralyzed, seem asleep to the agile and necessary strategic changes, immersed in their routines. They are only awakened to action

by external and intense stimuli as an example; a market crisis, a directive from top management, the disruption of a competitor – Externalities'. However, organizational response, when it occurs, is reactive, slow, and focused on a process of "putting out the fire" rather than proactive strategic reorientation – a analogous to reduced cognitive function. And, significantly, once the pressure subsides, the stimulus ceases, the organization and its actors tend to return to their lethargy and inertial tendency, without the change being truly internalized and legitimized (Bipp & Walczok, 2024). We use this analogy, which is particularly powerful to describe passive resistance and the difficulty of strategic mobilization in the face of the Hybridization Crisis.

Figure 23: Elements of the Hybridization Crisis



Source: Research (2025).

To understand the depth of this Crisis of Hybridization, especially with regard to the "crisis of identity" and meaning that it engenders, it is necessary to go beyond the observation of stupor and inertia. It is necessary to analyze a fundamental and paradoxical pattern that emerges in the narratives about the daily interaction with AI: a persistent tendency to anthropomorphize algorithmic systems. This phenomenon, in which systems are referred to as my interns or work partners, works as a powerful mechanism for legitimizing and internalizing the new reality. However, it is precisely here that what this thesis calls the Anthropomorphic Paradox is revealed: The same humanization that facilitates the acceptance of AI is the one that masks its true nature, intensifying the risks of reification and fomenting the crisis of meaning that will be explored below.

4.3.1 The Anthropomorphic Paradox and the Crisis of Meaning in Hybrid HRM

An in-depth analysis of the interviews about everyday interaction with AI and PA reveals a subtle but fundamental pattern: a persistent tendency towards the anthropomorphization of algorithmic systems. This is not just a combination of terms or words, but a deep-seated psychological and technical phenomenon to which individuals apply human-like characteristics, intentions, and behaviors to non-human agents , where respondents refer to AI as “my interns”, “best friends”, a “guide” with “near-human language”, or a “facilitator” who can “find ways”. This thesis argues that this represents a profound process of the social construction of reality, with paradoxical and even risky consequences for the future of People Management (Nass & Moon, 2000; Tondu, 2012).

As highlighted by the distinction proposed by Tondu (2012), anthropomorphism can be defined in two ways: as a psychological predisposition of human beings to anthropomorphize and as a technical construction rooted in the physical and behavioral constitution of robots and other interfaces. In our research, we realized that the AI algorithms studied here are capable, even without a physical manifestation, of having a high power to affect human behavior. This influence seems to be linked to how directly and intensely one interacts with these tools and, in doing so, increases the acceptance of such tools as quasi-human. So we found a direct relationship: the more anthropomorphism there seems to be in the AI tool – whether it's conveyed through communication skills, simulated “personality”, or simply ease of use – the more willing users are to accept it and interact with it (Tondu, 2012).

From the perspective of SCRT, this anthropomorphization functions as a powerful mechanism of legitimation and internalization. From the viewpoint of ANT, it is an act of translation and enrollment par excellence. The impulse to treat machines as social actors is not a conscious choice but a primal, automatic response. The foundational “Media Equation” theory established that the human brain did not evolve to differentiate between real and mediated social actors, leading to what its pioneers, Clifford Nass and Youngme Moon, (2000) describe as a state of mindlessness in human-computer interaction. They argue:

> “[…]To the extent that media are social, the human brain does not differentiate between mediated social actors and non-mediated social actors. The social brain is not a modern brain. All of the cues that are relevant to social interaction in the real world are present in media; as

> a consequence, individuals apply the full range of social rules and expectations to computers. This mindlessness is a basic orientation to the world that is extremely difficult, if not impossible, to overcome (Nass & Moon, 2000, p.95).

This insight is critical: the humanization of AI is not a deliberate strategy but an unconscious default. It is this very mindlessness that opens the door to the Anthropomorphic Paradox: the same process that facilitates the acceptance of AI is what masks its true nature and enhances its risks. By treating the algorithm as a quasi-subject, we risk reifying its outputs, treating its results as objective facts while forgetting they are fallible constructions. The warning from E06 that you cannot; "[...] just delegate to the AI" and that one must "analyze on top of the AI" reveals a critical awareness of this danger. The concern of E09 about AI providing misinformation or masking "people's real knowledge" exposes the fundamental flaw in blindly trusting this "new partner".

The humanization of AI obliterates the perception of its "black box". Comfortable with the conversational interface, we tend to forget that its "agency" is algorithmic, not empathetic, purely statistical and lacks the contextual understanding that underpins true human judgment. As highlighted by (Bipp & Walczok, 2024):

> "[…] This selective blindness creates the conditions for a subtle yet corrosive dehumanization of the employees themselves. When the decision to hire or promote is increasingly based on the outputs of an algorithmic 'partner' whose logic is opaque, the subjectivity and complexity of the human individual are rendered secondary. This process is not merely theoretical; it manifests practically through what organizational psychologists term work intensification – the perceived need to work at an increasing speed and reduce idle time. […]Recent experimental research has powerfully demonstrated the paradoxical nature of modern work arrangements, showing that while high work scheduling autonomy is a desirable resource, it causally leads to higher work intensification, which in turn hinders work engagement. This provides empirical support for the notion that the very flexibility enabled by algorithmic systems can become a mechanism of

> objectification, pressuring employees into a state of relentless optimization (Bipp & Walczok, 2024, p. 8**).**

As organizational scholars have noted, when managerial systems rely exclusively on quantifiable metrics and automated processes, employees report higher levels of objectification, feeling treated as interchangeable instruments or "cogs in the machine" rather than as whole persons (Schweitzer & De Cremer, 2023). When the decision to hire or promote is increasingly based on the outputs of an algorithmic partner whose logic is opaque, the subjectivity and complexity of the human individual are rendered secondary.

Therefore, this identity crisis, one of the pillars of the Hybridization Crisis, that this thesis suggests, is intensified by this paradox. The crisis is not just about what HR professionals will do in the future, but about how they will relate to the very notion of human in their management. The greatest challenge for HRM in the age of algorithmic inevitability will not be to implement AI, but to resist the seduction of its apparent humanity. The real strategic task will be to maintain a posture of humanistic skepticism, using AI as a powerful tool but never as an oracle or a substitute for judgment, ethics, and empathy. The survival of HRM as a genuinely people-focused area will depend on its ability to be the critical and conscious guardian of this frontier, ensuring that, in the new hybrid reality, technology serves to broaden humanity at work, and not, paradoxically, to diminish it.

4.3.2 Practical Implications: Navigating the Hybridization Crisis

The findings of this thesis, by unveiling the complex emergence of a hybrid HRM and its accompanying crisis, project pragmatic implications and critical paths for professionals and organizations. The following recommendations aim to offer a compass for navigating the tensions and potentialities of this new socio-material reality. The first and most fundamental implication is the need to cultivate critical analytical literacy or data literacy (Khan, 2024). As this thesis has shown, the objectivity of PA is not raw data but a constructed and mediated reality. Therefore, the training of HR professionals must foster a deep understanding of how data is generated, what biases it may carry, and how algorithms translate human complexity into quantifiable outputs. Only with this criticality is it possible to dialogue with the AI "black box" and avoid the trap of accepting its conclusions as absolute truths.

Second, to prevent this new analytical order from unraveling into distrust, it is crucial to prioritize robust and transparent data and AI governance. The challenges with fragmented systems and the lack of a clear owner for projects are symptoms of a structural crisis. The

response lies in establishing clear policies and defining roles and responsibilities for the entire data lifecycle. This governance is the process of institutionalization that confers legitimacy and stability to the socio-technical network (Forsten-Astikainen et al., 2017; O'Neil, 2016).

The third implication lies in deliberately fostering human-machine collaboration as a path to co-intelligence. The dominant view must be one of symbiosis, not substitution. It is necessary to design processes that leverage the strengths of both: the algorithmic capacity to process data and the human capacity for contextual interpretation and ethical judgment. This means creating explicit mechanisms for human oversight and the contestation of algorithmic decisions, ensuring that machine efficiency is always guided by human wisdom(Mollick, 2024).

Consequently, HR leadership must take the lead in managing hybrid change. HR leaders are uniquely positioned to act as the primary translators of this new reality, promoting open dialogue, managing resistance, and investing in the necessary reskilling. It is their responsibility to ensure that technology serves the strategic and, fundamentally, human purposes of the organization (Ferrar & Green, 2021; Rasmussen et al., 2023). Lastly, the most pressing recommendation is not to deify technology, but to exercise and maintain a healthy and humanistic skepticism. Experience, empathy, and ethical judgment – intrinsically human attributes – remain irreplaceable. The Hybridization Crisis can only be successfully navigated if organizations resist the seduction of a purely technological solution and reaffirm, in their practices and culture, the non-negotiable value of the human dimension.

4.4 Limits and Limitations of Research

All rigorous scientific research is defined as much by the territory it explores as the boundaries it consciously establishes. In this section, as an exercise in transparency and academic rigor, we outline the limits and limitations of this thesis. We understand by limits the deliberate methodological and theoretical choices that circumscribed the scope of the research to allow a specific deepening. The limitations, in turn, are the consequences inherent to these choices, which define the horizons of validity and the applicability of the findings presented here.

A central limitation of this investigation was its empirical anchoring in the unique context of the BPI and from the perspective of a particular group of actors: the Human Resources Directors and Business Partners. This choice was not fortuitous but strategic. With the most possible density, we sought to capture the Translation of Reality, as PA demands

Objectivity and the emergence of the Hybridization Crisis from the perspective of those at the forefront of strategic decision-making on human capital. We explored the tensions, meanings, and power dynamics in a highly complex sector by limiting the focus to this universe. However, this intentional choice implies that the voice of the collaborators – directly affected and sometimes datafied by the PA systems – was tangential, investigated by the constituted research design. This limit is not a failure but a deliberate frontier of study, which opens a critical and indispensable avenue for future research, without which the understanding of the phenomenon will remain incomplete.

Derived from these limits, the first and most evident limitation emerges: the nature of our findings is peculiarly and contextually anchored. The present thesis does not aim at the statistical generalization of its results, which would be inconsistent with its qualitative and constructionist epistemological matrix. The value of this research lies in the analytical and theoretical generalization: in the ability of the "Hybridization Crisis" model and the "Reality Translation" analysis to offer a rich conceptual framework for the interpretation of analogous phenomena in other organizational contexts, which should, however, be investigated in their particularities.

A second limitation of a temporal nature comes from the object of study itself. The speed of evolution of Artificial Intelligence and People Analytics is exponential. This study, therefore, should be understood as a temporal portrait, a snapshot of a target in vertiginous movement. The specific tools, platforms, and algorithms mentioned here may become obsolete in a short space of time. However, as researchers, we believe that the human and organizational dynamics that this thesis has uncovered – the processes of institutionalization and legitimation, the crises of meaning and agency, the ethical and governance challenges – have a more perennial relevance. They constitute the social grammar underlying the Hybridization Crisis, which will continue to manifest even as specific technologies change. Acknowledging these frontiers and limitations does not diminish the robustness of the findings but, on the contrary, strengthens their validity by honestly situating them in their context of production, inviting the academic community to expand and deepen the paths that this thesis sought to explore.

4.5 Suggestions for Future Research

Robust research is defined by the answers it offers and the quality of the new questions it can raise. This thesis's frontiers and findings, therefore, do not represent an endpoint but a

fertile field from which diverse and promising paths for future investigations sprout. The paths suggested here represent the logical and necessary developments to deepen and expand the understanding of the hybridization phenomenon in contemporary management.

The nature of this study as a temporal portrait of the Hybridization Crisis and Algorithmic Inevitability invites, almost as a logical consequence, the realization of longitudinal investigations. Following, over time, how this sixth organizational crisis evolves within companies, what coordination and governance strategies are developed to navigate or absorb it, and how it eventually resolves or transforms itself would offer a dynamic validation and refinement indispensable to the model proposed here. Contrastingly, contextual expansion with intersectoral and intercultural comparative studies is necessary. Investigating how the adoption and causalities of PA vary in industries with different logics, from pharmaceuticals to cultural contexts with different conceptions of privacy and authority, would allow us to test the robustness and adaptability of our theoretical framework, revealing the universal and particular nuances of organizational hybridization.

Having deliberately focused on the strategic perspective of managers, it is now imperative that future research is dedicated to exploring the experience of the other pole of this relationship: employees. Giving a voice to those directly affected by algorithmic management is crucial for a complete and humanized understanding of the phenomenon. Qualitative research that explores their perceptions, feelings of autonomy or control, experiences with datafication, and their forms of resistance or appropriation of new technologies is fundamental. This understanding of lived experience is the necessary condition for a deeper dive into the specific ethical dilemmas that emerge from the application of PA. Analyses focused on critical sub-processes of HRM – such as algorithmic bias in recruitment, perceived fairness in automated performance appraisal, or the implications of career development personalization, can, under the employee's perceptual lens, unveil the complex moral, functional, and mental consequences of work hybridization.

Finally, in addition to diagnostic and critical analysis, academia contributes to the construction of solutions to the challenges it identifies. In this sense, a more prescriptive and interdisciplinary research strand can be dedicated to developing and testing AI and PA governance models. This endeavor requires collaboration between researchers in management, law, ethics, and computer science to create frameworks that not only exist on paper but effectively balance the search for effectiveness and competitiveness with safeguarding ethics and protecting employees' rights. Investigating how to build mechanisms

of algorithmic transparency, explainability, and contestability that are technically feasible, organizationally practical, and legally sound is, perhaps, one of the most urgent and relevant research challenges for building a future of work that is truly hybrid, fair, and sustainable, and co-created.

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APPENDIX A – INTERVIEW GUIDE

Initial presentation.

The Problematic

The challenge of transforming HRM into an era based on big data.

The Problem

The exponential growth of data, as well as the ability to analyze them through the application of algorithms, has generated important changes in the daily life of HRM. The current algorithmic potential tends to produce significant – even disruptive – changes in the form and content of HRM's work. Such transformations are portrayed by the ability to automate, or create process automation, generate greater precision and personalization of decision-making processes, and be predictive in some organizational scenarios, as well as in HR practices and policies. Such a context is reflected in the following research problem: How has the current algorithmic inevitability changed the everyday reality of HRM?

The Research Question

This research project focuses on the subfield of Strategic Human Resource Management, with an emphasis on People Analytics, classifying it as a potential organizational actor. This actor (PA) has an innovative potential, based on algorithmic technology, and can promote greater centrality to HR professionals, as well as enable a strategic role of HRM, and transform the organizational social reality. Embedded technologies in PA act as active actors, contributing to organizational transformation, integration and innovation. In view of this, the following research question is presented:

How People Analytics, a new "organizational actor", and an opportunity for competitive evolution based on data, has influenced and resignified the social reality of HRM in the Brazilian pharmaceutical industry?

General Objective

To evaluate the potential of People Analytics as an "organizational actor" that transforms and resignifies the daily reality of HRM in the Brazilian pharmaceutical industry

Specific Objectives:

1. To understand how PA and AI are experienced and signified in this context of BPI HRM.

2. To analyze the daily changes in and in HRM processes, resulting from the application of AI and/or PA tools.

3. Explore (im)possibilities of adopting PA/AI in HRM, considering the protagonism of human and non-human actors.

Participant Identification:

Name:

Function:

Time in Role:

My Professional Story:

Team size:

I answer to:

Organization Identification:

Type of Capital:

Company Size:

Industry Type:

Type of Management:

HR Size:

Market of Operation:

No. of Employees:

Introduction Questions:

1. Could you introduce yourself and talk a little about your experiences in the HR area?

2. How do you analyze the current context of exponential evolution of analytical tools for HRM?

3. How do you analyze the adoption of People Analytics in your organization, in relation to the current level of development of applying this method (PA)?

4. Does People Analytics have the potential to transform HRM when implemented? What characteristics would be preponderant?

5. How has PA fostered innovative processes in your organization, promoting the materialization of abstractions through data?

6. How has PA influenced organizational stakeholders, based on the proposed results and insights?

7. Can people analytics be considered an agent of organizational change transforming intuition through rationalization? Why? Can you give an example?

8. In your analysis, has the "power to discover patterns in the data" of People Analytics contributed to reshaping or resignifying the organizational social reality through new HRM practices and/or policies?

9. Do you consider that decision-making, assisted by People Analytics, has been more agile and assertive? Example.

10. In your opinion, can PA alter the potential of subjectivity or human intuitive actions by algorithmic rationality in HRM?

11. In your analysis People Analytics, as a vector of vertical and horizontal integration for HRM, does it tend to generate greater organizational centrality evidencing the role of the CHRO?

12. Could you highlight the main challenges you face when integrating People Analytics into organizational strategies?

13. In your assessment, does People Analytics tend to generate objectification, rationalization, dehumanization, or ethical and legal challenges in its use? How to overcome this challenge?

14. In your analysis can PA be transformed into a method of surveillance and control, exacerbating dysfunctions in management?

15. What are the skills and competencies you consider essential for HR professionals in the context of implementing People Analytics projects, improving organizational competitiveness?

16. How have you used the potential of People Analytics for analysis of the organizational network (ONA).

17. Algorithmic agency tends to be increasingly present in this transformation, how do you assess that the process of co-creation and constitution of collaborative networks with the implementation of AP is being developed?

18. In your opinion, does PA as PA influence organizational diversity (under the broad aspect of the concept)? Could you give an example?

19. In your organization, is the implementation (was) being carried out with internal or external staff? What factors were fundamental for this decision?

20. In your opinion, does PA have the "power" to strengthen collaborative networks instead of organizational silos? How have you dealt with this challenge?

Closing

21. Did I leave any point without the necessary approach that we should resume?

22. Is there a question that I didn't ask, that any of you would like to ask, that could be relevant to the discussion?

APPENDIX B – CODEBOOK NVIVO 15® -

Codes\\Step 5_ Refining, Defining and Naming Themes

NAME	DESCRIPTION
OG_ To analyse the potential of People Analytics as an organizational actor that transforms and resignifies the reality of HRM at BPI	
OE1_ To understand how PA and AI are experienced and signified in the context of BPI's HRM	
1 Reality Translation	Aspects of how everyday reality influenced by algorithmization and AI is being perceived and conceived.
1.1_ Uncertainties	How BPI HR professionals perceive the challenges and risks in adopting and implementing PA in their organizational context
1.1.1_ Groups Formation	What are the processes that enable and disable the congruence of organizational actors?
Sponsor	Is there an actor who advocates in favour of this process?
Conflict in the Roles	How roles are being defined
Leader for PA	There is a person responsible for innovation processes in the organization
Technical Support for Analysis	Is there advanced technical support – experts?
A Single Platform	How are HR technologies established?
1.1.2_Atitude and Action	How are actions and experiences perceived in relation to technologies
Acceptance of Technology	Portrait of the organization in the face of new technologies
Technology Adoption	Is there acceptance for PA?
Hands-on Analytical Experience	Is there experienced perceived or reported?
My Professional Career	Where did I come from?
Value of Technology Implementation	An advantage is perceived in the implementation of new technologies?
1.1.3_Making Credibility	How are the values of the technologies adopted and/or implemented constituted?
Challenge of Legitimation	Are there legitimatory processes in the changes that are taking place?
Effectiveness	Some real effectiveness is shown
Functionality	Arguments about how systems tend to be more functional
Perceived ROI in AI	Feedback analysis of some project has already been carried out
Validation	There is validation of innovations regardless of their size?

Value of Technology Implementation	What is the perceived value of technology in this implementation
1.1.4_Analytical Cognition	Analysing the intention of adoption, as well as maturity and analytical mindset of the interviewees, power and attitude
Adoption Intention	Is there an intention to expand?
Analytical Maturity	What is HR analytical maturity
HR Commercial Mindset	There is a quantitative proposition
Power of Attitude	There is perceived action
Power of Subjectivity	AI Insights vs Human Heuristics
Human Protagonism	Who is the protgonism with
Decision making	AI effectively collaborates in decisions
1.2_Problematization (protagonism)	It analyses how data is perceived, if there is an algorithmic mentality that expands the Organizational vision through more strategic HR.
1.2.1_Protagonism	a way for HR to maintain and expand its role
Data	
Data collection	It represents how the data collection process is perceived by respondents
Data access	Insight into data access
Legal Basis	The concern with the legal and juridical guidelines of the organization
Confidence	Data as a reliable source
Data as a Strategic Instrument	Insight into how analytics processes drive strategic decisions
Data as a means of integration	Data generates possibilities for interaction
Data Structuring	Challenge in the process of structuring integrated data systems
Data management	Who cares?
Data production	Who produces?
Data quality	Perception of the effectiveness of the data
Validation	Data as part of the process of validating and convincing the argumentation
Analytical Maturity	Who analyses?
Human Protagonism	As the interactions evidenced?
1.2.2_ Analytical Mentality	how to increase analytical aspects in HRM
PA with HR	Should HR take care of this type of project?
PA in HR vs Other Areas	Should HR give in this responsibility?
HR Processes	What HR processes look like
Robust	
Current HR System	How they are computerized

Holistic 1.2.3_Visão of the Organization	Enables HR to impute broad issues aligned with organizational objectives
Conflict in the Roles	What are the conflicts
Leader for PA	
Localized Solutions	
Sponsor	
Technical Support for Analysis	
A Single Platform	
1.2.4_ A Strategic HR	provides vision, propositions and more strategic actions
Demand Direction for Analytics	How HR receives or requests AI-managed processes
Strategy as Imposition	
Me and the others	View of how I (HR) is doing in relation to other organizations
Evolution of Technology in HR	Perception of the evolution of current HRM
Development factor	There is a central factor of development
HR Commercial Mindset	Language alignment between areas
Strategic HR	Is HR strategic?
Globalized HR	How it is globalized
Innovative HR	It is perceived as innovative
More Integrated HR	How are the HR integration processes in everyday life
Transactional HR	Perceives itself as transactional
1.3_Criating Synergies (interests)	Coordinating HR and Organizational Interests
Communication	How is HR communication
Co-responsibility	There is co-responsibility
Data as a Strategic Instrument	Insight into how analytics processes drive strategic decisions
More Integrated HR	
1.4_Enrollment (recruiting humans and non-humans)	How AI empowers HR
Individualized Action	Individual attitude of the manager in adoption and/or implementation
Streamline Processes	Make processes more agile
AI Effectiveness	Reports on how small implementations have changed everyday reality
Enabler	AI as an enabler of understanding and performing tasks
Tool	A tool for everyday support
Influencer	Ability to change human conceptions
Insights Production	Breadth of analysis and insight generation
Human Protagonism	Conflict between rationality and subjectivity

Decision making	Instrument for structuring decision-making processes
Value of Technology Implementation	Assuming value creation through AI
OE2_ Analyse the daily changes in and in HRM processes resulting from the application of AI and PA	
2_ An Objectivity Required	The influence of AI on the processes of construction of quantified reality
2.1_Institucionalizando	How changes become new routines
Streamline Processes	AI's ability to streamline processes (tool)
Implemented Automation	The actions that respondents consider to be automated processes
Communication	Power of AI to facilitate communication processes
Changes	Evidence on AI-driven changes
Rejection of change	Maintaining the Status Quo
HR and Change	Acceptance, engagement, and capillarization of change
Strategic HR	Link with the business and value generation
Imposition	Achievement in top down order, not out of necessity or opportunity
Innovative HR	How much HRM is engendered in this perspective
More Integrated HR	HRM's current ability to integrate
2.2_ Building Legitimacy	Creation of symbology, conceptual and practical mechanisms
AI Effectiveness	Reports on how small implementations have changed everyday reality
Functionality	Arguments about how systems tend to be more functional
Impacts of dysfunctions	Human errors influenced by poorly constituted interactions
Analytical Maturity	Search for a more quantitative HR
Perceived ROI in AI	Quantifiable results reported
Insights Production	(tool) that produces insights in an agile and strict way
Decision making	Facilitator in the decision-making process
Validation	The Sponsor's work
OE3_ Explore (im)possibilities of AI adoption in HRM, considering human and non-human protagonism.	
3_ EVERYDAY RELATIONSHIPS IN A HYBRID ENVIRONMENT	The performance of human and non-human actors constituting new social relations
3.1_Interações of the Actors	Interactions between organizational actors
Silo Knowledge	Difficulty in sharing knowledge
Consequences of Silos	Restrictions imposed by an individual actor
Co-responsibility	Co-creation attitude

Daily life as a challenge	The shackles of rigid organizational processes
Management of daily demands	Everything that demands daily attention
Individualism	It portrays the lack of collaboration between people
3.2_ Impediments to the Status Quo	The limits to the transformation of the organizational reality
Daily life as a challenge	The shackles of organizational processes, rework
Functionalist Focus	Reports on HR's functionalist DNA
Management of daily demands	Everything that demands daily attention
Power of Subjectivity	AI Insights vs Human Heuristics
3.3_ Multifaceted Structure	The changes that the translation process unveils about social reality
Structural Systems Dilemma	It portrays the multiplicity of control systems in HR
Multiple Information Systems	HR systems separate from organizational ERP
Senior Management Support	The fundamental conceptual and financial support of senior management
3.4_ A new identity for HRM	Propositions on the new perspectives for HR
Risks	
Reliance on AI	Fear of human dependence on AI
Lack of Agility	Without centralization, without protagonism, change is slow and fragmented
Innovation only in Image	Rhetoric very different from Practical Action
Legal Rigidity	HR policies and practices being limited by Brazilian law and compliance
Safety	LGPD issues
Technological Volatility	How up to date can I be?